Topics in String Theory

Supplementary course, 2011 Winter

Lectures by Leonard Susskind

Companion edition by LazyingArt LLC with Video2Book

About These Notes

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Contents

Chapter 1

Lecture 1: Topics in String Theory

Overview. A challenge to reductionism is sharpened through bosonization in $1+1$ dimensions, electric–magnetic duality, D-branes, strong-coupling transitions, emergent geometry, and the combinatorial landscape of string vacua. The same physical system may admit radically different elementary descriptions; which description looks simple depends on the coupling and the geometry.

1.1 Reductionism, Complexity, and the Question of Simplicity

Susskind opens with a deliberately familiar philosophical slogan: big things are made of little things, and little things are made of littler things. He then adds the more ambitious version of the same idea, the one that really matters for fundamental physics: as one descends through deeper layers, the objects should become not only smaller but also simpler. His metaphor is that houses are made of bricks, and nobody mistakes a house for a brick. The brick is smaller, but it is also simpler.

The first move of the lecture is to confront that expectation with the actual state of particle physics. If reductionism were delivering increasing simplicity in that strong sense, one might expect present-day elementary particle physics to be close to economical. Instead, Susskind emphasizes that even before quantum gravity enters, the bookkeeping already looks messy. In the lecture these are rough counts, but the roughness is part of the point:

$$\begin{aligned} N_{\text{species}} &\sim 75, \\ N_{\text{independent}} &\sim 10\text{–}20, \\ N_{\text{parameters}} &\sim 20. \end{aligned} \tag{1.1}$$

Even this is not the end of the story. The Standard Model does not contain gravity, does not explain dark matter, does not supply the extra structure needed for inflation, and carries a severe fine-tuning problem. Adding the usual repair mechanisms only increases the count. Supersymmetry is the lecture's example. At the time of the lecture, many physicists hoped that the Large Hadron Collider would expose superpartners; Susskind stresses that the corresponding extension would introduce more than a hundred additional parameters rather than immediately simplifying the bookkeeping. His schematic message is

$$N^{\text{extended}}_{\text{parameters}} \sim 200. \tag{1.2}$$

These numbers are deliberately rough and historical. They serve the argument that a deeper description need not look simpler at the level of its particle inventory and measured constants.

This quantitative complaint is not yet Susskind's main thesis. It is the prelude. The deeper claim, and the one that drives the rest of the lecture, is that modern quantum field theory and string theory do not obviously support a unique, coupling-independent distinction between the bricks and the houses. The next examples are chosen to show that the very meaning of "elementary" may depend on which variables are efficient.

1.2 Bosonization: A First Breakdown of Fixed Fundamentality

Susskind's first theoretical example comes from 1 + 1-dimensional quantum field theory: one dimension of space, one of time. He announces at once that he is not going to derive the equivalence in detail; the point is not the formal proof but the conceptual lesson.

One starts with a field theory of fermions. The fermion field is denoted by ψ. Two nearby fermions can form a bosonic composite, and the corresponding bosonic field is denoted by ϕ. At first sight the hierarchy seems straightforward: ψ is basic, while ϕ is a composite or effective variable. In the lecture the boson is introduced exactly this way, as a field that creates bound pairs of fermions.

What was discovered, however, is that the same theory can be rewritten from the opposite starting point. One may begin with the bosonic field ϕ, and then the fermion no longer appears as a point-like constituent. Instead it appears as a kink of the bosonic field, an extended configuration in which ϕ interpolates smoothly between two asymptotic values. The minimum mathematical content of the lecture is this asymptotic structure:

$$\phi(x \to -\infty) = \phi_-, \qquad \phi(x \to +\infty) = \phi_+. \tag{1.3}$$

A convenient schematic representative is

$$\phi(x) \approx \frac{\phi_+ + \phi_-}{2} + \frac{\phi_+ - \phi_-}{2} \tanh\left(\frac{x}{\ell}\right), \tag{1.4}$$

where ℓ is the width of the transition region.
[1]

The belt analogy in the lecture is precise enough to be worth keeping. A flat belt with no twist is a configuration with no kink. If one rotates one end by 2π, the belt returns to the same local orientation far away, but in between it must pass through an extended twisted region. That twisted region is the kink. In the bosonic description, the fermion is represented by precisely such an extended object.

The point becomes sharper once a coupling is introduced. Susskind calls it G in this example. The statement is not that the two descriptions are vaguely related, but that the efficiency of the two descriptions depends on the parameter regime:

$$G \ll 1 \Rightarrow \ell \text{ small, tight kink; } \psi \text{ efficient,} \tag{1.5}$$
$$G \gg 1 \Rightarrow \ell \text{ large, broad kink; } \phi \text{ efficient.} \tag{1.6}$$

For small G, the kink is tight and the fermion looks like the simple object. As G increases, the kink broadens and the fermion becomes less point-like. The bosonic variables then become the more economical ones.

[1] Editorial clarification: The hyperbolic-tangent profile is a standard smooth representative of the kink drawn in the lecture, not a formula derived there. The precise bosonization dictionary depends on the model; the conceptual point is the exchange between a fermionic excitation and a topological soliton of a bosonic field.

Schematic update rule.

1. Begin with a fermionic theory described by ψ.
2. Identify bosonic composites described by ϕ.
3. Rewrite the same theory entirely in terms of ϕ.
4. In the bosonic description, the fermion reappears as a kink of width ℓ.
5. For $G \ll 1$, ℓ is small and the fermionic description is simplest.
6. For larger G, ℓ grows and the bosonic description becomes simplest.

This is the first explicit counterexample to the fixed "houses and bricks" picture. The lecture's punchline is not merely that composite objects exist, but that what counts as the elementary variable can change with the coupling.

1.3 Electric Charge, Magnetic Monopoles, and Weak versus Strong Coupling

Susskind next asks whether an analogous story appears in more familiar particle physics. The example is electric charge versus magnetic monopole. An ordinary charged particle, such as the electron, is an electric monopole in this language: a localized source of electric field. The magnetic monopole is introduced by the bar-magnet analogy. If one could somehow isolate the end of a bar magnet without the invisible connecting bar, the field lines would look exactly like those of an electric charge, but magnetic rather than electric.

The relevant control parameter is the fine-structure constant α. In the lecture Susskind treats it heuristically as the square of the electric charge:

$$\alpha \sim e^2. \tag{1.7}$$

He also gives a rough operational interpretation: it measures how readily an electron radiates, and he summarizes that strength by saying that an accelerated electron radiates on the order of "1/100" photons in the relevant sense. In the lecture's own rough numerical language,

$$\alpha \approx 10^{-2}. \tag{1.8}$$

This is not intended as a precision convention; it is a pedagogical way of encoding that electric charge is weak.

[2]

When $\alpha \ll 1$, the electron is the simple object. Its electric field is weak, its vacuum polarization cloud is dilute, and it looks almost point-like. The magnetic monopole is then the opposite: its field is enormous, its surroundings are violently distorted, and it is broad, complicated, and heavy.

[2] Editorial clarification: The weak–strong relation has a precise ancestor in Dirac quantization. In one common convention, electric and magnetic charges obey $eg_m = 2\pi n$. Thus a weak electric coupling corresponds to a strong magnetic one, although exact numerical factors and protected mass relations depend on the theory.

Susskind summarizes the situation in precisely this weak–strong style:

$$\begin{aligned} \alpha \ll 1 \Rightarrow &\text{ simple electron;} \\ &\text{ heavy, extended monopole,} \\ \alpha \gg 1 \Rightarrow &\text{ simple monopole;} \\ &\text{ heavy, structured electron.} \end{aligned} \tag{1.9}$$

The lecture does not derive a monopole mass formula. It does not need one. The point is qualitative. As α increases, the electron's field strengthens and its structure grows, while the monopole shrinks and becomes lighter. Far beyond unity, the useful description is exchanged: the monopole is simple and the electron complicated.

That is the field-theory version of the earlier lesson from bosonization. The distinction between basic and composite is not absolute. It depends on the coupling.

Question from the audience. If reductionism is not a reliable guide, why do physicists rely on calculus, which seems to divide a system into arbitrarily small pieces? [a]

Response. Calculus asserts smooth variation; it does not identify the fundamental objects of a theory. Quantum fields fluctuate on every distance scale, producing an endless hierarchy rather than a final smooth layer at which one can decide once and for all what is elementary.

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[a] Lecture timestamp: 00:21:21.

The exchange continues with the suggestion that the hierarchy resembles a fractal. Susskind accepts the analogy, then sharpens it: quantum fields fluctuate on every scale and are therefore never smooth in the classical sense. This excessive variation is part of what makes quantum field theory mathematically difficult; ordinary calculus must be used with renormalized quantities rather than with a naively smooth microscopic field.

1.4 Strings, D-Branes, and the Return of Weak–Strong Duality

The next transition in the lecture is deliberately naive. If string theory is supposed to be the theory of the fundamental building blocks, then surely the answer must be simple: the building blocks are strings. Susskind states that expectation openly, and then spends the next part of the lecture dismantling it.

1.4.1 Why strings are not the whole story

The first complication is the unavoidable presence of D-branes. A Dp-brane is a p-dimensional object on which strings can end. The letter D comes from Dirichlet boundary conditions, but the name is far less important than the role: these objects are part of the theory and cannot be discarded. They are not optional ornaments.

The simplest examples are:

- a D0-brane, which is point-like,
- a D1-brane, or D-string, which is string-like,
- an F-string, the ordinary fundamental string.

A D0-brane is a heavy point-like site on which any number of F-strings can end. A D1-brane is a heavy cable-like object, much thicker and more massive per unit length than an F-string, but likewise a place where F-strings can end. At weak coupling the lecture does not present these objects as equally elementary alternatives. The D-branes are introduced as heavy, structured, composite-looking objects that the theory nevertheless forces upon us.

[3]

Notation. Susskind uses G and g somewhat loosely in different parts of the lecture. Here G is reserved for the bosonization example, while g denotes the string coupling.

The string coupling g is defined operationally, not axiomatically. It is the parameter that measures the probability that a fluctuating string splits and forms daughter strings. This is why Susskind compares it directly to the role of α in QED. The lecture also pauses to identify tension with energy per unit length:

$$T = \frac{E}{L} = \frac{M}{L}. \tag{1.10}$$

For an F-string and a D-string, the standard tension formulas make the weak–strong comparison explicit:

$$T_{\mathrm{F1}} = \frac{1}{2\pi\alpha'}, \qquad T_{\mathrm{D1}} = \frac{1}{2\pi\alpha' g_s}, \qquad \frac{T_{\mathrm{D1}}}{T_{\mathrm{F1}}} = \frac{1}{g_s}. \tag{1.11}$$

[4]

1.4.2 F-strings and D-strings

The weak–strong story then repeats itself in string language. At small coupling the F-string is light, thin, and close to the original intuitive image of the fundamental object. The D-string is thick, heavy, and full of F-string structure. As the coupling grows, the F-string splits more readily; the cloud of daughter strings proliferates around it, and it becomes heavier and more complicated. At the same time the D-string has increasing difficulty emitting the now-heavy F-strings, so its surrounding structure thins out and it becomes lighter and more line-like. The lecture's weak–strong map is therefore

$$g \ll 1 \Rightarrow \begin{array}{c}\text{F-string light and thin;}\\ \text{D-string heavy and structured,}\end{array} \tag{1.12}$$

$$g \sim 1 \Rightarrow \text{F-string and D-string comparable,} \tag{1.13}$$

$$g \gg 1 \Rightarrow \begin{array}{c}\text{D-string light and thin;}\\ \text{F-string heavy and structured.}\end{array} \tag{1.14}$$

Susskind remarks that around $g \sim 1$ the two descriptions cease to be sharply distinct. Beyond that point they effectively exchange roles. This is the string-theoretic version of the electron–monopole interchange, and its accepted name is S-duality.

[3]Editorial clarification: The odd-brane theory used in this part of the lecture is Type IIB string theory. Its stable D-branes have odd spatial dimension, including D1- and D3-branes. Susskind postpones these names because the duality pattern, not the taxonomy, is the immediate subject.

[4]Editorial clarification: These tension formulas supply the conventional normalization behind the lecture's qualitative cable-and-thread comparison. They are an editorial addition; the lecture uses only the scaling with coupling.

The same exchange can be followed after a string closes into a loop. At weak coupling, a small F-string loop is a light particle-like excitation, while the corresponding D-string loop is heavy and complicated. At strong coupling their roles reverse: the D-string loop becomes the light description, and among the resulting light closed-string excitations is the graviton. The lecture uses this example to insist that even the identity of a familiar particle may depend on which side of the duality is used.

The lecture pushes the point one step further. In the idealized theory the coupling is not merely a number to be dialed by hand; it behaves like a field. Schematically,

$$g = g(x). \tag{1.15}$$

Different regions of space can therefore favor different dual descriptions. In one region F-strings are the light variables; in another region D-strings are. In between, nothing is especially simple. That is why, for Susskind, string theory does not restore the simple reductionist picture. It radicalizes the ambiguity.

[5]

Question from the audience. If the coupling varied enough that D-strings became lighter than F-strings in another region, would the change affect observable spectra? [a]

Response. Yes. That is one reason the exactly symmetric, mathematically controlled model being discussed cannot be identified directly with our world. Its lessons concern what consistent quantum theories can do, not a literal spectrum for nature.

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[a]Lecture timestamp: 00:34:15.

Question from the audience. Is there an analogous dual description for a D2-brane? [a]

Response. Yes, but it is not another D-string-like exchange. The even-brane case will lead to a membrane interpretation and an additional spatial dimension, which is the next construction.

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[a]Lecture timestamp: 00:34:39.

Question from the audience. Does a larger probability for a string to split account for the increased internal complexity? [a]

Response. Yes. To make the mechanism concrete, return to QED: an electron emits photons, photons fluctuate into charged pairs, and those charged lines radiate again. Increasing the coupling makes that hierarchy less dilute.

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[a]Lecture timestamp: 00:34:53.

1.4.3 A question-driven return to quantum electrodynamics

A student then asks whether the larger splitting probability simply means a more complicated structure “upstream.” Susskind answers by returning to quantum electrodynamics and drawing the electron’s worldline again. This return matters: it shows that the string story is not an isolated curiosity but another instance of the same weak–strong pattern.

[5]Editorial clarification: In perturbative string theory the local coupling is set by the dilaton, $g_s(x) = e^{\Phi(x)}$. Writing $g = g(x)$ is therefore not merely an analogy, although realistic compactifications generally generate a potential that constrains such variations.

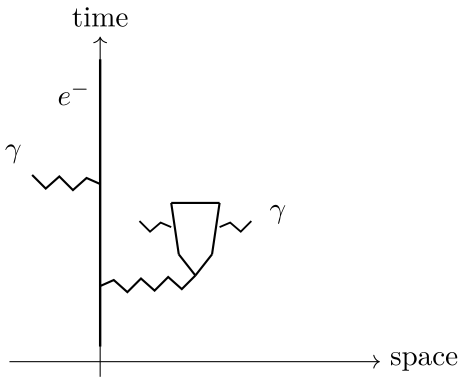

Figure 1.1: Schematic QED hierarchy used in the lecture: a point electron emits photons, photons create pairs, and the daughter charged lines radiate again, producing structure at successively shorter scales.

The lecture's point is that even a point electron is accompanied by a hierarchy of processes:

1. an electron emits and absorbs photons,
2. a photon can create an electron–positron pair,
3. the daughter charged lines can themselves emit photons,
4. the hierarchy continues to smaller scales.

For weak coupling this cloud is dilute, so the electron looks almost point-like in the laboratory. For stronger coupling it proliferates, and the electron becomes a structured object. The string story is then presented as a direct analogue of this.

The same later discussion also motivates Susskind's heuristic energy bookkeeping. He insists that the relevant energy is not merely the sum of free-particle contributions. One must also include interaction energy:

$$E_{\text{tot}} = E_e + E_\gamma + E_{\text{int}}. \tag{1.16}$$

Here E_{int} is the crucial term. It is what lets the mass of the dressed electron include the energy associated with its surrounding field and virtual structure. The lecture does not turn this into a formal renormalized self-energy derivation. The point is heuristic but essential: the object's mass already includes the interaction energy required to support the complicated cloud around it.

Question from the audience. Where does the energy for the virtual photons and electron–positron pairs come from if an electron does not contain enough free energy to create a real pair? [a]

Response. One never starts with a bare electron and later attaches its Coulomb field. The observed electron is a dressed state, and its mass already includes field and interaction energy. In QED the total energy is not the free electron energy plus the free photon energy alone; the interaction term is indispensable. Calling the intermediate pairs virtual is useful, but the energy bookkeeping belongs to the complete interacting state.

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[a] Lecture timestamp: 00:37:01.

Question from the audience. When a high-energy probe reveals this structure, has the probe created it rather than discovered something already present? [a]

Response. That interpretive distinction can be debated. Operationally, sufficiently energetic scattering has a calculable, nontrivial final state. At weak coupling most events still look simple, while roughly an α-sized fraction reveal additional radiation or pairs.

[a] Lecture timestamp: 00:38:59.

Question from the audience. When the D1-brane is said to be more massive than the F-string, is its tension also larger? [a]

Response. Yes. For an extended string, "mass" here means energy per unit length, which is precisely the tension.

[a] Lecture timestamp: 00:39:37.

The reason this figure belongs here, even though the weak–strong comparison was introduced earlier, is that in the lecture Susskind redraws the contrast at this later question-driven moment while discussing interaction energy and proliferating structure.

1.5 Even-Brane Theories and the Emergent Extra Dimension

Up to this point the lecture has been telling a strong story about exchange: fermion versus boson, electron versus monopole, F-string versus D-string. The next step is more surprising. Susskind switches to a different version of string theory, one with even-dimensional D-branes rather than odd-dimensional ones. The distinction matters because in this theory there is no D-string to exchange with the F-string. One has D0-branes, D2-branes, D4-branes, and so on.

The strong-coupling limit therefore cannot simply be a relabeling of one string as another. According to the lecture, something stranger happens: an extra spatial dimension becomes manifest. Susskind deliberately states this in the most dramatic form first — "a new dimension materializes" — and then immediately translates it into the more careful statement: a very small compact direction expands until it can no longer be ignored.

The bookkeeping is part of the point, so he pauses over it:

$$\begin{aligned} \text{10-dimensional string theory} &= 9 \text{ space} + 1 \text{ time}, \\ \text{11-dimensional theory} &= 10 \text{ space} + 1 \text{ time}. \end{aligned} \tag{1.17}$$

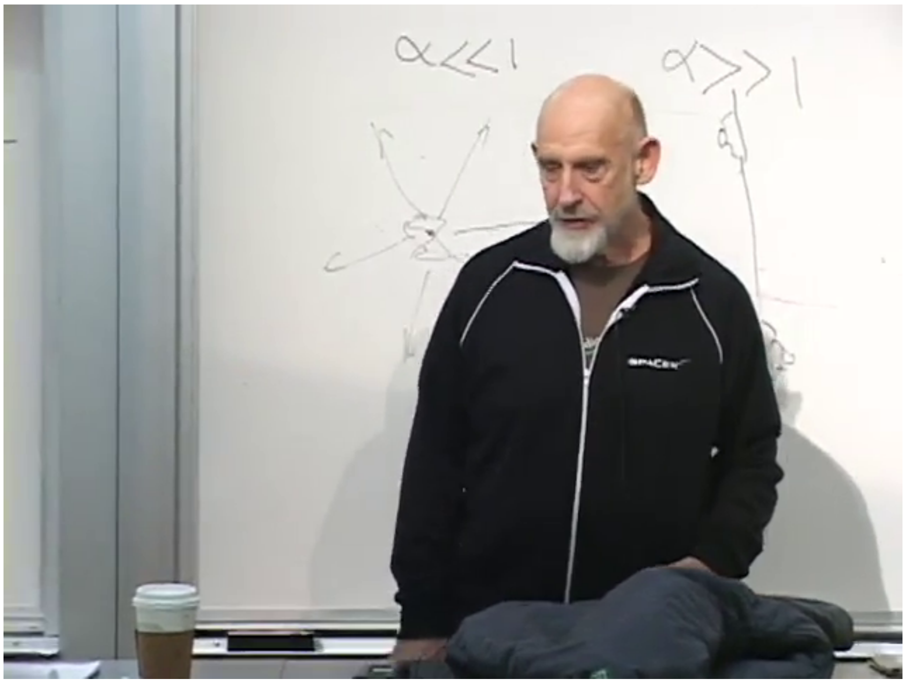
The lecture's claim is that the strong-coupling limit of the even-brane theory is effectively eleven-dimensional.

[6]

1.5.1 The peanut-butter sandwich

To picture the compact direction, Susskind uses the "peanut-butter sandwich" model: two infinitely thin pieces of bread, with real space between them. The special rule is that going out one side

[6] Editorial clarification: The even-brane theory is Type IIA string theory, and its strong-coupling completion is M-theory. In standard units the emergent-circle radius and eleven-dimensional Planck length scale as $R_{11} = g_s \ell_s$ and $\ell_{11} = g_s^{1/3} \ell_s$. These names and formulas make explicit the modern dictionary behind the lecture's geometric account.



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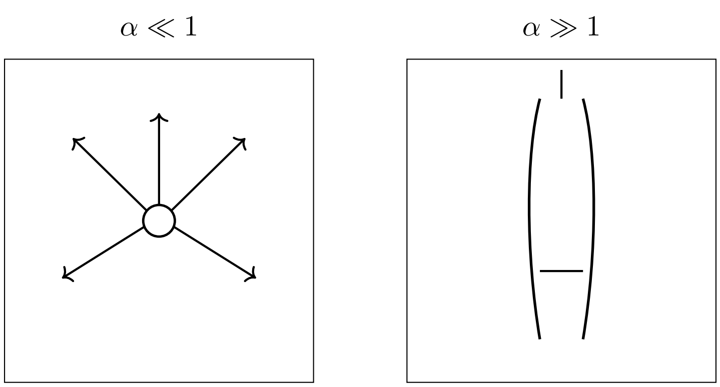

Figure 1.2: The weak–strong QED comparison during the interaction-energy discussion. The diagram isolates the two regimes visible around Susskind: a dilute weak-coupling cloud and a proliferating strong-coupling one.

returns one to the other. In a conventional notation that the lecture describes pictorially rather than formally, one may write

$$y \sim y + 2\pi R. \tag{1.18}$$

If R is very small, observers living in the sandwich effectively describe their world as lower-dimensional. The key strong-coupling update is then

$$g \uparrow \qquad \Longrightarrow \qquad R_{\text{compact}} \uparrow . \tag{1.19}$$

That is what Susskind means when he says that a new dimension of space appears: not that space is created from nothing, but that a previously tiny compact direction expands and becomes visible in the effective theory.

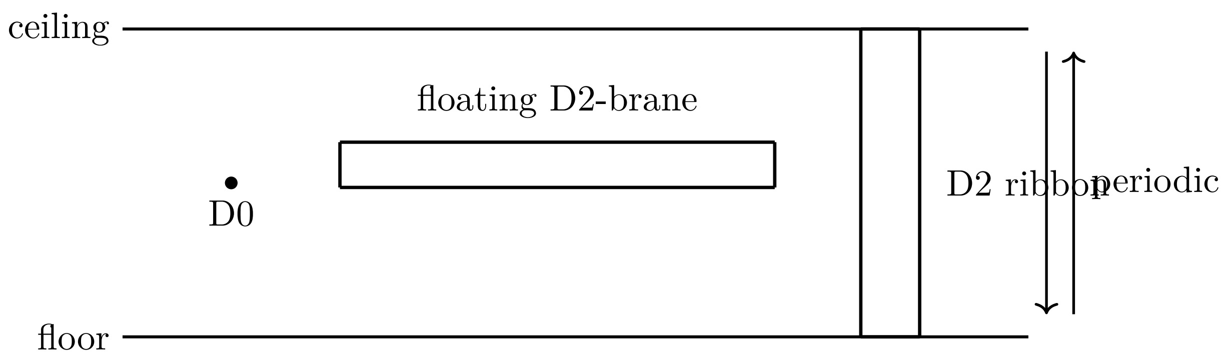

Figure 1.3: The sandwich compactification. The upper and lower boundaries are periodically identified; one D2-brane floats between them like a carpet, while another stretches from floor to ceiling and looks string-like when the compact direction is small.

1.5.2 What becomes of the D0- and D2-branes?

The lecture now asks what happens to the D0-branes and the strings. The answer is asymmetric. As g becomes large, the D0-branes get lighter and lighter; in the higher-dimensional interpretation they join the tower of particles carrying momentum in the newly manifest direction. Thus a heavy, complicated object in the weak-coupling picture becomes an ordinary light excitation in the strong-coupling picture.

The mass relation displays the identification:

$$M_{D0}(n) = \frac{n}{g_s \ell_s} = \frac{n}{R_{11}} = M_{\text{KK}}(n). \tag{1.20}$$

[7]

The D2-branes explain the fate of the strings. There are two configurations emphasized in the lecture:

1. a D2-brane floating between floor and ceiling like a carpet in the peanut butter;

[7] Editorial clarification: Susskind calls the D0-branes "the gravitons" of the eleven-dimensional theory. More precisely, their bound states are the Kaluza–Klein momentum modes of the eleven-dimensional supergravity multiplet, whose massless sector contains the graviton. The distinction preserves the lecture's physical identification while avoiding a literal one-particle overstatement.

2. a D2-brane stretched from floor to ceiling like a ribbon, extended along the visible direction.

When the sandwich is thin, the ribbon looks effectively one-dimensional. It is the object that low-energy observers call a string. As the compact direction opens up, that same object becomes unmistakably membrane-like. Because it now contains more material in the vertical direction, it becomes heavy and unstring-like.

The logic of the transition.

1. Start with a very thin compact direction, so the floor-to-ceiling ribbon behaves effectively like a string.
2. Increase g, and the compact direction expands.
3. The D0-branes become light and are reinterpreted as Kaluza–Klein momentum states in the higher-dimensional theory.
4. The string-like D2 ribbon becomes a visibly extended membrane and grows heavy.

This is the lecture's hinge from duality to emergent geometry. Strong coupling does not merely exchange one object for another; it changes which geometry is the efficient one.

Susskind pauses over how strange this sounds. The web of proposed identifications required difficult mathematical consistency relations, some of which first appeared to physicists as conjectures and were later established by mathematicians. His claim is not that every realistic string vacuum is understood, but that the duality web itself has survived unusually stringent checks.

If the coupling varies from place to place in this idealized setting, the efficient geometry can vary with it. One region may look nearly ten-dimensional, with light strings and heavy D0-branes; another may open into the eleven-dimensional description, with heavy wrapped membranes and light Kaluza–Klein modes. The local change of language is part of the same duality web.

The transcript briefly renders the name of these relations as "inequalities." In context the intended word is *dualities*: relationships between opposite ends of parameter space. The correction matters because the lecture's conclusion is precisely that no unique list of fundamental constituents is known. Strings, branes, particles, and higher-dimensional geometry can transform into one another; either no member of the web is absolutely fundamental, or some deeper organization remains unknown.

1.6 D3-Branes, Wrapped Membranes, and a Geometrical Picture of Duality

After the break Susskind explicitly describes the subject as an "enormous web" of interconnected theories and ideas. The next examples are not new isolated effects; they are bridges tying together the previous ones.

1.6.1 A D3-brane worldvolume

The first bridge uses a D3-brane. A D3-brane has three spatial dimensions, and an observer confined to it can easily mistake it for ordinary space. The brane can bend and fluctuate, while strings ending on it can join, split, and scatter. To an observer who sees only directions along the brane, each endpoint looks like a particle and the interactions among endpoints look like an ordinary quantum field theory.

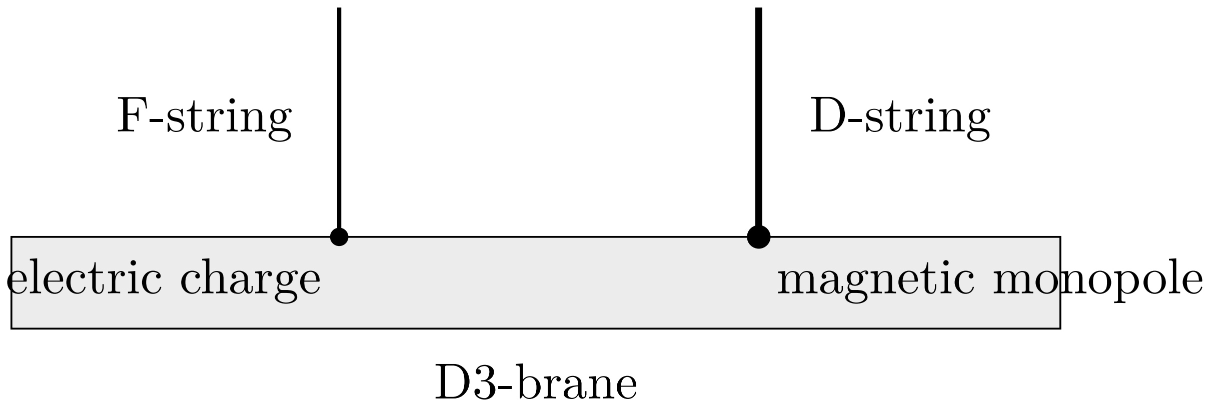

Figure 1.4: D3-brane worldvolume interpretation. An F-string endpoint appears as an electric charge; a D-string endpoint appears as a magnetic monopole.

This is one of the lecture's most satisfying identifications. An F-string ending on the D3-brane appears to the worldvolume observer as an electric charge. A D-string ending on the same brane appears as a magnetic monopole. In other words, the duality between F-strings and D-strings becomes, on the D3-brane worldvolume, the electric–magnetic duality of ordinary field theory. What was earlier an analogy is now an explicit dictionary.
[8]

1.6.2 Two compact directions and asymmetric wrapping

Susskind then returns to the higher-dimensional geometry and compactifies not one but two small directions. Since one cannot draw eleven dimensions on a blackboard, he compresses the seven large directions not used in the sketch into one visible horizontal direction, then draws the two compact ones explicitly. He insists on drawing the compactification asymmetrically, because one compact direction must be shorter than the other if the effective strings are to have different tensions.

A membrane may wrap either compact direction and look like a string from the non-compact directions. If R_1 and R_2 are the two circle radii, its effective tensions are

$$\begin{aligned} T_{(1)} &= 2\pi R_1 T_{\mathrm{M2}}, \\ T_{(2)} &= 2\pi R_2 T_{\mathrm{M2}}, \\ \frac{T_{(1)}}{T_{(2)}} &= \frac{R_1}{R_2}. \end{aligned} \tag{1.21}$$

Thus unequal circle sizes produce unequal string tensions. With the shorter wrapping identified as the light F-string and the longer one as the heavy D-string, one may write schematically

$$g_s \sim \frac{R_{\mathrm{short}}}{R_{\mathrm{long}}}, \qquad \frac{T_{\mathrm{D1}}}{T_{\mathrm{F1}}} \sim \frac{1}{g_s}. \tag{1.22}$$

[8]Editorial clarification: For a stack of D3-branes, the low-energy worldvolume description is four-dimensional $\mathcal{N} = 4$ supersymmetric Yang–Mills theory. Its electric–magnetic duality is the field-theory realization of Type IIB S-duality; a single brane gives the Abelian limit used pictorially in the lecture.

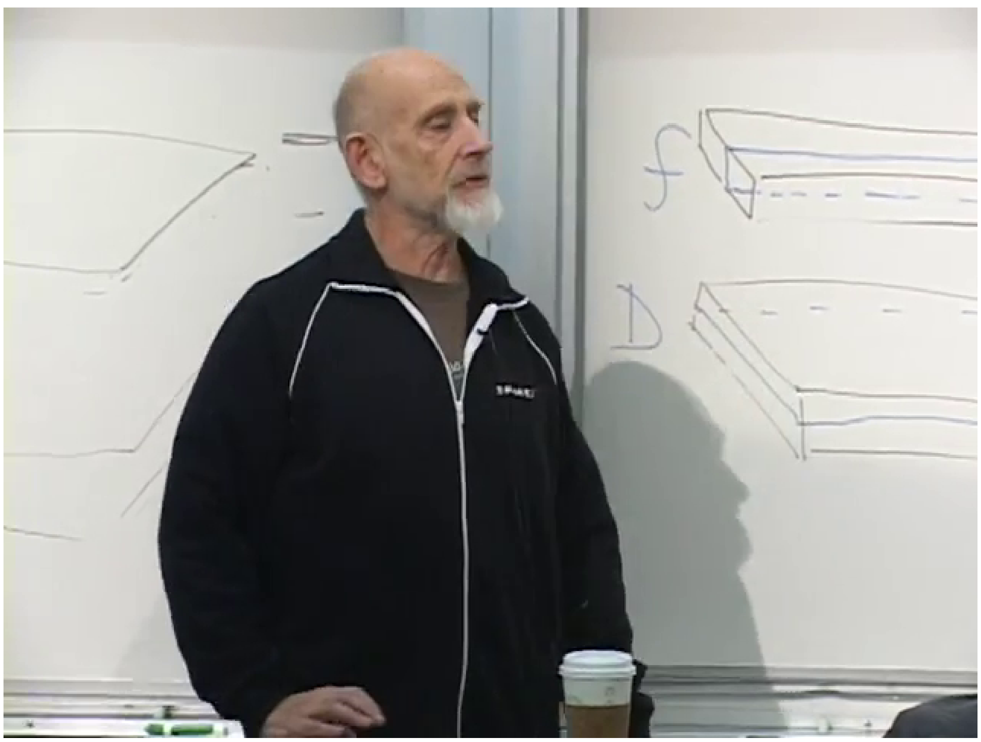

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Figure 1.5: The completed board state for the two wrapped membranes. The short and long compact cycles produce the F- and D-string-like ribbons drawn at the right.

Question from the audience. Can the coupling in this picture be understood as the ratio of the two compact dimensions? [a]

Response. Yes. The weak–strong parameter is geometrized as an aspect ratio. Making one direction short produces a small coupling; exchanging the two directions exchanges the light and heavy strings.

[a]Lecture timestamp: 01:10:13.

[9]

The wrapped-membrane picture can be summarized without the perspective distortion of the blackboard drawing:

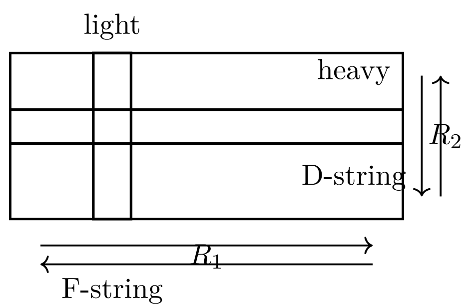


Figure 1.6: Two wrapped membrane configurations both look string-like from the non-compact directions, but unequal compact scales make one effective string lighter than the other.

This is the geometric version of the earlier duality story. Squeezing one direction while stretching the other makes one wrapped object heavier and the other lighter, so the roles of F-string and D-string can interchange.

Question from the audience. If the compact directions are cyclic, are the wrapped objects infinitely wide? [a]

Response. No. A compact circle has finite circumference even though one can traverse it repeatedly without reaching an edge. The top and bottom of the cut-open drawing represent the same point, just as the endpoints of an interval can represent one circle point.

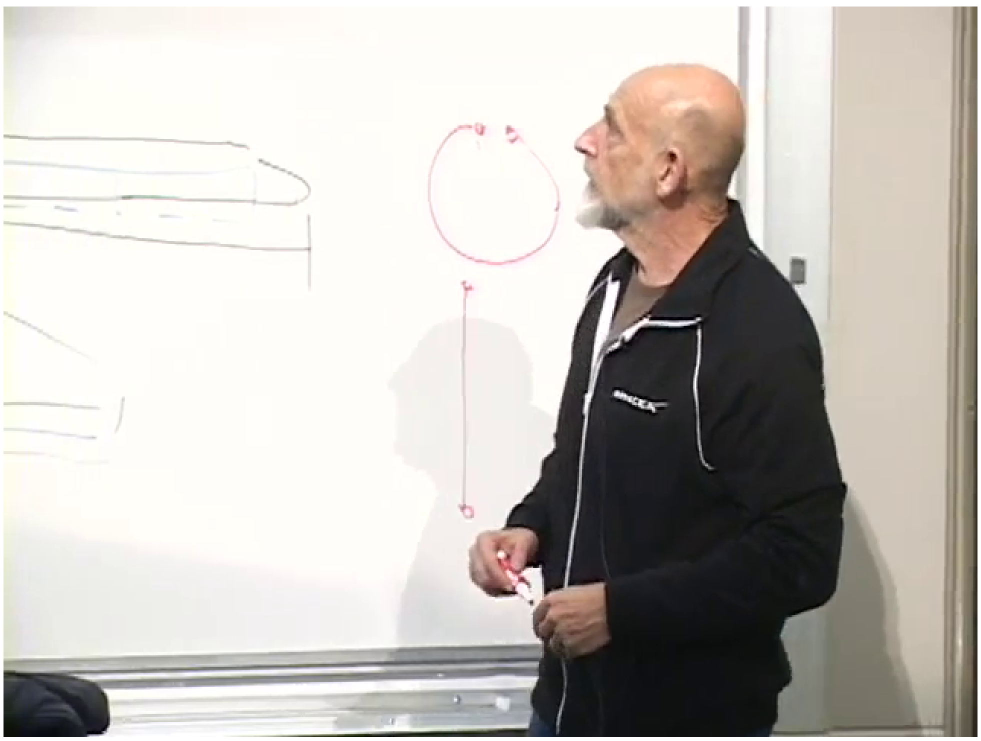
[a]Lecture timestamp: 01:10:41.

Question from the audience. Why is the compact path a circle rather than a helix? [a]

Response. At each fixed point in the non-compact direction, the compact direction is a circle. A helix appears only if one advances in the non-compact direction while simultaneously winding around the circle. The blackboard interval is simply a circle cut open, with its two endpoints identified.

[a]Lecture timestamp: 01:12:23.

[9]Editorial clarification: The lecture temporarily calls this ratio α, reusing the symbol previously used for the QED fine-structure constant. We write g_s here to keep the two meanings distinct. In the full M-theory construction the Type IIB coupling is encoded in the complex structure of the compact two-torus; the precise numerator depends on how its cycles are labelled.



Lecture frame: 01:12:25.000.

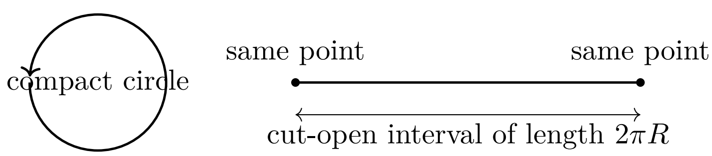


Figure 1.7: A compact circle and its cut-open blackboard representation. Crossing one endpoint returns at the other; moving around the circle while also moving in a non-compact direction would instead trace a helix.

Question from the audience. Does this geometry itself explain how a D-brane can look like a collection of F-strings when probed closely? [a]

Response. The connection is present, but it is not transparent in this elementary picture; seeing it requires more geometric machinery. The diagram is meant to organize the duality, not to derive every microscopic constituent description.

[a]Lecture timestamp: 01:12:49.

Question from the audience. Could the long and short compact directions exchange as one moves through space, or as time evolves? [a]

Response. Either is possible in the idealized model. Squashing one cycle while expanding the other interchanges which wrapped string is light. This is another reason "fundamental string" is only a name: the preferred description can change from region to region or with time.

[a]Lecture timestamp: 01:13:35.

1.7 Idealization, Specially Symmetric Points, and the Landscape

The last movement of the lecture steps back from the elegant duality examples and asks what they do, and do not, tell us about the real world.

First, Susskind warns that the precisely controlled dualities are features of highly idealized supersymmetric theories, in which bosons and fermions occur with exactly matched masses. His analogy is to circular orbits in Newtonian mechanics: one studies them first because they are symmetric and mathematically tractable, not because actual planetary motion is exhausted by perfect circles. In exactly the same way, supersymmetric string theories are easier to analyze because they are more symmetric than nature appears to be.

The idealization is nonetheless informative. A perfect circular orbit is not a realistic planetary system, but it displays features that survive in less symmetric orbits. Likewise, the supersymmetric examples show consistent ways in which quantum mechanics, gravity, and microscopic structure can fit together. Susskind's deliberately sharp formulation is that the tractable string theories are not, literally, the theory of our world; their value is that they reveal phenomena likely to reappear in less symmetric and less tractable settings.

1.7.1 A construction kit with many outcomes

D-branes are only part of the available construction kit. The lecture also names fluxes, orbifolds, and orientifolds. One first chooses a compact geometry, then chooses which branes and fluxes occupy its cycles. Even the simple sandwich changes its lower-dimensional physics when one inserts one floating D2-brane, two of them, or three. Adding or subtracting such ingredients changes the effective particles, interactions, and parameters.

Suppose, in the lecture's schematic count, a six-dimensional compact space has roughly 500 cycles and each cycle admits on the order of 10 choices of wrapped branes, fluxes, or related structures. Then the number of constructions grows like

$$N_{\text{vacua}} \sim 10^{500}. \tag{1.23}$$

The lecture's counting logic is elementary:

$$10 \times 10 \times \cdots \times 10 = 10^{500} \tag{1.24}$$

with 500 factors. The DNA analogy makes the same point in a more familiar register:

$$N_{\mathrm{DNA}}(500) = 4^{500}. \tag{1.25}$$

During the DNA comparison, an audience member checks that there are four alternatives along one strand: A, C, G, and T. The correction reinforces rather than changes the argument. A short list of local choices, repeated many times, produces a vast number of global arrangements. The exact exponent is not the point, and Susskind calls 10^{500} an old underestimate; the point is combinatorial proliferation.

This gives the final inversion of the opening reductionist expectation. Complicated particle physics need not arise from complicated microscopic rules. It may arise because our vacuum is one complicated arrangement of many simple ingredients, just as a complicated organism can be encoded by a sequence built from four bases.

Question from the audience. After equivalent descriptions related by duality are identified, would complete information select only one or a few of these possible vacua as descriptions of our world? [a]

Response. Presumably it would select a restricted set, but many distinct vacua may look extremely similar at accessible energies. The quoted landscape count is meant to remain large even after dual descriptions are identified, and irrelevant high-energy details may function like "junk DNA."

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[a] Lecture timestamp: 01:26:11.

1.7.2 Symmetric points and generic complexity

Within the landscape there are specially symmetric points. In the geometric picture an equal-aspect configuration,

$$R_1 = R_2, \tag{1.26}$$

makes the two wrapped descriptions equally simple. In the electric–magnetic analogy this is represented schematically by

$$m_{\mathrm{monopole}} = m_e. \tag{1.27}$$

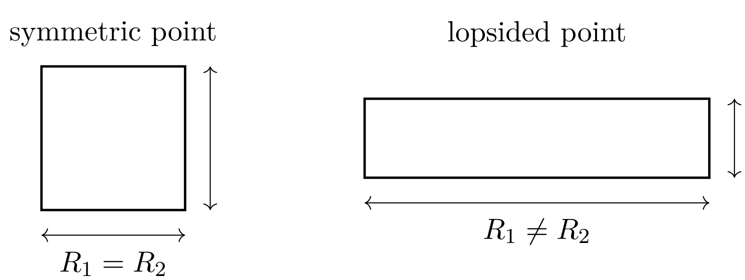

Figure 1.8: An equal-aspect compactification treats the two wrapped strings symmetrically; a lopsided compactification makes one description much lighter than the other.

Question from the audience. Are there vacua in the landscape where the construction becomes much simpler because the F- and D-string descriptions coincide? [a]

Response. Yes, there are specially symmetric points. Our world does not appear to sit at one: the familiar electric particle and a magnetic monopole do not have equal masses. Why nature occupies a lopsided point is not known.

[a]Lecture timestamp: 01:27:21.

[10]

Susskind then recalls an analogy from communication theory. Random codes can be extraordinarily good, while the codes easiest to describe and decode have extra mathematical structure and historically performed less well. Coding theory later found highly effective structured constructions, but the analogy remains: anything compactly named or described occupies a very special, low-complexity corner of the full space of possibilities. Likewise, the string vacua that are easiest to calculate may be atypically simple.

The appropriate questions may therefore change. Instead of demanding the exact compactification in every microscopic detail, one may ask which features are common or generic across a broad ensemble. Susskind presents this as a research problem, not a settled conclusion, and rejects ideological arguments in place of calculation.

1.7.3 Cosmological questions

Question from the audience. Is a compact direction of enormous circumference observationally different from an infinite direction, and do we know that our three large spatial dimensions are infinite? [a]

Response. We do not know that they are infinite. A closed three-sphere can expand while remaining compact. Measurements of near-flatness provide only a lower bound on the curvature radius, just as surveying a nearly flat field can bound the Earth's size without proving that its surface is infinite. Only a detected curvature or global identification would determine the finite scale.

[a]Lecture timestamp: 01:30:03.

Question from the audience. If the coupling can vary through space in the idealized model, how could such a variation occur in the real world? [a]

Response. Not as the unrestricted smooth variation used in the symmetric example. The realistic possibility to be discussed later is variation by discrete transitions between regions with different effective laws. Whether such jumps occur is a cosmological question.

[a]Lecture timestamp: 01:32:43.

[11]

[10]Editorial clarification: The equal electron–monopole mass is a schematic feature of a suitably protected self-dual model, not a prediction of ordinary nonsupersymmetric QED. It expresses the lecture's symmetric-point idea rather than a claim about the observed electron.

[11]Editorial clarification: This response should be read as the lecture's preview of flux vacua and domain transitions, not as a general theorem forbidding every smoothly varying scalar field. Observations strongly constrain variations of dimensionless couplings, while stabilization mechanisms determine whether continuous moduli remain available in a given model.

The recording does not preserve the final audience question. The surviving response says that the proposed phenomenon can occur for some kinds of black holes but not at the horizon of an ordinary Schwarzschild black hole, whose horizon is locally an ordinary place for a freely falling observer. Because the question cannot be recovered reliably, no more specific claim is attributed to the exchange.

1.8 From Bricks to a Web of Descriptions

The hope that deeper layers of physics must be simpler has proved fragile. In $1 + 1$-dimensional bosonization, the same theory can be organized either around fermions or around bosons, with the fermion reappearing as a kink. In quantum electrodynamics, weak and strong coupling exchange the descriptive roles of electron and magnetic monopole. In string theory, F-strings and D-strings repeat the same pattern, and in the even-brane theory strong coupling goes further still: a compact direction expands, D0-branes become Kaluza–Klein momentum states, and effective strings are reinterpreted as wrapped membranes in a higher-dimensional geometry. Modern quantum field theory and string theory therefore do not furnish a unique once-and-for-all answer to the question "which are the bricks and which are the houses?" They furnish a web of descriptions, each efficient in its own regime, together with a huge landscape of possible constructions whose generic output is complexity rather than simplicity.

Chapter 2

Geometry, Information, and the Black-Hole Area Law

Overview. The announced destination is cosmology, but the shortest road to it begins with a return to special relativity. A metric tells us the geometry of spacetime, and the condition of zero proper time tells us how light moves. We shall carry that one rule into the Schwarzschild geometry, distinguish the horizon from the singularity, and then ask what becomes of information when a black hole radiates away. The answer begins with ordinary entropy and ends with Bekenstein's one-bit argument: a black hole stores information in units of horizon area rather than volume.

Susskind briefly postpones a promised accounting of string theory's successes and failures. It is late, he says, and substance is more useful than an audit. The substance tonight is the minimum geometry and thermodynamics needed to formulate the black-hole information problem.

2.1 One Rule for Light

Special relativity treats space and time as one geometric object. For neighboring events separated by dt, dx, dy, dz, Susskind uses the mostly-minus convention and writes

$$d\tau^2 = dt^2 - \frac{1}{c^2}\left(dx^2 + dy^2 + dz^2\right). \tag{2.1}$$

The opposite signs of the temporal and spatial pieces distinguish Minkowski geometry from four-dimensional Euclidean geometry. The factor $1/c^2$ merely reconciles units; by measuring time and distance compatibly we may set $c = 1$.

For timelike separation $d\tau^2 > 0$, and $d\tau$ is the proper time recorded by a clock following the trajectory. Spacelike separations instead have $d\tau^2 < 0$. The case needed tonight lies exactly between them. A light ray follows a *null* trajectory,

$$d\tau^2 = 0. \tag{2.2}$$

Restricting first to motion along the x-axis gives

$$0 = dt^2 - \frac{dx^2}{c^2}, \tag{2.3}$$

$$c\,dt = \pm dx, \tag{2.4}$$

$$\frac{dx}{dt} = \pm c. \tag{2.5}$$

The signs describe right- and left-moving rays. The numerical value assigned to c changes when centimeters are replaced by meters or kilometers; the invariant statement is not a number but the null condition.

General relativity complicates the metric, not this rule. With coordinates x^μ, $\mu = 0, 1, 2, 3$,

$$ds^2 = g_{\mu\nu}(x)\,dx^\mu dx^\nu, \tag{2.6}$$

where repeated indices are summed and the coefficients may vary from point to point. To find the locally allowed directions of light, set this quadratic form to zero.

2.2 Spherical Coordinates and the Schwarzschild Metric

A nonrotating black hole is spherically symmetric, so Cartesian coordinates conceal the useful structure. Let r measure radial distance and let θ, ϕ locate a direction on a unit sphere. Its line element is

$$d\Omega^2 = d\theta^2 + \sin^2\theta\,d\phi^2. \tag{2.7}$$

On a sphere of radius r, angular displacements therefore contribute $r^2 d\Omega^2$. Flat spacetime in spherical coordinates is

$$d\tau^2 = dt^2 - \left(dr^2 + r^2 d\Omega^2\right), \qquad (c = 1). \tag{2.8}$$

Susskind does not derive the black-hole solution from Einstein's equations. He writes it down and asks us to inspect it. The board makes the intended comparison explicit.

For a black hole of mass M, define the Schwarzschild radius

$$R_s = \frac{2GM}{c^2}. \tag{2.9}$$

The exterior line element is

$$d\tau^2 = \left(1 - \frac{R_s}{r}\right) dt^2 - \frac{dr^2/c^2}{1 - R_s/r} - \frac{r^2}{c^2} d\Omega^2. \tag{2.10}$$

Equivalently, after setting $c = 1$,

$$ds^2 = \left(1 - \frac{2GM}{r}\right) dt^2 - \frac{dr^2}{1 - 2GM/r} - r^2 d\Omega^2. \tag{2.11}$$

Far away, $R_s/r \ll 1$, and the flat spherical metric returns. This asymptotic check matters more than memorizing the formula, though Susskind recommends the latter as material for cocktail parties.
[1]

[1] Editorial clarification: The board alternates between $d\tau^2$ and ds^2 and does not display every factor of c consistently. Equation (2.10) follows the lecture's time-coordinate convention, in which t has units of time. The standard form with $x^0 = ct$ is algebraically equivalent.

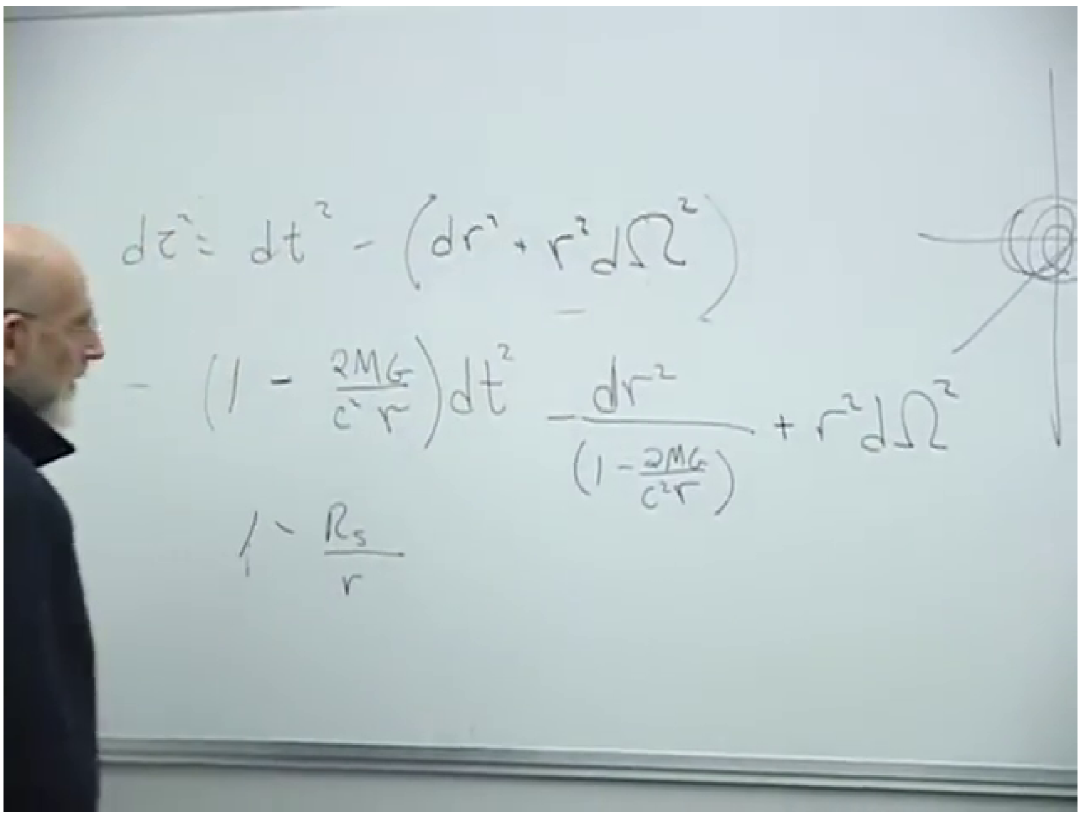

Figure 2.1: Flat spherical spacetime above the Schwarzschild deformation. The repeated factor $1 - 2GM/(c^2 r) = 1 - R_s/r$ and the radial sketch are visible together.

Lecture frame: 00:19:12.520.

2.2.1 Horizon is not singularity

Two radii look suspicious in the coordinates. At $r = R_s = 2GM$ in natural units, the coefficient of dt^2 vanishes and that of dr^2 diverges. At $r = 0$, curvature itself becomes singular. These are not two versions of the same event:

$$\begin{aligned} r = R_s &: \quad \text{horizon,} \\ r = 0 &: \quad \text{curvature singularity.} \end{aligned} \tag{2.12}$$

Tidal forces give the physical distinction. A gravitational field varies across an extended body: the feet may be pulled more strongly than the head, stretching the body while transverse directions are squeezed. At the horizon those tidal forces are finite, and at the horizon of a sufficiently massive black hole they can be very small. Nothing locally catastrophic need occur there. At $r = 0$, by contrast, the tidal forces diverge. The alarming coefficients at $r = R_s$ signal a defect of Schwarzschild coordinates; the singularity at $r = 0$ is physical.

[2]

2.3 Radial Light Near the Horizon

Now return to the one rule. A radial ray keeps its angular position fixed, so $d\Omega = 0$, and a light ray has $ds^2 = 0$. In $c = 1$ units, Equation (2.11) gives

$$0 = \left(1 - \frac{2GM}{r}\right) dt^2 - \frac{dr^2}{1 - 2GM/r}. \tag{2.13}$$

Multiplication by $1 - 2GM/r$ and a square root yield two branches,

$$dr = \pm \left(1 - \frac{2GM}{r}\right) dt, \tag{2.14}$$

$$\frac{dr}{dt} = \pm \left(1 - \frac{2GM}{r}\right). \tag{2.15}$$

The plus sign is outgoing and the minus sign ingoing. Far away their coordinate speeds approach ± 1. At the horizon both satisfy $|dr/dt| \to 0$: in Schwarzschild coordinate time an outgoing ray appears to stand still there, while an ingoing ray takes infinite coordinate time to arrive.

Inside the horizon the future light cones tip so far inward that both future-directed radial branches lead toward decreasing r. No signal originating there can escape to the exterior, and every future-directed causal path reaches the singularity.

Question from the audience. If the speed of light is invariant, what exactly does it mean to say that light slows down near the horizon?[a]

Response. The displayed quantity is dr/dt in a particular coordinate system. Numerical speed already changes when meters are replaced by centimeters; here the effective radial and temporal calibrations vary with position. The universal statement is that every light ray follows a null trajectory, $ds^2 = 0$.

[a]Lecture timestamp: 00:31:44.

[2]Editorial clarification: The finiteness of curvature at $r = R_s$ can be checked from the Kretschmann scalar $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/(c^4r^6)$, which is finite at the horizon and diverges only as $r \to 0$. This invariant was not written in the lecture, but makes its tidal-force distinction precise.

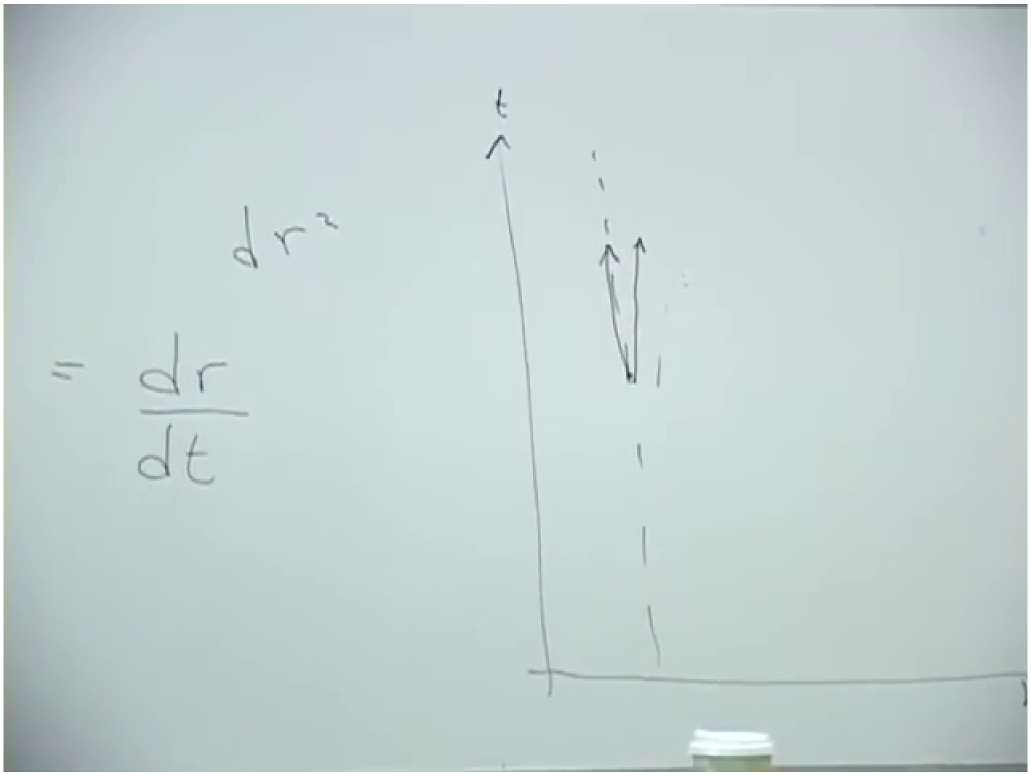

Figure 2.2: The completed t-r blackboard sketch: radial null rays become nearly vertical as the dashed horizon is approached.

Lecture frame: 00:30:25.000.

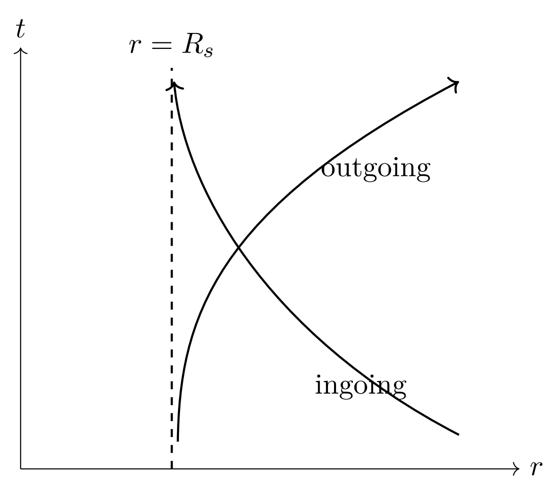

Figure 2.3: A cleaned coordinate-time version of the board sketch. The apparent slowing is a statement about Schwarzschild t, not about a local failure of light to move at c.

[3]

Question from the audience. Does the same null-speed statement hold regardless of the material medium through which light travels?[a]

Response. No. The available recognition preserves no longer explanation of this answer.

[a]Lecture timestamp: 00:33:24.

[4]

Question from the audience. What does an incoming light ray look like on the same t-r diagram?[a]

Response. It is the reverse-looking branch: from large r it bends toward the horizon and approaches it asymptotically. It takes an infinite amount of the coordinate time t to arrive.

[a]Lecture timestamp: 00:33:32.

[5]

2.4 Why Evaporation Creates a Problem

The horizon deserves much more study: how it forms, what an infalling observer sees, and whether anything locally unusual occurs there. Susskind defers those questions and jumps to quantum mechanics. From the outside, matter and all its distinguishing information become squeezed ever closer to the horizon and unavailable. That alone need not destroy information; it may simply hide it.

Hawking's result makes the issue acute. A settled classical black hole looks like the dullest object imaginable: spherical, dark, and unable to send a message. But "black" has two meanings. It can mean emitting nothing, or it can mean emitting thermal black-body radiation. A quantum black hole is black in the second sense. It has a temperature, glows, and loses energy. Since

$$E = Mc^2, \qquad R_s = \frac{2GM}{c^2}, \tag{2.16}$$

radiation decreases both its mass and its horizon radius. If evaporation eventually removes the hole, what becomes of the information that appeared to remain at its horizon? It might be destroyed, trapped in a tiny remnant, or encoded in the outgoing radiation. The tension among those possibilities is the information problem, and string theory will later help to sharpen its resolution.

No calculation of quantum fields in curved spacetime is attempted tonight. Instead Susskind follows Bekenstein's simpler route: first understand entropy as information, then calculate how much information a black hole can hide.

[3]Editorial clarification: A freely falling local observer measures vacuum light at c; the vanishing Schwarzschild-coordinate speed at the horizon is not a local observable.

[4]Editorial clarification: The null-speed claim concerns propagation in vacuum spacetime. Interactions with matter can give a wave packet a smaller effective speed; that is a property of the medium, not a change in the local causal speed c.

[5]Editorial clarification: An infalling observer reaches and crosses the horizon in finite proper time; only Schwarzschild t diverges.

2.5 Information That Is Present but Hidden

Imagine dividing a box into cells small enough that each cell is either occupied or empty. A binary string—one answer per cell—distinguishes the configuration from every other allowed configuration. More realistic matter requires questions about particle species and other degrees of freedom, but the principle is unchanged: information is measured by the minimum number of yes–no questions required to specify the state.

Question from the audience. Is this only a description of a static system?[a]

Response. It is an instantaneous description, at one moment of time. Dynamics then maps that state to another state.

[a]Lecture timestamp: 00:45:49.

The next principle is deeper: closed-system evolution does not merge two distinct microscopic states into one. If two configurations differ now, their evolved configurations remain distinguishable in principle. Classical Hamiltonian evolution is invertible and quantum evolution is unitary.[6]

The bathtub makes the difference between loss and inaccessibility concrete. Two streams of drops could encode two different Morse-code texts. After the water settles, both tubs look alike. The visible ripples have gone, yet a complete microscopic measurement would still find different molecular configurations. The message has spread into degrees of freedom too small and numerous to track.

Entropy is this hidden information. Relative to the macroscopic questions we can ask—temperature, total energy, composition, water level—many microscopic distinctions remain possible but inaccessible. In the lecture's bit-counting language,

$$S = \text{number of hidden bits.} \tag{2.17}$$

The precise statistical formula is $S = k_B \ln \Omega$, but the qualitative definition is enough for the black-hole argument.

2.6 Temperature as the Cost of Hiding a Bit

Temperature is familiar to measure and surprisingly indirect to define. Susskind chooses entropy and energy as the primitive concepts. Suppose a computer bit is reset to a standard value. The physical device remains, but the distinction between its former zero and one states disappears from the computer. It cannot disappear from the complete world; it is transferred to microscopic degrees of freedom in the environment, which heat by a small amount. This is Landauer's principle.

If one entropy unit is one bit and temperature is measured in energy units, the lecture writes

$$\Delta S = 1, \tag{2.18}$$

$$\Delta E = T\,\Delta S, \tag{2.19}$$

$$dE = T\,dS. \tag{2.20}$$

[6]Editorial clarification: Dissipative effective equations can erase distinctions only because unobserved environmental degrees of freedom have been omitted; the complete closed state retains them.

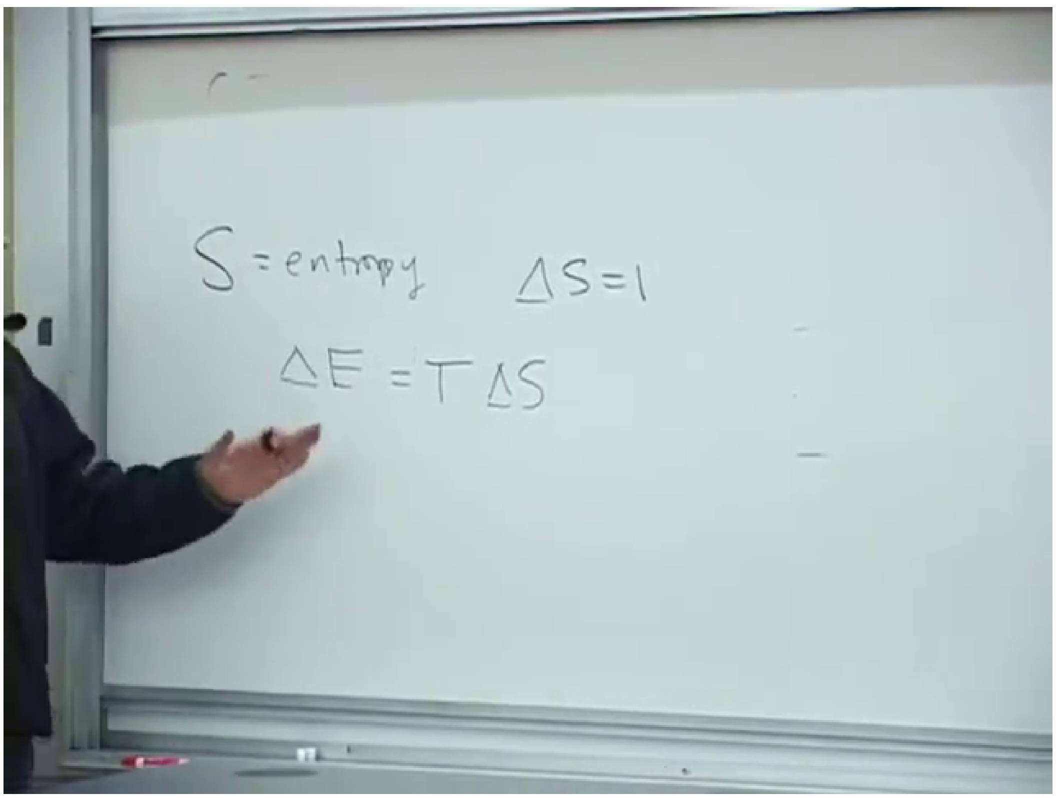

Figure 2.4: The board's information-theoretic thermodynamics: S is entropy, one hidden bit has $\Delta S = 1$, and its minimum energy cost is written $\Delta E = T\Delta S$.

Lecture frame: 00:58:40.000.

[7]

An audience estimate puts the minimum temperature rise from erasing roughly 100 gigabytes into a glass of water near 10^{-15} degrees. Susskind accepts the point rather than the exact number: at room temperature one bit carries an extraordinarily small minimum energy, and real computers dissipate much more because erasure is not performed reversibly.

Question from the audience. Is the hard-drive example only an analogy, or is erasure heating a real law?[a]

Response. It is a real physical constraint, Landauer's rule. Resetting a memory cell removes the old distinction from the computer, so that information must be carried into the environment. Ordinary operations are far less efficient than the ideal minimum because they also drive circuits, move charges, and create unrelated heat.

[a]Lecture timestamp: 01:00:02.

Question from the audience. Does hiding one bit cost more energy at a higher temperature?[a]

Response. Yes. The relation $dE = T\,dS$ says exactly that: for the same entropy transfer, a hotter heat bath requires the larger energy transfer. One may equally regard this energy-per-hidden-bit relation as the definition of temperature.

[a]Lecture timestamp: 01:02:07.

Question from the audience. If one writes over a previously stored bit, has that old information been lost?[a]

Response. It has left the computer, but it has not vanished from the closed physical system. The overwritten distinction has been dispersed into the environment, where it contributes inaccessible information and therefore entropy.

[a]Lecture timestamp: 01:02:43.

The identification of thermodynamic entropy with missing microscopic information goes back to Boltzmann; Shannon gave the corresponding information-theoretic language in the twentieth century. Susskind remarks that this explanation was strangely absent from the thermodynamics courses he and members of the audience encountered in the 1950s. Computer science made the connection harder to overlook, but did not originate its physical content.

2.7 Putting One Bit into a Black Hole

We can now build a black hole conceptually, bit by bit, just as a bathtub can be filled drop by drop. The question is differential: when one additional bit is hidden, how much does the horizon grow? Begin with a hole of mass M and radius R_s, then send in the simplest object whose only recoverable distinction is whether it was absorbed.

[7]Editorial clarification: In conventional units the reversible erasure cost of one binary bit is $k_B T \ln 2$. Equation (2.20) uses $k_B = 1$ and treats the lecture's entropy unit as one bit while suppressing the associated $\ln 2$. These conventions do not affect the scaling argument that follows.

Question from the audience. Doesn't the information require infinite coordinate time to enter the hole?[a]

Response. It becomes exponentially compressed near the horizon and unavailable to the exterior in a short practical time, even though Schwarzschild coordinate time assigns the actual crossing an infinite value. For the outside accounting it can therefore be treated as stored at the horizon.

[a]Lecture timestamp: 01:08:07.

A photon is a natural candidate, but a generic photon carries more than one bit. Its polarization may encode a binary choice. A sharply localized arrival point supplies many further yes–no questions: upper or lower hemisphere, left or right half, and so on. Quantum mechanics limits localization to roughly a wavelength, so the way to suppress positional information is to use the longest wavelength the hole can still absorb.

Question from the audience. Is there really an upper wavelength beyond which the photon cannot enter?[a]

Response. Yes. A wave much larger than the hole is predominantly scattered rather than absorbed. The useful marginal choice is a wavelength comparable to the horizon radius: long enough to wash out the point of entry, yet short enough to have a substantial absorption probability.

[a]Lecture timestamp: 01:12:15.

Question from the audience. Does the direction from which the photon arrives itself add information?[a]

Response. For a wavelength of order the whole horizon, the field varies so slowly across the hole that the arrival direction cannot be resolved with fine precision. This is why the long-wavelength limit approximates a single yes–no datum rather than a sharply located impact.

[a]Lecture timestamp: 01:13:28.

Thus choose, at order of magnitude,

$$\lambda \sim R_s = \frac{2GM}{c^2}. \tag{2.21}$$

Photon kinematics gives

$$E = cp, \qquad p = \hbar k = \frac{h}{\lambda}, \qquad \Delta E \sim \frac{\hbar c}{R_s}. \tag{2.22}$$

Here $k = 2\pi/\lambda$ and $h = 2\pi\hbar$. The lecture deliberately ignores the 2π, as well as order-one absorption factors.

Question from the audience. Should momentum be written $\hbar/\lambda$, or h/λ?[a]

Response. With λ the ordinary wavelength, the exact relation is $p = h/\lambda = 2\pi\hbar/\lambda$; with wave number $k = 2\pi/\lambda$, it is $p = \hbar k$. The present estimate keeps only the scale, so the factor 2π is intentionally discarded.

[a]Lecture timestamp: 01:16:00.

The energy cost of hiding one bit is the temperature scale. Equations (2.21) and (2.22) therefore give

$$k_B T_H \sim \frac{\hbar c}{R_s} \sim \frac{\hbar c^3}{GM}. \tag{2.23}$$

The exact Hawking result is

$$k_B T_H = \frac{\hbar c^3}{8\pi GM}. \tag{2.24}$$

The elementary argument gets the dependence on $M, G, c, \hbar$, but not the coefficient.

2.8 One Bit Occupies One Planck-Scale Patch

The same photon tells us how the horizon grows. Its energy changes the mass by

$$\Delta M = \frac{\Delta E}{c^2} \sim \frac{\hbar}{cR_s}. \tag{2.25}$$

Since $R_s = 2GM/c^2$,

$$\Delta R_s = \frac{2G}{c^2}\Delta M \sim \frac{G\hbar}{c^3 R_s}, \tag{2.26}$$

where all order-one factors have again been suppressed. Multiplying by R_s,

$$R_s \Delta R_s \sim \frac{G\hbar}{c^3}. \tag{2.27}$$

Because $A = 4\pi R_s^2$, its differential is $dA = 8\pi R_s\, dR_s$. Therefore

$$\Delta A \sim \frac{G\hbar}{c^3} = \ell_P^2, \qquad \ell_P = \sqrt{\frac{G\hbar}{c^3}}. \tag{2.28}$$

During the live calculation, factors of c are restored halfway through and an audience member catches the missing c^{-2} in the Schwarzschild radius. That correction is exactly what produces the c^{-3} in Equation (2.28).

Question from the audience. Did restoring c in only part of the calculation lose two powers of it?[a]

Response. Yes. The dimensional relation is $R_s = 2GM/c^2$, not simply $2GM$. Restoring that factor makes the area scale $G\hbar/c^3$.

[a]Lecture timestamp: 01:21:18.

Using $G \simeq 6.7 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$, $\hbar \simeq 1.05 \times 10^{-34}\,\mathrm{J\,s}$, and $c^3 \simeq 2.7 \times 10^{25}\,\mathrm{m^3\,s^{-3}}$,

$$\ell_P^2 \simeq 2.6 \times 10^{-70}\,\mathrm{m}^2, \qquad \ell_P \simeq 1.6 \times 10^{-35}\,\mathrm{m}. \tag{2.29}$$

The lecture's spoken powers wander while the estimate is done at the board; the physical conclusion is stable. This scale is far below direct experimental resolution.

The memorable image is of impenetrable bits with sharp elbows, accumulating on the horizon and pushing one another aside. Each new bit claims approximately one Planck-area patch. It is a metaphor for the area accounting, not a claim that literal square tiles form the horizon.

Question from the audience. If the first photon is still approaching when another photon enlarges the horizon, doesn't the old one become located inside the new horizon?[a]

Response. The simple static picture is not precise enough to settle that question. The event horizon in a dynamical spacetime must be defined globally, and its location changes as matter arrives. In the exterior description used here, the information remains associated with the growing horizon. A careful definition of that horizon is deferred.

[a]Lecture timestamp: 01:26:39.

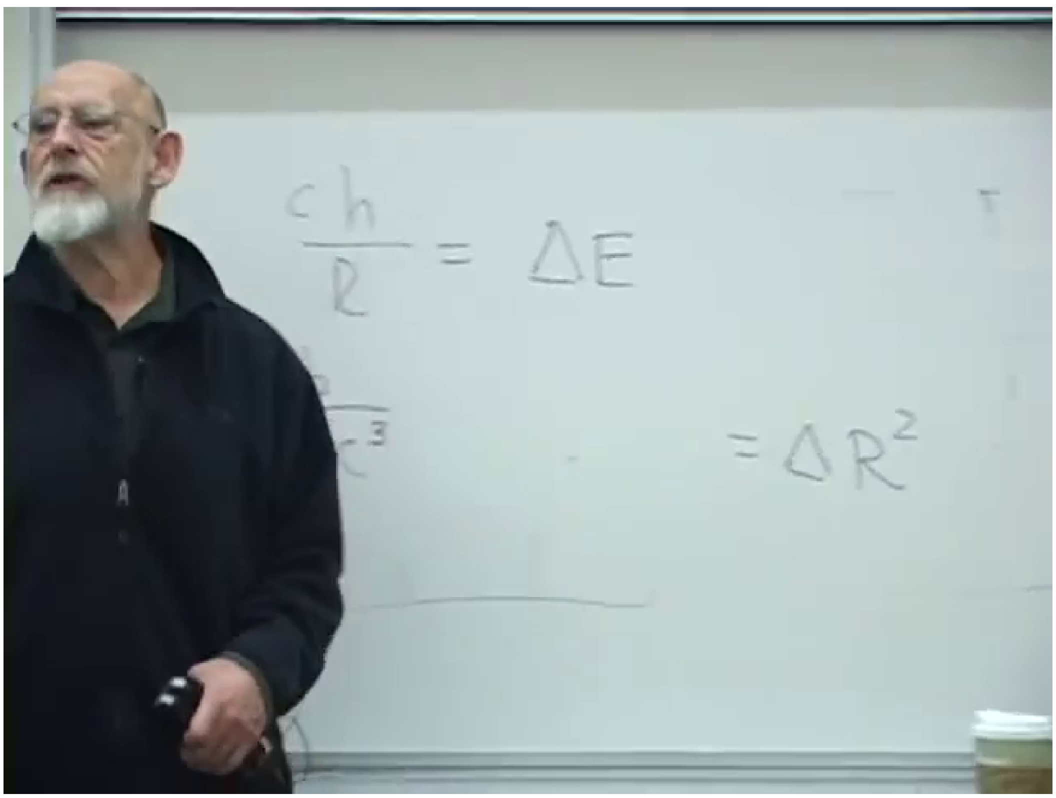

Figure 2.5: The completed scaling argument on the board: the one-bit energy $\Delta E \sim \hbar c/R_s$ leads to a radius-squared, hence area, increment of order $G\hbar/c^3$.

Lecture frame: 01:22:40.000.

Question from the audience. Would the same issue arise when a huge mass is added, for example when two black holes collide?[a]

Response. Yes; then the dynamical character is impossible to ignore. The lecture postpones the full merger geometry until the horizon has been defined more carefully.

[a]Lecture timestamp: 01:27:03.

Question from the audience. Is a Planck area defined only on a flat patch, rather than on a spherical horizon?[a]

Response. It is an area scale on the spherical surface, not a requirement that the horizon be flat. The planar-versus-spherical distinction is not significant for this counting.

[a]Lecture timestamp: 01:27:30.

[8]

For a hole eight kilometers across, the marginal photon is correspondingly a long radio wave of roughly that scale. An audience analogy compares the size-independent area increment with adding the same extra circumference to a string wrapped around a basketball or around Earth: in each case the radial gap created by a fixed added circumference is the same. The exact geometries differ, but both examples isolate a change that is independent of the object's original size.

2.9 The Area Law

Adding one bit at a time changes the horizon area by a universal microscopic amount. The number of bits that can be hidden must therefore be proportional to A/ℓ_P^2, not to the volume. Bekenstein inferred the proportionality; Hawking's calculation fixed its coefficient:

$$S_{\mathrm{BH}} = \frac{k_B A c^3}{4G\hbar} = \frac{k_B A}{4\ell_P^2}. \tag{2.30}$$

If entropy is counted in binary bits rather than thermodynamic units, divide S_{BH}/k_B by $\ln 2$.

Black holes therefore have more entropy than ordinary systems of comparable size, by an enormous margin. Yet the answer is still smaller than a naive volume count might suggest. The independent information is bounded by a surface measure. This inversion—a three-dimensional gravitating region described by information scaling as a two-dimensional boundary—is the seed of the holographic principle.

Question from the audience. Where does the factor 1/4 in the entropy formula come from?[a]

Response. Bekenstein's argument fixed the dependence on area and the fundamental constants but left an unknown coefficient; his estimates produced possibilities such as logarithms of two. Hawking's quantum-field calculation fixed the coefficient to precisely 1/4.

[a]Lecture timestamp: 01:30:30.

[8]Editorial clarification: When the horizon curvature radius is enormous compared with ℓ_P, each Planck-size neighborhood is locally almost flat.

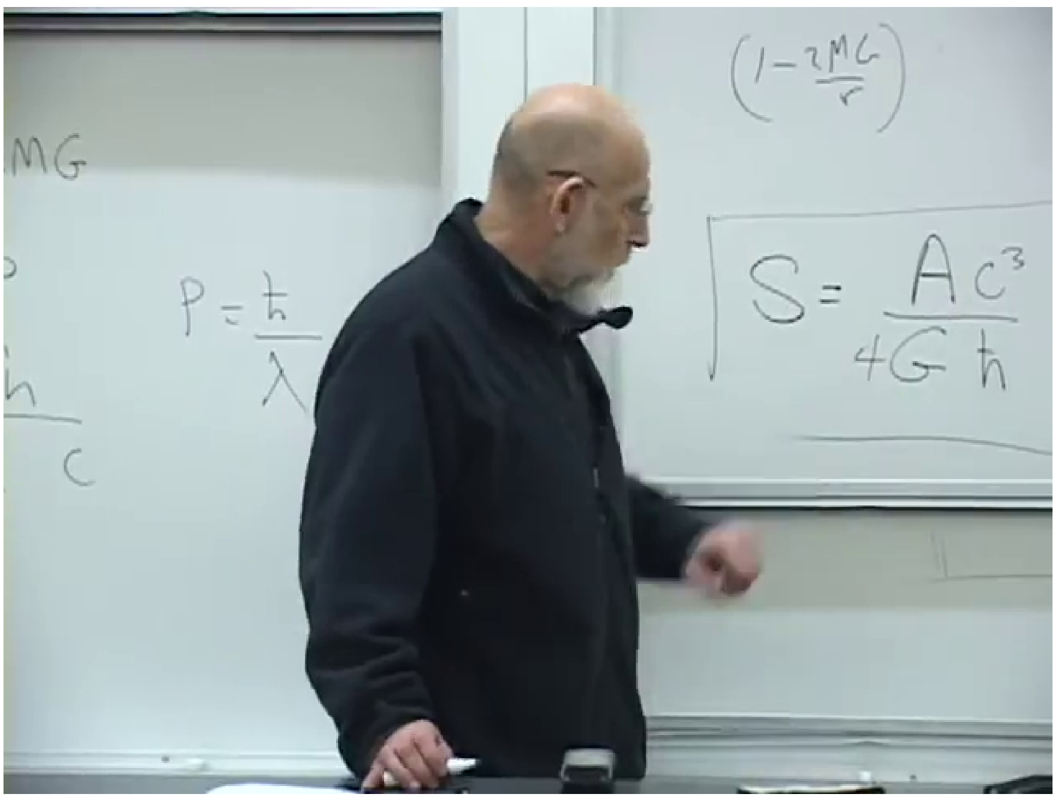

Figure 2.6: The final board formula, including Hawking's factor 1/4: $S \propto Ac^3/(4G\hbar)$ in the lecture's entropy units.

Lecture frame: 01:31:02.000.

The position of $\hbar$ contains the final surprise. One might expect a quantum entropy to be proportional to $\hbar$ and vanish in the classical limit. Instead $\hbar$ is in the denominator. As $\hbar \to 0$, a wave can carry arbitrarily fine information at arbitrarily small energy cost, so the entropy capacity diverges. Quantization does not add a tiny entropy to an otherwise classical hole; it prevents the information capacity from being infinite.

Question from the audience. Why does $\hbar$, whose units are angular momentum, also set the relation between ordinary momentum and wavelength?[a]

Response. $\hbar$ has units of action, equivalently momentum times distance. Since $p\lambda$ has those same units, the quantum relation can be $p \sim \hbar/\lambda$ up to the 2π convention. Angular momentum $\mathbf{r} \times \mathbf{p}$ carries the same dimensions for the same reason.

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[a] Lecture timestamp: 01:33:23.

The lecture stops here, not because the subject is exhausted but because this is the right boundary. Geometry has told us why the horizon hides information; thermodynamics has told us how to count hidden information; quantum mechanics has supplied a finite area per bit. The next question is no longer whether a black hole has microscopic states, but how those states encode and eventually return what fell in.

Chapter 3

The Horizon in Flat Coordinates and the Thermodynamics of Evaporation

Overview. The Schwarzschild metric appears to become singular at the horizon, although a freely falling observer should notice nothing exceptional there. We resolve that tension by converting the near-horizon metric to flat Minkowski space in hyperbolic polar coordinates. The same geometry then separates Alice's infalling description from Bob's accelerated exterior description. The second half develops entropy and temperature, derives Hawking's inverse-mass temperature and the M^3 evaporation law, and closes by asking what information conservation requires of the microscopic theory of gravity.

3.1 The Puzzle at $r = 2GM$

Begin with a very large black hole. Its horizon can have such a small curvature that an observer falling through it sees, over any modest laboratory-sized region, nearly empty flat spacetime. No wall is struck, and no local alarm announces the crossing. Yet the Schwarzschild metric seems to say that something singular occurs at precisely that radius. The purpose of the first half of the lecture is to make these two statements compatible.

We use the mostly-plus convention. In flat spacetime,

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2, \tag{3.1}$$

so a timelike worldline has $ds^2 < 0$ and proper time satisfies $d\tau^2 = -ds^2$. This is the overall sign reversal of the convention used in the preceding lecture; no observable depends on that choice.

With $c = 1$, the Schwarzschild line element is

$$\begin{aligned} ds^2 &= -\left(1 - \frac{2GM}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2GM}{r}} + r^2 d\Omega_2^2, \\ d\Omega_2^2 &= d\theta^2 + \sin^2\theta\, d\phi^2. \end{aligned} \tag{3.2}$$

The Schwarzschild radius is $R_s = 2GM$, or $R_s = 2GM/c^2$ when c is restored. Far away, $r \gg R_s$, both radial coefficients approach their flat-space values. At $r = R_s$, however, the coefficient of dt^2 vanishes and the coefficient of dr^2 diverges. That looks dramatic. The question is whether the geometry is dramatic or only the coordinates.

3.2 Two Coordinate Lessons

3.2.1 The harmless origin of polar coordinates

On an ordinary Euclidean plane,

$$ds^2 = dx^2 + dy^2. \tag{3.3}$$

Introduce polar coordinates

$$x = \rho \cos\theta, \qquad y = \rho \sin\theta. \tag{3.4}$$

A small displacement has radial length $d\rho$ and transverse length $\rho\, d\theta$, hence

$$ds^2 = d\rho^2 + \rho^2 d\theta^2. \tag{3.5}$$

At $\rho = 0$, the angular coefficient vanishes. This does not make the center of the plane singular. It says only that every value of θ names the same point there. A peculiar metric coefficient may therefore diagnose a peculiar coordinate system rather than peculiar physics.

3.2.2 Hyperbolic polar coordinates

Now replace the Euclidean plane by $1 + 1$-dimensional Minkowski spacetime,

$$ds^2 = -dT^2 + dX^2. \tag{3.6}$$

In the right-hand wedge $X > |T|$, define

$$X = \rho \cosh\omega, \qquad T = \rho \sinh\omega, \qquad \rho > 0. \tag{3.7}$$

The ordinary identity $\cos^2\theta + \sin^2\theta = 1$ is replaced by

$$\cosh^2\omega - \sinh^2\omega = 1. \tag{3.8}$$

Thus a line of constant ρ is the hyperbola

$$X^2 - T^2 = \rho^2. \tag{3.9}$$

Unlike the periodic angle θ, the rapidity-like coordinate ω runs from $-\infty$ to $+\infty$. At large positive ω, $\cosh\omega$ and $\sinh\omega$ become nearly equal and the hyperbola approaches the future null ray $X = T$.

Differentiating Equation (3.7) gives

$$dX = \cosh\omega\, d\rho + \rho \sinh\omega\, d\omega, \tag{3.10}$$

$$dT = \sinh\omega\, d\rho + \rho \cosh\omega\, d\omega. \tag{3.11}$$

The cross terms cancel, leaving

$$ds^2 = -\rho^2 d\omega^2 + d\rho^2. \tag{3.12}$$

Equation (3.12) still describes perfectly flat spacetime. Its time coefficient vanishes at $\rho = 0$, just as the angular coefficient of Equation (3.5) vanishes at the origin. The coordinate chart degenerates there; spacetime does not.

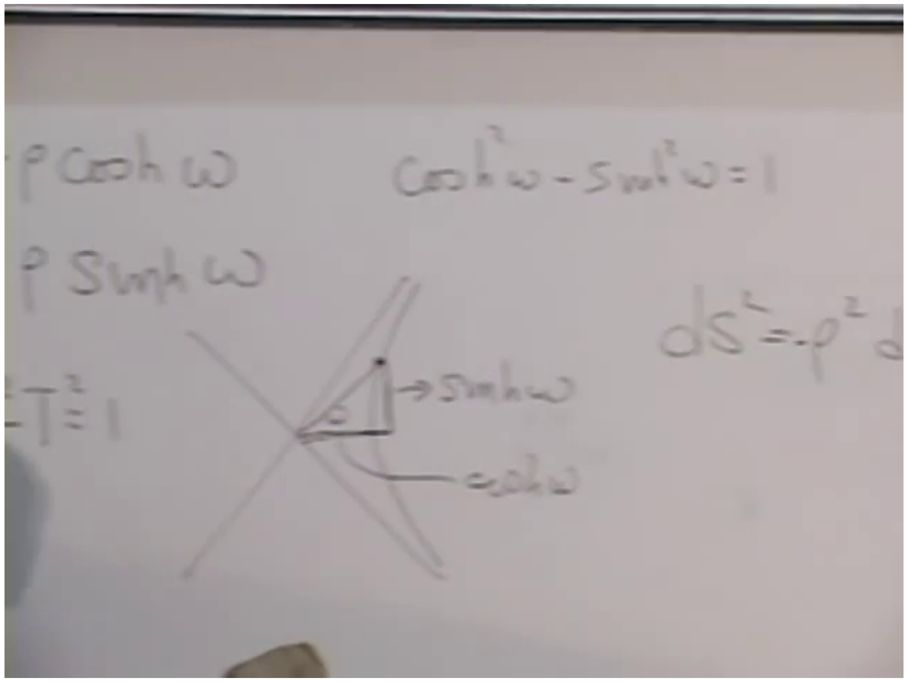

Figure 3.1: The completed hyperbolic-coordinate board: $X = \rho\cosh\omega$, $T = \rho\sinh\omega$, the unit-hyperbola construction, and the flat metric in the form $ds^2 = -\rho^2 d\omega^2 + d\rho^2$.

Lecture frame: 00:16:55.000.

3.3 Zooming In on the Schwarzschild Horizon

Return to the radial-time part of Equation (3.2):

$$ds^2 = -\frac{r-2GM}{r}\,dt^2 + \frac{r}{r-2GM}\,dr^2. \tag{3.13}$$

We now inspect only a narrow neighborhood of $r = 2GM$. In factors that vary slowly, r may be replaced by $2GM$. The factor $r - 2GM$ must be retained, because it vanishes and changes sign at the horizon. Therefore

$$ds^2 \approx -\frac{r-2GM}{2GM}\,dt^2 + \frac{2GM}{r-2GM}\,dr^2. \tag{3.14}$$

Choose a new radial coordinate ρ so that the second term is simply $d\rho^2$:

$$d\rho^2 = \frac{2GM}{r-2GM}\,dr^2. \tag{3.15}$$

Outside the horizon, taking the positive square root and integrating from $r = 2GM$ gives

$$d\rho = \sqrt{\frac{2GM}{r-2GM}}\,dr, \tag{3.16}$$

$$\rho = 2\sqrt{2GM}\sqrt{r-2GM}, \tag{3.17}$$

$$\rho^2 = 8GM(r-2GM), \qquad r - 2GM = \frac{\rho^2}{8GM}. \tag{3.18}$$

Substitution into Equation (3.14) yields

$$ds^2 \approx -\frac{\rho^2}{16G^2M^2}\,dt^2 + d\rho^2. \tag{3.19}$$

Finally rescale Schwarzschild time,

$$\omega = \frac{t}{4GM}, \qquad d\omega^2 = \frac{dt^2}{16G^2M^2}, \tag{3.20}$$

and obtain

$$\boxed{ds^2 \approx -\rho^2 d\omega^2 + d\rho^2.} \tag{3.21}$$

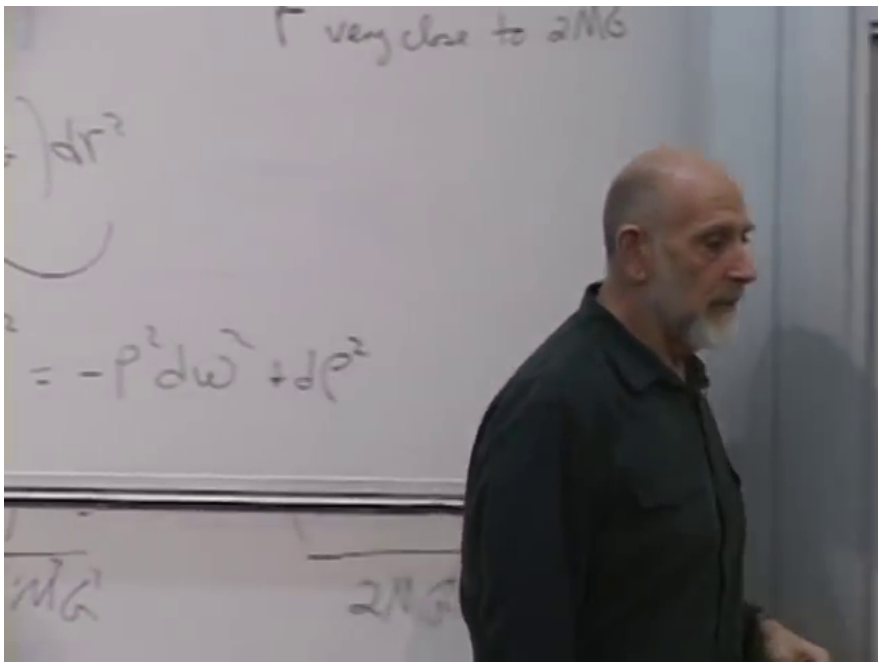

Figure 3.2: The board after the time rescaling: the near-horizon Schwarzschild metric has become the same hyperbolic-polar metric as flat Minkowski spacetime.

Lecture frame: 00:31:35.000.

The approximation improves without limit as the inspected neighborhood shrinks toward the horizon. It proves the local statement that was wanted: for a sufficiently large hole, the horizon is ordinary spacetime written in an extraordinary chart. Curvature is not literally zero for a finite-mass Schwarzschild hole, but it is finite at the horizon and can be made arbitrarily small there by increasing M.[1]

Question from the audience. For $r < 2GM$, Equation (3.18) makes this exterior ρ imaginary. Does that signal a physical problem?[a]

Response. It reproduces the same coordinate failure in a different form. The present chart was constructed for the exterior wedge; extending through the horizon requires another chart. The failure of this particular ρ is not a physical event at the horizon.

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[a] Lecture timestamp: 00:32:30.

[1] Editorial clarification: The local curvature scale is of order R_s^{-2}. The Rindler metric is the leading term in a local near-horizon expansion, not a claim that the entire Schwarzschild geometry is globally flat.

Question from the audience. In the near-horizon rewrite, how does the approximated line follow from the preceding exact line?[a]

Response. Only the slowly varying occurrences of r have been set equal to $2GM$. The small difference $r - 2GM$ is left intact. That is the approximation, and it becomes increasingly accurate as r approaches $2GM$.

[a]Lecture timestamp: 00:39:21.

3.4 What the Coordinates Mean

The transformation identifies

$$\rho = 0 \quad \Longleftrightarrow \quad r = 2GM. \tag{3.22}$$

The exterior $r > 2GM$ is the right Rindler wedge $X > |T|$. Constant-ρ hyperbolae represent observers who remain at fixed Schwarzschild radius. An outgoing light ray emitted anywhere in that wedge crosses every larger-ρ worldline, so an exterior observer can communicate outward. A point beyond the future null boundary cannot send a light signal back into the right wedge. That null boundary is the horizon.

Question from the audience. Does the hyperbolic-coordinate patch shown here always describe the region outside the horizon?[a]

Response. Yes. In this construction $r > 2GM$ gives $\rho^2 > 0$ and places the event in the right exterior wedge. The null edge $\rho = 0$ is the horizon; the present coordinates do not continue as real exterior coordinates beyond it.

[a]Lecture timestamp: 00:35:22.

Question from the audience. Does a constant-ρ hyperbola represent one time or one distance?[a]

Response. It represents one fixed proper distance from the horizon in the near-horizon approximation, and therefore one fixed Schwarzschild radius. Motion along the hyperbola is the passage of time for the observer stationed there.

[a]Lecture timestamp: 00:37:21.

Along a fixed-ρ trajectory, Equation (3.12) gives

$$d\tau = \rho\, d\omega. \tag{3.23}$$

Thus ω is not itself the observer's proper time. It is an angular time common to the family of hyperbolae; each observer multiplies it by that observer's ρ, just as an ordinary angle is multiplied by a circle's radius to give arc length.

Question from the audience. Is ω the proper time of the accelerated observer?[a]

Response. Almost. At fixed ρ, the proper-time increment is $\rho\, d\omega$. The factor of ρ is essential.

[a]Lecture timestamp: 00:51:45.

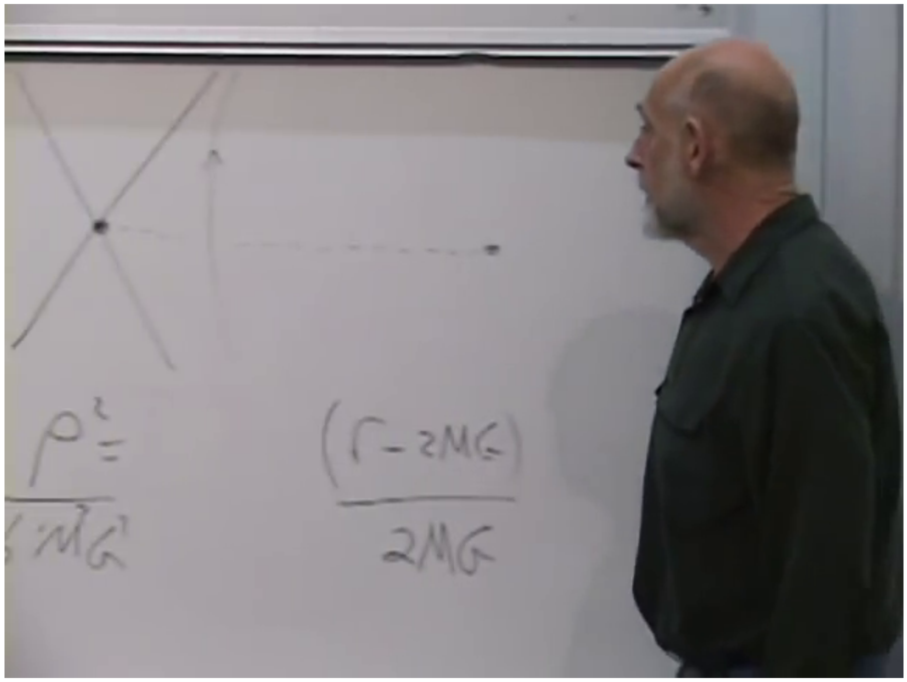

Figure 3.3: The near-horizon causal board: crossed null boundaries, constant-radius hyperbolae in the exterior wedge, and the coordinate relation between ρ and $r - 2GM$.

Lecture frame: 00:34:35.000.

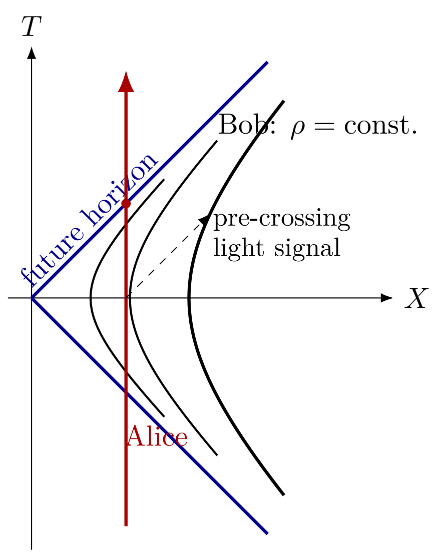

Figure 3.4: A typeset companion to the board. Bob follows an accelerated constant-ρ hyperbola; Alice follows a locally inertial line through the future horizon. The null boundary limits Bob's causal access without becoming a local obstacle for Alice.

3.5 Alice Falls and Bob Hovers

Let Bob maintain a fixed distance from the horizon while Alice releases his hand and falls. Bob's worldline is a hyperbola and is therefore accelerated. Alice's local worldline is approximately straight. She crosses the null boundary in finite proper time and encounters no local discontinuity.

The causal descriptions are asymmetric. After Alice crosses, light from her cannot reach Bob, while light emitted by Bob can still reach Alice for a time. Bob receives signals emitted by Alice at successively earlier stages of her finite infall. Their arrival times stretch farther and farther apart, their frequency redshifts, and their intensity fades. In Schwarzschild time he never sees the crossing itself; Alice appears to approach and flatten against the horizon asymptotically. Her nose does not flatten while the back of her head remains ordinary: signals from every part are compressed into the same limiting layer.

Question from the audience. Is Alice able to see Bob after crossing because Bob's photons travel inward toward her?[a]

Response. Yes. The one-way causal relation is the point: Bob's inward signals can enter, whereas Alice's later outward signals cannot escape back to Bob.

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[a]Lecture timestamp: 00:42:48.

Question from the audience. If Alice's feet cross first while her eyes remain outside, does she temporarily lose sight of her own feet?[a]

Response. There is no visual blip during an ordinary fall. Light always takes time to travel from feet to eyes; by the time the relevant signal arrives, her eyes have crossed too. Only a violent attempt to hold her head outside while her feet continue inward would create a different and destructive situation.

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[a]Lecture timestamp: 00:43:03.

Holding position is the noninertial choice. Bob needs a rocket, just as a person standing on the floor needs the floor's upward force to avoid free fall. The closer Bob hovers to the horizon, the larger the required proper acceleration; in the local Rindler approximation its magnitude is $a = 1/\rho$ in units $c = 1$.[2]

Question from the audience. Are we accelerating while standing here in the lecture room?[a]

Response. Yes, relative to a freely falling frame. The floor supports us; without it, or without a jet pack, we would fall.

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[a]Lecture timestamp: 00:45:40.

Question from the audience. Will Bob eventually see Alice within one Planck length of the horizon, and what happens after that?[a]

Response. That is where the classical geometric description ceases to be sufficient. The Planck length contains $\hbar$, so answering what Bob resolves there requires quantum mechanics.

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[a]Lecture timestamp: 00:49:24.

[2]Editorial clarification: Restoring units gives $a = c^2/\rho$ for the local Rindler observer. An exactly static Schwarzschild observer has a proper acceleration that diverges as the horizon is approached.

Question from the audience. The constant-radius curves seem to meet the horizon at a right-angle-like coordinate corner. Can a derivative be taken there?[a]

Response. Something singular happens to this coordinate chart at that point, but not to local physics. A chart that covers the other wedges removes the artificial corner. For the present exterior discussion, the Rindler picture already contains the needed causal facts.

[a]Lecture timestamp: 00:51:12.

It is therefore unhelpful to ask, without naming an observer, whether Alice *really* crosses. Alice's clock reaches the horizon in finite proper time. Bob's received signal sequence never contains the crossing event. Physics asks what each observer measures and how their records can be compared.

Question from the audience. Is the sharp statement that Bob can see only a finite portion of Alice's future, even though his own observation time is unbounded?[a]

Response. Yes. Alice assigns a finite proper time to the visible portion of her fall. Bob receives its increasingly delayed and redshifted signals forever in the idealized eternal geometry, but he does not see her continue aging beyond the horizon crossing.

[a]Lecture timestamp: 00:53:33.

3.6 Acceleration Horizons Are Already Enough

Nothing in this causal structure requires a black hole. A rocket that accelerates uniformly forever follows a Rindler hyperbola in flat spacetime and has a null boundary beyond which no signal can catch it. For that observer there is an acceleration horizon, even though an inertial observer regards the same region as ordinary Minkowski space.

Question from the audience. Is this phenomenon specific to black holes, or does it occur for any horizon?[a]

Response. The same phenomenon occurs for a uniformly accelerated observer. Such an observer has an acceleration horizon and cannot see events beyond it. Locally, the black-hole horizon is not physically different from that case.

[a]Lecture timestamp: 00:55:25.

This is the equivalence principle at work: understand uniform acceleration first, then recognize the same local structure in a gravitational field. In retrospect, one could have anticipated gravitational horizons from acceleration horizons even before black holes were accepted. Einstein himself did not accept the physical formation of black holes, despite having supplied the theory from which their geometry follows. The horizon is boring for the freely falling observer and consequential for the observer who insists on remaining outside. That is a tension between valid descriptions, not yet a paradox.

3.7 Returning to Heat and Entropy

After the break, put on the thermodynamic hat. In the previous construction, a black hole was filled like a bathtub, one drop at a time, except that each drop was a photon carrying roughly one bit. Each bit increased the horizon area by a quantity of order a Planck area. The exact entropy is

$$S_{\mathrm{BH}} = \frac{k_B A}{4\ell_P^2}. \tag{3.24}$$

The lecture jokes that the factor 1/4 could have been avoided had Planck defined his length differently. The serious point is that dimensional analysis fixes the scale but not the numerical coefficient; Hawking's calculation fixes the coefficient.[3]

Temperature feels elementary because fingers and thermometers register it directly. Entropy often feels more mysterious. The lecture reverses that hierarchy: entropy and energy are treated as the basic variables, while temperature is derived from how much energy must change when entropy changes:

$$dE = T\,dS. \tag{3.25}$$

In the lecture's one-bit units, $\Delta S = 1$, so the minimum energy associated with adding or erasing a bit is called T. Erasing carelessly—with the logical equivalent of a sledgehammer—can dissipate arbitrarily more energy. The definition concerns the least dissipation possible when the missing information is transferred into microscopic environmental degrees of freedom. In conventional units, one binary bit carries $\Delta S = k_B \ln 2$, and reversible erasure dissipates at least $k_B T \ln 2$.[4]

3.7.1 The one-photon estimate

Choose a photon whose wavelength is comparable to the Schwarzschild radius:

$$\lambda \sim R_s = \frac{2GM}{c^2}. \tag{3.26}$$

Its energy is, up to the 2π convention distinguishing h from $\hbar$,

$$\Delta E \sim \frac{\hbar c}{\lambda}. \tag{3.27}$$

That is the energy cost of the added bit, so

$$k_B T_H \sim \frac{\hbar c}{R_s} \sim \frac{\hbar c^3}{GM}. \tag{3.28}$$

The heuristic argument cannot determine factors such as 2 and π. The exact Hawking result is

$$\boxed{k_B T_H = \frac{\hbar c^3}{8\pi GM}.} \tag{3.29}$$

Two facts are immediate. First, the temperature vanishes with $\hbar$, so it is a quantum effect. Second,

$$T_H \propto \frac{1}{M}. \tag{3.30}$$

A large black hole is cold. For one solar mass,

$$T_H \approx 6 \times 10^{-8}\ \mathrm{K}, \tag{3.31}$$

which is of the same order as the lecture's 10^{-8} K estimate and far below the cosmic microwave background temperature of about 2.7 K.

[3] Editorial clarification: The Planck length has a standard definition $\ell_P = \sqrt{G\hbar/c^3}$. The factor 1/4 is physical once this convention and the usual thermodynamic entropy are fixed; the remark about redefining it is a classroom joke about conventions.

[4] Editorial clarification: This is the Landauer normalization suppressed by the lecture's choice to measure both entropy and temperature in energy-based natural units.

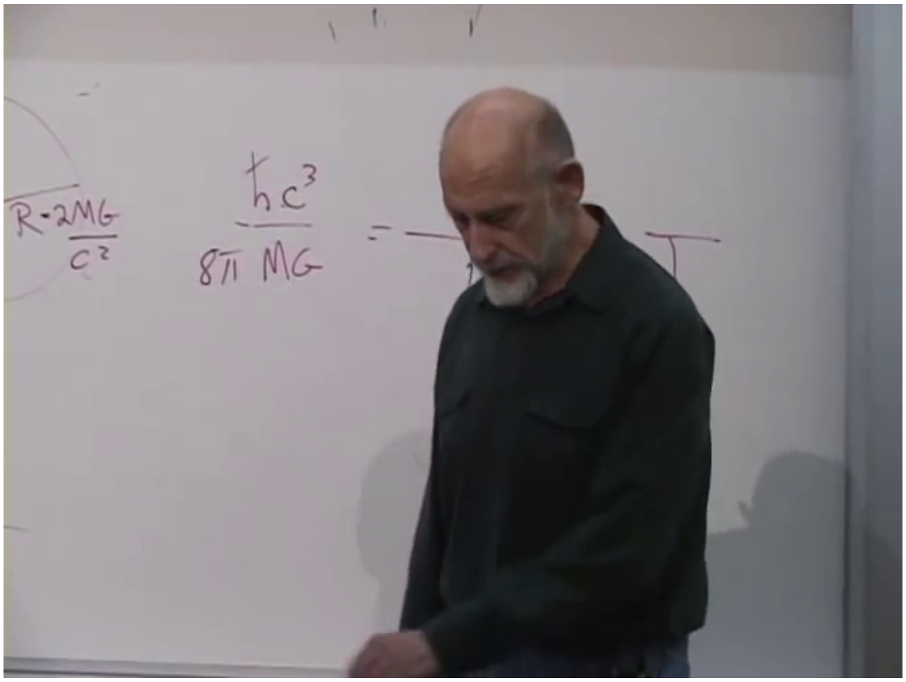

Figure 3.5: The completed temperature board: $R_s = 2GM/c^2$, the one-bit scale $\hbar c/R_s$, and the exact 8π Hawking normalization.

Lecture frame: 01:07:48.000.

3.7.2 Negative heat capacity

Put such a black hole in a warmer environment. It absorbs energy, its mass increases, and Equation (3.29) says that it becomes colder. It then absorbs still more readily. Put it in a colder environment and the reverse runaway begins: it radiates, loses mass, becomes hotter, and radiates faster. Formally,

$$C_{\mathrm{BH}} = \frac{dE}{dT_H} = c^2 \frac{dM}{dT_H} < 0. \tag{3.32}$$

Question from the audience. Is this what is meant by negative specific heat?[a]

Response. Exactly. A self-gravitating system can heat up while losing energy. A star does something analogous: as it radiates, reduced pressure lets it contract, and gravitational compression raises its temperature.

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[a] Lecture timestamp: 01:14:08.

An isolated black hole is not thereby unphysical. In today's universe, a stellar hole is cooler than its surroundings and predominantly absorbs. As the universe expands and the background cools, the balance can eventually reverse and evaporation can dominate. The simple equilibrium with an effectively infinite heat bath is unstable.[5]

[5] Editorial clarification: Boundary conditions matter. A black hole can be stabilized in a finite reflecting cavity or, for appropriate large holes, in anti-de Sitter spacetime; the instability described here is the asymptotically flat, ordinary-bath case.

3.8 The Evaporation Law

For the scaling derivation, imagine being without a reference book and choose the units in which the remembered equations are shortest:

$$c = \hbar = G = k_B = 1. \tag{3.33}$$

These are Planck units. The characteristic scales quoted in the lecture are

$$\ell_P \sim 10^{-35}\ \text{m}, \qquad t_P \sim 10^{-43}\ \text{s}, \qquad m_P \sim 10^{-8}\ \text{kg}. \tag{3.34}$$

The Planck mass is roughly a visible dust mote, whereas the Planck length and time are extraordinarily small. Since $c = 1$, t_P is simply the light-crossing time of ℓ_P.

Treat the hole as a black body, meaning a thermal emitter rather than an object that must remain perfectly dark. It can emit electromagnetic radiation and, in principle, other available species including gravitons. The radiated power is proportional to the emitting area because radiation leaves through the surface. The Stefan–Boltzmann scaling is

$$L = -\frac{dE}{dt} \sim AT^4. \tag{3.35}$$

In Planck units $E = M$, $R_s \sim M$, $A \sim R_s^2 \sim M^2$, and $T \sim R_s^{-1} \sim M^{-1}$. Therefore

$$-\frac{dM}{dt} \sim R_s^2 \left(\frac{1}{R_s}\right)^4 \sim \frac{1}{R_s^2} \sim \frac{1}{M^2}, \tag{3.36}$$

$$-M^2 \frac{dM}{dt} \sim 1, \tag{3.37}$$

$$-\frac{d(M^3)}{dt} \sim 1. \tag{3.38}$$

Ignoring order-one coefficients, the lifetime is

$$\boxed{t_{\text{evap}} \sim M^3.} \tag{3.39}$$

The schematic Stefan–Boltzmann treatment suppresses greybody factors and the number of particle species. For a neutral, nonrotating hole in the simplest massless-emission approximation, restoring constants gives

$$t_{\text{evap}} = \frac{5120\pi G^2 M_0^3}{\hbar c^4}. \tag{3.40}$$

This coefficient is an editorial completion; the lecture is deriving only the robust M_0^3 dependence.

3.8.1 Solar, mountain, and kilogram scales

The Sun has mass $M_\odot \sim 10^{30}\,\text{kg} \sim 10^{38} m_P$. Cubing and multiplying by one Planck time gives the lecture's deliberately rough chain

$$t_{\text{evap}} \sim 10^{114} t_P \sim 10^{71}\ \text{s} \sim 10^{64}\ \text{yr}. \tag{3.41}$$

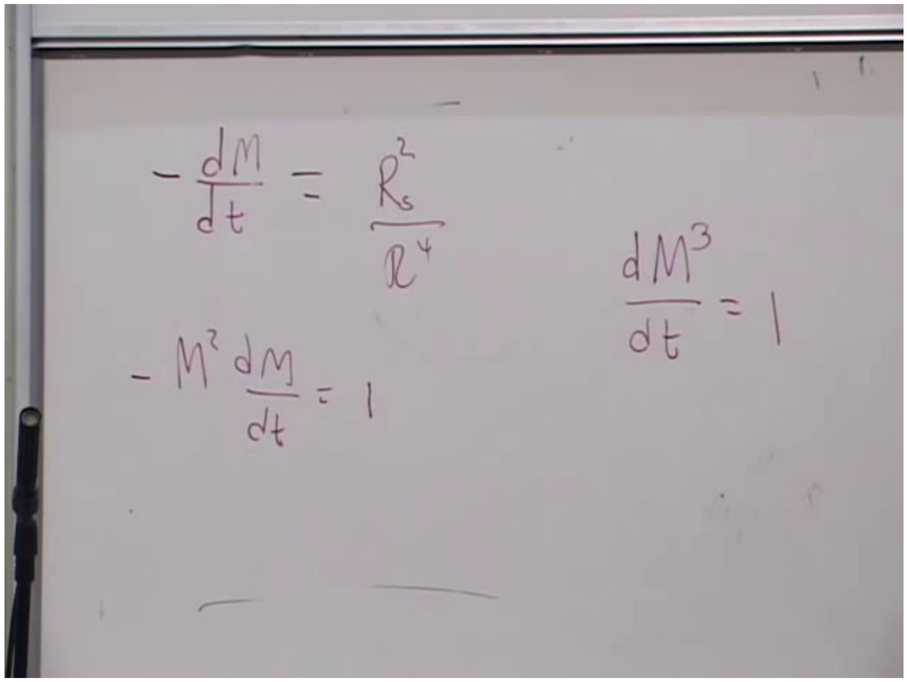

Figure 3.6: The unobstructed evaporation board: luminosity gives $-dM/dt \sim 1/M^2$, hence a uniform decrease of M^3 and a lifetime proportional to the initial mass cubed.

Lecture frame: 01:23:20.000.

Either estimate is fantastically longer than the roughly 10^{10}-year age of the universe. Restoring the coefficient in Equation (3.40) raises the standard solar-mass estimate to about 2×10^{67} years.[6]

The class next tries a *mountain-mass* hole. It settles on the rough classroom value $M \sim 10^{10}\,\text{kg}$, hence $M \sim 10^{18} m_P$, and obtains

$$t_{\text{evap}} \sim 10^{54} t_P \sim 10^{11}\ \text{s} \sim 10^{4}\ \text{yr}. \tag{3.42}$$

This preserves the arithmetic actually carried through at the board. The verbal geometry used to motivate that mass—roughly $100\,\text{km}^3$ of rock—would in fact be much heavier, and the exact lifetime coefficient would also change the result substantially.[7]

Question from the audience. Would a black hole with a mass of only a kilogram be extremely small?[a]

Response. Yes. Since radius is proportional to mass and a solar mass corresponds to a few kilometres, a one-kilogram Schwarzschild radius is about 10^{-27} m. Such a hole would also be extremely hot and short-lived.

[a] Lecture timestamp: 01:29:14.

[6] Editorial clarification: The difference is expected: the blackboard calculation drops all numerical constants and rounds the Planck scales by powers of ten. Its purpose is the mass-cubed scaling, not precision lifetime astronomy.

[7] Editorial clarification: At ordinary rock density, $100\,\text{km}^3$ is of order 10^{14} kg, not 10^{10} kg. For 10^{10} kg, Equation (3.40) gives a lifetime of order 10^6 years. These corrections do not alter the lecture's lesson that lifetime falls as the cube of mass.

Question from the audience. Do these evaporation estimates assume zero ambient temperature?[a]

Response. They apply when the surroundings are cooler than the hole, so that net energy flows outward. A hotter environment instead feeds the black hole. The mass whose Hawking temperature matches the present background is left as an exercise.

[a]Lecture timestamp: 01:29:53.

For orientation, setting $T_H = 2.7\,\mathrm{K}$ gives M of order $10^{22}\,\mathrm{kg}$ and a Schwarzschild radius of order $10^{-4}\,\mathrm{m}$.[8]

3.9 What Must Be Under the Metric?

Question from the audience. What mechanism makes the photons come out of the black hole?[a]

Response. Before giving a microscopic mechanism for the radiation, one must identify the degrees of freedom responsible for the entropy. The thermodynamic argument constrains what must exist without yet displaying those variables explicitly.

[a]Lecture timestamp: 01:30:53.

The analogy is fluid mechanics. Hydrodynamics describes smooth density and velocity fields without following every molecule. Entropy nevertheless reveals that many hidden microscopic configurations underlie the same smooth state. In the same way, classical general relativity describes a smooth metric but does not expose the numerous microscopic degrees of freedom counted by black-hole entropy. The area law and Hawking radiation are clues that Einstein's theory is an effective macroscopic description.

Question from the audience. If the black hole radiates energy, is that the same as saying that it radiates information, so that information swallowed during growth returns during evaporation?[a]

Response. That is the requirement: if the complete evolution conserves information, the outgoing radiation must eventually distinguish the possible states that formed the hole. This must still be reconciled with the causal picture in which matter crosses the horizon.

[a]Lecture timestamp: 01:33:27.

Energy flux by itself is not enough to specify information. A perfectly thermal density matrix can carry energy while forgetting the detailed initial state. In a unitary description, the required information is encoded in fine correlations among the emitted quanta.[9]

[8]Editorial clarification: This evaluates the exercise posed in the lecture; it is not part of the spoken derivation.

[9]Editorial clarification: This distinction makes explicit the modern form of the information problem. It sharpens rather than changes the question raised in the lecture.

3.10 Information Conservation as a Structural Principle

Question from the audience. Is conservation of information truly a fundamental principle?[a]

Response. It is among the most fundamental. If distinct initial states could evolve into exactly the same final state, the laws would erase their own past. The usual conservation laws rely not only on symmetries but also on a reversible Hamiltonian or unitary form of evolution.

[a]Lecture timestamp: 01:34:13.

Imagine, as the lecture does, a fictional fundamental law that simply makes every sliding eraser come to rest. Different initial velocities all end in the same state. The final eraser no longer distinguishes its histories: information has been destroyed. Momentum and mechanical energy also disappear. Real friction does not do this at the microscopic level; momentum and energy pass into the table, air, and thermal degrees of freedom.

Even a perfectly uniform infinite tabletop would possess translation symmetry, yet the fictitious damping law could still make the eraser stop. Translation symmetry alone is therefore not enough to derive momentum conservation from such a non-Hamiltonian rule. Noether's theorem assumes an action principle and the associated reversible dynamics. In the language used here, symmetry plus information-preserving dynamics gives the conservation law.

Question from the audience. Is information conservation more fundamental than causality?[a]

Response. They are different principles at a comparable foundational level. Causality must first be stated precisely—for example, that signals cannot propagate backward outside the permitted causal structure—but neither principle can simply replace the other.

[a]Lecture timestamp: 01:37:00.

Question from the audience. Is the phase-space area preservation discussed in Hamiltonian mechanics the classical version of this idea?[a]

Response. Yes. Liouville's theorem says Hamiltonian flow preserves phase-space volume. A cloud of possible initial states cannot collapse into a smaller volume and erase distinctions. In quantum mechanics the corresponding statement is unitarity: inner products, and hence distinctions between orthogonal states, are preserved.

[a]Lecture timestamp: 01:37:38.

Classically,

$$\frac{d}{dt}\,\mathrm{Vol}_\Gamma = 0, \tag{3.43}$$

for a phase-space region transported by Hamiltonian flow. Quantum mechanically,

$$\begin{aligned} |\psi(t)\rangle &= U(t)\,|\psi(0)\rangle\,, \\ U^\dagger U &= I, \\ \langle\psi(t)\,|\,\phi(t)\rangle &= \langle\psi(0)\,|\,\phi(0)\rangle\,. \end{aligned} \tag{3.44}$$

These are precise forms of the claim that microscopic distinctions survive time evolution.

3.11 Which Way Does Entropy Depend?

The final exchange returns to the information-theoretic entropy. Given probabilities p_i for microscopic alternatives,

$$S = -k_B \sum_i p_i \ln p_i \tag{3.45}$$

in thermodynamic units, or without k_B for dimensionless Shannon entropy.

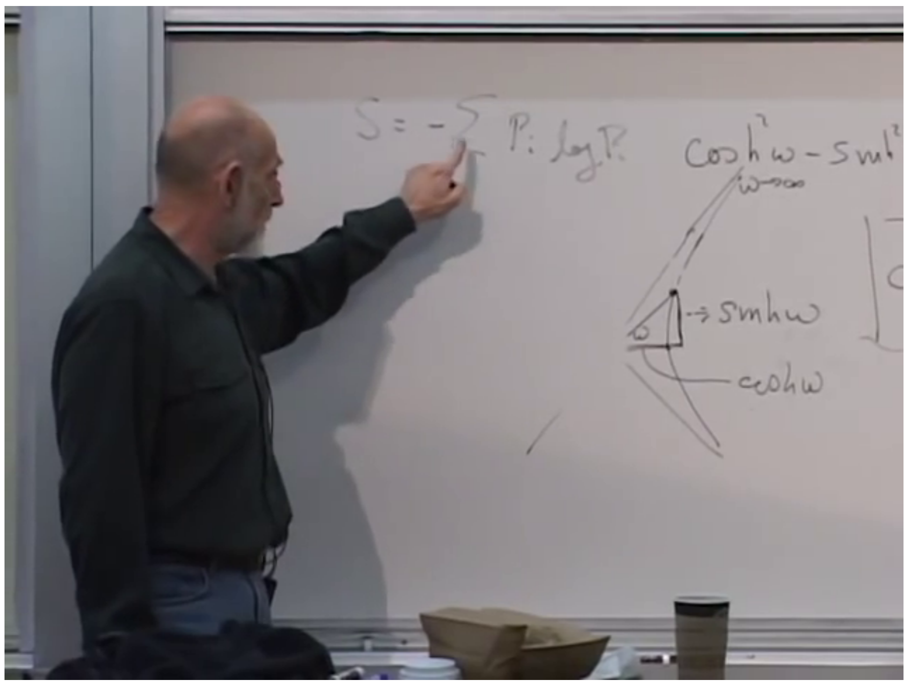

Figure 3.7: The closing board formula $S = -\sum_i p_i \log p_i$, written while distinguishing a probability distribution from the single entropy value computed from it.

Lecture frame: 01:40:32.000.

Question from the audience. Does maximizing the entropy define the probability distribution, rather than the probabilities defining the entropy?[a]

Response. A probability distribution determines one entropy value through Equation (3.45); one number cannot reconstruct all the p_i. Maximizing entropy can select a distribution only after normalization and other constraints, such as fixed mean energy, have also been supplied.

[a] Lecture timestamp: 01:38:33.

That distinction ends the lecture at the right frontier. Smooth geometry has led to a horizon; the horizon has led to thermodynamics; thermodynamics has exposed hidden degrees of freedom; and information conservation now demands that those degrees of freedom evolve without erasing the distinctions that entered the hole. The mechanism has not yet been supplied. The geometry and the bookkeeping have instead made clear what any successful microscopic theory must explain.

Chapter 4

Black Holes, White Holes, and the Global Causal Diagram

Overview. A horizon is locally ordinary and globally decisive. We begin with the near-horizon Schwarzschild metric, follow it outward with the Euclidean cigar analogy, and then ask what Alice and Bob can actually see. The singularity and the time-reversed white-hole region enlarge that local picture into the eternal Schwarzschild spacetime. Penrose compactification makes the causal structure finite and readable. The final step turns from an eternal solution to a black hole made by collapse, where Newton's shell theorem and Birkhoff's theorem supply the pieces to be joined.

4.1 From the Horizon to Infinity

Near a sufficiently large Schwarzschild horizon, curvature is weak over a small laboratory and the radial-time geometry is approximately flat. The convenient radial variable is not the Schwarzschild coordinate r, but the proper distance ρ measured outward from the horizon along a constant-time spatial slice:

$$\rho = 0 \quad \Longleftrightarrow \quad r = 2GM. \tag{4.1}$$

The convenient time coordinate is the dimensionless quantity

$$\omega = \frac{t}{4GM}, \tag{4.2}$$

where $c = 1$. Thus ω is Schwarzschild time measured in units of twice the Schwarzschild radius.

With the mostly-plus convention, the near-horizon radial-time metric is

$$ds^2 \approx -\rho^2 d\omega^2 + d\rho^2. \tag{4.3}$$

The angular two-sphere has been suppressed because the present argument concerns only time and radial distance.

Equation (4.3) has the same relation to Minkowski spacetime that polar coordinates have to a Euclidean plane. For

$$T = \rho \sinh \omega, \qquad X = \rho \cosh \omega, \tag{4.4}$$

one obtains

$$-dT^2 + dX^2 = -\rho^2 d\omega^2 + d\rho^2. \tag{4.5}$$

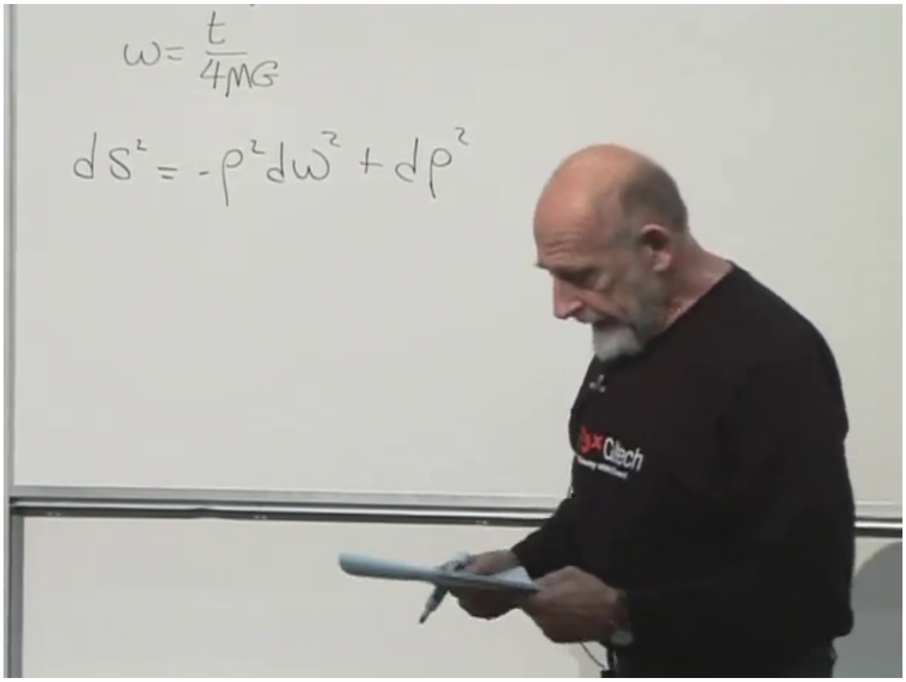

Figure 4.1: The near-horizon metric and the rescaled Schwarzschild time.

Lecture frame: 00:05:46.800.

Lines of constant ρ are hyperbolae $X^2 - T^2 = \rho^2$, while ω runs from the remote past to the remote future. The two null lines $X = \pm T$ meet at the bifurcation point of the reduced diagram. In four dimensions that point represents a two-sphere, and the future and past null sheets form a bifurcate Killing horizon.[1]

4.1.1 The far region

The radial-time part of the Schwarzschild metric is

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2GM}{r}}. \tag{4.6}$$

Far from the hole, $r \gg 2GM$, the metric approaches

$$ds^2 \approx -dt^2 + dr^2. \tag{4.7}$$

There r and ρ differ only by an inessential additive correction, so with $dt = 4GM\, d\omega$,

$$ds^2 \approx -(4GM)^2 d\omega^2 + d\rho^2. \tag{4.8}$$

The radial coefficient is simple in both limits. The time coefficient is what changes:

$$ds^2 \approx \begin{cases} -\rho^2 d\omega^2 + d\rho^2, & \rho \ll GM, \\ -(4GM)^2 d\omega^2 + d\rho^2, & \rho \gg GM. \end{cases} \tag{4.9}$$

[1]Editorial clarification: Calling only the crossing point the bifurcate horizon is useful shorthand in the radial diagram. More precisely, the two horizon branches intersect on a bifurcation two-sphere.

Along a hovering trajectory, $d\rho = 0$. Close to the horizon,

$$d\tau = \rho \, d\omega, \tag{4.10}$$

whereas far away $d\tau \approx 4GM \, d\omega = dt$. For a fixed increment of the exterior time coordinate, less and less proper time elapses on a clock held closer to the horizon.

4.1.2 The cigar

Spacetime is difficult to picture, so replace the timelike direction by an ordinary angular direction. Write

$$\begin{aligned} d\ell^2 &= F(\rho)^2 d\theta^2 + d\rho^2, \\ F(\rho) &\sim \begin{cases} \rho, & \rho \to 0, \\ 4GM, & \rho \to \infty. \end{cases} \end{aligned} \tag{4.11}$$

Near $\rho = 0$, this is the plane in polar coordinates. Far away, the circumference no longer grows with radius:

$$C_\infty = \int_0^{2\pi} 4GM \, d\theta = 8\pi GM. \tag{4.12}$$

The surface therefore rounds off at one end and approaches a cylinder at the other. It looks like a cigar.

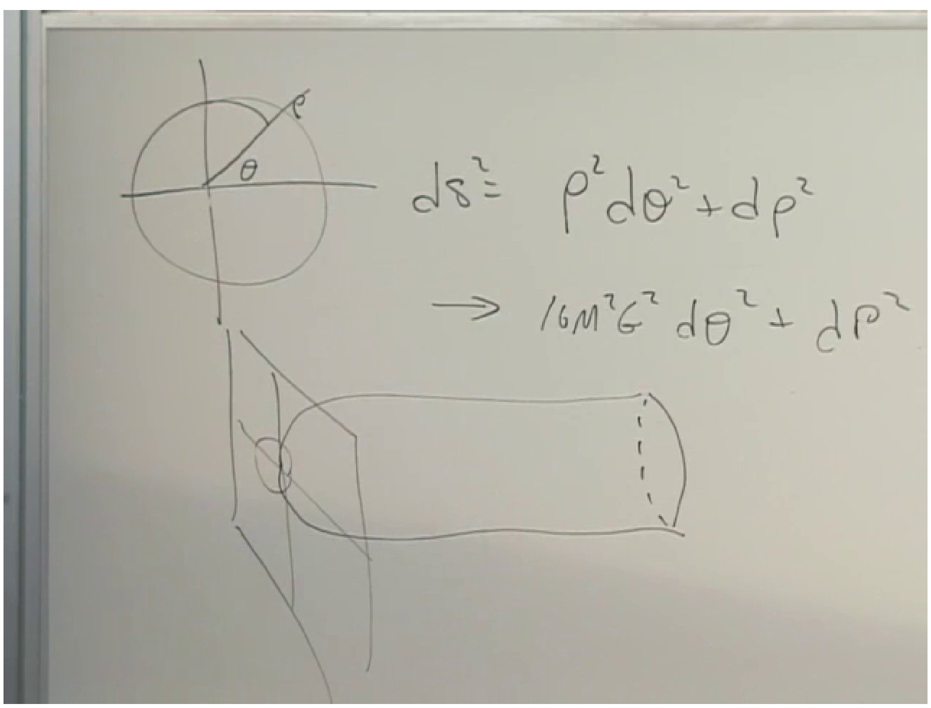

Figure 4.2: The Euclidean comparison: a polar tip joined to an asymptotic cylinder of circumference $8\pi GM$.

Lecture frame: 00:14:05.000.

The larger the mass, the larger the asymptotic girth and the gentler the bend near the tip. A small hole corresponds to a tightly curved cigar; a large hole has a broad, weakly curved tip. This is a picture of the interpolation in Equation (4.9), not an embedding of the Lorentzian black hole itself.[2]

4.2 Reading the Horizon Causally

Outside the hole, $\rho > 0$, a stationary observer follows a constant-ρ hyperbola. Far away, only modest acceleration is needed to resist gravity; close to the horizon, the required proper acceleration grows without bound. Once a worldline crosses the future null boundary, no future-directed causal signal from it can reach the exterior.

Question from the audience. Are light rays straight lines on this diagram?[a]

Response. Radial light rays are drawn at 45°, unless a different convention is stated. An ingoing ray crosses the horizon; an outgoing ray that is already inside cannot cross back to the exterior.

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[a]Lecture timestamp: 00:20:21.

Question from the audience. Does an ingoing light ray stay on the horizon?[a]

Response. No. A ray aimed inward crosses the future horizon and continues into the black-hole interior. Only a horizon generator itself remains on the null boundary.

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[a]Lecture timestamp: 00:21:12.

4.2.1 Bob watches Alice

Let Bob hover outside at fixed radius while Alice falls inward. Bob sees Alice only by receiving outward-going light from her. Suppose she waves periodically according to her own clock. Signals emitted later in her fall take progressively longer Schwarzschild times to reach Bob, and he receives fewer waves per unit time. No signal emitted after horizon crossing can return to him.

The Rindler coordinates expose the limiting behavior. Define the outgoing null coordinate

$$U = T - X = -\rho e^{-\omega}. \tag{4.13}$$

Approaching the future horizon means $U \to 0^-$, which requires $\omega \to +\infty$ for fixed exterior ρ. A finite crossing event for Alice is therefore pushed to infinite exterior time. Successive wave crests arrive with exponentially increasing separation, so their frequency is exponentially redshifted.[3]

At first Bob receives optical light, then infrared, then radio waves of enormous wavelength. The useful statement is not that Alice remains as a permanently visible photograph on the horizon. She becomes dimmer, redder, and finally undetectable.

[2]Editorial clarification: The precise geometric object is obtained by continuing Schwarzschild time to imaginary time. Smoothness at the tip then fixes the Euclidean-time period to $8\pi GM$, the same scale that later determines the Hawking temperature. The lecture uses only the visual analogy here.

[3]Editorial clarification: This null-coordinate calculation supplies the mathematical form of the redshift described qualitatively in the classroom. It uses only the local Rindler approximation.

Question from the audience. Does Bob see Alice frozen at one exact place outside the horizon?[a]

Response. In the classical exterior-coordinate description, images received at later and later times come from events ever closer to the horizon. The limiting image is frozen there, but its radiation is redshifted until it effectively disappears. Quantum black-hole physics has not yet entered this account.

[a]Lecture timestamp: 00:26:31.

Question from the audience. As Alice fades, does she also look compressed?[a]

Response. Yes. Relative motion produces longitudinal Lorentz contraction in the exterior description. The important observational effect here is still the redshift: her internal oscillations appear slower and the received wavelength grows.

[a]Lecture timestamp: 00:28:34.

4.2.2 Alice watches Bob

Alice's proper time remains regular at the horizon. She continues to receive inward-going signals from Bob after crossing, because those signals follow the same causal direction as she does. She cannot reply to the exterior once she is inside. The horizon is one-way, not a wall encountered symmetrically from both sides.

Question from the audience. Does Alice see Bob's light blueshifted as she falls through?[a]

Response. In the picture being used, she sees Bob receding and his signals shifted toward the red; nothing singular occurs at her crossing. The precise frequency shift depends on both worldlines and on when Bob emits, so the invariant statement is that her measured frequency remains finite at a regular horizon.

[a]Lecture timestamp: 00:30:29.

Question from the audience. Does the same causal restriction apply to gravitons?[a]

Response. It applies to genuine gravitational radiation just as it applies to photons. No radiative disturbance emitted from behind the future horizon can reach Bob.

[a]Lecture timestamp: 00:31:15.

Question from the audience. When Alice falls in, does her energy increase the mass and radius of the hole?[a]

Response. Yes. Once her energy is included in the black hole, its mass rises and its Schwarzschild radius $R_s = 2GM$ rises with it. The dynamical formation problem will make that statement geometric.

[a]Lecture timestamp: 00:31:43.

4.2.3 Why the exterior still has gravity

A black hole is dark but not gravity-free. A stationary gravitational field is not a stream of on-shell gravitons continually escaping from the interior. It is the already established exterior geometry. If the source changes, the change in the field propagates causally; a distant observer cannot detect that change before a light signal could arrive.

Question from the audience. If gravitons cannot escape, why does the black hole exert gravity at all?[a]

Response. The static field is already present outside. One may loosely describe it using virtual gravitons, but those are not detectable particles carrying fresh messages from the interior. What is limited by the speed of light is a change in the field, including gravitational radiation.

[a]Lecture timestamp: 00:32:04.

Question from the audience. How can virtual gravitons be reconciled with the statement that a field change travels at light speed?[a]

Response. Virtual quanta are bookkeeping elements in a perturbative calculation, not signal-carrying particles with measurable trajectories. Gauge-invariant disturbances and information propagate causally. The same distinction appears between a static electromagnetic field and electromagnetic radiation.

[a]Lecture timestamp: 00:34:02.

The classroom language that virtual gravitons can exceed the speed of light should therefore not be read as superluminal communication.[4]

4.3 Can Alice Still See Her Feet?

Let Alice fall feet first. Her feet cross the horizon before her head. When she looks down, she never sees the simultaneous event on the foot worldline; light takes time to travel, so she sees her feet at a slightly earlier event. There is a continuous sequence of such past events before and after her head crosses. Nothing locally discontinuous happens merely because one part of her body has passed $r = 2GM$.

The dangerous experiment is to reverse course after the feet are inside. To keep her head outside, Alice must accelerate outward. Her feet, already beyond the future horizon, cannot follow an exterior hovering trajectory. The separation in the accelerated frame grows without bound, and a material body cannot remain rigid across that causal boundary. She must either fall freely with her feet or be torn apart while attempting to hold the rest of herself outside.

This stretching is not a divergent tidal force at a large horizon. It is the consequence of demanding incompatible worldlines for an extended body.[5]

[4]Editorial clarification: Off-shell virtual particles need not satisfy the on-shell dispersion relation, but they are not observables and cannot be used to transmit information. In general relativity the clean description is simpler: the stationary exterior metric exists, while changes of the gravitational field have causal propagation.

[5]Editorial clarification: A sufficiently large Schwarzschild horizon has small tidal curvature. The pathology in this thought experiment comes from the attempted acceleration and rigidity constraint, not from a local curvature singularity at the horizon.

4.4 The Singularity Is in the Future

Return to Equation (4.6). Outside,

$$1 - \frac{2GM}{r} > 0, \tag{4.14}$$

so t is timelike and r is spacelike in Schwarzschild coordinates. Inside,

$$1 - \frac{2GM}{r} < 0, \tag{4.15}$$

and those coordinate roles reverse: decreasing r is a timelike direction. This does not mean that space physically turns into time at the horizon. It means that the Schwarzschild chart degenerates there and that its labels have a different causal character inside.

The genuine singularity is at

$$r = 0. \tag{4.16}$$

It is not a vertical worldline representing a place one might avoid. It is a spacelike future boundary. Once Alice enters the black-hole region, every future-directed timelike or null path ends there. The curvature invariant

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48G^2M^2}{r^6} \tag{4.17}$$

diverges as $r \to 0$, whereas it is finite at $r = 2GM$.[6]

4.5 Time Reversal and the White Hole

The Schwarzschild metric is independent of t, and dt^2 is unchanged by $t \mapsto -t$. Its maximal extension is therefore symmetric under time reversal. Alongside the future singularity and future horizon, the mathematical solution contains a past singularity and a past horizon.

Time-reverse Alice's fall. She now emerges from the past singularity, crosses the past horizon outward, and appears in Bob's exterior. That region is called a white hole. It is not an escape route from the future black-hole interior: the future and past horizons are different null boundaries.

Question from the audience. If Alice cannot cross outward through the black-hole horizon, how can the reversed Alice come out?[a]

Response. She does not cross outward through the future horizon. She crosses outward through the past horizon, exactly as the time reverse of ordinary infall. The future black-hole region still has no outward causal path.

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[a]Lecture timestamp: 00:50:24.

Question from the audience. Must Bob's entire history also be run backward in the white-hole picture?[a]

Response. No. The extended geometry admits a history in which Bob remains in the exterior while Alice emerges from the past horizon. The puzzle is not mathematical consistency; it is the extraordinary initial condition represented by the past singularity.

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[a]Lecture timestamp: 00:52:15.

[6]Editorial clarification: Equation (4.17) is the standard Schwarzschild Kretschmann scalar. It makes invariant the distinction drawn geometrically in the lecture between the regular horizon and the curvature singularity.

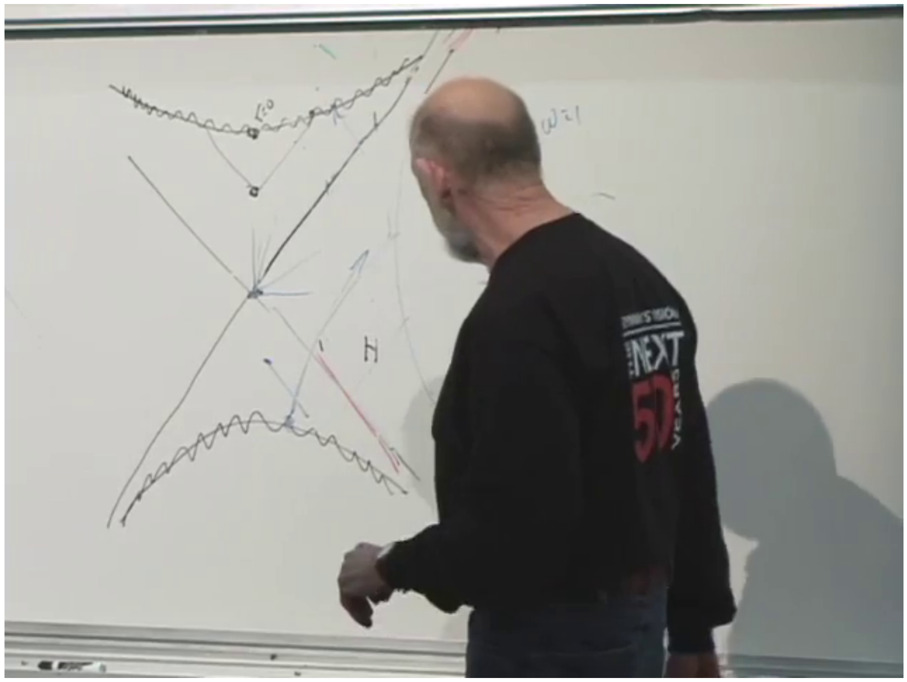

Figure 4.3: The developed causal sketch: crossed horizons, future and past singularity curves, and observer trajectories accumulated during the discussion.

Lecture frame: 01:00:19.818.

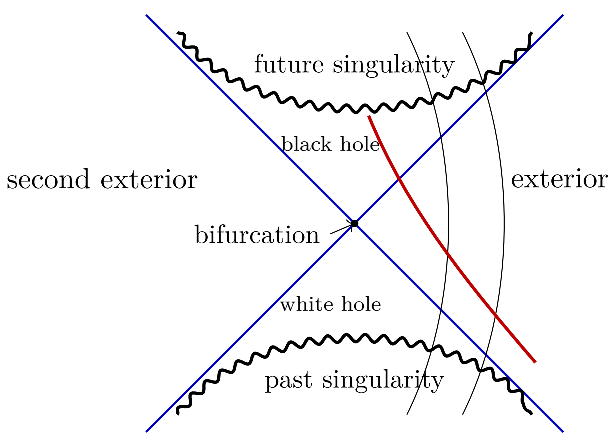

Figure 4.4: A cleaned uncompactified causal picture. Null horizons cross at the bifurcation surface; the spacelike singularities lie in the future and past wedges.

4.5.1 Possible is not typical

Microscopic mechanics is time-reversal invariant to an excellent approximation, yet macroscopic histories have a preferred direction. A bomb easily explodes; scattered fragments almost never reconvene to rebuild it. A ball easily rolls from an unstable summit; a randomly thrown ball almost never comes to rest precisely there. A racked triangle of billiard balls readily scatters, while scattered balls do not spontaneously arrange themselves into the rack. An ice cube melts in hot water, while the water almost never expels a newly assembled ice cube.

The distinction is entropy. A low-entropy macrostate occupies a tiny region of phase space; a high-entropy macrostate contains an enormous number of microscopic arrangements. The equations permit the reversed trajectory, but preparing its initial microstate requires fantastically precise correlations.

Question from the audience. Is not each exact scattered configuration just as unlikely as the exact time-reversed one?[a]

Response. Yes, for individually specified microstates. The asymmetry appears when we compare macrostates. A scattered arrangement includes vastly more microstates than a precisely assembled triangle, so almost every microscopic state compatible with the first description remains macroscopically disordered.

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[a]Lecture timestamp: 00:56:55.

A white hole that emits a fully organized Alice is the gravitational version of such an entropy-lowering history. It is allowed by the reversible equations but requires an extraordinarily special past. The horizon itself carries entropy and hidden microscopic information, so an Alice-sized fluctuation is not logically forbidden; it is overwhelmingly suppressed.

Question from the audience. Would such a fluctuation be quantum mechanical or thermal?[a]

Response. Both kinds can contribute. In an ordinary hot system the dominant language is thermal fluctuation; for a black hole, quantum and thermal effects are intertwined. Either way, the central suppression is entropic.

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[a]Lecture timestamp: 00:59:48.

4.6 Proper Distance and Penrose Compactification

Before drawing the global spacetime, we should make the radial jargon precise. Proper time is what a clock measures along a timelike worldline. Proper distance is what a chain of local meter sticks measures along a chosen spacelike curve. For the Schwarzschild constant-t slice,

$$d\rho = \frac{dr}{\sqrt{1 - 2GM/r}}. \tag{4.18}$$

Thus r is an areal coordinate, not the direct meter-stick distance from the horizon.

Now begin with radial flat spacetime. Keep only t and $\rho \geq 0$. The line $\rho = 0$ is the spatial origin, and a radial light ray entering the origin continues outward on the opposite physical side, which appears as a reflection in the reduced half-plane.

A Penrose diagram maps all finite and infinite values of t and ρ into a finite region while preserving null directions. Its two rules are:

1. compactify the entire spacetime into a finite domain;
2. keep radial light rays at 45°.

Lengths and ordinary angles are distorted; causal relations are not.

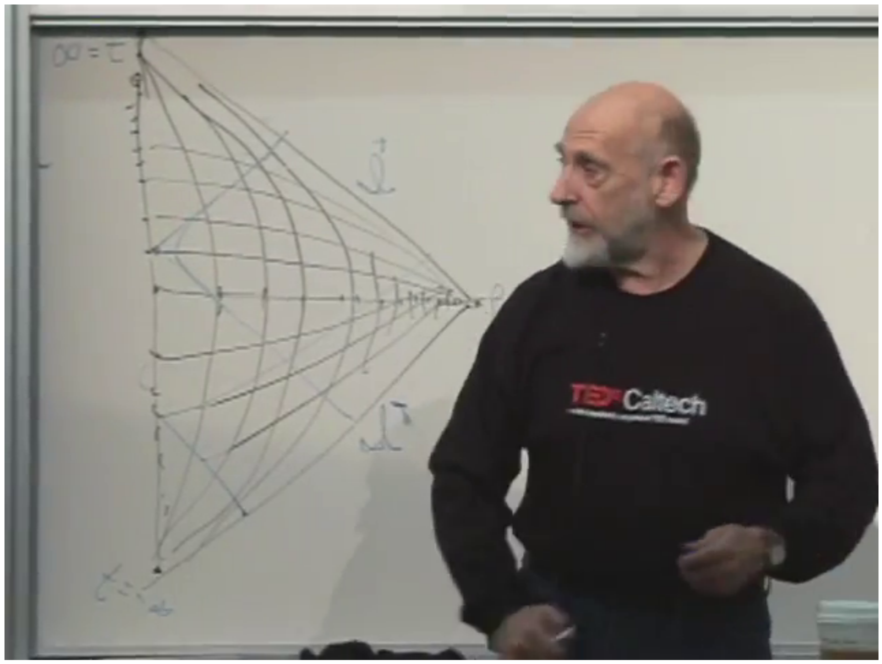

Figure 4.5: The completed compactification grid for radial Minkowski spacetime. Infinite time lies at the upper and lower left endpoints, and infinite radius is compressed to the right tip.

Lecture frame: 01:14:44.040.

The endpoints have distinct causal meanings:

$$\begin{aligned} i^- &: \text{ past timelike infinity,} \\ i^+ &: \text{ future timelike infinity,} \\ i^0 &: \text{ spatial infinity,} \\ \mathcal{I}^- &: \text{ past null infinity,} \\ \mathcal{I}^+ &: \text{ future null infinity.} \end{aligned} \tag{4.19}$$

The symbols $\mathcal{I}^-$ and $\mathcal{I}^+$ are pronounced scri minus and scri plus.

Question from the audience. Should a null ray pass through the corners of every distorted grid cell, and therefore curve slightly?[a]

Response. If the full coordinate grid were drawn exactly, its intersections would be consistent with the null curve. The essential construction is conformal: radial null directions remain at 45°, even though the constant-t and constant-ρ grid lines bend and crowd.

[a]Lecture timestamp: 01:11:08.

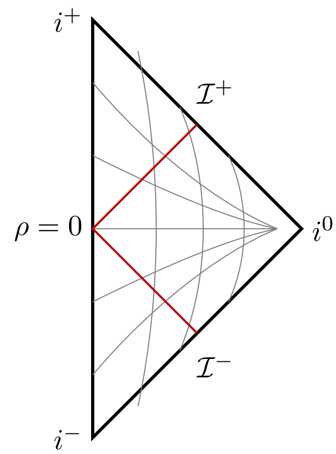

Figure 4.6: Radial Minkowski spacetime in a finite triangle. The red null ray enters from $\mathcal{I}^-$, reaches the origin, and leaves through $\mathcal{I}^+$.

Question from the audience. How is a light ray drawn if it misses the origin?[a]

Response. It reaches a nonzero distance of closest approach and returns outward. Once angular motion is retained, its projected radial path need not be a single straight 45° line.

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[a]Lecture timestamp: 01:11:45.

Question from the audience. Then why is that light ray not at 45°?[a]

Response. The 45° rule was stated for radial light rays. A null ray carrying angular momentum uses part of its motion in the angular directions, so its projection onto the t-ρ plane is not radial.

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[a]Lecture timestamp: 01:12:49.

4.7 The Eternal Schwarzschild Diagram

The compactified diagram has four regions. Its two side diamonds are asymptotically flat exteriors. The upper triangle is the black hole interior and ends at the future spacelike singularity. The lower triangle is the white hole region and begins at the past spacelike singularity. Together these regions form the maximally extended Schwarzschild geometry.

An ingoing radial light ray from $\mathcal{I}^-$ crosses the future horizon and ends on the singularity. Every causal curve in the upper triangle shares that fate. The second exterior is often described as the other mouth of an Einstein–Rosen bridge, but it is not traversable: any attempted causal trip from one exterior to the other enters the black-hole region and reaches the singularity.

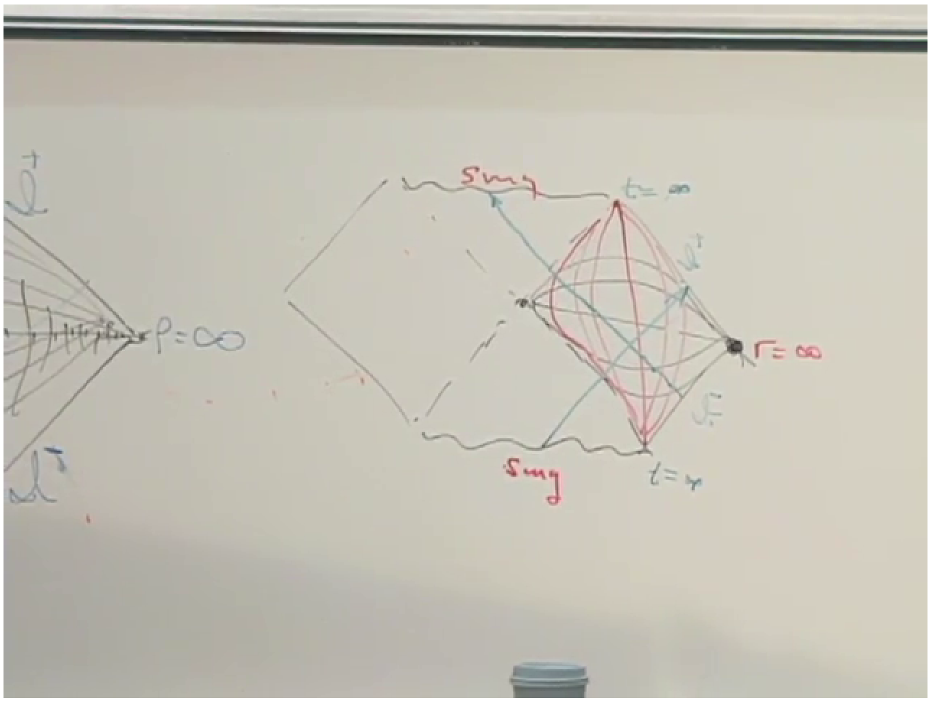

Figure 4.7: The completed eternal Schwarzschild Penrose diagram, with singularities, horizons, null trajectories, and one asymptotic boundary marked on the board.

Lecture frame: 01:21:30.000.

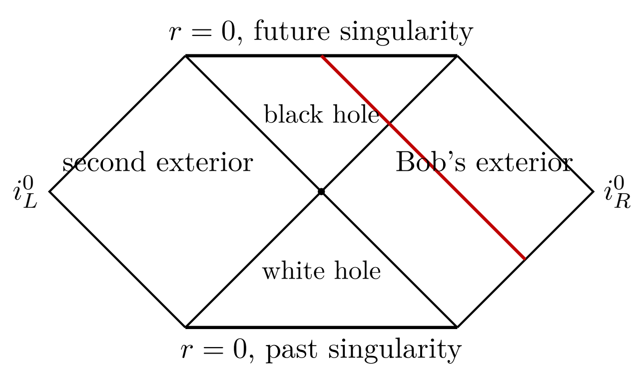

Figure 4.8: The four regions of eternal Schwarzschild. An ingoing radial light ray crosses the future horizon and terminates at $r=0$.

Question from the audience. Are the two exterior worlds connected by a wormhole, and can one signal through it?[a]

Response. The eternal geometry contains an Einstein–Rosen bridge on suitable spatial slices, but no causal signal crosses from one exterior to the other. The bridge pinches off too quickly; even light entering a horizon ends at the future singularity.

[a]Lecture timestamp: 01:22:21.

This diagram describes a hole that has existed for all time. Real astrophysical holes form from collapsing matter. Their causal diagram has no past white-hole singularity and no second asymptotic exterior. To construct that geometry, we need to understand spherical shells.

4.8 Preparing a Black Hole by Collapse

Imagine a diffuse spherical shell of incoming particles. They may be photons: energy gravitates, so a sufficiently concentrated pulse of radiation can form a black hole. Before collapse the central region resembles empty flat spacetime; after enough energy is compressed, the exterior resembles Schwarzschild. The task is to join those regions across the moving shell.

4.8.1 Newton's shell theorem

For a spherically symmetric Newtonian shell of total mass M, Gauss's law for gravity reads

$$\oint \mathbf{g} \cdot d\mathbf{A} = -4\pi G M_{\text{enc}}. \tag{4.20}$$

Spherical symmetry makes $\mathbf{g} = g(r)\hat{\mathbf{r}}$, so

$$g(r) = \begin{cases} 0, & r < R_{\text{shell}}, \\ -\dfrac{GM}{r^2}, & r > R_{\text{shell}}. \end{cases} \tag{4.21}$$

Inside, no mass is enclosed and the gravitational field vanishes. Outside, the shell acts exactly like a point mass M at the center. The potential inside is constant rather than zero, but no local Newtonian force remains.[7]

The conclusion remains true while a perfectly spherical shell contracts, provided its total mass is conserved. Only the location of the boundary between the inside and outside solutions changes.

[7]Editorial clarification: The field-line argument given in the classroom is captured precisely by Gauss's law. The distinction between zero field and constant potential matters when comparing clock normalizations in relativity.

4.8.2 Birkhoff's theorem

The relativistic counterpart is Birkhoff's theorem: every spherically symmetric vacuum region is locally a portion of the Schwarzschild family. For a shell with no mass at its center,

$$ds^2_{\text{inside}} = -dT^2 + dR^2 + R^2 d\Omega_2^2, \tag{4.22}$$

$$ds^2_{\text{outside}} = -\left(1 - \frac{2GM}{r}\right) dt^2 + \frac{dr^2}{1 - 2GM/r} + r^2 d\Omega_2^2. \tag{4.23}$$

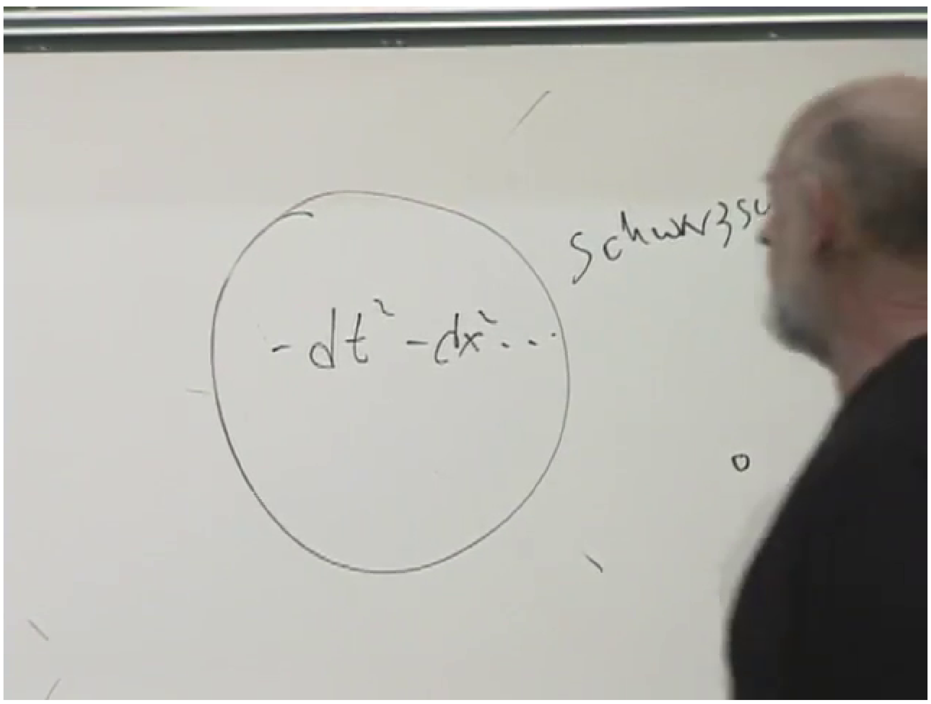

Figure 4.9: The shell-matching board: flat spacetime inside the circle and the Schwarzschild solution outside.

Lecture frame: 01:34:15.000.

Question from the audience. What metric appears outside a spherical shell in general relativity?[a]

Response. The Schwarzschild metric, with mass parameter equal to the shell's total conserved mass. The empty central region is locally Minkowski spacetime.

[a]Lecture timestamp: 01:33:17.

Question from the audience. If the shell contracts, why does the mass parameter not change?[a]

Response. Because total energy is conserved. The shell's radius and kinetic or binding contributions may change, but the conserved exterior mass is fixed for an isolated spherical system. The moving shell changes where the flat and Schwarzschild regions are joined.

[a]Lecture timestamp: 01:34:33.

Flat interior does not mean that an interior clock necessarily has the same normalization as a clock at infinity. Matching across the shell can multiply the interior time coordinate by a constant, and a moving shell requires the corresponding junction conditions.[8]

We now possess the two pieces needed for formation: Minkowski spacetime inside the contracting shell and Schwarzschild spacetime outside it. As the shell moves inward, their interface traces a world tube. Following that tube through the would-be Schwarzschild radius will replace the unphysical past half of Figure 4.8 with a regular pre-collapse region and reveal where the event horizon of a real black hole begins.

[8]Editorial clarification: Birkhoff's theorem fixes each vacuum region locally. The global relation between their coordinates is determined by matching the induced metric and extrinsic curvature across the shell. Those details are deferred, exactly as the collapse construction is deferred to the next lecture.

Chapter 5

Building a Black Hole and the Near-Horizon Puzzle

Overview. We shall build a black hole from three pieces: the causal diagram of flat spacetime, the Schwarzschild diagram, and Birkhoff's theorem for a spherical shell. The construction reveals that an event horizon is a global boundary and can begin in a region that is still locally flat. We then turn from causal geometry to entropy and Hawking temperature. A thermometer held near the horizon becomes arbitrarily hot, while a freely falling observer expects no local catastrophe. The closing argument asks what an outside experiment can actually establish without creating the disturbance it is meant to detect.

5.1 A Black Hole at the Temperature of Empty Space

There was supposed to have been homework: build a black hole and bring it to class for show and tell. No one had one to display, but a calculation found through Wolfram supplied a gentler opening problem. What radius and mass would give a black hole a Hawking temperature near the 3 K temperature of the cosmic microwave background?

Question from the audience. What was calculated on the website?[a]

Response. The radius of a black hole whose temperature is about 3 K. The point is not a precision calculation; it is an estimate of the physical scale.

[a]Lecture timestamp: 00:00:39.

Such a hole would momentarily have the same temperature as its surroundings, but the equilibrium would be unstable. A Schwarzschild black hole has negative heat capacity. If it becomes slightly hotter than the bath, it emits, loses mass, and becomes hotter still. If it becomes slightly colder, it absorbs, gains mass, and becomes colder still.[1]

The estimate begins with the characteristic wavelength of 3 K radiation.

[1]Editorial clarification: This is the precise stability argument behind the deliberately loose classroom wording. A canonical heat bath cannot provide stable equilibrium for an isolated Schwarzschild black hole.

Question from the audience. What wavelength belongs to radiation at about three kelvin?[a]

Response. The room offered a centimeter, 21 centimeters, and even 10 meters before settling on the microwave range. The useful order-of-magnitude answer for the argument was about a centimeter, small enough for the jokes about a microwave oven and a breadbox.

[a] Lecture timestamp: 00:02:03.

The typical Hawking quantum has wavelength comparable, up to numerical factors, with the Schwarzschild radius:

$$\lambda_{\mathrm{H}} \sim R_s, \qquad R_s = \frac{2GM}{c^2}. \tag{5.1}$$

Taking the classroom value $\lambda_{\mathrm{H}} \sim 1\,\mathrm{cm}$ gives $R_s \sim 1\,\mathrm{cm}$. Since a solar-mass black hole has a radius of a few kilometers and R_s is proportional to M, the corresponding mass is roughly

$$M \sim 10^{-5} M_\odot, \tag{5.2}$$

which is planetary in scale. This is why the lecture arrives at an Earth-like estimate.

Question from the audience. Does that agree with the numerical result from Wolfram?[a]

Response. The reported result was nearer the mass of the Moon. That sounded small compared with the rough estimate, but either answer makes the intended point: the object is not something one could carry in a pocket.

[a] Lecture timestamp: 00:05:53.

Keeping the constants explains the discrepancy. From

$$T_H = \frac{\hbar c}{4\pi k_B R_s}, \tag{5.3}$$

one finds, at $T_H = 3\,\mathrm{K}$,

$$R_s \simeq 6.1 \times 10^{-5}\,\mathrm{m}, \qquad M \simeq 4.1 \times 10^{22}\,\mathrm{kg}. \tag{5.4}$$

That is of lunar rather than terrestrial mass. The microwave wavelength, the factor 4π, and the choice of which black-body wavelength to call typical account for the gap between the quick estimate and the numerical value.[2]

The arithmetic has served its purpose. We now want to build a black hole, mathematically and in imagination. The full field equations are too hard for the present purpose. Pictures will supply the basic facts.

5.2 The First Working Part: Flat Spacetime

Use polar coordinates centered on an observer and suppress the angles. The reduced coordinates are (t, r), with $r \geq 0$. A point in this reduced diagram is not a point of four-dimensional spacetime; except at $r = 0$, it represents an entire two-sphere S^2.

[2] Editorial clarification: Equation (5.4) restores the numerical factors omitted in the lecture's dimensional estimate. The estimate itself is retained because it is the route used in class.

Radial light rays move at 45°. An ingoing ray reaches the origin, and an outgoing ray leaves it along the other null direction. The infinite half-plane can be compressed into a finite Penrose wedge by a conformal transformation. The transformation is not unique. Its essential properties are that infinity is brought to a finite boundary and radial null directions remain at 45°. In the language of the blackboard, we smush the diagram.

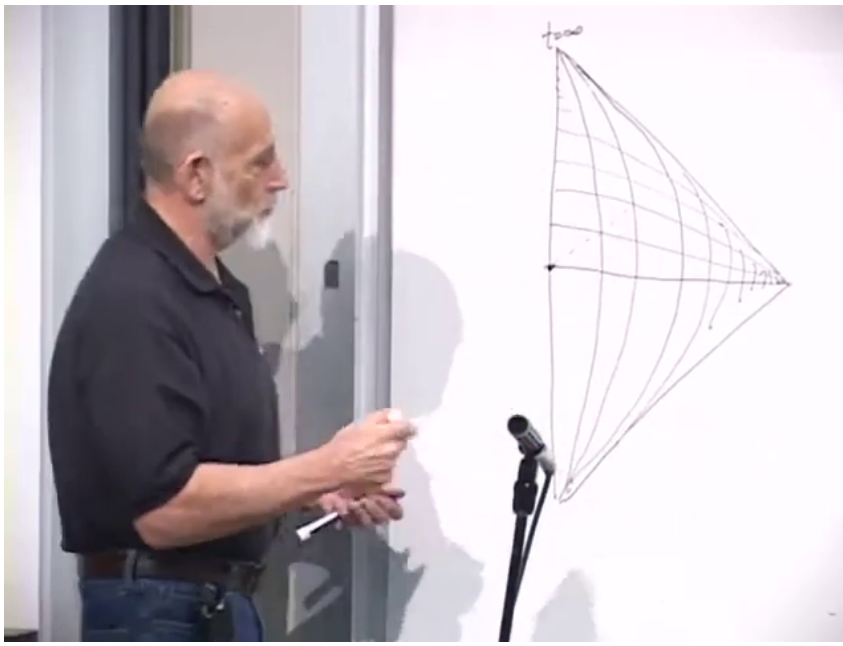

Figure 5.1: The upper half of compactified radial Minkowski spacetime. The vertical edge is $r = 0$, while the curve families crowd toward compactified infinity.

Lecture frame: 00:11:41.560.

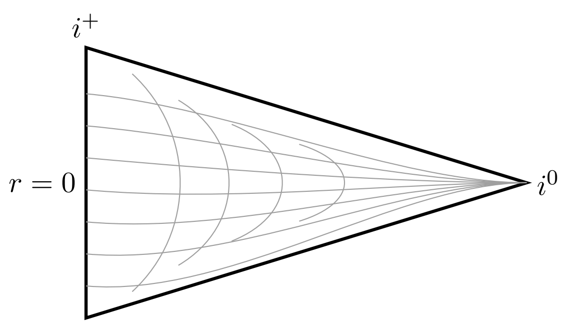

Figure 5.2: A clean version of the upper wedge. The curves are qualitative constant-r and constant-t families, not an exact coordinate grid.

Worldlines at fixed r, vertical before compactification, bow and accumulate near future timelike infinity. Constant-t slices likewise crowd toward spatial infinity. The crowding is a coordinate effect, not a physical compression of matter.

The 45° rule applies directly to radial light. A ray with angular momentum may miss the origin, reach a nonzero distance of closest approach, and return outward. Far from the origin its projected motion looks almost radial; only near closest approach does the reduced (t, r) trajectory visibly bend. This completes the first working part.

5.3 The Second Working Part: Eternal Schwarzschild

The maximally extended Schwarzschild solution has two exterior diamonds, a future black-hole region, and a past white-hole region. Its upper and lower wavy boundaries are the future and past singularities. For a black hole made by collapse, most of this structure will soon be discarded, but the ordinary right exterior is exactly the piece we need.

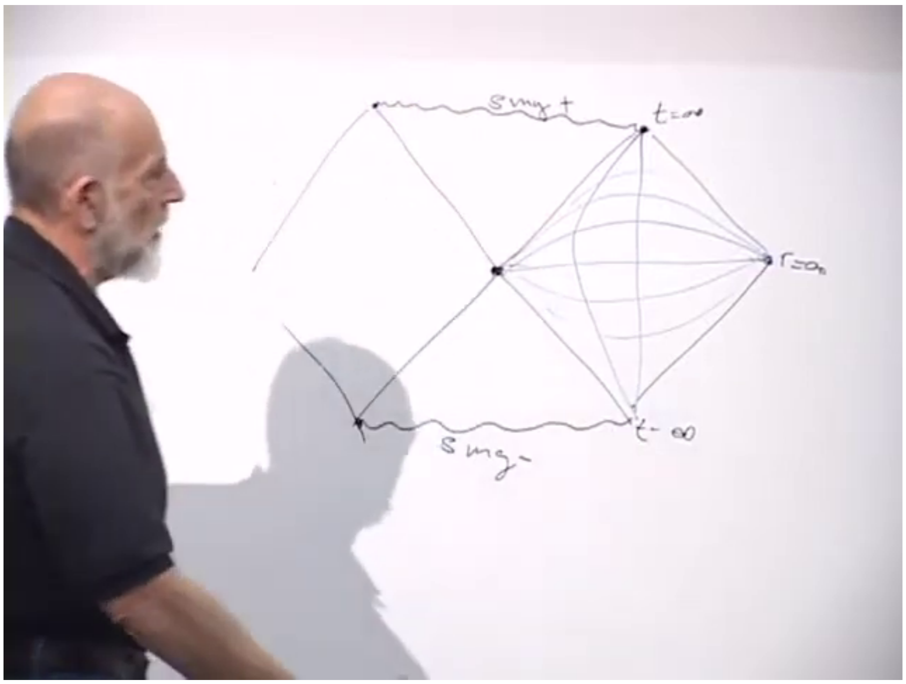

Figure 5.3: The eternal Schwarzschild diagram, with the ordinary exterior and its schematic coordinate grid on the right.

Lecture frame: 00:16:55.975.

Far from the hole, the exterior looks increasingly like compactified flat space. Lines such as $r = 6GM$ and $r = 100GM$ approach the behavior of ordinary constant-radius worldlines. The inner boundary of the exterior is at

$$r = 2GM \tag{5.5}$$

in units $c = 1$.

Near the center, the comparison with flat space fails. A radial ray in Minkowski space reaches $r = 0$ and re-emerges on the opposite physical side. A future-directed ray inside Schwarzschild does not bounce. It ends on the spacelike singularity at $r = 0$. That causal difference is what the collapse construction must preserve.

5.4 The Theorem That Lets Us Join the Pieces

The missing ingredient first appears in Newtonian gravity. For a spherically symmetric mass distribution, Gauss's law gives

$$\oint \mathbf{g} \cdot d\mathbf{A} = -4\pi G M_{\text{enc}}, \tag{5.6}$$

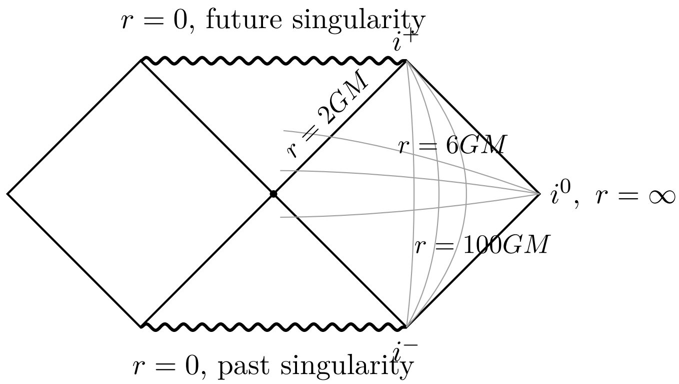

Figure 5.4: The right exterior resembles the flat wedge at large radius. Its inner null edge is $r = 2GM$ when $c = 1$; the other labeled radii are schematic examples.

and therefore

$$\mathbf{g}(r) = -\frac{GM_{\text{enc}}(r)}{r^2}\,\hat{\mathbf{r}}. \tag{5.7}$$

Matter outside the sphere of radius r contributes no net field there. For a hollow shell of total mass M,

$$\mathbf{g} = \begin{cases} 0, & r < R_{\text{shell}}, \\ -\dfrac{GM}{r^2}\hat{\mathbf{r}}, & r > R_{\text{shell}}. \end{cases} \tag{5.8}$$

Birkhoff's theorem is the relativistic replacement. Every spherically symmetric vacuum region is locally a Schwarzschild region with a constant mass parameter. The empty interior of a spherical shell has mass parameter zero and is locally Minkowski; the exterior has mass parameter M and is Schwarzschild:

$$\begin{aligned} M_{\text{enc}} &= 0 && \text{inside the shell,} \\ M_{\text{enc}} &= M && \text{outside the shell.} \end{aligned} \tag{5.9}$$

The statement remains useful while the shell moves, because the vacuum regions on either side remain spherically symmetric.

[3]

For a laboratory image, imagine a sphere densely covered with synchronized lasers, as at a large inertial-confinement facility, all fired inward at once. In the idealization there is no apparatus, only a perfectly spherical shell of light. The shell carries energy E, so the exterior mass parameter is

$$M = \frac{E}{c^2}, \tag{5.10}$$

or simply $E = M$ in natural units. Because the shell moves at light speed, it is an ingoing null line on the reduced diagram. Each point on that line represents a whole spherical wavefront, not one photon trajectory.

[3]Editorial clarification: Flat inside is a local statement. A static shell can change the normalization of the interior time coordinate even though the interior curvature vanishes. For a moving or null shell, the two vacuum metrics must be matched across the shell; the idealized construction is a thin-shell junction, closely related to the ingoing Vaidya geometry. None of these qualifications changes the causal cut-and-paste argument used here.

5.5 Sewing the Collapse Spacetime

Start with flat spacetime and draw the ingoing null shell. The region inside the shell is correct: Birkhoff's theorem says it is flat. The region outside is wrong, because the exterior must already know the shell's total mass through its Schwarzschild geometry.

Now draw the same shell in the eternal Schwarzschild diagram. There the region outside the shell is correct. The white hole, the second exterior, and the inappropriate part of the black-hole interior are wrong for a collapse beginning in ordinary empty space, so erase them.

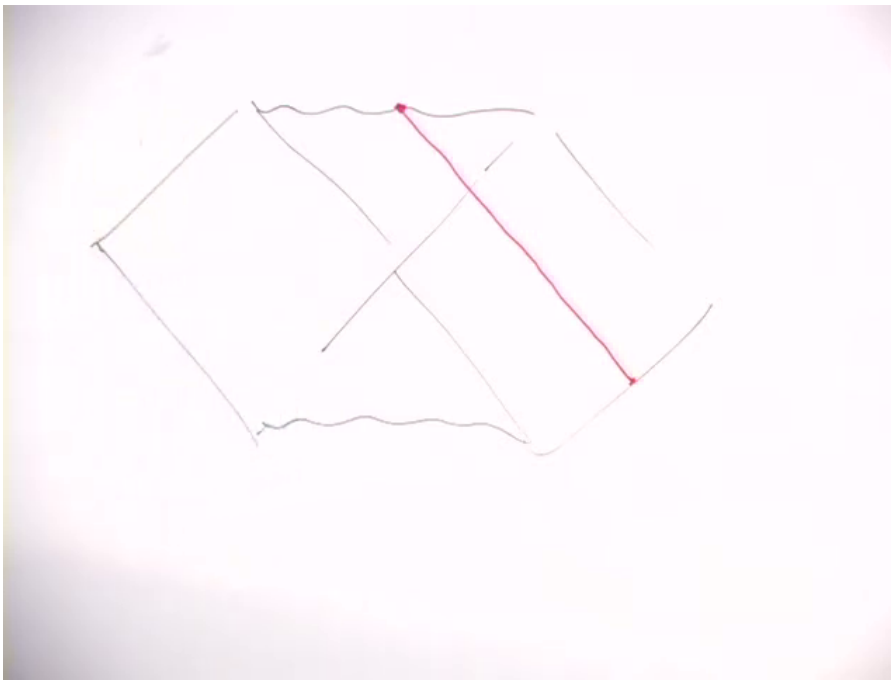

Figure 5.5: The null shell, in red, drawn across the Schwarzschild diagram while the correct exterior portion is being isolated.

Lecture frame: 00:32:00.000.

The shell reaches $r = 0$ in the flat piece. In the Schwarzschild piece it reaches the beginning of the future singularity. These are the same endpoint. Pasting the flat interior to the Schwarzschild exterior along the shell gives the simplest spacetime of black-hole formation.

Question from the audience. At the upper-left joining point, is the flat-space endpoint really the same as the point on the Schwarzschild side?[a]

Response. Yes. The flat-space shell shrinks to $r = 0$, and the Schwarzschild singularity is also at $r = 0$. The two points being identified have the same geometric meaning: the two-sphere has shrunk to zero size.

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[a]Lecture timestamp: 00:34:50.

This construction answers a question left by the eternal solution. A black hole made from collapse has no past white hole and no second asymptotic universe. It still has a future singularity.

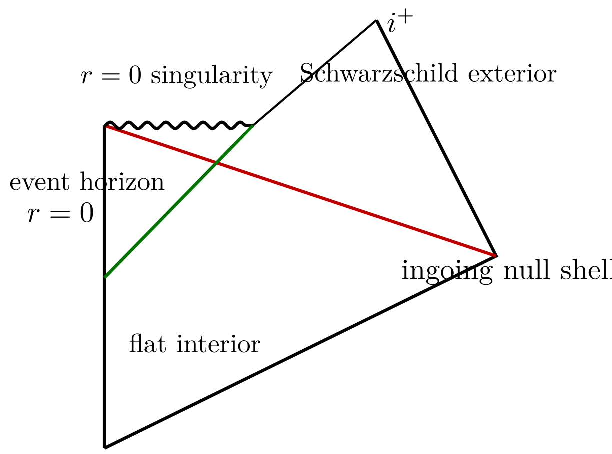

Figure 5.6: The collapse geometry. The red shell joins a flat interior to a Schwarzschild exterior. The green event horizon begins earlier in the flat region and ends at the future boundary of the black-hole region.

5.6 The Horizon Is a Global Boundary

The old label $r = 2GM$ identified the horizon in the stationary Schwarzschild geometry, but it is not the definition needed in a dynamical spacetime.

Definition 5.1. The future event horizon is the boundary of the set of events that can send a future-directed causal signal to future null infinity.

On the collapse diagram, trace the outgoing null ray that just fails to escape. Rays just outside it reach infinity; rays just inside it end at the singularity. That separator is the true horizon.

The startling feature is now unavoidable. The horizon extends backward into a region where the shell has not yet arrived. The geometry there is locally flat, and an observer crossing that line detects no curvature pulse, wall, or flash. Nevertheless, the observer is already unable to reach infinity, because the future shell will close every possible escape route. Event horizons are teleological in precisely this global sense: locating one requires knowledge of the spacetime's entire future.[4]

The drain analogy isolates the causal idea. Imagine a rower near a plugged drain in a motionless lake. If the plug remained forever, the rower could leave. If someone is certain to pull it, the later current may make escape impossible even though no water is moving yet. A frozen Niagara Falls makes the same point: calm water does not guarantee a future route to safety.

Question from the audience. Is the plug pulled because the spherical shell of light becomes more and more concentrated?[a]

Response. The concentration creates the black hole, but the drain story is being used for a narrower purpose: one can already be beyond the point of no return before anything locally dramatic occurs.

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[a]Lecture timestamp: 00:43:13.

[4]Editorial clarification: No local experiment can determine that a smooth event horizon has just been crossed. Other notions, such as apparent or trapping horizons, are more local and need not coincide with the event horizon during collapse.

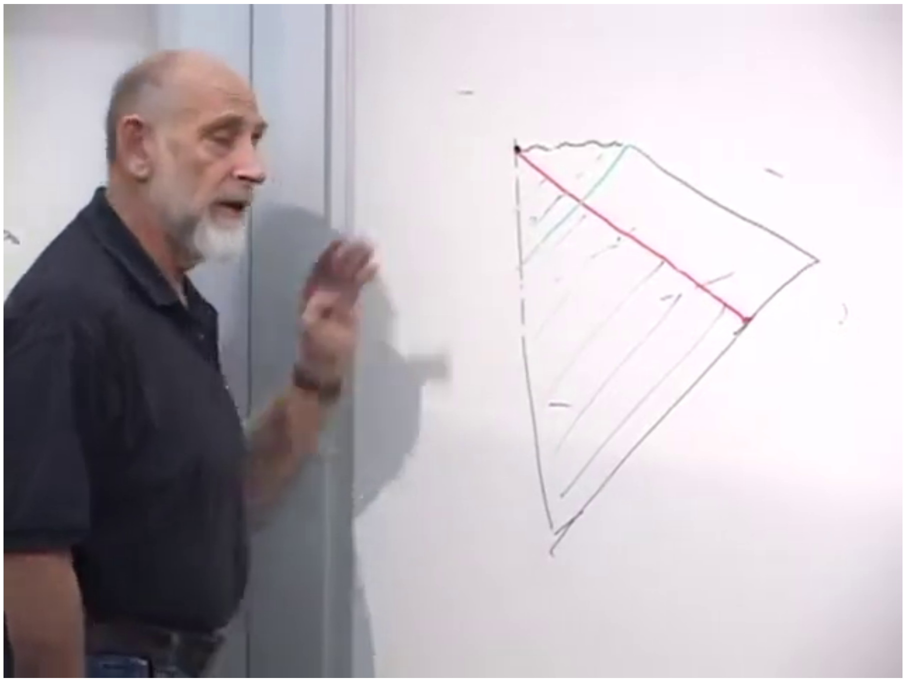

Figure 5.7: The completed board construction. The incoming shell is red and the event horizon is green; part of the green line lies in the still-flat region below the shell.

Lecture frame: 00:39:10.000.

5.6.1 A horizon made of growing spheres

Every point on the reduced horizon represents an S^2. At the horizon's first point that sphere has zero area. Moving upward along the horizon, the sphere grows until its areal radius reaches

$$R_s = \frac{2GM}{c^2}. \tag{5.11}$$

It remains at that size for a stationary hole and grows again if more energy falls in.
[5]

5.6.2 What the distant observer can reconstruct

The long discussion that follows separates an observer already inside the global horizon from the history by which that observer came to be there.

[5]Editorial clarification: In the idealized spherical collapse diagram, the horizon generators meet at a formation point, or crease, on $r = 0$. It is this limiting two-sphere that has zero area; the statement should not be read as a finite Schwarzschild horizon suddenly shrinking to a point.

Question from the audience. What is the status of the person trapped before the shell arrives? Is that person outside the horizon from the distant observer's point of view?[a]

Response. No. Once the worldline is behind the event horizon, its future reaches the singularity and no new signal from it reaches the distant observer. The person did not begin existence there, however. Earlier particles, ancestors, or other precursor energy entered from a region that was visible.

[a]Lecture timestamp: 00:44:03.

The distant observer may therefore see the precursor matter approaching the horizon without ever receiving a signal from beyond it. There is no contradiction in seeing through the incoming shell: it is made of light, and the observer can also see the shell itself and infer that a black hole has formed.

Question from the audience. Does the line of sight pass through the collapsing shell, and can the outside observer still know that the hole formed?[a]

Response. Yes. The observer sees the incoming shell and can receive light whose path crosses it while that light is still able to escape. What looks strange is that the precursor matter approaches the newly determined horizon even before the shell reaches the center.

[a]Lecture timestamp: 00:46:51.

Question from the audience. Does this continue the rule that, from outside, matter looks as though it has been compressed onto the horizon?[a]

Response. Yes. In the exterior description it remains associated with the horizon, becoming ever more redshifted, until the later quantum process of black-hole evaporation changes the story.

[a]Lecture timestamp: 00:47:49.

This outside description motivates the next subject. If information appears plastered onto a two-dimensional surface, it is less surprising that black-hole entropy scales with that surface's area.

5.7 Entropy and the Temperature Seen from Infinity

Near a large horizon we may replace the spherical surface by a local plane, just as we use a flat-Earth approximation in a small laboratory. Two facts from the earlier discussion are then brought together:

$$\begin{aligned} S_{\mathrm{BH}} &\propto A_H, \\ \Delta A_H &\sim \ell_P^2 \quad \text{per added bit, schematically.} \end{aligned} \tag{5.12}$$

The mnemonic of one bit per Planck area expresses the area law; it is not a microscopic derivation.

The Hawking temperature measured far from a Schwarzschild black hole is

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}. \tag{5.13}$$

With $R_s = 2GM/c^2$,

$$T_H = \frac{\hbar c}{4\pi k_B R_s}. \tag{5.14}$$

Thus larger holes are colder.

Question from the audience. Can temperature depend on position?[a]

Response. In ordinary conversation it plainly can: New York and Stanford need not have the same temperature. The sharper question concerns a system in gravitational thermal equilibrium. There, local thermometer readings can vary with height even though the redshifted equilibrium temperature is constant.

[a]Lecture timestamp: 00:52:43.

Temperature is read through local kinetic energies. In a dilute atmosphere above a heated Earth, molecules lose kinetic energy as they climb and gain it as they fall. Photons are redshifted upward and blueshifted downward. A low thermometer and a high thermometer therefore need not register the same local energy scale.

For a static spacetime, the equilibrium relation is

$$T_{\mathrm{loc}}(r)\sqrt{-g_{tt}(r)} = T_\infty, \tag{5.15}$$

the Tolman law. The Hawking temperature in Equation (5.13) is T_∞, the reading normalized at infinity.

5.8 The Hot Static Atmosphere Near the Horizon

Let ρ be proper distance outward from the horizon. Close to a large Schwarzschild horizon, Equation (5.15) becomes

$$k_B T_{\mathrm{loc}}(\rho) \simeq \frac{\hbar c}{2\pi\rho}, \tag{5.16}$$

or, in units $c = \hbar = k_B = 1$,

$$T_{\mathrm{loc}}(\rho) = \frac{1}{2\pi\rho}. \tag{5.17}$$

A tiny thermometer held on a cable at fixed ρ reports an increasing temperature as it is lowered. An atom maintained in that static atmosphere becomes excited, then ionized, then dissociated; at still larger local energies even nuclei and hadrons cease to remain intact. Lowering a person feet first gives the memorable diagnosis: a hot foot.

The word *held* is essential. This divergence belongs to a detector kept stationary by acceleration. It describes the thermal atmosphere redshifted relative to the exterior time. A freely falling detector follows a different trajectory and, for a sufficiently large hole in a regular quantum state, need not encounter a divergent bath at the horizon.[6]

The ionization threshold makes the conflict concrete.

Question from the audience. How much energy is required to ionize hydrogen?[a]

Response. The class gives 13 electron volts, correctly identifying the 13.6 eV ground-state ionization energy. The accompanying estimate of a few thousand kelvin is too small, but only the existence of a finite threshold is needed for the argument.

[a]Lecture timestamp: 01:06:53.

[6]Editorial clarification: This distinction separates the Tolman and acceleration temperatures seen by static observers from the regular local experience of a freely falling observer. The lecture deliberately turns that contrast into an operational puzzle rather than resolving it by terminology alone.

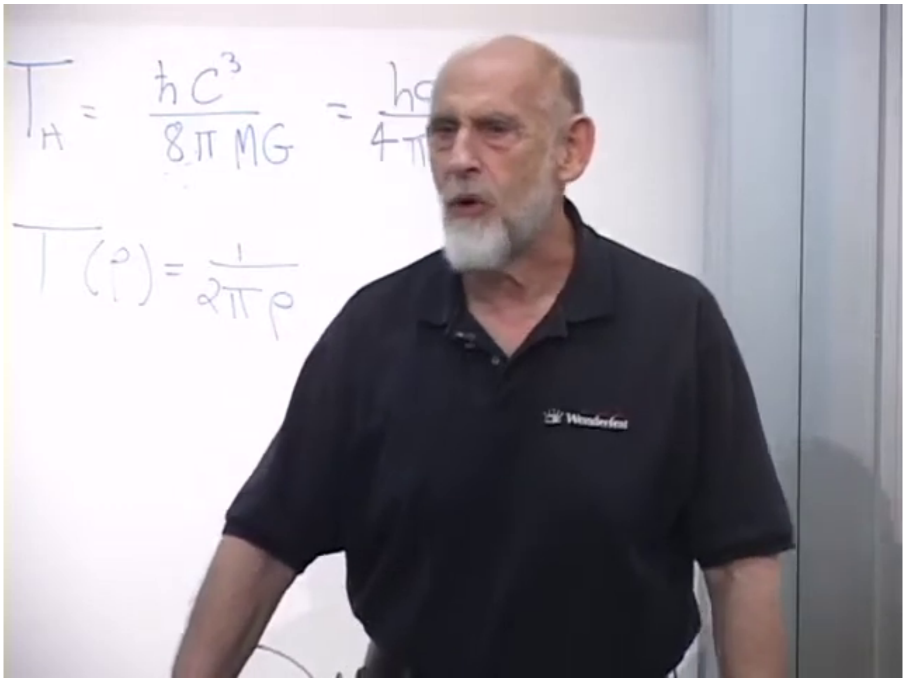

Figure 5.8: The asymptotic Hawking temperature and the near-horizon local temperature $T(\rho) = 1/(2\pi\rho)$ on the board.

Lecture frame: 01:02:40.000.

Indeed,

$$\frac{13.6\,\mathrm{eV}}{k_B} \simeq 1.58 \times 10^5\,\mathrm{K}. \tag{5.18}$$

There is therefore some small proper distance ρ_{ion} at which the static temperature reaches the ionization scale. A static extrapolation says the atom should already be ionized there, outside the horizon. A freely falling atom, by contrast, expects no local wall and can cross the horizon in finite proper time. Can an outside observer perform an experiment that decides between these descriptions?

5.9 A Heisenberg Microscope

Before applying the question to a black hole, recall the photon microscope. To locate a particle with resolution Δx, the probe wavelength must obey

$$\lambda \lesssim \Delta x. \tag{5.19}$$

One photon then carries momentum of at least

$$p_\gamma \gtrsim \frac{\hbar}{\Delta x}. \tag{5.20}$$

Scattering that photon creates a recoil of the same order. If one first measures position and immediately checks velocity, the second measurement sees a state changed by the first. The experiment creates the condition it was intended to exclude.

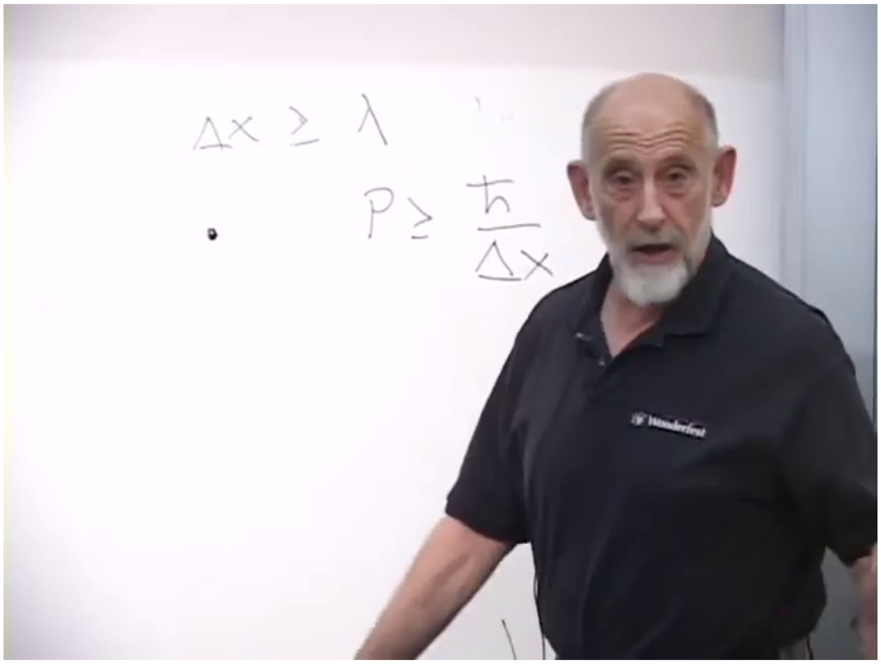

Figure 5.9: The microscope estimates $\Delta x \gtrsim \lambda$ and $p_\gamma \gtrsim \hbar/\Delta x$.
Lecture frame: 01:11:50.000.

This is not merely a limitation of classical optics. Classical waves could be made arbitrarily weak; quantum mechanics requires at least one quantum in a successful detection event. Improving the spatial resolution therefore forces a finite momentum transfer.

5.10 The Near-Horizon Experiment

Now let an atom fall through the layer beginning at ρ_{ion}. Relative to the static frame it is moving inward at nearly c, so the available interval before horizon crossing is of order

$$\Delta\tau \sim \frac{\rho_{\text{ion}}}{c}. \tag{5.21}$$

After crossing, no result can be sent back to the exterior. The measurement must distinguish an intact atom from an ionized one during this short interval.

Resolving the atom within the layer requires

$$\lambda \lesssim \rho_{\text{ion}}. \tag{5.22}$$

The probe energy is consequently bounded by

$$E_\gamma = cp_\gamma \gtrsim \frac{\hbar c}{\rho_{\text{ion}}}. \tag{5.23}$$

Apart from the factor 2π, this is the same scale as Equation (5.16):

$$E_\gamma \gtrsim 2\pi k_B T_{\text{loc}}(\rho_{\text{ion}}). \tag{5.24}$$

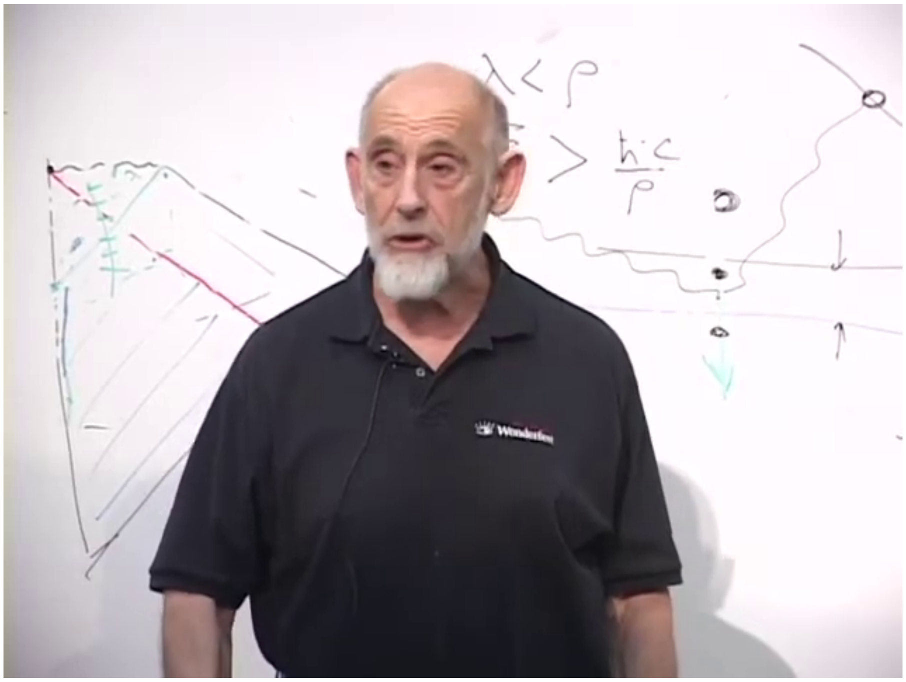

Figure 5.10: The near-horizon estimate on the board: $\lambda < \rho$ forces a local probe energy of order $\hbar c/\rho$.

Lecture frame: 01:18:30.000.

The photon energetic enough to resolve the event is also energetic enough to ionize the atom.

Question from the audience. Would that high-energy photon ionize the atom even in its freely falling frame?[a]

Response. Yes. The atom is struck by a real energetic quantum and recoils or ionizes in its own local frame. The measurement does not reveal an undisturbed fall; it replaces it with a disturbed one.

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[a]Lecture timestamp: 01:19:07.

Question from the audience. What if the atom simply falls and no one performs the experiment?[a]

Response. Then this particular disturbance is absent. If an outside observer does perform the high-resolution experiment, the atom is in trouble. The argument concerns what can be verified from outside, not a proof that an unmeasured infaller burns.

---

[a]Lecture timestamp: 01:19:39.

5.10.1 Can a thermometer or transmitter do better?

Moving the apparatus does not remove the information problem.

Question from the audience. Could one measure farther away, infer the gradient, or let a freely falling thermometer record the temperature and recover it later?[a]

Response. To learn the near-horizon record, something must interact with the thermometer and return information in quanta. Turning the thermometer around requires violent acceleration, while reading it remotely requires emission or scattering. Either route changes the physical system whose undisturbed history was to be tested.

[a]Lecture timestamp: 01:19:51.

Question from the audience. Is the high probe energy needed for small resolution the same difficulty as getting the reflected observation away from the near-horizon region?[a]

Response. They are two sides of the same redshifted scale. The photon can be low in energy when it finally reaches infinity because it loses energy climbing out, but it must have high local frequency where it resolves the atom.

[a]Lecture timestamp: 01:21:51.

[7]

Question from the audience. Could the object emit the information itself rather than being struck by an external photon?[a]

Response. Yes, but the local emission still needs the bandwidth required to identify what happened during the short interval. Emission also gives the object recoil. Replacing reflected light by a transmitter does not produce a cost-free record.

[a]Lecture timestamp: 01:22:51.

Question from the audience. The falling atom has a preferred direction. Is its directed kinetic energy being called temperature?[a]

Response. No. Throwing a baseball does not make the baseball hot. Bulk infall energy and thermal agitation are distinct. The temperature in the argument belongs to the surrounding static radiation atmosphere, not to the center-of-mass motion of the atom.

[a]Lecture timestamp: 01:23:10.

Question from the audience. Could a co-falling laboratory use a magnetic field to test whether the atom is ionized?[a]

Response. A laboratory can perform a local test. The difficulty is communicating its result back outside before crossing. That communication again requires an interaction and outgoing quanta with the necessary local resolution.

[a]Lecture timestamp: 01:23:50.

[7]Editorial clarification: A photon emitted just outside the horizon need not possess a large energy merely to escape. The decisive requirement here is that it encode the event within a proper time and distance of order ρ; gravitational redshift then relates that large local frequency to a much smaller asymptotic one.

Question from the audience. Would a transmitter built into the thermometer need so much local energy that it destroys the apparatus?[a]

Response. That is the same obstruction in engineering form. Near enough to the horizon, the signal energy required on the relevant local scale is comparable to the thermal scale being tested.

[a]Lecture timestamp: 01:24:57.

Question from the audience. Is the high-temperature region so thin that there is no time to carry out the measurement leisurely?[a]

Response. Exactly. Its proper thickness is of order ρ, and the crossing time is of order ρ/c. One may describe the limitation as spatial resolution or temporal bandwidth; either description demands high local momentum or energy.

[a]Lecture timestamp: 01:25:15.

The corresponding time-bandwidth estimate is

$$\Delta E \, \Delta\tau \gtrsim \hbar. \tag{5.25}$$

Here this should be read as a Fourier or operational resolution bound, not as an operator uncertainty relation identical in status to $\Delta x \, \Delta p \geq \hbar/2$. The lecture invokes time-energy uncertainty heuristically; Equation (5.23), derived from spatial resolution, is the firmer part of the argument.

Question from the audience. Could one use lower-energy, lower-resolution probes on many atoms and recover the answer statistically?[a]

Response. This is a sensible question, and it is left open in the classroom. Each atom provides only one passage, and any recovered signal must still carry enough information to identify the short near-horizon event. The lecture appeals to the uncertainty principle but does not supply a complete statistical no-go proof.

[a]Lecture timestamp: 01:26:26.

5.11 What Has and Has Not Been Established

The argument does not predict that a freely falling person must feel the divergent static temperature.

Question from the audience. Are we saying that we can predict the falling person will not feel the temperature?[a]

Response. No. The result is narrower: an exterior experiment designed to confirm that the atom did not experience the high-temperature disturbance creates a disturbance of the same scale. That is all this operational analysis establishes.

[a]Lecture timestamp: 01:27:38.

Thus the conclusion is not a firewall theorem, nor a proof that the infaller is burned, nor a proof that nothing happens. It is a limit on exterior verification: *resolving the undisturbed near-horizon event requires a probe that disturbs it.* If the experiment is performed, its quanta produce the effect it was meant to test. If it is not performed, this argument alone assigns no additional experimentally accessible fact. The comparison with the two-slit experiment is deliberate: asking which slit was taken without a path measurement and asking what an exterior apparatus certifies without an interaction are both attempts to attach an operational answer where the proposed evidence cannot be obtained.

Chapter 6

Long Strings and the Black-Hole Correspondence

Overview. A black hole has an entropy, so it must possess microscopic degrees of freedom. We shall build the simplest string-theoretic account of those degrees of freedom: first count the states of a long excited string, then vary the string coupling until the string and black-hole descriptions meet. The final equality is deliberately left as the problem to be completed in the next lecture.

6.1 Entropy Demands Microscopic Carriers

To say that a system has a large entropy is to say that it can hide a large amount of microscopic information. A bathtub of hot water may have an entropy of order 10^{30}. That statement already tells us that there are enormously many microscopic degrees of freedom, whether or not we choose to inspect them molecule by molecule. Thermodynamics can determine the entropy by measuring energy and temperature, but it does not identify the objects that carry it.

Fluid dynamics makes the distinction familiar. It describes density, velocity, pressure, and macroscopic flow without referring to atoms. Add thermodynamics and one discovers that a fluid has entropy. The entropy announces a microscopic structure that the coarse-grained equations do not display; it invites a new question rather than answering it.

Black holes present the same challenge in a sharper form. Bekenstein's proposal, completed by Hawking's calculation, assigns entropy proportional to horizon area. General relativity determines the geometry and its thermodynamics, but it gives no inventory of the microscopic states. What are the small, numerous objects whose configurations account for the area law? In string theory there is a concrete candidate. The argument developed here is intentionally qualitative: it gets the parametric dependence right while suppressing numerical constants and the formal machinery of a full state count.

6.2 Three Scales and One Notational Hazard

Gravity supplies a distinguished length, the Planck length,

$$\ell_p = \sqrt{\frac{G_N \hbar}{c^3}}. \tag{6.1}$$

Question from the audience. Which combination of G_N, $\hbar$, and c gives the Planck length?[a]

Response. It is $\sqrt{G_N\hbar/c^3}$, about 10^{-33} centimeters. We shall immediately set $\hbar = c = 1$, so the relation becomes much simpler.

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[a]Lecture timestamp: 00:06:08.

In natural units,

$$\hbar = c = 1, \qquad G_N = \ell_p^2. \tag{6.2}$$

Thus Newton's constant has dimensions of area. This is why an area divided by G_N is dimensionless and can be an entropy.

String theory has a second length, ℓ_s. Heat a string, or put it in a highly excited state, and it wiggles on many scales. The characteristic shortest scale in this elementary picture is the string length. One may regard ℓ_s as the size associated with one elementary unit of string oscillation.

The third quantity is the dimensionless string coupling g_s. It controls splitting and joining: a string can pinch into two strings, or an excited string can emit a closed-string quantum.

Question from the audience. What is the physical meaning of the string coupling?[a]

Response. It controls the chance for a string interaction, such as splitting when a string crosses itself. More precisely, g_s is the interaction amplitude; a corresponding probability or rate begins at order g_s^2.

---

[a]Lecture timestamp: 00:09:20.

The emitted massless closed-string excitation can be a graviton. In that respect g_s plays a role analogous to electric charge: it is the factor attached to an interaction vertex.[1]

6.3 Why Gravity Knows About g_s^2

Consider first the exchange picture for electromagnetism. One charged object emits a photon with a factor of e, and a second absorbs it with another factor of e. The exchange amplitude therefore contains e^2. The picture is not a literal classical projectile story, but it correctly displays how the coupling enters.

The string-theory version replaces the photon by a graviton and the electric charge by g_s. One factor accompanies emission and another absorption, so the gravitational strength scales as g_s^2.

At long distance the same interaction must reproduce Newton's force law,

$$F_{\text{grav}} \sim G_N \frac{M_1 M_2}{r^2}. \tag{6.3}$$

[1]Editorial clarification: The lecture alternates between calling g_s a probability and an amplitude. Perturbatively it is the amplitude-level coupling; probabilities and rates contain its square.

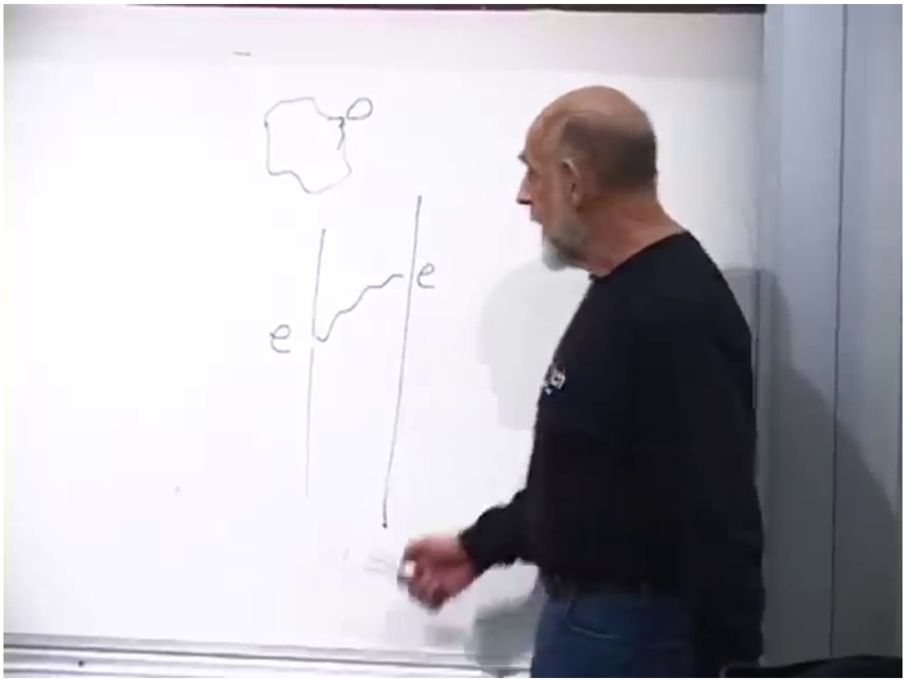

Figure 6.1: The completed exchange sketches on the board: a coupling is attached at emission and again at absorption.

Lecture frame: 00:11:30.000.

The masses and inverse-square dependence describe the sources and propagation; the point needed here is the coupling multiplying them.

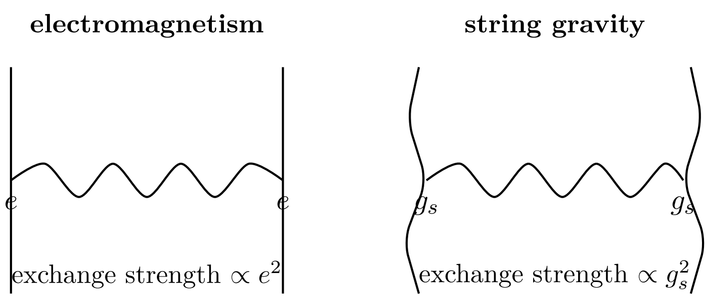

Figure 6.2: The coupling-counting analogy. Each end of the exchanged quantum contributes one interaction factor.

This observation almost determines Newton's constant. The coupling g_s is dimensionless, while in four dimensions G_N has dimensions of length squared. The available microscopic length is ℓ_s, hence

$$G_N \sim g_s^2 \ell_s^2. \tag{6.4}$$

Combining this with Equation (6.2) gives

$$\ell_p \sim g_s \ell_s. \tag{6.5}$$

Equation (6.4) is the simplified relation used throughout this lecture.[2]

[2]Editorial clarification: In a compactified string theory the effective four-dimensional Newton constant also

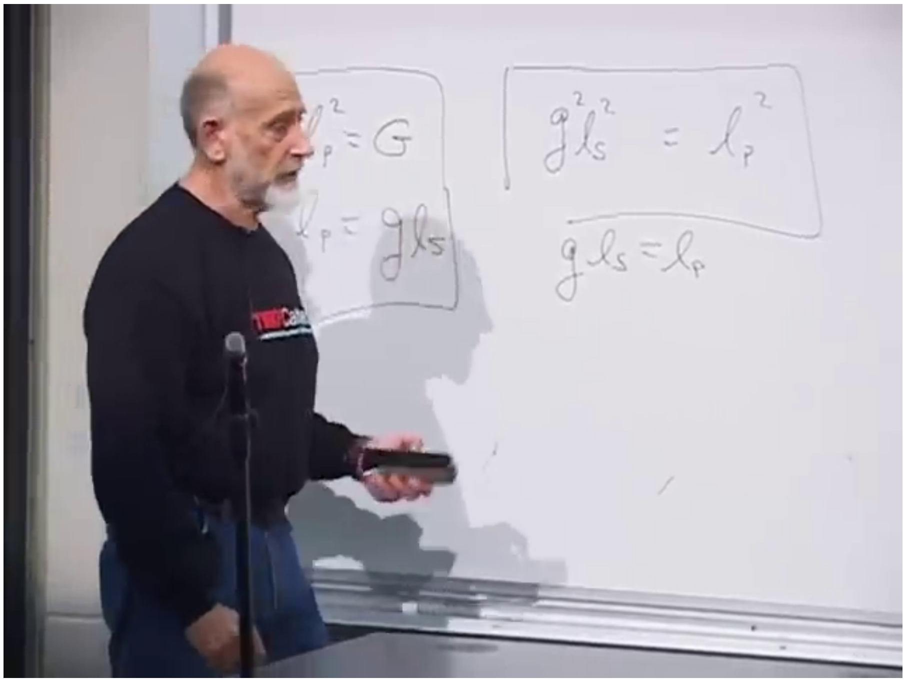

Figure 6.3: The basic relations retained on the board: $\ell_p^2 = G_N$ and $\ell_p \sim g_s\ell_s$.

Lecture frame: 00:15:45.000.

Question from the audience. Is g_s greater or less than one, and which length is larger?[a]

Response. We work at weak coupling, $g_s < 1$. Then $\ell_p \sim g_s\ell_s$ implies that the string length is larger than the Planck length. Strong coupling requires a different description and is not needed here.

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[a]Lecture timestamp: 00:16:00.

6.4 The Black-Hole Count

Suppressing Boltzmann's constant and numerical factors, the Bekenstein–Hawking entropy is

$$S_{\mathrm{BH}} = \frac{A}{4G_N} sim \frac{A}{G_N} sim \frac{A}{\ell_p^2}. \tag{6.6}$$

The factor $1/4$ is exact in the semiclassical formula, but it will play no role in this scaling argument.

Question from the audience. Does the subscript "BH" mean black hole or Bekenstein–Hawking?[a]

Response. For our purposes it can remind us of both. The important statement is that the entropy is the horizon area measured in Planck areas.

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[a]Lecture timestamp: 00:16:51.

For a four-dimensional Schwarzschild black hole,

$$R_s = 2G_N M, \qquad A = 4\pi R_s^2. \tag{6.7}$$

depends on the compactification volume and order-one conventions. The lecture suppresses those data to isolate the correspondence scaling.

Dropping 2 and 4π as the lecture does,

$$S_{\mathrm{BH}} \sim \frac{R_s^2}{G_N} \sim \frac{(G_N M)^2}{G_N} \tag{6.8}$$
$$\sim M^2 G_N \sim M^2 \ell_p^2. \tag{6.9}$$

This is the form we shall compare with a string count.

6.5 A Long String as a Random Walk

Begin with a small string and pump energy into it. A generic highly excited string does not remain a tidy loop. It becomes a long, tangled object, like fishing line after a badly snarled reel. The tangle has entropy because a huge number of configurations look macroscopically alike.

To count them, replace space by a lattice whose links have length ℓ_s. Sites are the vertices, links are the lines, and the elementary squares are plaquettes. A string is represented by a connected sequence of links. This model is deliberately crude, but its exponential state count captures the desired scaling.

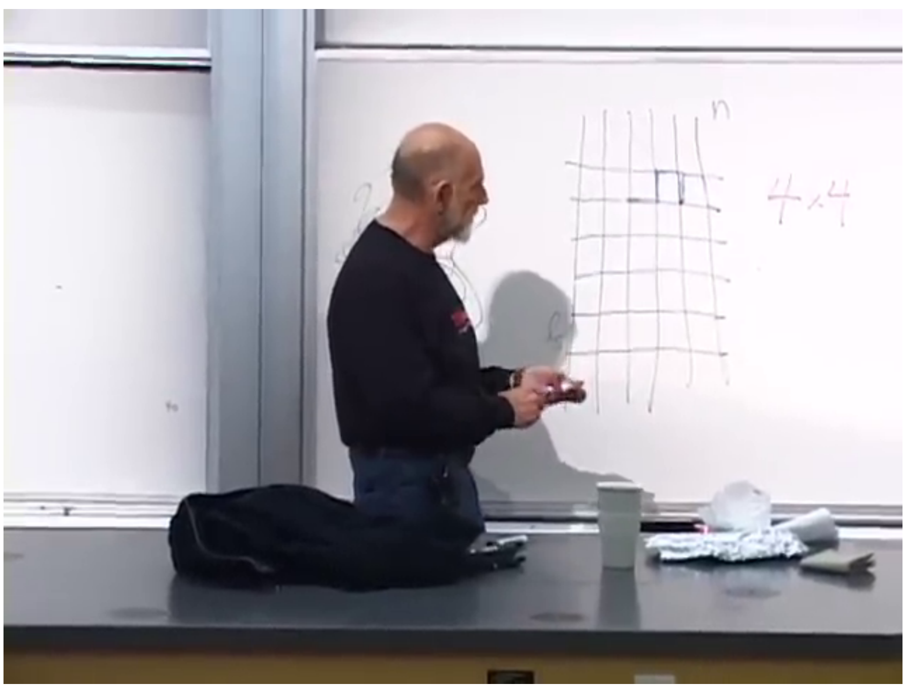

Figure 6.4: The two-dimensional lattice model. The blue segment marks one step of the string; the four possible continuations are counted at each site.

Lecture frame: 00:23:50.000.

Question from the audience. How many choices are there at each step of the toy walk?[a]

Response. Four on the square lattice, since immediate backtracking is allowed. In three dimensions there would be six. Another lattice changes the ordinary number but not the fact that the number of walks grows exponentially with the number of links.

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[a] Lecture timestamp: 00:22:43.

If the string has n links, the two-dimensional toy count is

$$\Omega_n \sim 4^n. \tag{6.10}$$

There is one exceptional straight walk, but it is only one state among 4^n. A typical walk folds repeatedly and occupies a tangled ball. For large n, almost every randomly generated configuration looks qualitatively like every other one.

The entropy is the logarithm of the number of macroscopically indistinguishable configurations:

$$S_{\mathrm{st}} = \log \Omega_n \sim n \log 4 \sim n. \tag{6.11}$$

Question from the audience. Why does the entropy become proportional to n rather than to 4^n?[a]

Response. Entropy is the logarithm of the number of states. Thus $\log(4^n) = n \log 4$, and the factor $\log 4$ is an order-one constant that we shall suppress.

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[a] Lecture timestamp: 00:25:39.

If L is the contour length obtained by following the string from one end to the other, then

$$n = \frac{L}{\ell_s}, \qquad S_{\mathrm{st}} \sim \frac{L}{\ell_s}. \tag{6.12}$$

Here L is not the diameter of the tangled ball. It is the much longer distance measured along the string.

We next trade contour length for mass. With $c = \hbar = 1$, mass has dimensions of inverse length. One string-sized link therefore carries mass of order

$$m_{\mathrm{link}} \sim \frac{1}{\ell_s}. \tag{6.13}$$

Question from the audience. Why must the mass of one link scale as $1/\ell_s$?[a]

Response. In these units mass is an inverse length, and ℓ_s is the only available length. Equivalently, a string tension of order $1/\ell_s^2$ multiplied by a link length ℓ_s gives the same result.

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[a] Lecture timestamp: 00:29:20.

The full string then has

$$M \sim n m_{\mathrm{link}} \sim \frac{n}{\ell_s} \sim \frac{L}{\ell_s^2}, \qquad L \sim M \ell_s^2. \tag{6.14}$$

Substitution into Equation (6.12) yields

$$S_{\mathrm{st}} \sim M \ell_s. \tag{6.15}$$

The lattice walk ignores world-sheet constraints and numerical coefficients, but the exponential growth of the highly excited string spectrum gives this same leading scaling.[3]

[3] Editorial clarification: The random-walk construction is a mnemonic for the high-level density of string states, not a literal discretization of a fundamental string.

6.6 The Mismatch That Starts the Argument

The two answers are now

$$S_{\mathrm{BH}} \sim M^2\ell_p^2, \qquad S_{\mathrm{st}} \sim M\ell_s. \tag{6.16}$$

They resemble one another, but they are not equal at fixed mass and arbitrary coupling. The black-hole expression is quadratic in M; the string expression is linear. A hot string cannot simply be declared to be a black hole without explaining how the two regimes meet.

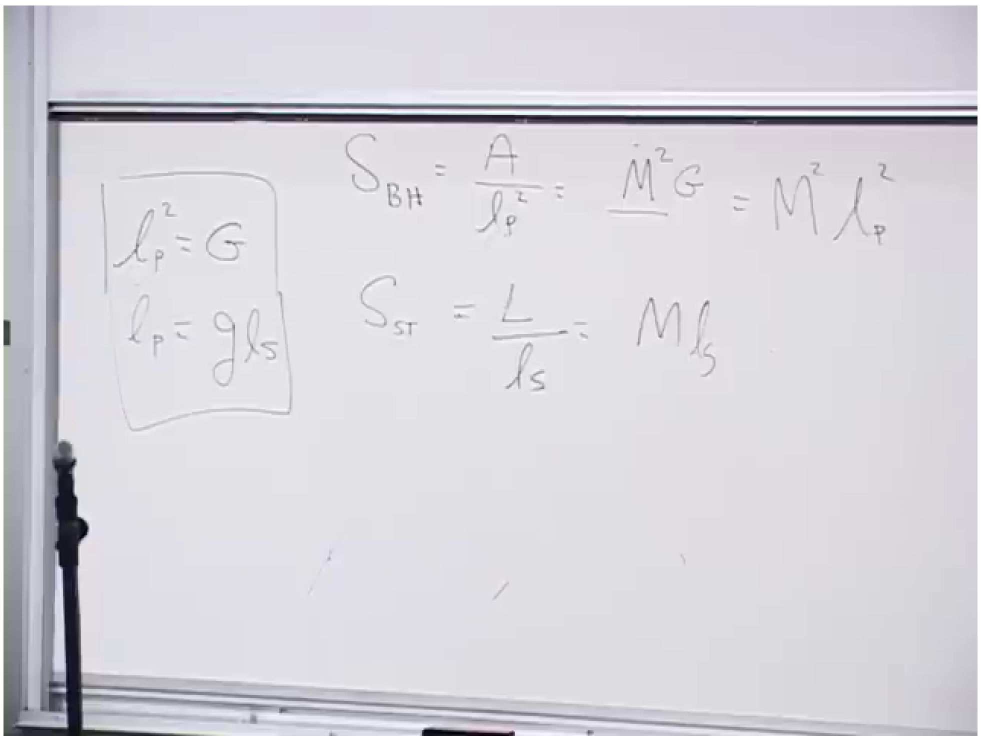

Figure 6.5: The unobstructed board comparison: black-hole entropy above, long-string entropy below, with the Planck–string relations retained at left.

Lecture frame: 00:33:20.000.

This is where the class paused. The suspense is useful: the mismatch is not a failure of the idea but the clue that one must compare the same state while changing the strength of gravity.

6.7 Turning the Coupling Dial

Hold ℓ_s fixed and imagine that g_s can be varied. At $g_s = 0$, strings neither split nor join and gravity vanishes because $G_N \sim g_s^2\ell_s^2$. A sufficiently energetic state is then a long, weakly interacting string.

Increase g_s slowly. String interactions become possible and, more importantly for the present argument, gravity strengthens.

Question from the audience. As g_s increases, is the first effect simply that the string splits into shorter strings?[a]

Response. Splitting and joining become possible, but hold ℓ_s fixed and follow the high-entropy state. The central effect is that G_N grows, so the long string's self-gravity pulls the tangled ball inward.

[a]Lecture timestamp: 00:36:29.

The object contracts and becomes denser. Its gravitational potential energy is negative, so contraction lowers its total energy and therefore its mass.

Question from the audience. Does the energy increase or decrease as self-gravity pulls the string together?[a]

Response. It decreases. The binding energy becomes more negative. The mass along the path need not remain fixed even though we continue to follow the same state.

[a]Lecture timestamp: 00:37:48.

Eventually the object's size is comparable to its Schwarzschild radius. Beyond that point the black-hole description is the natural one. A larger initial excitation reaches this condition at a smaller coupling because a more massive object requires less gravitational strength to form a horizon.

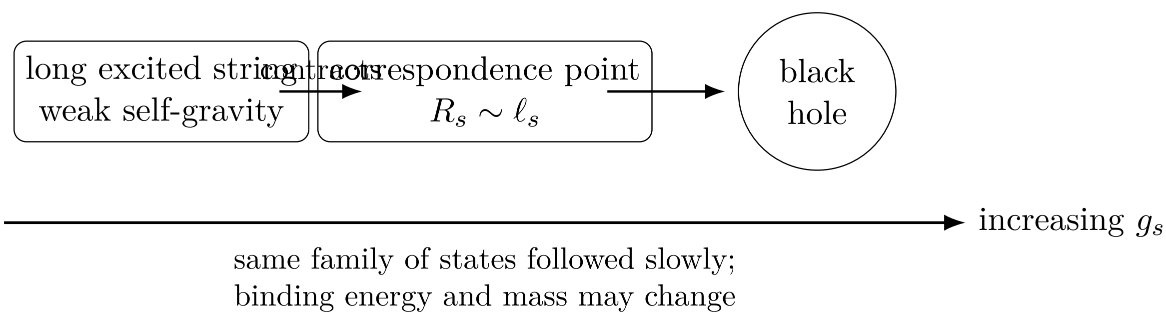

Figure 6.6: The correspondence path at fixed ℓ_s. Reversing the arrow takes the black-hole state back to a long-string state.

Question from the audience. Should we imagine g_s between zero and one?[a]

Response. Yes. The argument remains in the weak-coupling regime. If g_s is very small, the state must be correspondingly massive before it becomes a black hole. We do not need to enter the poorly controlled $g_s > 1$ regime.

[a]Lecture timestamp: 00:40:18.

The slow change is reversible in the thought experiment.

Question from the audience. If we turn the coupling back down until gravity can no longer hold the black hole together, what does it become?[a]

Response. A long string. At vanishing gravity everything in this construction must be described in string states. Turning the dial one way maps string to black hole; turning it back maps black hole to string.

[a]Lecture timestamp: 00:41:37.

At fixed total energy and within a fixed volume, the dominant high-entropy configuration is essentially one long string rather than an enormous number of short strings. That statement is not derived here. It is a result of the statistical mechanics of strings near the Hagedorn regime and will be revisited.

6.8 What the Outside Observer Sees

Where is the string when the object is described as a black hole? In the outside description it does not abruptly disappear. Schwarzschild time sends the apparent crossing to arbitrarily late time, so the stringy degrees of freedom are represented as accumulating on or near the horizon.

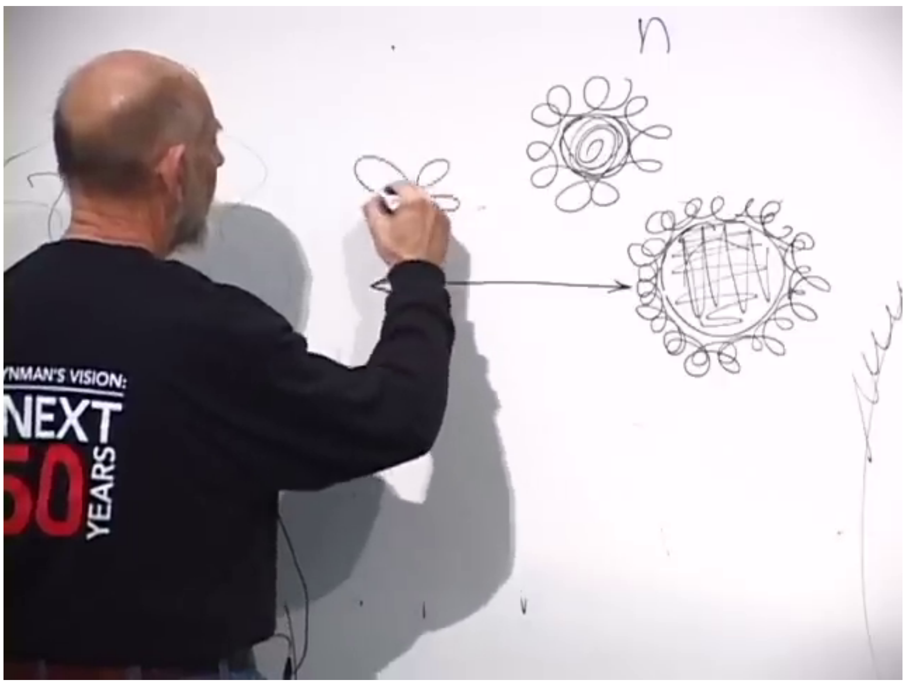

Figure 6.7: The board comparison between a macroscopic horizon, where string-scale structure is a thin fuzz, and a smaller horizon whose radius approaches the loop scale.

Lecture frame: 00:45:30.000.

For $R_s \gg \ell_s$, the black hole is macroscopic and the string-scale structure is only a thin layer relative to the horizon radius. As the radius decreases toward ℓ_s, the distinction between a smooth black hole and an extended string loses meaning. This is an outside, stretched-horizon description; it should not be confused with a coordinate-independent claim that infalling matter never crosses a regular horizon.[4]

The crossover occurs when

$$R_s \sim \ell_s. \tag{6.17}$$

Below the string scale, the Schwarzschild geometry cannot resolve the structure that is supposed to support it. The loops fluctuate away from the would-be horizon, and the state is better called a string.

[4]Editorial clarification: The lecture uses the exterior description familiar from black-hole complementarity. The phrase “on the horizon” denotes degrees of freedom resolved by the outside description, not a separate derivation of their covariant location.

6.9 The Correspondence Point

Use $R_s \sim G_N M$ in Equation (6.17):

$$G_N M \sim \ell_s. \tag{6.18}$$

Then substitute $G_N \sim g_s^2 \ell_s^2$:

$$M g_s^2 \ell_s^2 \sim \ell_s, \qquad M g_s^2 \ell_s \sim 1. \tag{6.19}$$

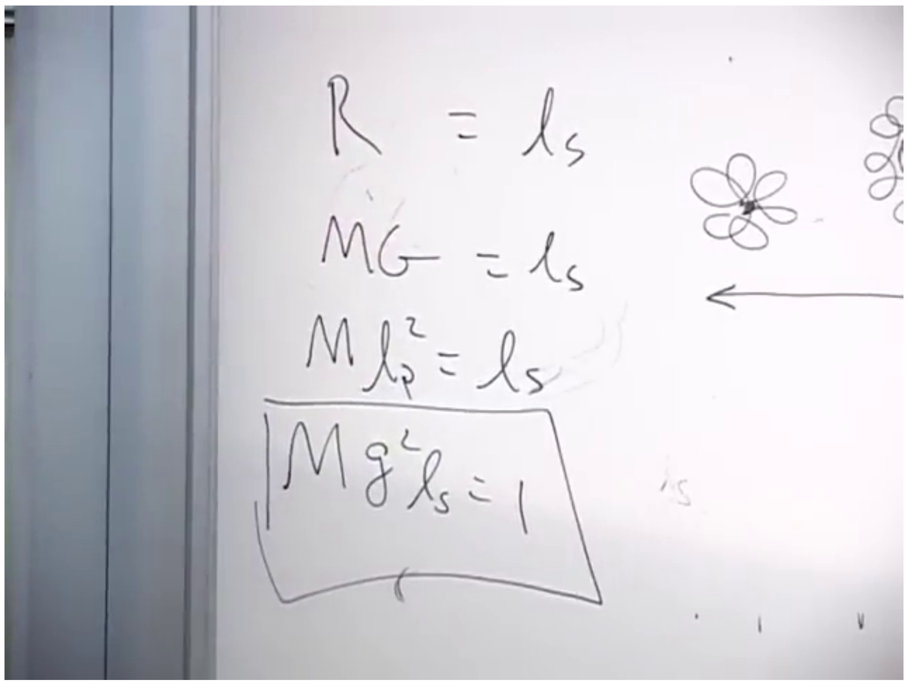

Figure 6.8: The completed transition algebra on the board: $R_s \sim \ell_s$ leads to the boxed dimensionless condition $M g_s^2 \ell_s \sim 1$.

Lecture frame: 00:49:50.000.

An audience member noticed what this condition is preparing. At the correspondence point,

$$S_{\mathrm{BH}} \sim M^2 \ell_p^2 \sim M^2 g_s^2 \ell_s^2 \tag{6.20}$$

$$= (M\ell_s)(M g_s^2 \ell_s) \sim M\ell_s \sim S_{\mathrm{st}}. \tag{6.21}$$

Question from the audience. Do the equations already imply that the black-hole and string entropies become equal?[a]

Response. Yes. That is exactly where the calculation is going. The ingredients have been placed on the board, but the lecture postpones their full assembly so that the logic of the correspondence can be reviewed carefully.

[a]Lecture timestamp: 00:50:18.

Equation (6.21) is the one-line algebraic preview implicit in the exchange. It establishes the scaling, not the exact coefficient 1/4. Exact microscopic matches require more controlled black holes and more detailed string constructions.[5]

[5]Editorial clarification: The lecture stops before presenting Equation (6.21) as its completed derivation. It is written

6.10 Why the Entropy Can Be Carried Across

One logical ingredient remains. Suppose a control parameter, such as a magnetic field or g_s, is varied sufficiently slowly while the system is isolated and the evolution is reversible. Energies, sizes, and binding may change, but the entropy is preserved:

$$\Delta S = 0. \tag{6.22}$$

The lecture connects this to conservation of information. Entropy measures information hidden by a coarse-grained description; a slow reversible change of the Hamiltonian rearranges the states without deleting that information.

Question from the audience. Does constant entropy require that the path avoid a phase transition?[a]

Response. Yes, one must be able to follow the state continuously. A finite system has no strictly nonanalytic thermodynamic phase transition of the infinite-volume kind, and if the change is slow enough the correspondence can be treated as a smooth crossover.

[a] Lecture timestamp: 00:54:01.

Adiabatic is being used here in the stronger quasistatic, reversible sense. An adiabatic but irreversible process can produce entropy; slowness and the ability to track the state are essential.[6]

Question from the audience. Why should one long string have more entropy than several shorter strings carrying the same total mass?[a]

Response. It is counterintuitive, but the explicit high-energy state count favors a long-string component in a fixed volume. The proof is deferred; for the present argument we need only that the black hole maps to the maximum-entropy string sector rather than to a dilute gas of very many short strings.

[a] Lecture timestamp: 00:54:44.

The operational strategy is therefore:

1. Begin with a black hole of mass M_0 at coupling $g_{s,0}$.
2. Lower g_s slowly while following the same high-entropy family of states.
3. Stop at $R_s \sim \ell_s$, where the black-hole description becomes a long-string description.
4. Count the string states using $S_{\rm st} \sim M\ell_s$.
5. Carry that entropy back along the reversible path to the original black hole.

String theory has produced much sharper entropy counts for charged, rotating, supersymmetric, near-extremal, and higher-dimensional black holes. The neutral Schwarzschild discussion here is the rough correspondence argument: it identifies the parametric density of states and the physical crossover, not an exact protected degeneracy.[7]

out here because the audience explicitly identifies the equality and every required relation has already appeared.

[6] Editorial clarification: The thermodynamic statement $\Delta S = 0$ requires a reversible adiabatic path. Quantum mechanically, the corresponding idea is to follow the same set of states as a parameter changes without uncontrolled level transitions.

[7] Editorial clarification: Exact agreement is strongest for special black holes whose state counts are protected as the coupling changes. The present lecture intentionally gives the simplest order-of-magnitude version.

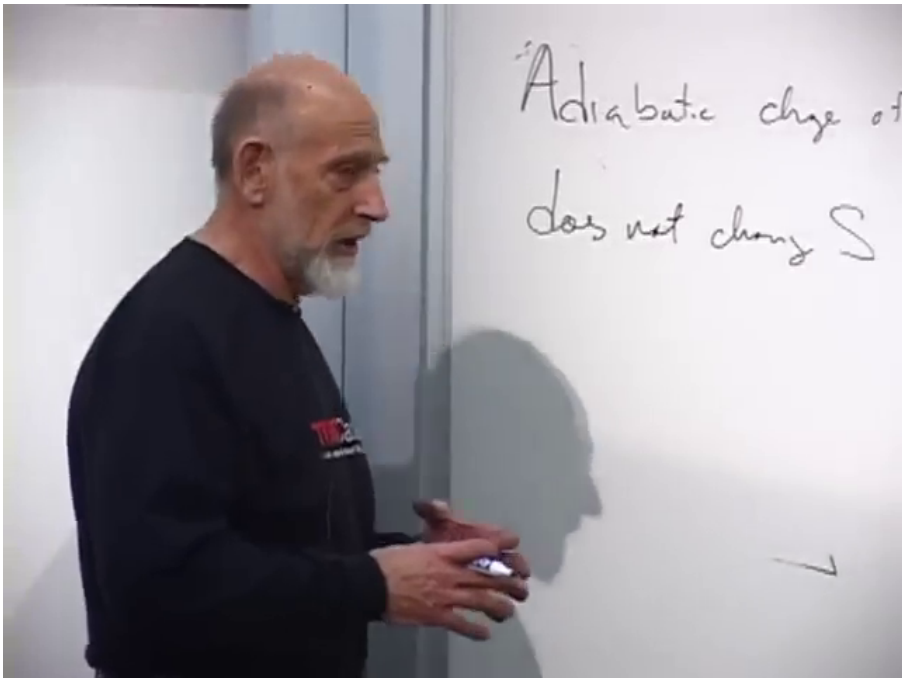

Figure 6.9: The principle retained on the board while the strategy is stated: an adiabatic change of g_s does not change S.

Lecture frame: 00:56:40.000.

Question from the audience. Could the state become two strings rather than exactly one when gravity is turned off?[a]

Response. There is a finite probability for a small number of long strings, and using two rather than one does not change the leading entropy scaling. What would change the argument is fragmentation into a huge gas of short strings; the high-energy statistical count does not favor that outcome.

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[a] Lecture timestamp: 00:58:25.

The black hole is the greatest entropy that can be packed into a region, and the long string is the corresponding high-entropy state when gravity is removed. Those two descriptions are now connected, but not yet developed into a precision calculation. The next step is to repeat the algebra carefully and ask how far the correspondence can be trusted.

Chapter 7

Black-Hole Entropy, Holography, and de Sitter Space

Overview. We complete the correspondence argument begun in the previous lecture. A highly excited string has an entropy proportional to its mass, while a Schwarzschild black hole has an entropy proportional to its area. The two formulas are connected not by identifying a strongly gravitating black hole with a free string, but by varying the string coupling slowly until the two descriptions meet. The area law then leads naturally to the holographic principle, and the final part of the lecture introduces expanding space and the de Sitter geometry whose horizon will be studied next.

7.1 Returning to the Counting Problem

There was one loose end in the long-string count. If a fixed total length is divided among several strings, identical strings of the same length are indistinguishable. Their locations in the box are nevertheless part of the state, and the box is chosen to have roughly the size occupied by the corresponding single long string.

Question from the audience. Are the short strings distinguished from one another, and do their positions count?[a]

Response. Strings with identical properties are indistinguishable, but their positions in the box are counted. The comparison is made at fixed volume, chosen to match the region occupied by the long string.

[a] Lecture timestamp: 00:00:37.

With that settled, we can state the target. We shall assemble a short list of equations and use them to obtain the scaling of black-hole entropy. No integral or infinite sum is required. The difficulty is not algebraic; it lies in knowing which quantities may be compared and at what stage of the argument.

7.2 Natural Units and the Planck Area

Set

$$c = \hbar = 1, \tag{7.1}$$

but keep Newton's constant explicit. The first consequence is that time and length have the same units. The photon relation

$$E = \frac{\hbar c}{\lambda} \tag{7.2}$$

and the mass-energy relation $E = mc^2$ then imply

$$[M] = L^{-1}. \tag{7.3}$$

Mass can therefore be quoted as an inverse length.

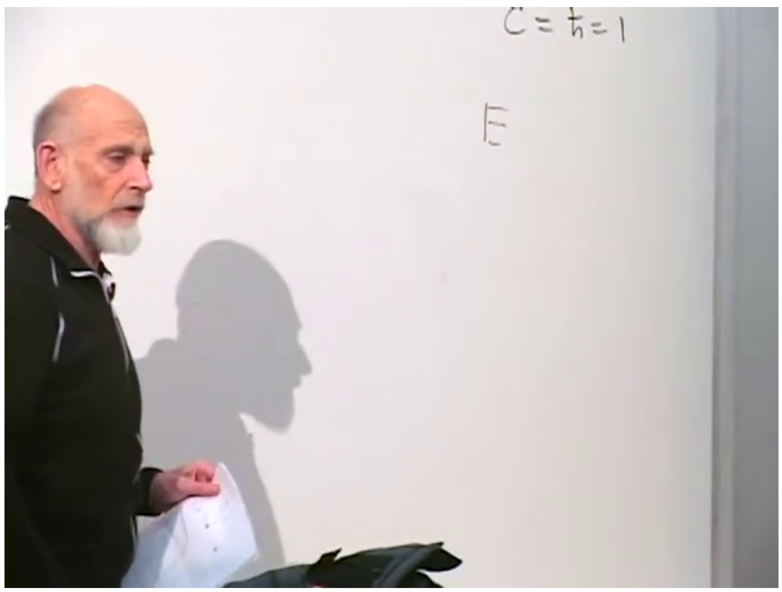


Figure 7.1: The natural-units convention at the start of the dimensional argument. The energy formula is completed in the equations above.

Lecture frame: 00:03:27.900.

To determine the units of G_N, use Newtonian acceleration only as dimensional bookkeeping:

$$a \sim \frac{G_N M}{R^2}. \tag{7.4}$$

Because T and L have the same units, $[a] = L/T^2 = L^{-1}$. Together with Equation (7.3), this gives

$$[G_N] = L^2. \tag{7.5}$$

Newton's constant is an area in these units. Defining the Planck length by

$$G_N = \ell_p^2, \tag{7.6}$$

we immediately understand why $A/G_N = A/\ell_p^2$ is dimensionless: it measures an area in Planck areas.

7.3 The String Coupling and the Strength of Gravity

Gravity must now be translated into string language. The useful picture is an exchange process. When a string crosses itself, it may split and rejoin. A small closed-string loop can be emitted by one object and absorbed by another; when that excitation is a graviton, the process mediates gravity. One interaction factor appears at emission and another at absorption, so the exchange strength contains g_s^2.

The coupling g_s is dimensionless. More precisely, it is an amplitude-level coupling, while a probability or rate begins at order g_s^2.[1] Since G_N has dimensions of area, a length scale is still missing. The fundamental string length ℓ_s, the scale of one elementary string wiggle, supplies it:

$$G_N \sim g_s^2 \ell_s^2, \qquad \ell_p \sim g_s \ell_s. \tag{7.7}$$

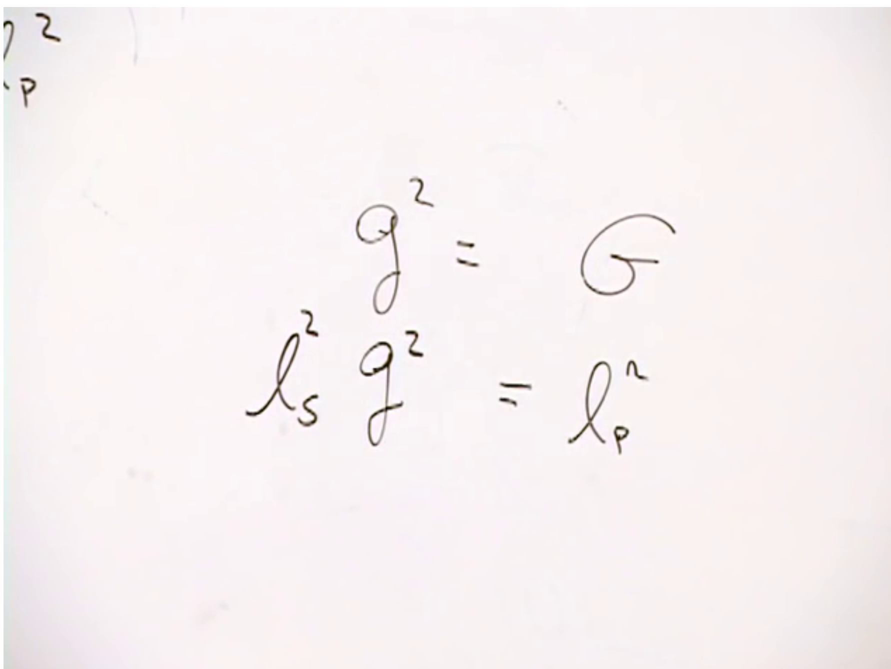


Figure 7.2: The coupling relation on the board: $G_N = \ell_p^2$ and $g_s^2 \ell_s^2 \sim \ell_p^2$.
Lecture frame: 00:12:15.000.

At weak coupling, $g_s \ll 1$, the Planck length is much smaller than the string length. Turning g_s down weakens gravity while leaving the elementary string scale fixed.[2]

7.4 Counting a Long String

Entropy counts the elementary decisions required to specify a state. Replace a highly excited string by a walk on a square lattice. At each vertex the next link may point in one of four directions.

[1] Editorial clarification: The lecture sometimes calls g_s a probability and then corrects this to the square root of a probability. The perturbative coupling belongs at amplitude level.

[2] Editorial clarification: For an effective four-dimensional theory, compactification data and numerical conventions also enter G_N. They are held fixed and suppressed in this scaling argument.

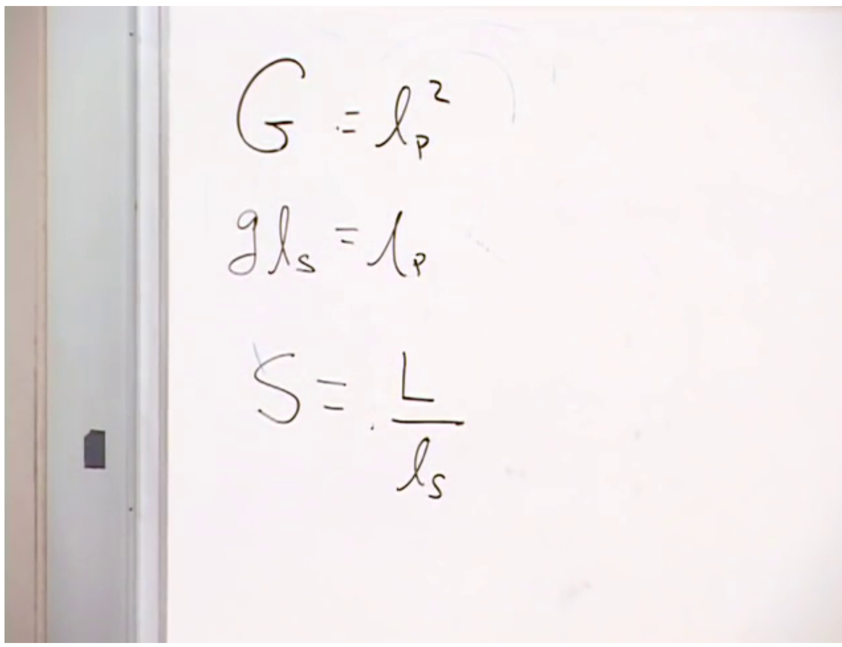
Whether the elementary choice is described as one four-way decision or two binary decisions changes only an order-one constant. What matters is the number of steps.

If L is the contour length of the string and each step has length ℓ_s, then

$$N = \frac{L}{\ell_s} \tag{7.8}$$

steps are required. The number of walks grows exponentially, schematically $\Omega_N \sim q^N$ with q of order unity, and hence

$$S_{\rm str} = \log \Omega_N \sim N \sim \frac{L}{\ell_s}. \tag{7.9}$$



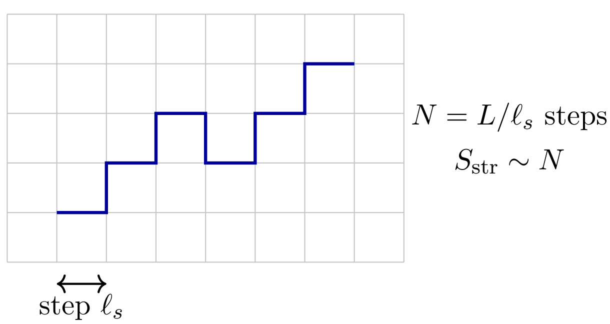


Figure 7.3: The lattice model on the board and its cleaned random-walk interpretation. The contour length, not the diameter of the tangled ball, determines the number of choices.

Lecture frame: 00:17:20.000.

The mass is counted from the same elementary segments. A segment of length ℓ_s has mass scale

$$m_{\rm seg} \sim \frac{1}{\ell_s}. \tag{7.10}$$

Question from the audience. Is the elementary segment being assigned the length ℓ_s?[a]

Response. Yes. Its characteristic length is ℓ_s, so in natural units its characteristic mass is $1/\ell_s$.

[a]Lecture timestamp: 00:18:56.

Adding the L/ℓ_s segment masses gives

$$M \sim \frac{L}{\ell_s}\frac{1}{\ell_s} = \frac{L}{\ell_s^2}, \qquad L \sim M\ell_s^2. \tag{7.11}$$

The kinetic and potential energies of the oscillating string have been compressed into this scaling relation; a detailed statistical-mechanical treatment gives the same dependence. Substitution into Equation (7.9) yields the string formula we shall need:

$$S_{\text{str}} \sim M\ell_s. \tag{7.12}$$

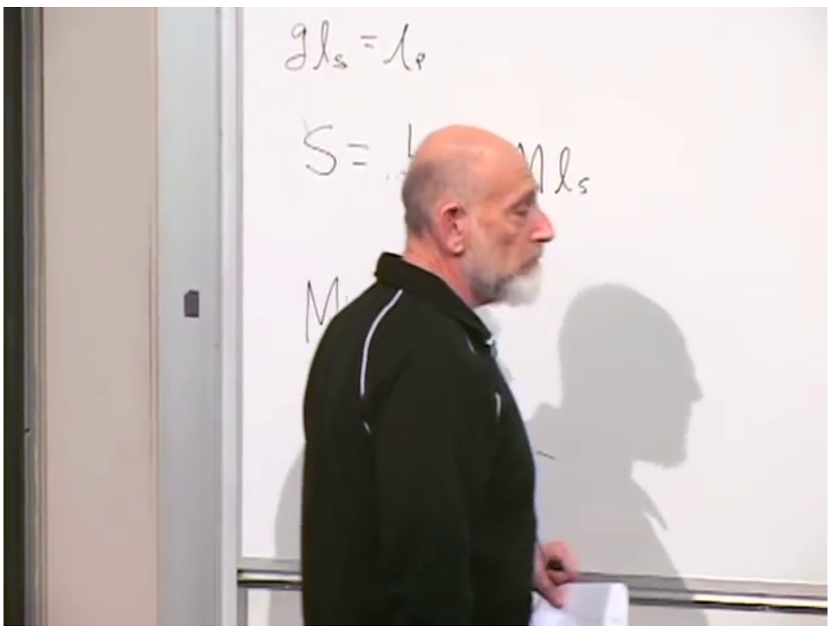


Figure 7.4: The string-side column assembled before the comparison: $\ell_p \sim g_s\ell_s$, $S_{\text{str}} \sim L/\ell_s \sim M\ell_s$, and $M \sim L/\ell_s^2$.

Lecture frame: 00:21:57.800.

7.5 Why a Free String Is Not Yet a Black Hole

On the gravitational side the exact semiclassical formula is

$$S_{\text{BH}} = \frac{A}{4G_N}. \tag{7.13}$$

Question from the audience. What numerical coefficient belongs in the denominator?[a]

Response. Four. The correspondence argument will recover the scaling but is too crude to determine that exact factor.

[a]Lecture timestamp: 00:23:28.

For a four-dimensional Schwarzschild black hole,

$$R_s = 2G_N M, \qquad A = 4\pi R_s^2. \tag{7.14}$$

Dropping numerical constants as the lecture does,

$$S_{\rm BH} \sim \frac{A}{G_N} \sim \frac{(G_N M)^2}{G_N} \tag{7.15}$$
$$\sim M^2 G_N \sim M^2 \ell_p^2. \tag{7.16}$$

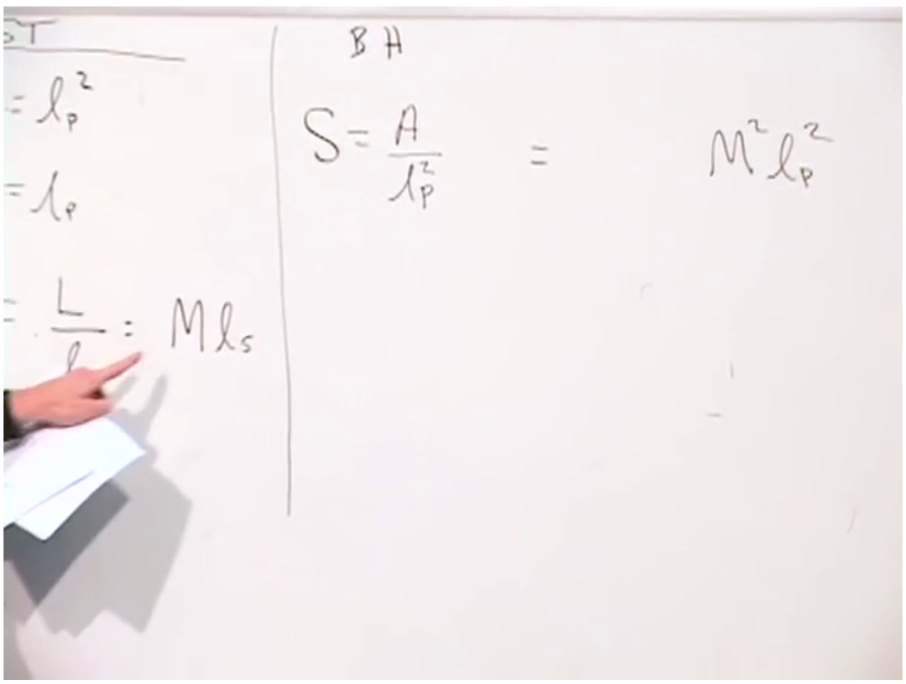


Figure 7.5: The completed comparison on the board: string entropy is a length in string units, whereas black-hole entropy is an area in Planck units.

Lecture frame: 00:25:35.000.

The resemblance between Equations (7.12) and (7.16) is suggestive, but they are not the same formula. It would be wrong simply to call the original black hole a free string and substitute. The free-string count omits the attractive interaction among the string's parts, precisely the interaction that compacts the string into a black hole and lowers its total energy through negative binding energy.

A star makes the error familiar. A hot gas of stellar mass placed in an oversized box is not a model of a star, even at the same temperature. Self-gravity changes the equilibrium size and the relation between energy and entropy. Following every self-interaction of a long string would be difficult, so we shall avoid doing so.

Question from the audience. If a small loop leaves a long string, its entropy decreases. Does graviton exchange therefore lower the entropy of the gravitating system?[a]

Response. Not in the equilibrium exchange process being used here. Virtual excitations are emitted and absorbed in both directions; the cartoon is not a literal loss of a permanent string fragment. Its only present purpose is to establish the two coupling factors and hence the g_s^2 dependence.

[a]Lecture timestamp: 00:28:48.

7.6 The Adiabatic Correspondence Game

Begin instead where gravity is negligible. At $g_s \to 0$, strings cross without splitting or joining and the self-gravitational attraction disappears. Increase g_s while holding ℓ_s fixed: gravity strengthens, a highly excited string contracts, and eventually it becomes a black hole. Reverse the dial slowly and the black hole must return to a string state. It may become a small number of long strings, but the high-energy counting from the previous lecture says that the dominant sector contains one long string rather than an enormous gas of short ones.

Question from the audience. What is held fixed while the coupling is changed?[a]

Response. The string length ℓ_s, and therefore the elementary string mass scale $1/\ell_s$, is held fixed. The dial changes g_s and hence $G_N \sim g_s^2 \ell_s^2$.

[a]Lecture timestamp: 00:32:19.

The change must be slow, isolated, and reversible. Under those conditions entropy is an adiabatic invariant:

$$\Delta S = 0. \tag{7.17}$$

Slowness alone is not enough if dissipative or uncontrolled transitions occur. Here “adiabatic” means that the same family of states can be followed continuously.[3]

Let the target black hole have mass M_0 and coupling $g_{s,0}$. Before using the area formula we are trying to explain, dimensional analysis says only that the entropy of a neutral, nonrotating black hole is some function of the dimensionless combination

$$S_{\mathrm{BH}} = f(M\ell_p) = f(M g_s \ell_s). \tag{7.18}$$

At fixed ℓ_s, an isentropic path therefore keeps

$$M g_s = M_0 g_{s,0} \tag{7.19}$$

constant, provided the same branch of states is followed.

As g_s decreases, negative binding energy is removed, so the mass M rises along this path. The underlying string configuration also becomes less tightly bound and spreads. The formal Schwarzschild radius, however, behaves as $R_s \sim M g_s^2 \ell_s^2 \sim g_s \ell_s$ on the adiabat and therefore decreases.[4]

[3]Editorial clarification: This is the reversible thermodynamic sense of adiabatic. An irreversible process can be adiabatic in the no-heat-flow sense and still produce entropy.

[4]Editorial clarification: The spoken discussion uses “grows” for the object released from gravitational binding. It should not be read as saying that the Schwarzschild radius grows along the derived adiabat.

Question from the audience. Does the object grow or shrink as gravity is weakened?[a]

Response. Its string structure is released and expands, while its mass increases because negative binding energy is removed. The radius assigned by the black-hole formula decreases until it reaches the string scale, where that black-hole description ceases to be useful.

[a]Lecture timestamp: 00:36:28.

7.7 Two Curves and Their Intersection

Plot mass vertically and string coupling horizontally. Equation (7.19) is a hyperbola, $M \propto g_s^{-1}$.

Question from the audience. What kind of curve has constant product Mg_s?[a]

Response. A hyperbola. It is the adiabat through the target point $(g_{s,0}, M_0)$.

[a]Lecture timestamp: 00:44:08.

The second curve marks the crossover between descriptions. A black hole cannot be resolved as a conventional Schwarzschild geometry once its radius is smaller than an elementary string wiggle. We therefore estimate the correspondence point by

$$R_s \sim \ell_s. \tag{7.20}$$

Using $R_s \sim MG_N$ and Equation (7.7), this becomes

$$Mg_s^2\ell_s^2 \sim \ell_s, \qquad Mg_s^2 \sim \frac{1}{\ell_s}. \tag{7.21}$$

Thus the transition curve falls as $M \propto g_s^{-2}$, more steeply than the adiabat.

At the intersection $(M_*, g_{s,*})$, the two equations are

$$M_* g_{s,*} = M_0 g_{s,0}, \tag{7.22}$$

$$M_* g_{s,*}^2 \sim \frac{1}{\ell_s}. \tag{7.23}$$

Square the first equation and divide by the second:

$$M_* \sim M_0^2 g_{s,0}^2 \ell_s. \tag{7.24}$$

Multiplying by ℓ_s,

$$M_*\ell_s \sim M_0^2 g_{s,0}^2 \ell_s^2 \sim M_0^2 G_{N,0}. \tag{7.25}$$

Question from the audience. When the two curve equations are divided, should ℓ_s end up upstairs or downstairs?[a]

Response. Upstairs. Dividing by $1/\ell_s$ multiplies by ℓ_s, which is why $M_* \sim M_0^2 g_{s,0}^2 \ell_s$.

[a]Lecture timestamp: 00:52:41.

At the crossover the string formula is valid, so

$$S_* \sim M_*\ell_s. \tag{7.26}$$

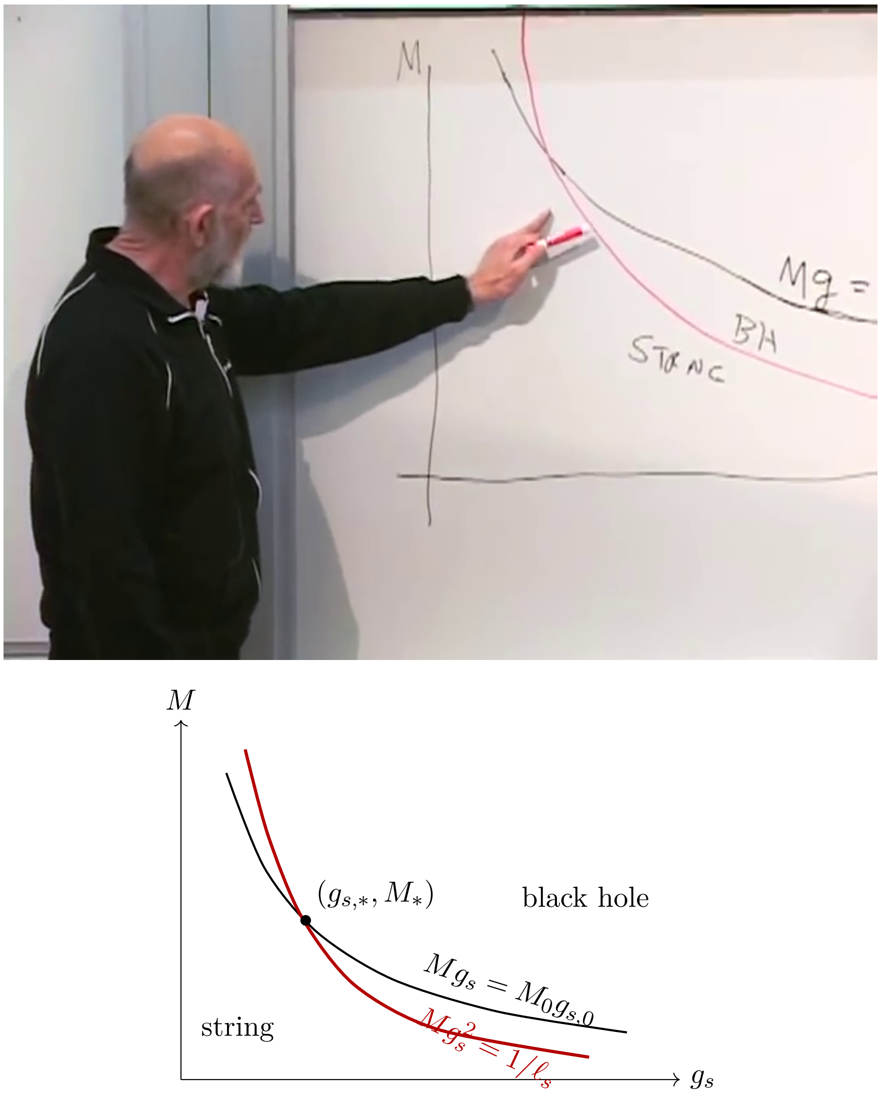


Figure 7.6: The board graph and its cleaned interpretation. The black adiabat intersects the steeper red correspondence curve at the point where the object becomes string-like.

Lecture frame: 00:49:35.000.

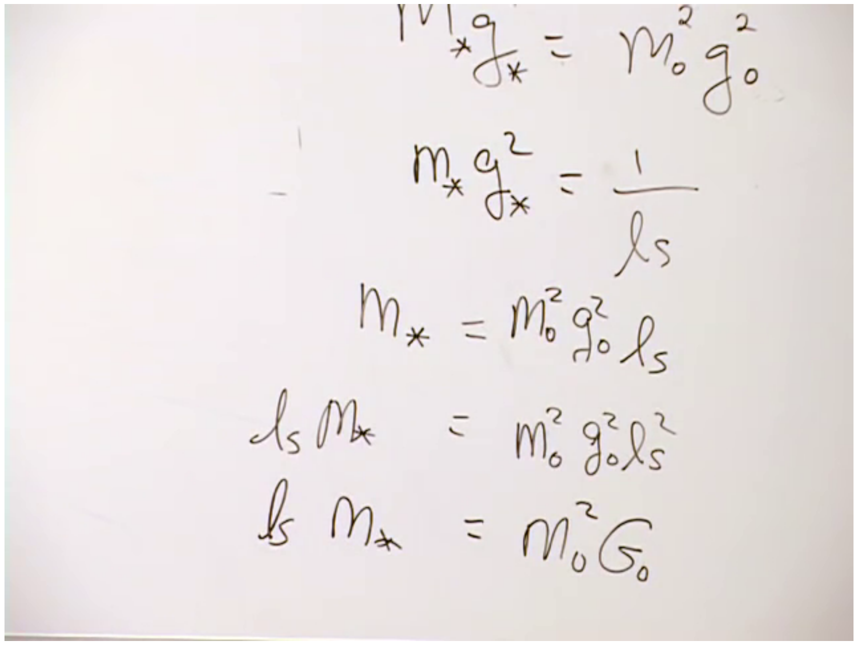


Figure 7.7: The completed crossover algebra, including $M_* \ell_s = M_0^2 g_{s,0}^2 \ell_s^2 = M_0^2 G_{N,0}$.
Lecture frame: 00:54:40.000.

Entropy was unchanged along the path. Combining this statement with Equation (7.25) gives the entropy of the original target black hole:

$$S_{\mathrm{BH}}(M_0) \sim M_0^2 G_{N,0} \sim M_0^2 \ell_{p,0}^2. \tag{7.27}$$

This is the desired area-law scaling.

7.8 What the Correspondence Does and Does Not Prove

The construction does not say that the original, strongly gravitating black hole is a free string. It moves the object to a regime where the string count is trustworthy, reads the entropy there, and carries that entropy back along a reversible path. The transition is a crossover rather than a sharply located phase boundary, so this estimate cannot determine the coefficient 1/4 in Equation (7.13).

The same logic nevertheless works parametrically for charged black holes, black objects in other dimensions, and extended black branes. More symmetric and supersymmetric examples permit protected state counts that retain exact information as the coupling changes; those methods recover numerical coefficients but add mathematical machinery not needed for the present conceptual argument.

Most importantly, the calculation supplies a microscopic interpretation. Entropy is evidence of hidden structure. Fluid mechanics describes density, pressure, and flow while omitting molecules; the fluid's entropy reveals that such microscopic degrees of freedom exist. General relativity similarly describes a horizon without displaying its microstates. The string count shows that a consistent quantum theory of gravity can contain enough states to account for the black-hole entropy.

Question from the audience. What would $S \sim M^3 \ell_p^3$ have meant instead of the quadratic result?[a]

Response. It would have looked like a volume law. Ordinary matter in a room or a cup of coffee has entropy proportional to volume; the quadratic answer is special because, through $R_s \sim M\ell_p^2$, it is proportional to horizon area.

[a]Lecture timestamp: 00:59:41.

The result is evidence that the Bekenstein–Hawking entropy is not merely a formal thermodynamic analogy. It points to microscopic gravitational degrees of freedom, even though the elementary derivation does not make those degrees of freedom visible one by one.

7.9 From Maximum Entropy to Holography

We now change the question. Instead of asking for the entropy of a specified state, ask for the greatest entropy compatible with a region.

Start with an ordinary lattice model of a room. Divide the volume into cells of size V_{cell}, and let each cell carry one yes-no question: occupied or empty. Susskind gives the imaginary atom the deliberately absurd name "meatonium" so that no realistic atomic details distract from the count. If

$$N = \frac{V}{V_{\text{cell}}}, \tag{7.28}$$

then the number of configurations and the maximum entropy are

$$\Omega = 2^N, \qquad S_{\max} = \log\Omega = N\log 2 \sim \frac{V}{V_{\text{cell}}}. \tag{7.29}$$

The actual room is far below this maximum. A state that explored all those cell choices chaotically would be extremely hot. What survives from the toy model is the ordinary expectation that the number of independent degrees of freedom, and hence maximum entropy, scales with volume.

Gravity overturns that expectation. Consider a spherical region with boundary area A, initial entropy S_{i}, and mass M below the mass M_{BH} of a black hole that would just fill the sphere. If M were larger, its horizon would already extend beyond the chosen boundary.

Now send in a shell with mass $M_{\text{BH}} - M$. Its composition is irrelevant; the lecture allows light, ordinary matter, or even Swiss cheese. Arrange for it to collapse. The final state is a black hole whose horizon just fills the region, with entropy

$$S_{\text{f}} = \frac{A}{4G_N}. \tag{7.30}$$

The second law requires

$$S_{\text{i}} \le S_{\text{f}} = \frac{A}{4G_N}. \tag{7.31}$$

Because a black hole of that size is an allowed state and saturates the inequality, the maximal entropy is

$$S_{\max} = \frac{A}{4G_N}. \tag{7.32}$$

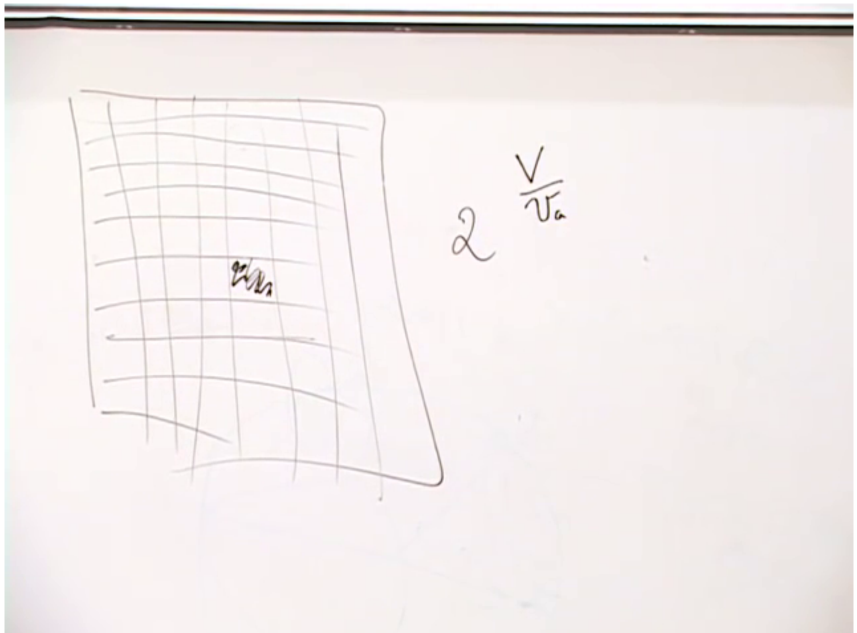


Figure 7.8: The room divided into cells, with two states per cell. This counts a maximum entropy, not the entropy of the orderly room occupied by the class.

Lecture frame: 01:05:20.000.

This is the holographic principle in its counting form: the maximum number of independent degrees of freedom in a gravitating region scales with the area of its boundary, approximately one order-unity degree of freedom per Planck area, rather than with the number of Planck volumes inside.[5]

The disparity is enormous. Local field-theory intuition suggests that choices made at different interior points should combine independently, producing a volume law. Gravity says that sufficiently many such choices would themselves collapse, and the black hole supplies the larger admissible entropy. The boundary count is therefore not a small correction to the volume count; it is a different organization of the Hilbert space.

Question from the audience. What happens for a very large universe with roughly uniform density?[a]

Response. That question requires cosmology rather than the static spherical construction. It motivates the immediate turn to expanding solutions of Einstein's equations.

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[a]Lecture timestamp: 01:15:48.

Question from the audience. Is the area bound connected with the uncertainty principle?[a]

Response. The lecture leaves this question open and moves next to cosmology.

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[a]Lecture timestamp: 01:16:03.

[5]Editorial clarification: The shell argument assumes the region can be enclosed and collapsed in the stated way. More general spacetimes require covariant entropy bounds, but the spherical argument captures the principle used in the lecture.

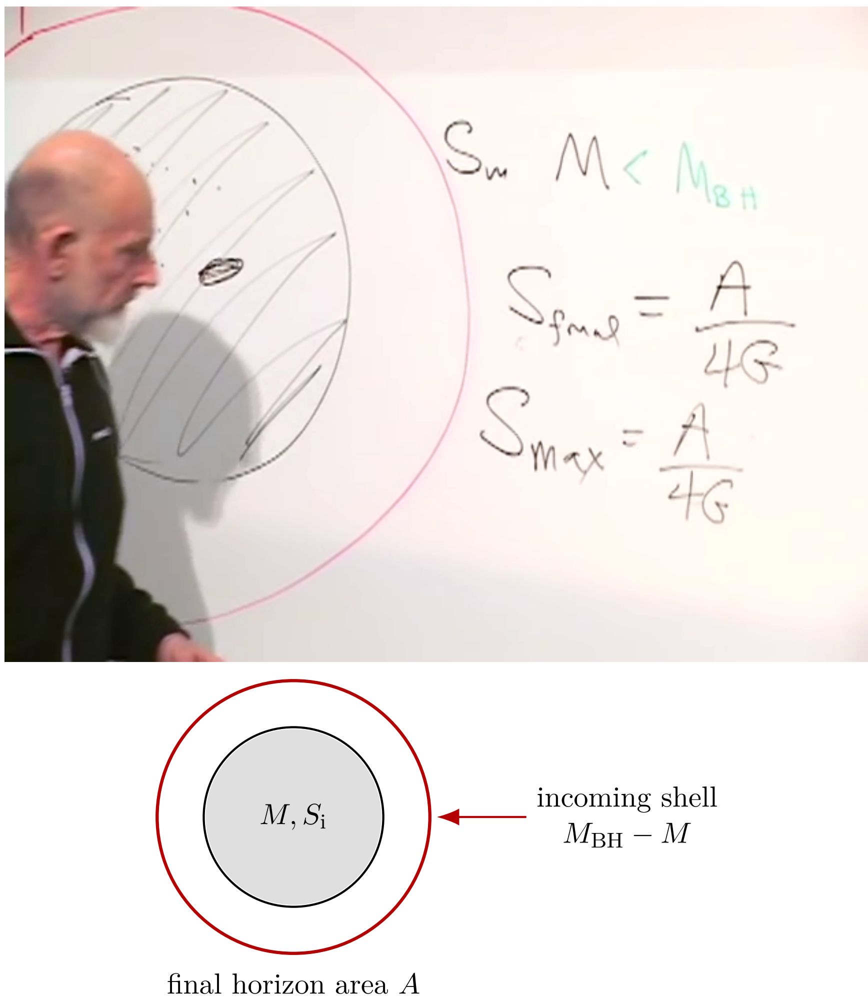

Figure 7.9: The shell construction. Enough additional mass is sent inward to form a black hole whose horizon just fills the selected region.

Lecture frame: 01:13:30.000.

7.10 Expanding Space

We shall not derive the cosmological solutions from Einstein's equations here; we shall write the solutions and learn how to read them. Spatial slices can be positively curved and closed, negatively curved and hyperbolic, or flat. For the present discussion take spatially flat Euclidean slices in a time-dependent spacetime.

Question from the audience. What is the spacetime analogue of ordinary Euclidean space?[a]

Response. Minkowski space. The cosmology considered here has flat spatial slices but is not Minkowski spacetime, because its spatial metric changes with time.

[a] Lecture timestamp: 01:19:37.

The familiar balloon picture describes a closed expanding surface one dimension lower. Susskind recalls trying the demonstration with a large marked balloon; the marker weakened the rubber and produced a literal bang. The mishap also exposes a conceptual defect in the analogy. As an ordinary balloon expands, its rubber molecules separate and the material thins until it breaks. Cosmological space does not acquire diluted local laws. To preserve the analogy one must imagine continually inserting new rubber molecules so that every small patch retains the same local properties. In a flat model, replace the balloon by a large sheet pulled uniformly outward and similarly replenished.

Mathematically, the expansion belongs in the metric:

$$\begin{aligned} ds^2 &= -dt^2 + a(t)^2 d\mathbf{x}^2, \\ d\mathbf{x}^2 &= \delta_{ij}\, dx^i dx^j = dx^2 + dy^2 + dz^2. \end{aligned} \tag{7.33}$$

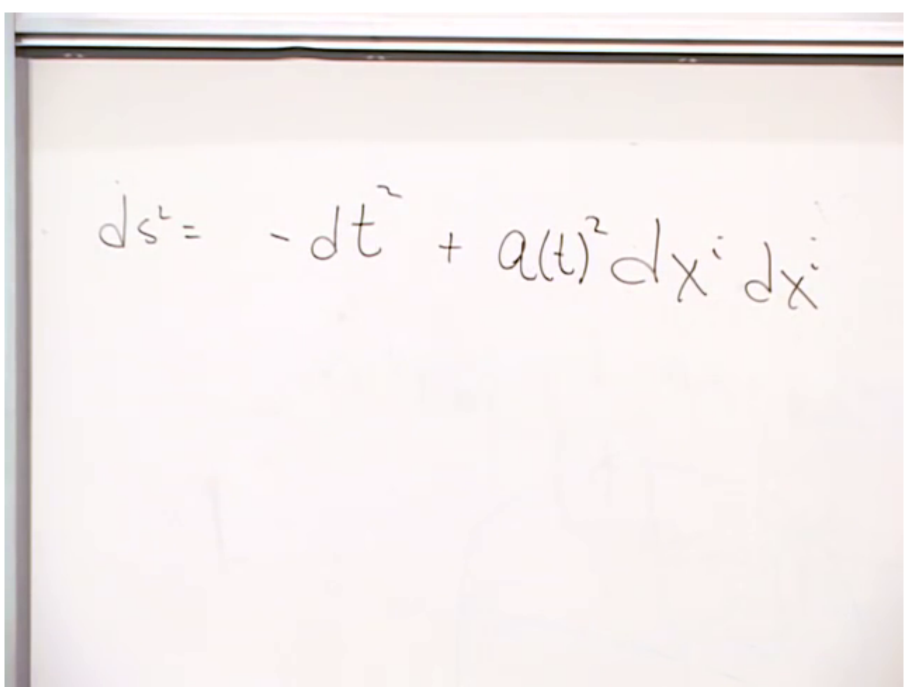

Figure 7.10: The spatially flat expanding metric as written on the board.
Lecture frame: 01:25:10.000.

Points that keep fixed spatial coordinates are called comoving. If two such points have coordinate separation Δx, their proper distance at time t is

$$D(t) = a(t)\Delta x. \tag{7.34}$$

Their coordinates do not move, but the metric distance between them changes. Differentiating at fixed Δx,

$$V = \dot{D} = \dot{a}\,\Delta x = \frac{\dot{a}}{a} D. \tag{7.35}$$

Define the Hubble parameter

$$H(t) = \frac{\dot{a}(t)}{a(t)}. \tag{7.36}$$

Then

$$V = H(t)D, \tag{7.37}$$

which is Hubble's law at a fixed cosmic time.

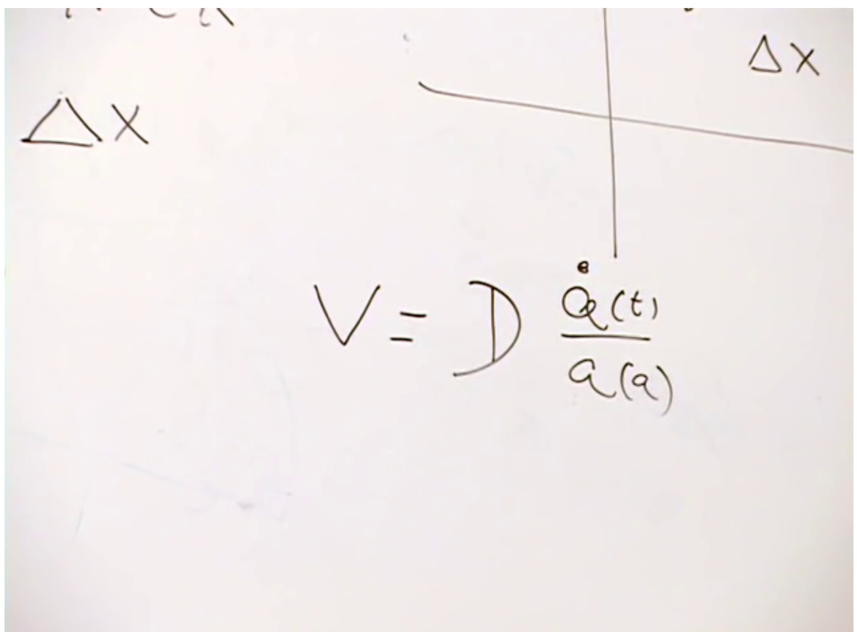

Figure 7.11: The equal-time recession law on the board: $V = D\,\dot{a}/a$.
Lecture frame: 01:29:10.000.

The name "Hubble constant" can mislead. In general $H(t)$ changes with time. What is spatially constant is the ratio V/D for every pair of comoving points on the same time slice.

Question from the audience. For distant galaxies, whose time appears in this relation? We observe them at an earlier epoch.[a]

Response. Equation (7.37) is an equal-time statement. An observation follows a past-directed light ray and samples the distant galaxy at an earlier time, so the relation between measured redshift and distance requires the full light-cone history rather than one instantaneous value of H.

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[a] Lecture timestamp: 01:31:10.

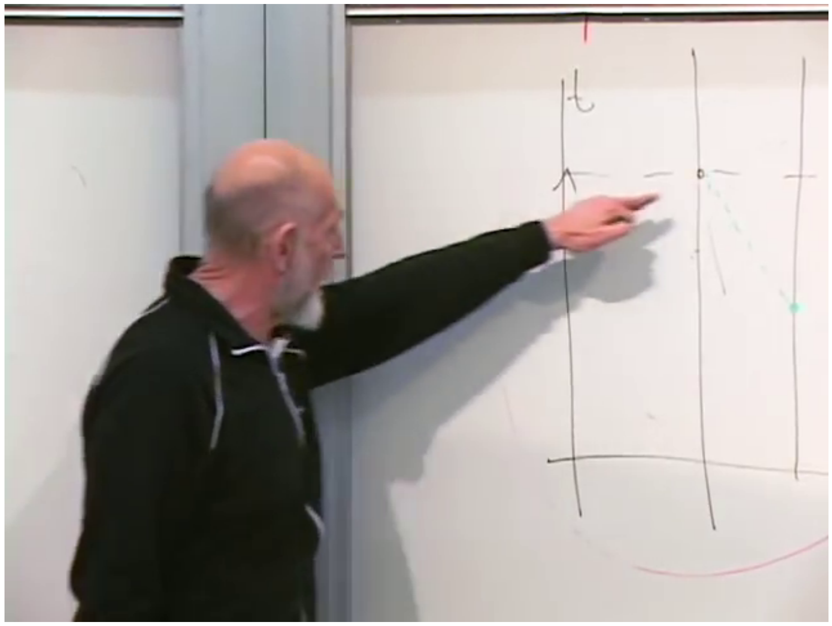

Figure 7.12: The spacetime sketch used to distinguish an equal-time separation from what an observer sees along a past light ray.

Lecture frame: 01:33:20.000.

7.11 The Hubble Scale and de Sitter Expansion

In units $c = 1$, Equation (7.37) assigns recession speed one at the distance

$$D_H = \frac{1}{H}. \tag{7.38}$$

More distant comoving points can have $V > 1$, or even $V \gg 1$, without violating special relativity. They do not pass a local observer faster than light; their proper separation grows because the intervening geometry expands.[6]

In many older decelerating cosmological models, $H(t)$ decreases and $1/H(t)$ grows, so regions once receding superluminally can later enter causal contact. The late-time picture discussed in the lecture instead approaches a positive constant H. Then $1/H$ remains fixed, and the expansion equation becomes

$$\frac{\dot{a}}{a} = H, \qquad \dot{a} = Ha. \tag{7.39}$$

Question from the audience. What are the units of a constant Hubble parameter?[a]

Response. Inverse time. Equivalently, with $c = 1$, it is inverse length. The quantity H^{-1} supplies both an expansion time and a distance scale; the lecture uses an order-of-magnitude example of about fifteen billion years.

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[a]Lecture timestamp: 01:38:24.

[6]Editorial clarification: For a general time-dependent cosmology, $1/H$ is the Hubble radius, not automatically an event horizon. It becomes the constant horizon scale in the de Sitter case discussed next.

The solution of Equation (7.39) is

$$a(t) = a_0 e^{Ht}. \tag{7.40}$$

Question from the audience. What function has a rate of change proportional to itself?[a]

Response. An exponential. Thus the scale factor is $a_0 e^{Ht}$, while the spatial coefficient $a(t)^2$ in the metric grows as e^{2Ht}.

[a]Lecture timestamp: 01:39:04.

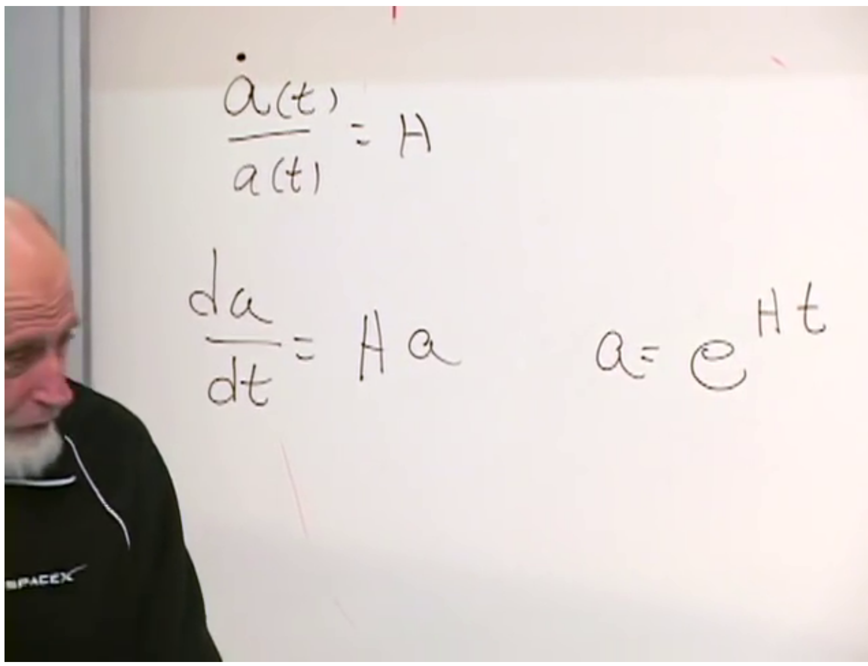

Figure 7.13: The constant-H equations and their exponential solution on the board.
Lecture frame: 01:39:30.000.

A spacetime with this exponential expansion is de Sitter space. Its constant Hubble scale produces an event horizon at finite proper distance. The geometry is in a sense the complement of the black-hole picture: an observer remains inside a cosmological horizon while sufficiently distant objects recede and can never send signals that catch the observer. A black-hole horizon hides an interior from an exterior observer; a de Sitter horizon hides the exterior of the observer's accessible patch. Unlike an ordinary evaporating black hole, the ideal de Sitter geometry is treated here as persisting. The comparison is only beginning here. The next lecture develops the horizon, its thermodynamics, and the puzzles it creates for cosmology.

Chapter 8

Vacuum Energy and de Sitter Horizons

Overview. Vacuum energy is unlike ordinary matter: it neither selects a preferred inertial frame nor dilutes as the universe expands. Once gravity is included, that apparently innocuous zero of energy changes the geometry of spacetime. We shall derive the matter-dominated law $a(t) \propto t^{2/3}$, contrast it with the vacuum-dominated law $a(t) \propto e^{Ht}$, and then rewrite de Sitter spacetime in conformal coordinates. The resulting causal diagram reveals an observer-dependent event horizon at fixed proper distance.

8.1 Quantum Fluctuations in the Vacuum

We begin with the question that motivates the entire lecture. Quantum field theory assigns energy to empty space. Does this mean that real electron–positron pairs are continually emerging from nothing?

Question from the audience. Do electron–positron production and annihilation actually occur in the vacuum, or is that merely a bookkeeping device?[a]

Response. Virtual pairs contribute to the vacuum energy, but they are not ordinary particles that can be watched emerging from empty space. They are fluctuations of the quantum ground state and persist only within the limits appropriate to a virtual process.

[a] Lecture timestamp: 00:00:06.

The harmonic oscillator gives the cleanest model. Classically it can remain motionless at the minimum of its potential. Quantum mechanically, exact position at the minimum and exact zero momentum are incompatible. The ground state compromises between the two and retains the zero-point energy

$$E_0 = \frac{1}{2}\hbar\omega. \tag{8.1}$$

This is the least energy the oscillator can have. It is not an excited state and does not represent a stream of observable particles.

A free field decomposes into normal modes, each behaving like an oscillator. The electromagnetic field in a cavity is a familiar example. Every mode contributes a term of the form in Equation (8.1); electron, quark, and other quantum fields contribute their own ground-state fluctuations. Adding these contributions produces what we call the vacuum energy. The notorious mismatch between naive microscopic estimates and the observed cosmological scale is real, but it is not the subject of this lecture.

Question from the audience. Will the discrepancy of roughly 120 orders of magnitude be discussed again?[a]

Response. Not here. The present goal is to understand cosmic horizons: how the cosmological equations produce accelerated expansion and why accelerated expansion creates event horizons.

[a]Lecture timestamp: 00:02:59.

Question from the audience. Is the ultraviolet–infrared relation from the previous lecture part of tonight's discussion?[a]

Response. It may be revisited later, but it is not part of the planned argument. The immediate path runs from vacuum energy to cosmology and then to horizons.

[a]Lecture timestamp: 00:03:59.

8.2 Why Vacuum Energy Is Lorentz Invariant

The next question appears elementary but exposes the defining property of vacuum energy. If empty space has a positive energy density, why does it not behave like an optical medium and alter the speed of light?

Question from the audience. Should a positive vacuum-energy density have a refractive effect, making light travel slightly more slowly?[a]

Response. Ordinary material can do that because it defines a preferred rest frame. Vacuum energy is different: it does not select such a frame, and every inertial observer obtains the same vacuum-energy density. Local measurements of light therefore retain the invariant speed c.

[a]Lecture timestamp: 00:04:35.

We must distinguish a symmetry of the laws from a symmetry of a particular state. The equations of physics may be translation invariant even though Stanford is not Berkeley and Earth is not interstellar space. Translating the *entire* physical arrangement leaves the laws unchanged; the actual arrangement of matter need not look the same at every location. Likewise, rotationally invariant equations do not make this classroom rotationally symmetric.

The same distinction applies to Lorentz transformations. A gas has a rest frame. An observer moving through it sees moving molecules and a different energy density, so the state of the gas is not Lorentz invariant even though its governing equations are. Ordinary matter, radiation baths, and laboratory apparatus all break the symmetry by selecting special motions or directions.

Vacuum energy is the exceptional form of matter that does not do so. Imagine a detector that reports the vacuum-energy density. It gives some value ρ_{vac} while at rest. Accelerate the detector and repeat the measurement: the result is unchanged. In relativistic notation this property is encoded by a stress tensor proportional to the metric,

$$T^{\text{vac}}_{\mu\nu} = -\rho_{\text{vac}} g_{\mu\nu}, \qquad p_{\text{vac}} = -\rho_{\text{vac}}. \tag{8.2}$$

[1]

The local qualification matters. A light ray passing an observer is measured with nearby clocks and rulers to move at c. Asking for the speed of a faraway light ray relative to us in an expanding geometry requires a choice of coordinates and a prescription for comparing distant events. That coordinate speed need not equal c, even though every local observer measures c.

Question from the audience. In this discussion, are vacuum energy and the cosmological constant synonymous?[a]

Response. Yes. They are two placements of the same term in Einstein's equations: it may be described as a cosmological constant on the geometric side or as Lorentz-invariant vacuum stress-energy on the source side.

[a]Lecture timestamp: 00:11:57.

8.3 What Dilutes and What Does Not

Fill a large region with nonrelativistic particles of mass m. On scales large enough to average over galaxies and clusters, suppose the distribution is homogeneous. If a comoving volume contains N particles, its matter density is

$$\rho_m = \frac{Nm}{V}. \tag{8.3}$$

When every linear dimension doubles, the volume grows by $2^3 = 8$, while N and m remain unchanged. Thus the density falls to one eighth of its former value. Ordinary matter dilutes because the same particles occupy a larger physical volume.

Radiation also dilutes, and more rapidly. Its number density falls with the volume while each photon's wavelength is stretched and its energy redshifted. We shall later record the result $\rho_r \propto a^{-4}$, but we will not solve the radiation-dominated cosmology tonight.

Vacuum energy is the exception:

$$\rho_{\text{vac}}(t) = \text{constant}. \tag{8.4}$$

It does not cluster, select a direction, select an inertial frame, or become less dense as space expands. If a comoving box grows, the vacuum energy contained in it grows in proportion to its volume.

[2]

8.4 Gravity Sees the Zero of Energy

In nongravitational laboratory physics, shifting every energy by the same constant usually changes nothing. Spectra depend on differences: the energy between an excited and an unexcited atom, or between a state with and without a particle. Empty space seems to offer no reference against which its absolute energy could be measured.

[1]Editorial clarification: The pressure relation is not derived on the board. It is the covariant statement behind the lecture's two defining properties: Lorentz invariance and constant density during expansion.

[2]Editorial clarification: Constant density does not mean constant total vacuum energy inside an expanding comoving box. Covariant stress-energy conservation, $\dot{\rho} + 3H(\rho + p) = 0$, is satisfied because vacuum energy has negative pressure $p_{\text{vac}} = -\rho_{\text{vac}}$. The work term supplies precisely the change in the box's total vacuum energy.

Gravity changes the question. Mass is energy, energy sources gravity, and therefore vacuum energy gravitates. A constant that was invisible in an atomic transition can curve spacetime.

Einstein's equation separates geometry from sources:

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}, \tag{8.5}$$

in units with $c = 1$. The Einstein tensor $G_{\mu\nu}$ is built from the metric $g_{\mu\nu}(x)$ and its derivatives. The stress tensor $T_{\mu\nu}$ contains energy density, momentum density, pressure, and stress from matter and nongravitational fields. Vacuum energy belongs on this source side just as ordinary matter does.[3]

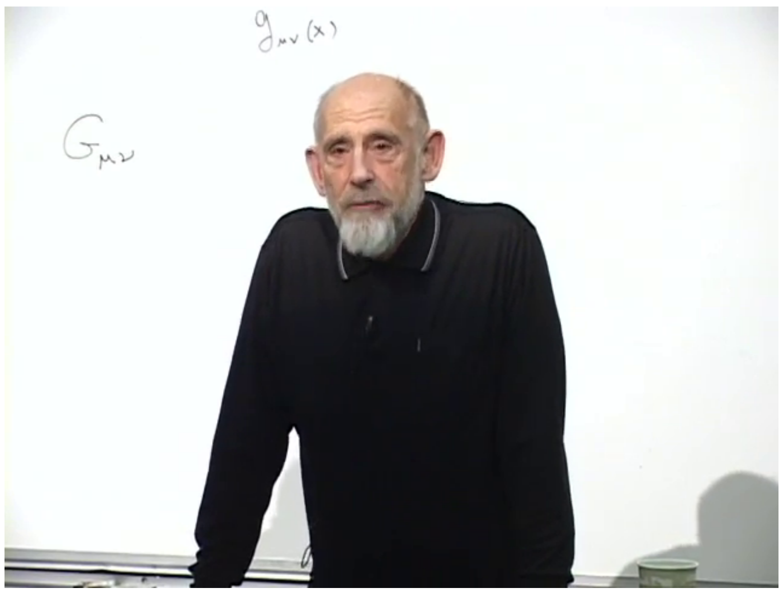

Figure 8.1: The geometric side of Einstein's equation, built from the spacetime metric $g_{\mu\nu}(x)$.
Lecture frame: 00:21:58.820.

8.5 Geometry Reduced to a Scale Factor

We now choose the simplest homogeneous expanding geometry with flat spatial slices:

$$ds^2 = -dt^2 + a(t)^2\, d\mathbf{x}^2, \qquad d\mathbf{x}^2 = dx^2 + dy^2 + dz^2. \tag{8.6}$$

The coordinate marks $\mathbf{x}$ are painted on an infinite rubber sheet. They need not move, but their physical separations do. Two comoving points separated by $\Delta\mathbf{x}$ have proper distance at fixed t

$$D(t) = a(t)|\Delta\mathbf{x}|. \tag{8.7}$$

Thus all the time dependence of this highly symmetric geometry is stored in one function, the scale factor $a(t)$. Constant a gives flat Minkowski spacetime after a spatial rescaling. Time-dependent a can leave each spatial slice flat while making the full spacetime curved.

[3] Editorial clarification: The spoken introduction briefly associates $8\pi/3$ with the tensor equation. The factor $1/3$ belongs to the spatially flat Friedmann equation below, not to Equation (8.5).

Question from the audience. For coordinate marks $x = 7$ and $x = 13$, is their coordinate separation 4?[a]

Response. No. It is $13 - 7 = 6$, so their physical separation is $6a(t)$. The correction is useful: x labels comoving position, while multiplication by $a(t)$ gives physical distance.

[a] Lecture timestamp: 00:27:40.

For this metric and a spatially uniform energy density, one component of Einstein's equation becomes the spatially flat Friedmann equation,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G_N}{3}\rho(t). \tag{8.8}$$

The ratio

$$H(t) \equiv \frac{\dot{a}(t)}{a(t)} \tag{8.9}$$

is the Hubble parameter. It is a definition at this stage and may vary with time.

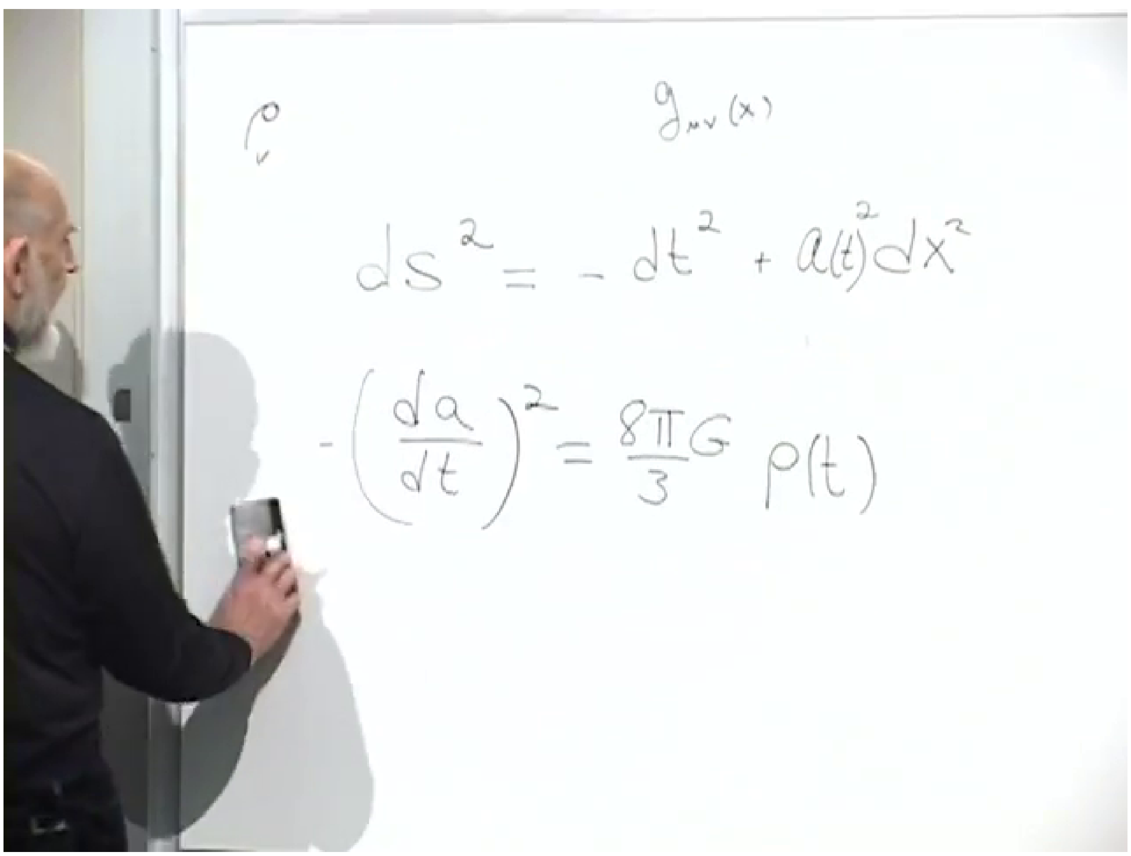

Figure 8.2: The flat expanding metric and Friedmann equation after the missing factor $1/a^2$ has been restored on the board.

Lecture frame: 00:30:30.000.

The energy-balance picture. For this energy-balance picture, Equation (8.8) is written schematically as

$$-\frac{3}{8\pi G_N}H^2 + \rho = 0. \tag{8.10}$$

The matter energy is positive, while the gravitational degree of freedom represented by the expanding scale factor carries a negative kinetic term. In this reduced problem, negative gravitational energy plus positive matter energy equals zero is a useful mnemonic.

[4]

8.6 Matter-Dominated Expansion

Return to a comoving mathematical box with coordinate side $\Delta x = 1$. Its physical side is $a(t)$, so its physical volume is $a(t)^3$. If it contains N nonrelativistic particles of mass m, then

$$\rho_m(t) = \frac{Nm}{a(t)^3} = \frac{\rho_0}{a(t)^3}. \tag{8.11}$$

The particles themselves do not expand; their mean separation does.

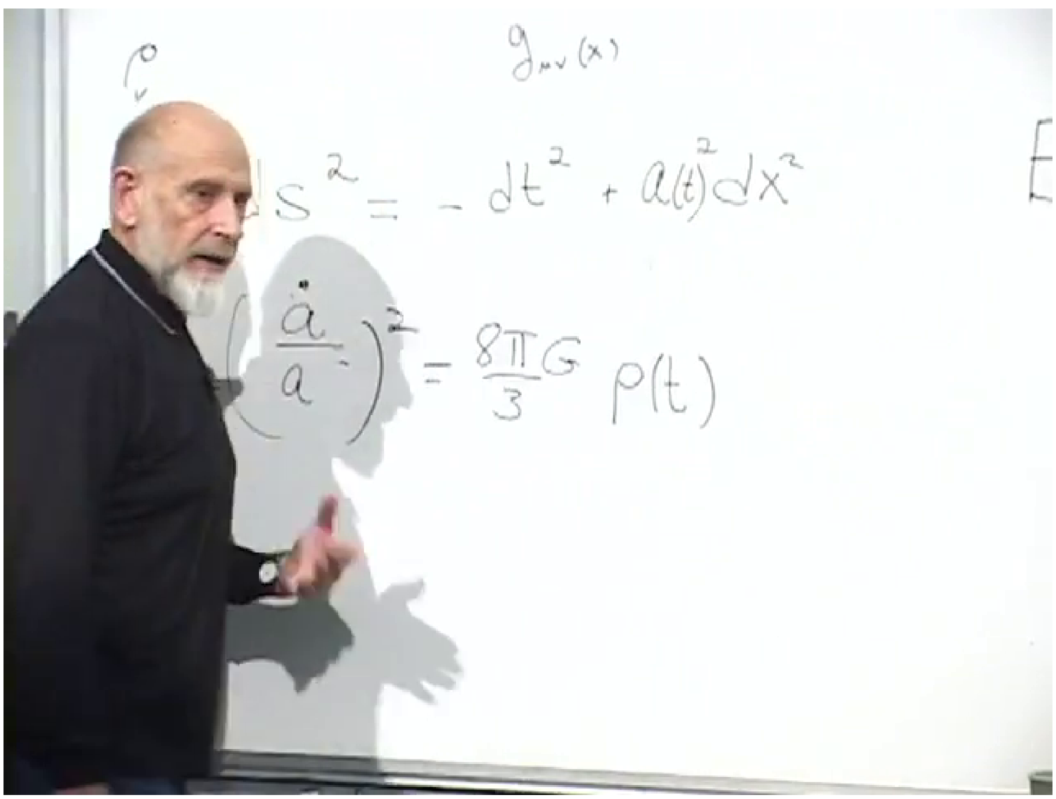

Figure 8.3: The metric and Friedmann equation as the comoving-box argument is assembled.

Lecture frame: 00:36:07.940.

Substitution into Equation (8.8) gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G_N \rho_0}{3a^3}. \tag{8.12}$$

[4]Editorial clarification: Equation (8.10) is the Hamiltonian constraint in a highly symmetric minisuperspace model, not a coordinate-independent theorem that every arbitrary universe possesses a globally conserved total energy equal to zero. The lecture uses it as a heuristic interpretation of the Friedmann constraint.

Write $K = 8\pi G_N \rho_0/3$ and choose the expanding branch:

$$\dot{a}^2 = \frac{K}{a}, \tag{8.13}$$

$$\dot{a} = \sqrt{K}\, a^{-1/2}, \tag{8.14}$$

$$a^{1/2}\, da = \sqrt{K}\, dt. \tag{8.15}$$

Question from the audience. How should the separated equation be solved?[a]

Response. Integrate both sides. The integral of $a^{1/2}\,da$ is $\frac{2}{3}a^{3/2}$, not $\frac{3}{2}a^{3/2}$; $\frac{3}{2}$ is the new power, while $\frac{2}{3}$ is the coefficient.

---

[a]Lecture timestamp: 00:38:15.

Accordingly,

$$\frac{2}{3}a^{3/2} = \sqrt{K}\,(t - t_B), \tag{8.16}$$

where t_B is an integration constant fixing the chosen origin of time. Apart from normalization and the time shift,

$$a(t) \propto (t - t_B)^{2/3}. \tag{8.17}$$

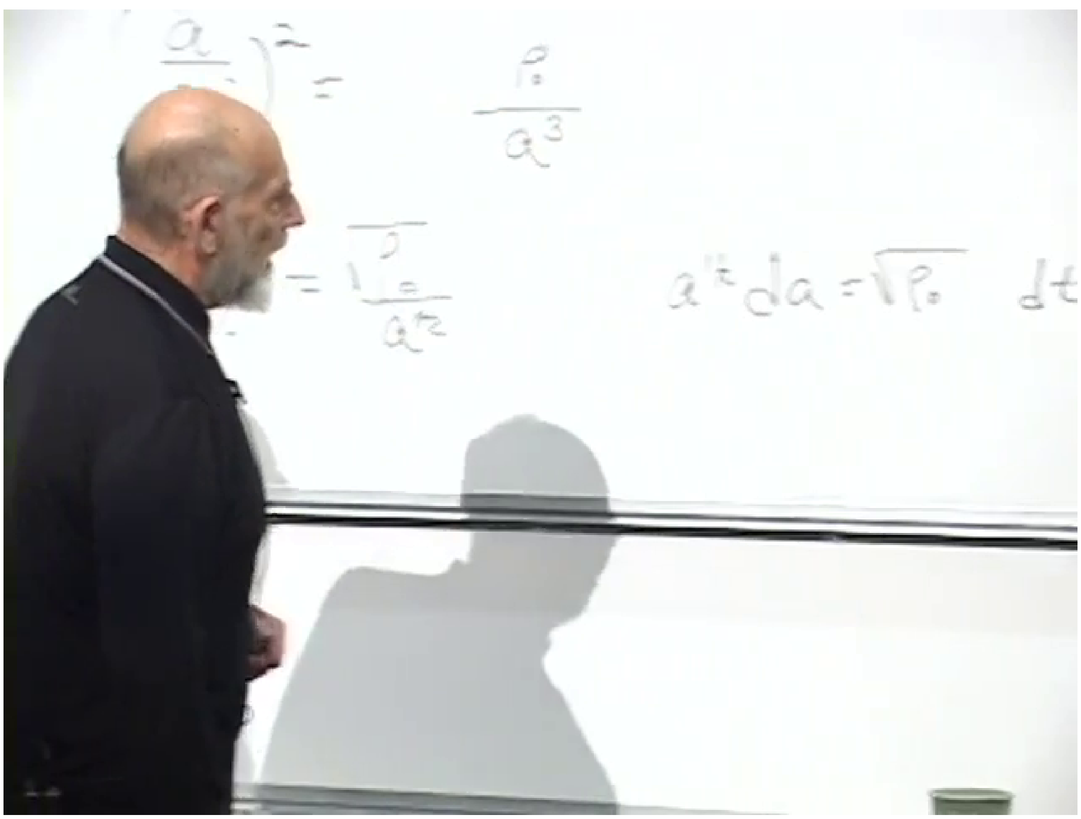

Figure 8.4: The matter-density substitution and separation of variables leading to $a^{1/2}da \propto dt$.
Lecture frame: 00:38:30.000.

The curve $t^{2/3}$ grows more slowly than a straight line. Its slope decreases, so a matter-dominated universe decelerates. In the Newtonian language used in the lecture, the particles began by separating, and their mutual gravitational attraction continually reduces the rate of separation.

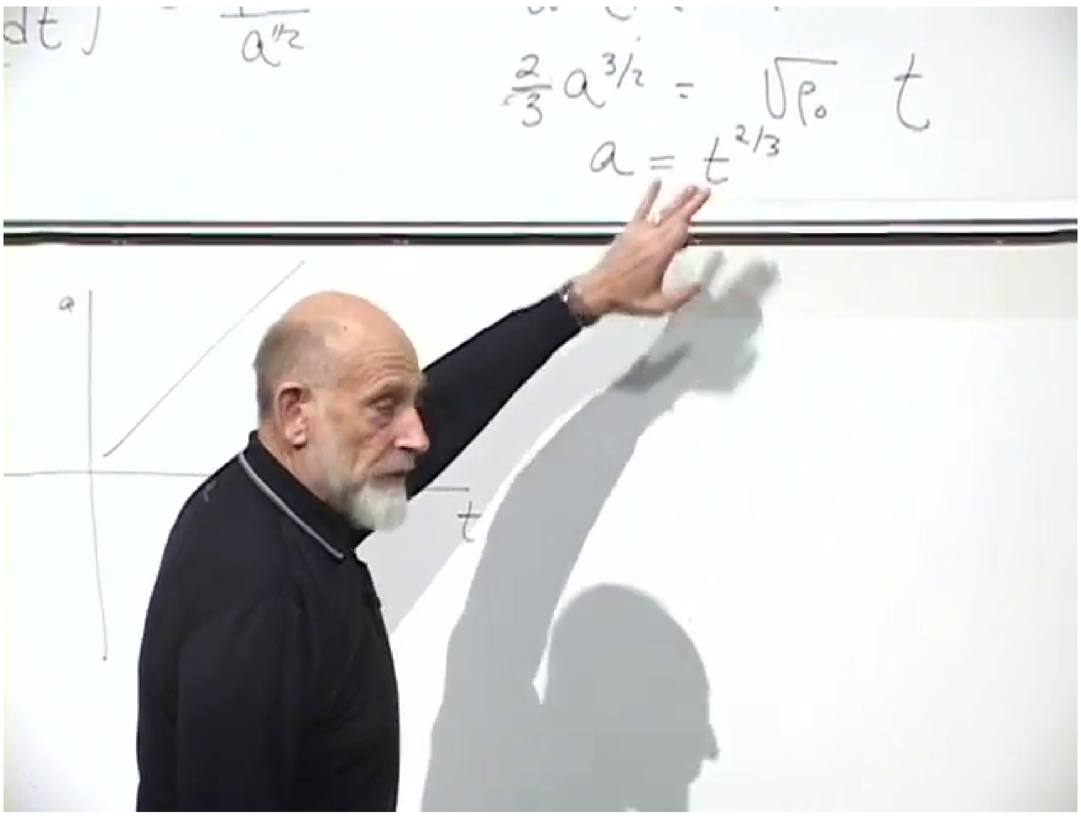

Figure 8.5: The board comparison of linear expansion with the concave-down matter law $a(t) \propto t^{2/3}$.
Lecture frame: 00:40:00.000.

For orientation, a radiation bath has

$$\rho_r \propto a^{-4}, \tag{8.18}$$

rather than a^{-3}: the volume dilution contributes three powers and cosmological redshift contributes one more. The corresponding scale factor can also be solved, but the lecture deliberately leaves that exercise aside.

8.7 Vacuum-Dominated Expansion

Now put only vacuum energy on the right-hand side of the Friedmann equation. Since ρ_{vac} is constant,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G_N}{3}\rho_{\text{vac}} \equiv H^2. \tag{8.19}$$

There are two uses of H to keep separate. Equation (8.9) defines the possibly time-dependent quantity $\dot{a}/a$. Equation (8.19) says that in a pure vacuum-energy universe its value is the constant

$$H = \sqrt{\frac{8\pi G_N \rho_{\text{vac}}}{3}}. \tag{8.20}$$

Taking the expanding branch yields

$$\dot{a} = Ha, \tag{8.21}$$

and therefore

$$a(t) = Ce^{Ht}. \tag{8.22}$$

The integration constant C fixes the arbitrary normalization of comoving coordinates. Growth proportional to the current value of a is exponential growth.

The Hubble law for comoving objects is

$$v_{\text{rec}} = HD. \tag{8.23}$$

In this pure de Sitter case, the proper distance at which the recession rate equals c is constant:

$$D_H = \frac{c}{H}, \tag{8.24}$$

or $D_H = H^{-1}$ in the $c = 1$ units used on the board. This fixed scale will reappear geometrically as the de Sitter event-horizon radius.[5]

The lecture offers a thought experiment for a vacuum-energy detector. Place two very light test bodies far from other sources, initially at rest relative to the comoving flow, and make their mutual attraction negligible. Their separation obeys the exponential expansion. The effect is far too small for a terrestrial apparatus, but cosmic distances and lookback time make the same physics observable.

[5] Editorial clarification: A Hubble radius is not automatically an event horizon in an arbitrary time-dependent cosmology. The identification works here because H is constant in exact de Sitter space.

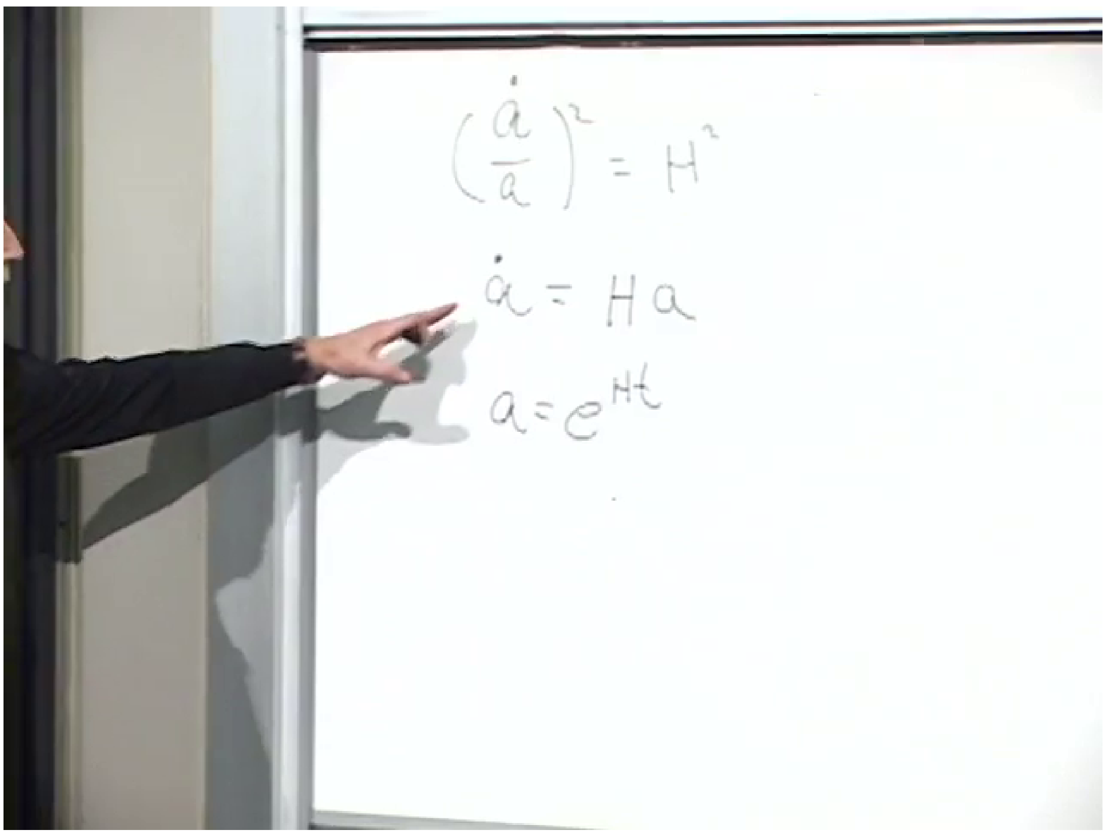

Figure 8.6: The vacuum-dominated equation $(\dot{a}/a)^2 = H^2$, its first-order form $\dot{a} = Ha$, and the exponential solution.

Lecture frame: 00:47:45.000.

8.8 From Matter Domination to Vacuum Domination

The real cosmological model considered in the lecture contains both terms:

$$H(t)^2 = \frac{8\pi G_N}{3}\left(\frac{\rho_0}{a(t)^3} + \rho_{\rm vac}\right). \tag{8.25}$$

At small a, the matter term dominates because a^{-3} is large, and the solution resembles $t^{2/3}$. As the universe grows, matter dilutes while vacuum energy remains fixed. Eventually the constant term wins and the scale factor bends upward toward e^{Ht}.

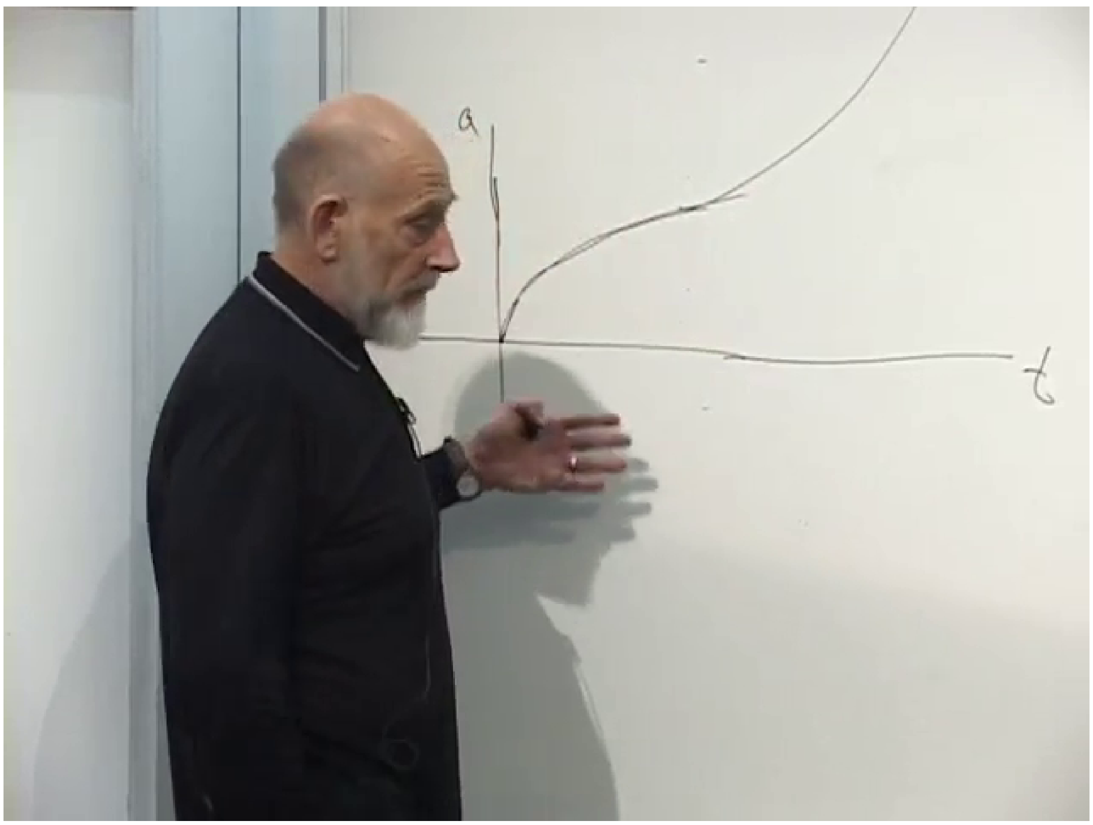

Figure 8.7: The qualitative transition drawn in the lecture: early matter-dominated deceleration followed by late vacuum-dominated exponential growth.

Lecture frame: 00:53:30.000.

Question from the audience. The exponential solution contains an integration constant. How do we know that it is not extremely small?[a]

Response. We measure it. Looking farther into the universe means looking farther into the past, so observations at different distances reconstruct the earlier portion of the expansion curve rather than merely its value today.

---

[a]Lecture timestamp: 00:52:41.

The lecture reports that the then-available late-time observations were consistent with an exponential branch at roughly the five-percent level and describes the evidence as convincing, though not as exact as a precision measurement such as the electron's properties. This should be read as the lecture's historical assessment rather than as a present-day precision statement.

Question from the audience. Does the upward-bending expansion curve eventually prevent us from tracing observations back to the Big Bang or the cosmic microwave background?[a]

Response. That is not pursued as a question about extrapolating the graph. The relevant invariant issue is the existence of an event horizon, and that is the causal question we now turn to.

[a]Lecture timestamp: 00:55:05.

8.9 Bound Systems and the Distant Future

If exponential expansion continues, every component that dilutes becomes negligible compared with ρ_{vac}. This does not mean that atoms, solar systems, or galaxies themselves are stretched apart. Systems held together strongly enough to remain bound do not follow the Hubble flow.

Question from the audience. Do galaxies that are not expanding today eventually expand when vacuum energy dominates?[a]

Response. Bound systems remain bound. The Milky Way and Andromeda belong to the nearby structure that will remain together, while sufficiently distant galaxies already moving with the expansion continue outward and eventually pass beyond our horizon.

[a]Lecture timestamp: 00:57:44.

The distinction between *attracted* and *bound* is essential. A rocket launched above escape velocity remains under Earth's attractive gravity, yet it does not return. Likewise, ordinary gravitational attraction may be the largest local force between two galaxies without their relative orbit being bound.

Question from the audience. Our galaxy is gravitationally associated with a group. Does accelerated expansion eventually overcome that attraction?[a]

Response. Only the genuinely bound part of the local structure stays together. A cluster as a descriptive label is not enough; binding means that the relative motion lies below escape. More distant galaxies can be attracted and nevertheless recede above escape speed.

[a]Lecture timestamp: 00:59:17.

For intuition, the lecture represents the relative radial acceleration schematically as an attractive inverse-square term plus a vacuum term linear in separation:

$$\ddot{r} \sim -\frac{G_N M}{r^2} + H^2 r. \tag{8.26}$$

The coefficients are time independent in the idealized de Sitter limit. The second term grows only because r grows; it does not arise from a gradual weakening of G_N.
6

[6]Editorial clarification: Equation (8.26) is a Newtonian relative-acceleration heuristic, not an additional force law fundamental to general relativity. In exact de Sitter space, neighboring freely falling worldlines have relative acceleration proportional to $H^2 r$.

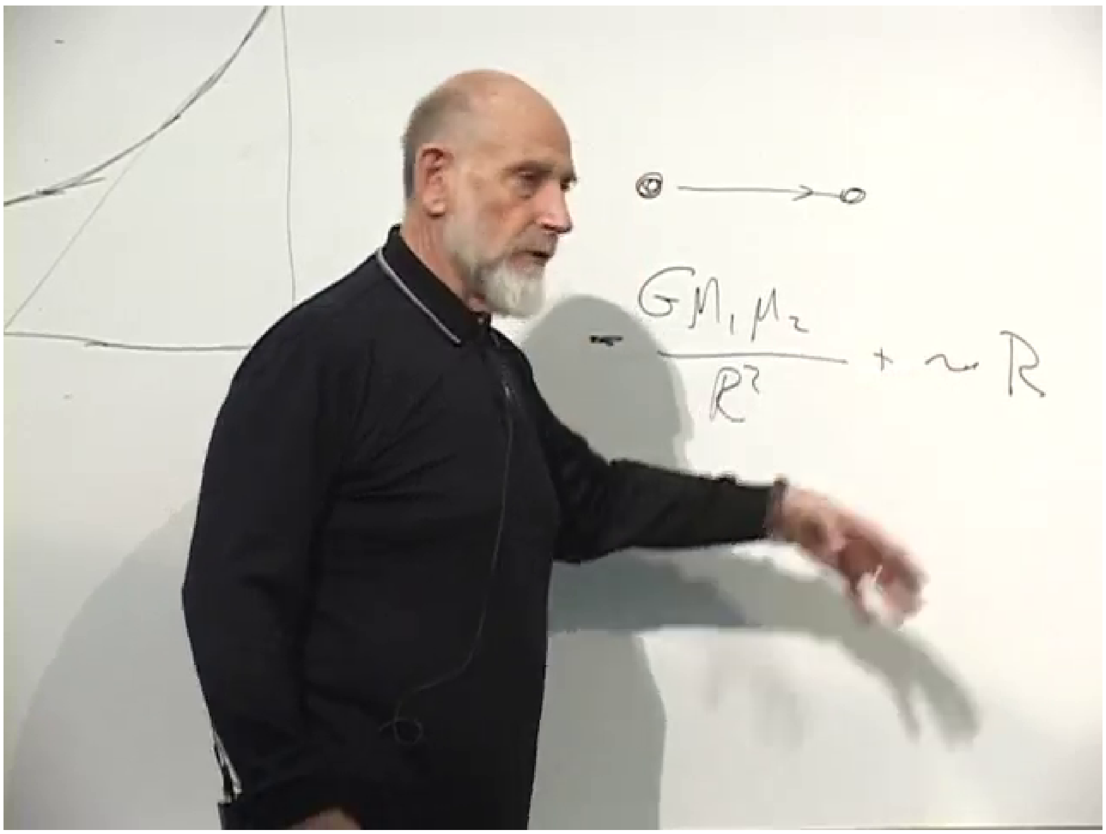

Figure 8.8: The board comparison between inverse-square attraction and the effective vacuum repulsion that grows linearly with separation.

Lecture frame: 01:04:10.000.

Question from the audience. Does the outward effect itself increase with time?[a]

Response. Not at a fixed separation. Its coefficient is constant. Freely receding objects move farther apart, and the term proportional to r is then larger because their distance is larger.

[a]Lecture timestamp: 01:01:18.

Question from the audience. Can the net effect be described as a gradual decrease of Newton's constant?[a]

Response. No. G_N remains constant. At a specified separation both coefficients are unchanged; only the relative importance of the $1/r^2$ attraction and the term proportional to r changes with distance.

[a]Lecture timestamp: 01:02:30.

Question from the audience. Were the same galaxies gravitationally bound in the early universe before vacuum energy became important?[a]

Response. Dominance of ordinary attraction does not by itself imply binding. Many galaxies were already separating at about escape speed or above it. Vacuum energy later determines whether the large-scale expansion decelerates toward recollapse or accelerates away, but largest force and bound orbit remain different statements.

[a]Lecture timestamp: 01:05:03.

Thus a far-future observer may inhabit a merged local structure while every unbound galaxy has become unobservable. Without historical records, that observer could reasonably infer an almost empty universe. This causal loneliness, rather than destruction of local objects, is the physical lead-in to the horizon.

8.10 What an Event Horizon Means

After the break, set

$$a(t) = e^{Ht} \tag{8.27}$$

and write the metric as

$$ds^2 = -dt^2 + e^{2Ht} d\mathbf{x}^2. \tag{8.28}$$

Question from the audience. What is the name of the spacetime described by this metric?[a]

Response. It is de Sitter spacetime, commonly shortened to de Sitter space. Its horizons matter because a vacuum-dominated universe approaches this geometry at late times.

[a]Lecture timestamp: 01:08:13.

Before drawing its causal structure, recall the role of a Penrose diagram. Infinite intervals are compressed into a finite picture while null rays remain at 45°. In flat spacetime, massive objects end at future timelike infinity, null rays end at null infinity, and an inertial observer allowed to wait arbitrarily long has no event horizon excluding a permanent region of spacetime.

A black-hole diagram supplies the contrast. An exterior observer can receive signals only from events in the causal past of the observer's future worldline. Even as the observation time tends to

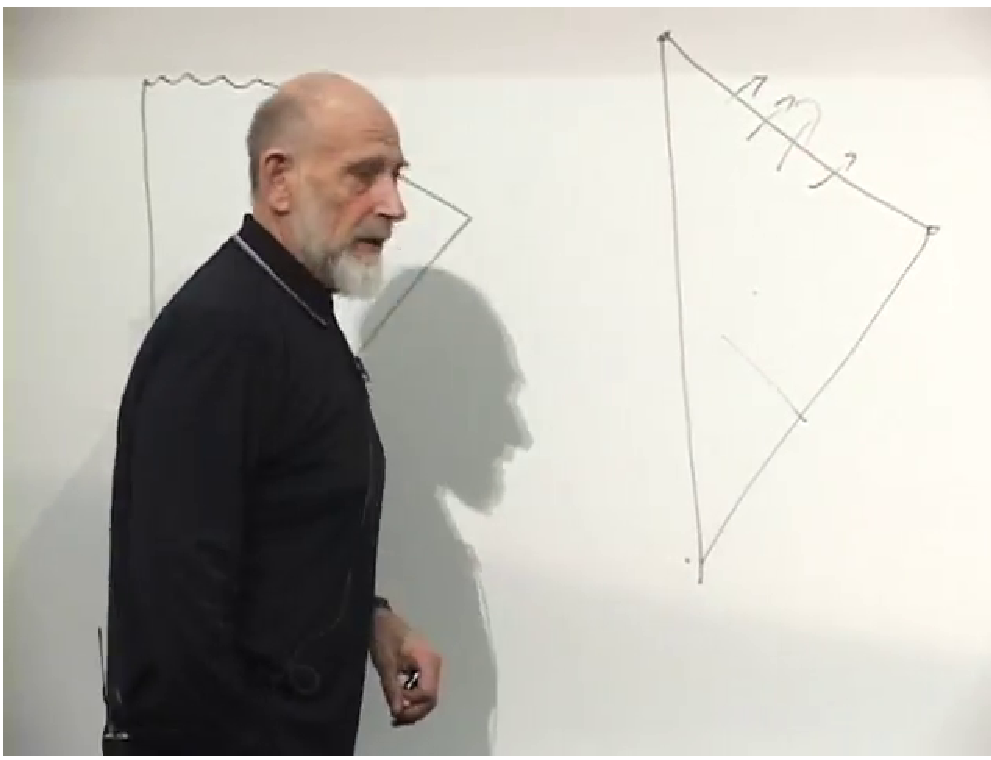

Figure 8.9: Flat-spacetime Penrose diagram used to distinguish timelike and null destinations before introducing a horizon.

Lecture frame: 01:10:35.000.

infinity, the interior region remains excluded. The boundary between events that can communicate with that observer and events that never can is the event horizon.

Definition 8.1. The future event horizon of an observer is the boundary of the set of events from which a future-directed causal signal can ever reach that observer's worldline.

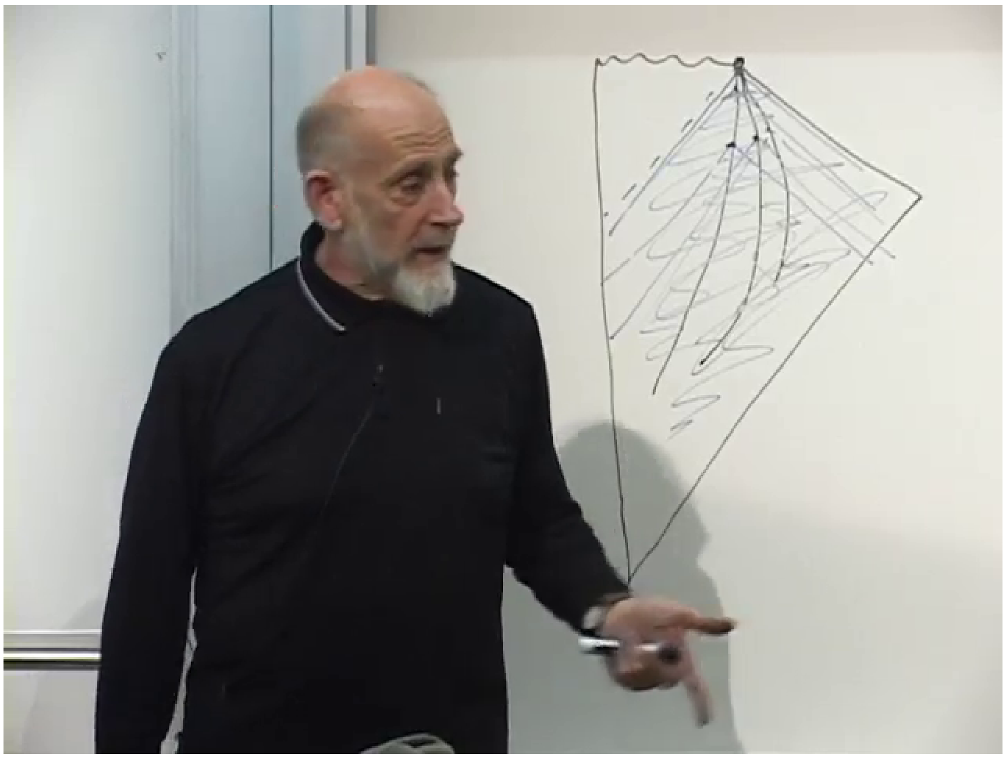

Figure 8.10: The black-hole comparison: backward light cones from arbitrarily late exterior observations never include the region beyond the event horizon.

Lecture frame: 01:13:50.000.

An event horizon is therefore a global causal boundary, not merely the distance at which an object happens to look faint today. To locate it, one must know which signals can arrive even after an arbitrarily long wait.

8.11 Compressing the Infinite Future

The spatial coordinates in Equation (8.28) already range over an infinite flat slice. We shall leave them alone and compress only time. Introduce conformal time T by requiring

$$dt^2 = e^{2Ht}dT^2. \tag{8.29}$$

Taking the future-directed square root and rearranging gives

$$dT = e^{-Ht}dt. \tag{8.30}$$

Question from the audience. What is the integral, including its sign?[a]

Response. It is $-H^{-1}e^{-Ht}$. An additive constant merely shifts the origin of T, so choose it to place the infinite future at $T = 0$.

[a]Lecture timestamp: 01:17:25.

Thus

$$T = -\frac{1}{H}e^{-Ht}, \qquad -\infty < T < 0. \tag{8.31}$$

As $t \to -\infty$, $T \to -\infty$. As the proper time $t \to +\infty$, $T \to 0^-$. An infinite future has been placed on the finite horizontal line $T = 0$; no observer reaches it in finite proper time.

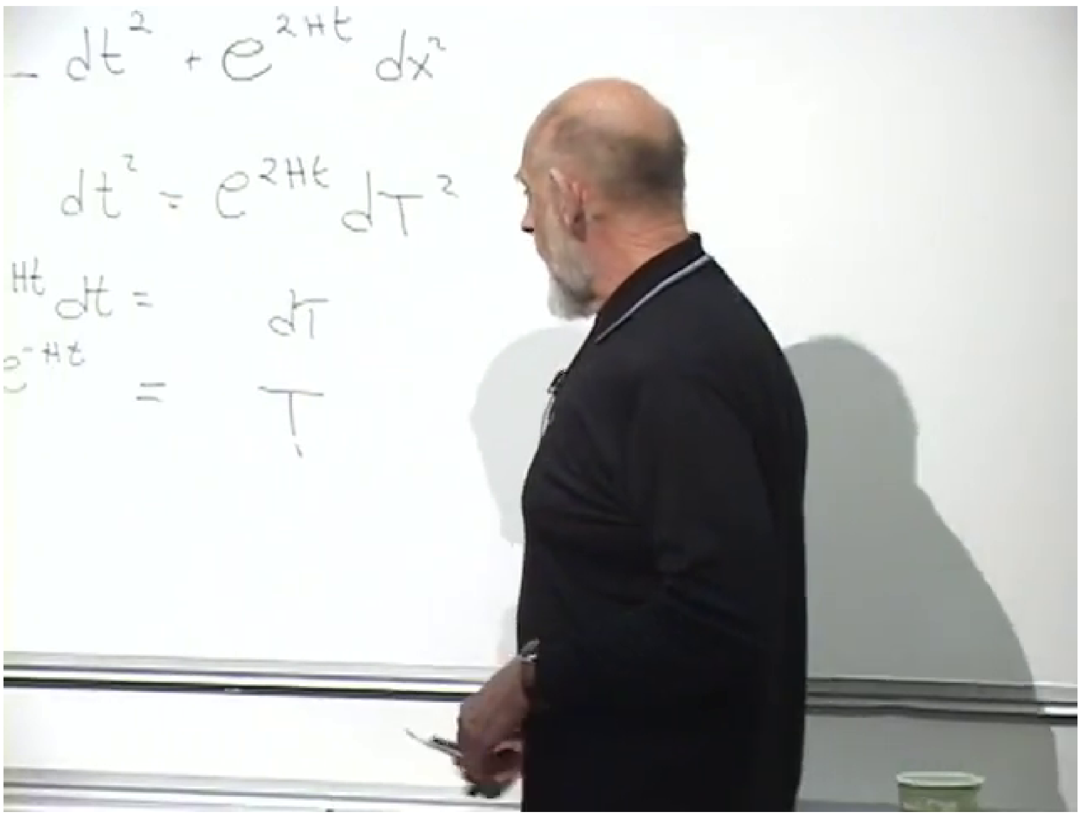

Figure 8.11: The conformal-time integral and the mapping of the infinite future $t \to +\infty$ to $T = 0^-$.

Lecture frame: 01:19:10.000.

From Equation (8.31),

$$HT = -e^{-Ht}, \qquad e^{2Ht} = \frac{1}{H^2T^2}. \tag{8.32}$$

Using $dt = e^{Ht}dT$ in Equation (8.28) then yields

$$ds^2 = \frac{1}{H^2T^2}\left(-dT^2 + d\mathbf{x}^2\right). \tag{8.33}$$

The expression in parentheses is the Minkowski metric. All the de Sitter geometry now sits in a common conformal factor that diverges as $T \to 0^-$.

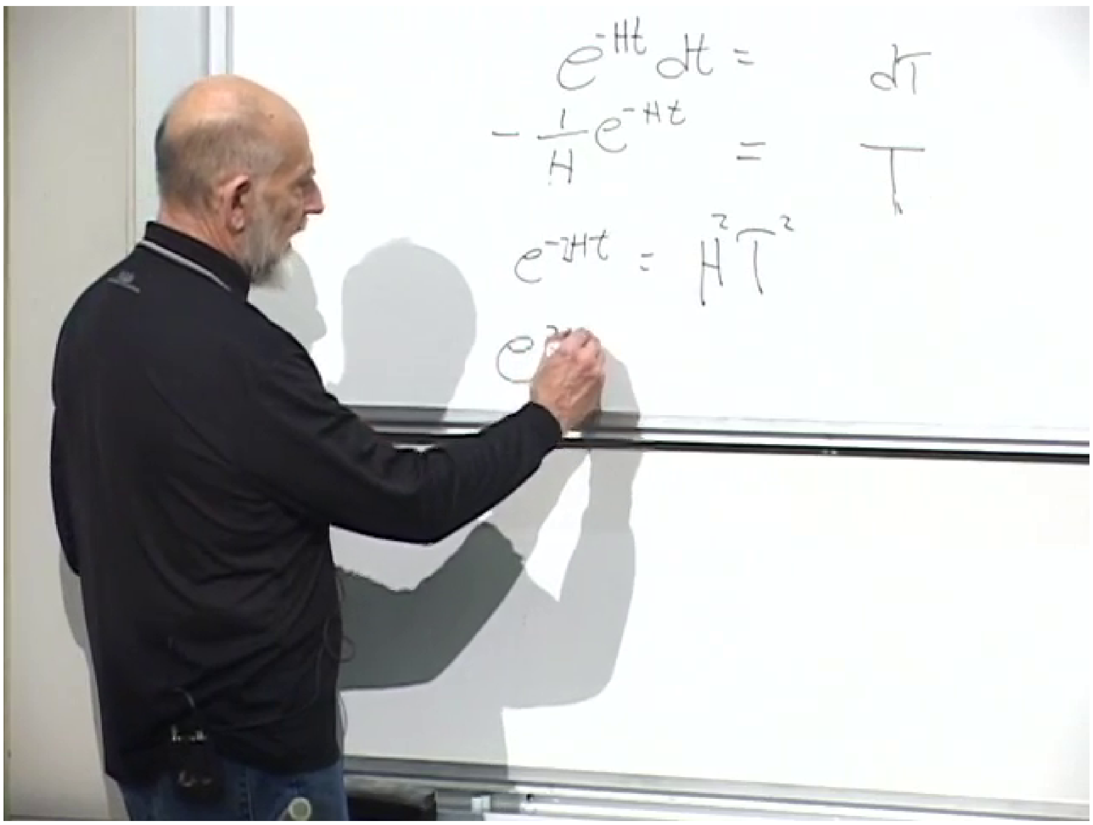

Figure 8.12: The algebra relating e^{2Ht} to $1/(H^2T^2)$, preparing the conformal form of the metric.

Lecture frame: 01:22:50.000.

Question from the audience. What condition determines the path of a light ray in this metric?[a]

Response. A light ray follows a null path, $ds^2 = 0$. The common conformal factor therefore cancels, leaving $dT = \pm dx$ along a chosen spatial direction.

[a] Lecture timestamp: 01:23:49.

Light rays are straight 45° lines in the (T, x) diagram:

$$\frac{dx}{dT} = \pm 1. \tag{8.34}$$

The coordinate picture is highly compressed near $T = 0$, but it preserves precisely what we need: which events can communicate.

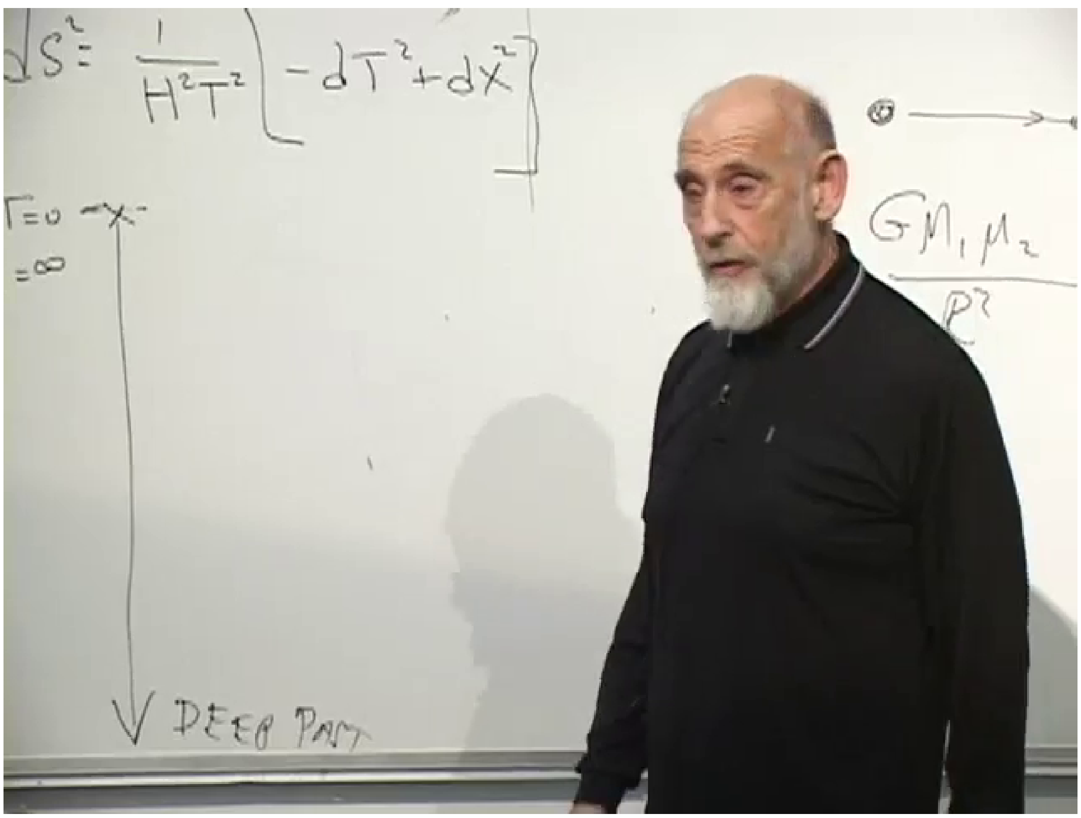

Figure 8.13: The conformal metric at the moment the null condition $ds^2 = 0$ is used to recover 45° light rays.

Lecture frame: 01:24:05.000.

8.12 Alice and Bob in de Sitter Space

Place Alice at fixed comoving position $x = 0$. Her worldline rises vertically and ends on the diagram at $T = 0$, although her wristwatch requires infinite proper time to get there. Draw the past-directed null rays from that endpoint. They bound the largest region from which Alice can ever receive a signal. Events outside it remain forever inaccessible.

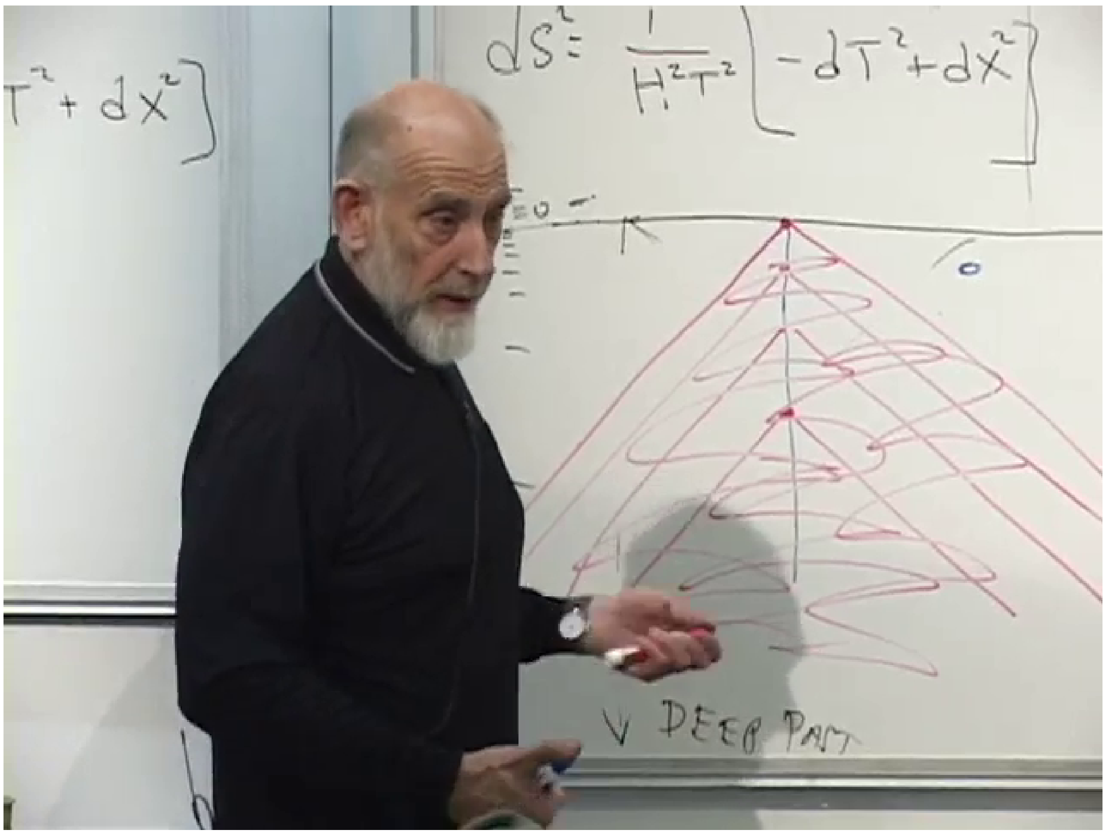

Figure 8.14: Alice's limiting past light cone. Its null boundary is her observer-dependent de Sitter event horizon.

Lecture frame: 01:28:50.000.

This horizon is private. A different observer has a different future endpoint and therefore a different limiting past light cone. Suppose Bob once lies within Alice's causal region and the two exchange messages. If Bob later follows the Hubble flow across Alice's horizon, signals emitted after that crossing cannot reach Alice, however long she waits. Bob has his own horizon and eventually loses the corresponding ability to receive late signals from Alice.

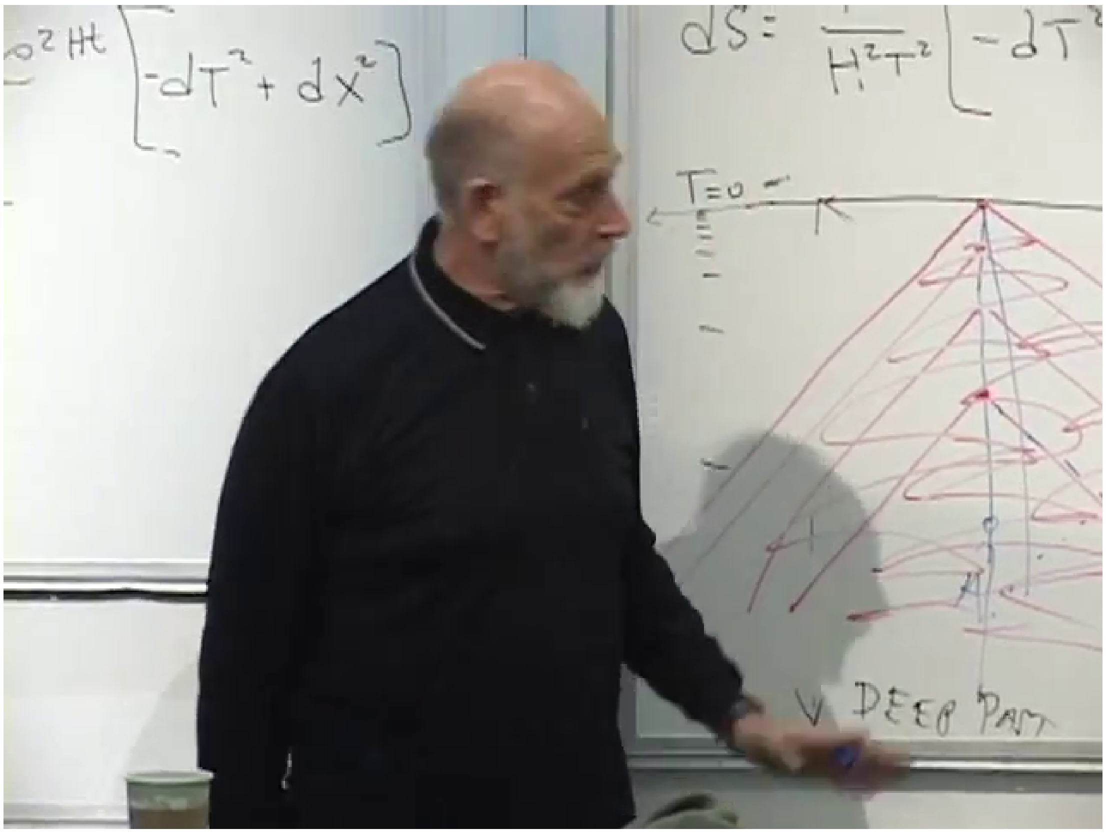

Figure 8.15: Alice and Bob with their distinct causal boundaries. A comoving Bob can leave Alice's causal past even though they exchanged signals earlier.

Lecture frame: 01:30:20.000.

If Alice and Bob hold hands, or more realistically remain bound at fixed proper separation, their comoving-coordinate separation shrinks as a^{-1}. Their curves approach each other in the conformal diagram, but this does not mean their proper distance vanishes. The growing conformal factor compensates for the shrinking coordinate gap, and communication remains possible indefinitely.

Question from the audience. Can Bob see Alice at the same time after their horizons separate them?[a]

Response. No observer ever sees a distant event at the same time in that literal sense; observation is always of an event in the past light cone. The sharper statement is that after horizon crossing, no newly emitted signal from the other worldline can arrive.

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[a]Lecture timestamp: 01:31:43.

In the far-future cosmological application, Alice represents our bound local remnant and Bob represents each unbound distant galaxy. The galaxies do not suddenly wink out at a finite instant on Alice's clock. Their later signals are stretched over longer intervals and shifted to longer wavelengths until they become physically undetectable.

Question from the audience. Does the theory really say that no distant galaxies will remain visible in the far future?[a]

Response. Unbound galaxies pass beyond the local observer's event horizon. Their final visible signals become increasingly delayed and redshifted, leaving only the bound local structure observable for practical purposes.

[a]Lecture timestamp: 01:33:38.

8.13 Redshift at the Horizon

One can still draw an old light signal from Bob reaching Alice at a very late conformal time, which can make it appear that she continues to see him. The physical content lies in the spacing and energy of those signals. Equal intervals of Bob's clock arrive separated by ever larger intervals of Alice's proper time. Frequencies and photon energies fall:

$$\nu_{\text{received}} \longrightarrow 0, \qquad E_{\text{received}} = h\nu_{\text{received}} \longrightarrow 0. \tag{8.35}$$

Bob's image slows, reddens, and fades. The limiting signal carries vanishing usable information.

Question from the audience. Is this like a black hole, where an outside observer never sees an infalling object cross the horizon?[a]

Response. Yes. In both cases the observer receives increasingly redshifted signals and never directly sees the crossing completed. The causal diagram says which later emissions can never arrive; the redshift explains how the last observable information fades.

[a]Lecture timestamp: 01:36:09.

The analogy prompts a question about temperature.

Question from the audience. How could Alice measure the temperature just above a horizon?[a]

Response. Operationally, she could lower a thermometer on a cable, keep it from crossing, and retrieve it. A de Sitter observer could imagine the analogous experiment near the cosmological horizon. Horizons are hot in the quantum theory, but the present lecture stops at the classical geometry and postpones that calculation.

[a]Lecture timestamp: 01:39:34.

No formula for de Sitter temperature is derived here. Hawking radiation enters only as a preview of the quantum correction to this classical picture.

8.14 The Fixed Horizon Radius

The conformal diagram can be translated back into the original time coordinate. Two comoving points at fixed coordinate separation satisfy

$$D(t) = e^{Ht}|\Delta x|, \tag{8.36}$$

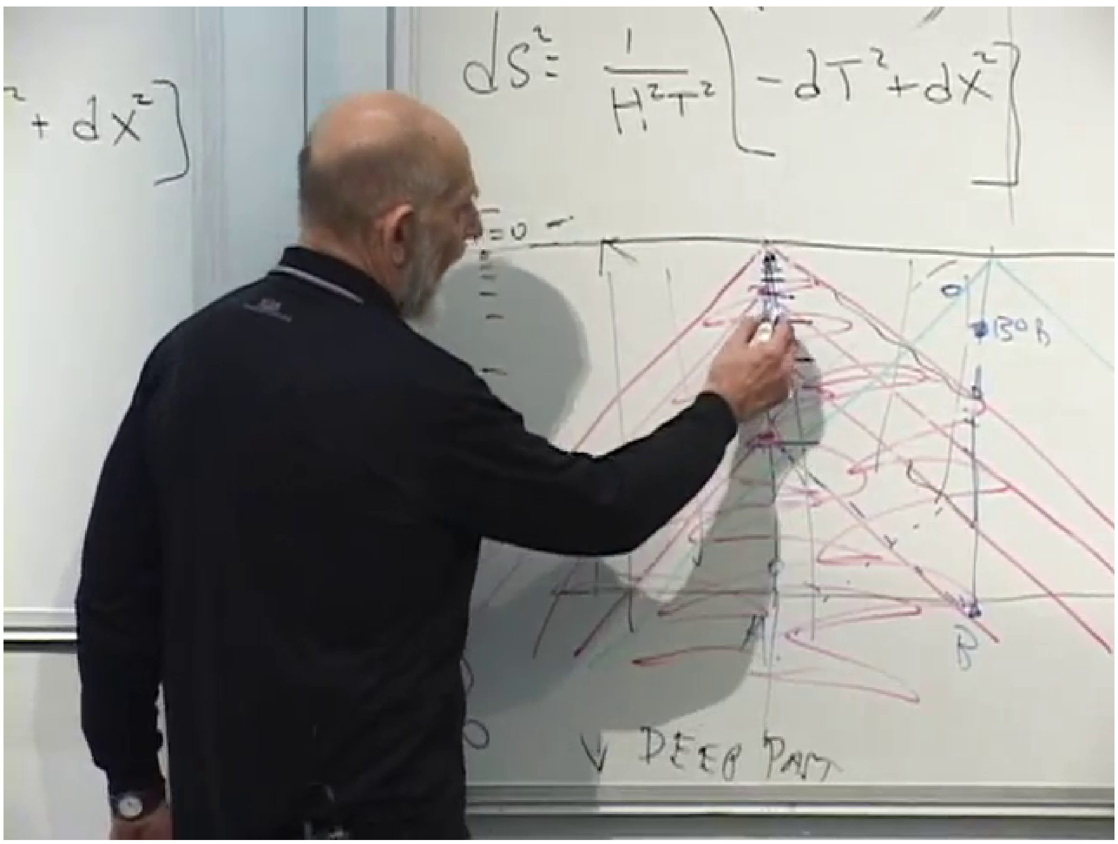

Figure 8.16: The late board state comparing de Sitter horizon redshift with the familiar external view of a black-hole horizon.

Lecture frame: 01:36:20.000.

so their proper distance grows exponentially.

Question from the audience. What does the same motion look like in the original, untransformed time coordinate?[a]

Response. For fixed comoving coordinates, the entire spatial metric grows with e^{2Ht}; hence proper separations grow like e^{Ht}. Undoing the coordinate transformation makes that expansion explicit.

[a]Lecture timestamp: 01:41:14.

The horizon behaves differently because its comoving coordinate distance from Alice is not fixed. From the 45° null line ending at $T = 0$, the horizon distance on a slice of constant $T < 0$ is

$$\Delta x_H = -T = \frac{1}{H} e^{-Ht}. \tag{8.37}$$

Multiplying by the scale factor gives

$$D_H(t) = a(t)\Delta x_H = e^{Ht} \frac{e^{-Ht}}{H} = \frac{1}{H}. \tag{8.38}$$

Restoring units, $D_H = c/H$. The coordinate gap shrinks, the metric scale grows, and their product remains constant.

Question from the audience. As time passes, does the distance from Alice to the horizon grow, shrink, or stay the same?[a]

Response. It stays the same. Its coordinate distance shrinks while the conformal factor grows, leaving the proper horizon radius fixed at H^{-1} in units with $c = 1$.

[a]Lecture timestamp: 01:42:42.

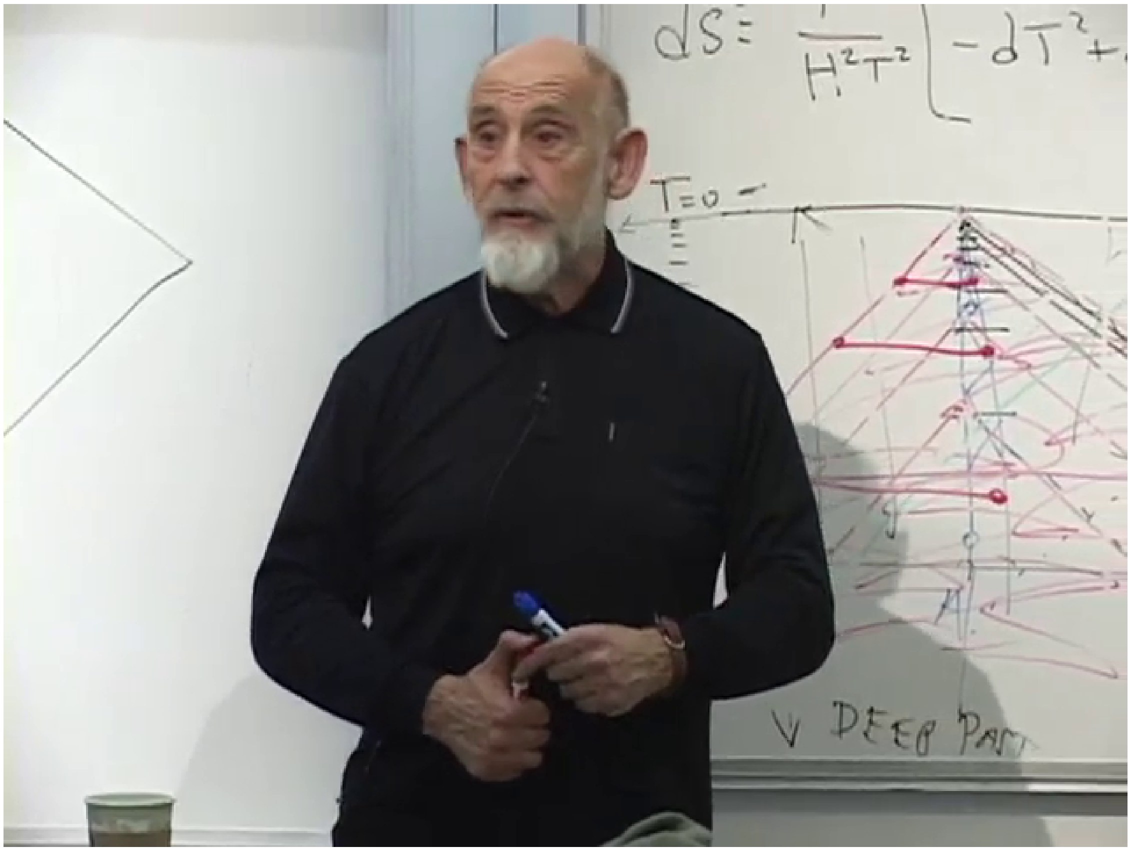

Figure 8.17: The final board state: successive horizontal slices have shrinking coordinate distance to the horizon but the same proper distance.

Lecture frame: 01:43:20.000.

Question from the audience. Does Alice continue to see the past projected onto the horizon because the old images never quite disappear?[a]

Response. Classically, the images persist only in the limiting sense that they become harder and harder to detect through unbounded redshift. That statement omits the Hawking radiation associated with the horizon, which will modify the quantum description.

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[a]Lecture timestamp: 01:43:45.

The lecture therefore closes at a precise boundary. Vacuum energy has produced exponential expansion; exponential expansion has produced a fixed, observer-dependent event horizon; and signals near that horizon have been driven to vanishing frequency. The horizon's temperature and Hawking radiation belong to what comes next. Assuming, as Susskind says, that we do not cross one another's horizons, we meet again in the next lecture.

Chapter 9

de Sitter Horizons and Cosmological Probability

Overview. A horizon is not merely a causal boundary. For a stationary observer it is a thermal system: a thermometer held a proper distance ρ from the horizon reads a temperature of order $1/(2\pi\rho)$, while gravitational redshift makes the radiation seen far away much colder. We shall carry this black-hole lesson into de Sitter space, derive the fixed horizon radius H^{-1}, and write the observer's static-patch metric. The second half of the lecture follows the thermodynamics to its unsettling consequences: finite de Sitter entropy, Poincare recurrences, Boltzmann brains, eternal inflation, and the measure problem. A final discussion separates the exact supersymmetric structures of string theory from the still-unknown framework that might describe our world.

9.1 What It Means for a Horizon to Be Hot

We begin before the formal lecture, with the question that determines the evening's direction.

Question from the audience. A cosmic horizon contains no black hole. By what mechanism can it nevertheless be hot?[a]

Response. The mechanism is the same as for a black-hole horizon. The useful question is operational: what does a thermometer held near the horizon record, and what radiation reaches an observer far from it?

[a] Lecture timestamp: 00:00:17.

The first experiment is local. Hold a thermometer at fixed proper distance ρ from any smooth, nonextremal horizon. Close enough to the horizon, the geometry has the universal Rindler form, and the thermometer reads

$$T_{\text{local}}(\rho) \simeq \frac{1}{2\pi\rho} \tag{9.1}$$

in units with $\hbar = c = k_B = 1$. Restoring constants gives

$$k_B T_{\text{local}}(\rho) \simeq \frac{\hbar c}{2\pi\rho}. \tag{9.2}$$

The divergence as $\rho \to 0$ does not mean that an infalling observer encounters an ordinary material furnace. It is the temperature assigned by an observer who remains stationary just outside the horizon and must therefore accelerate ever more violently.

The opening conversation also separates three notions that are easily conflated: energy, temperature, and entropy.

Question from the audience. If photons are sent into a black hole, do we still measure the same entropy, or does only the energy change?[a]

Response. Once a photon becomes inaccessible behind the horizon, additional information has been hidden. In the simplified bit language used here, one photon can hide one additional bit and thereby increase the entropy. Energy, temperature, and entropy are related, but they are not interchangeable.

---

[a] Lecture timestamp: 00:01:22.

[1]

The near-horizon region continually emits and reabsorbs high-frequency quantum fluctuations. A horizon makes those fluctuations thermal for the stationary observer. Yet this raises an immediate puzzle. If the region is arbitrarily hot, why is a distant observer not blinded by arbitrarily energetic photons?

9.2 Redshift and the Hawking Temperature

An ordinary hot wall remains a source of high-frequency radiation even when we stand far from it. A horizon is different because its radiation must climb through a gravitational redshift. The photon's speed remains c; what decreases is its energy,

$$E = \hbar\omega, \qquad \omega_{\text{received}} < \omega_{\text{emitted}}. \tag{9.3}$$

The piece-of-chalk analogy used in the lecture is useful only up to this point. A thrown stone slows as it rises, whereas a photon continues at light speed while its frequency and energy decrease.

For a Schwarzschild black hole, the temperature measured by an observer at infinity is

$$T_H = \frac{1}{8\pi M} \tag{9.4}$$

in units $G_N = \hbar = c = k_B = 1$. In ordinary units,

$$T_H = \frac{\hbar c^3}{8\pi G_N M k_B}. \tag{9.5}$$

A larger black hole is colder. The lecture explains this by tracing a typical photon backward. Thermal photons have characteristic energy $E_\gamma \sim T_H$. A photon that reaches infinity with energy of order M^{-1} had a much larger local energy when it began near the horizon. The redshift between the near-horizon layer and infinity is more severe for a larger hole, leaving a lower asymptotic temperature.

[1] Editorial clarification: For a physical black hole the entropy change is fixed thermodynamically by $dS = dE/T_H$; it is not universally one bit for every photon, regardless of energy. The one-bit statement is the lecture's information-theoretic idealization.

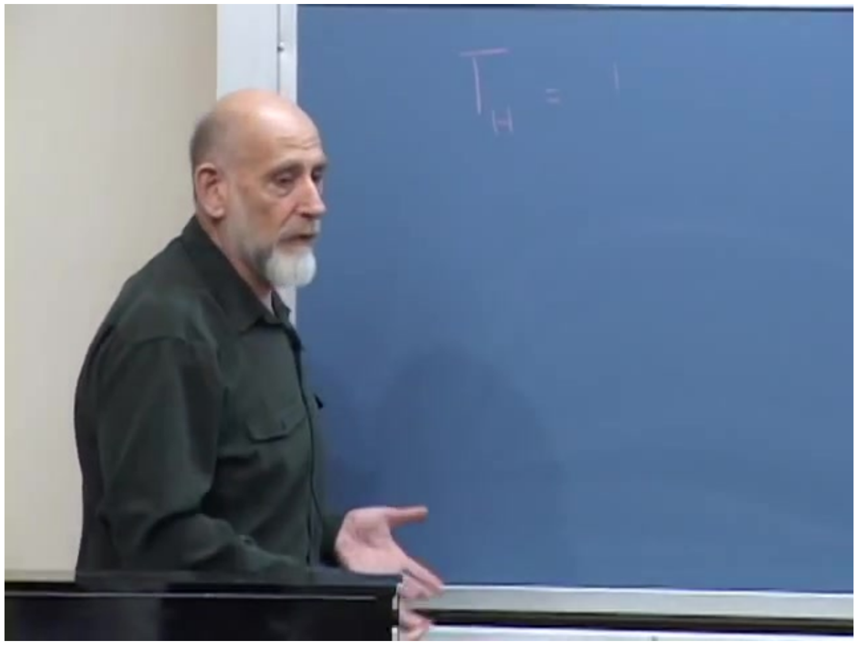

Figure 9.1: The Hawking temperature $T_H = 1/(8\pi M)$ in natural units.

Lecture frame: 00:08:12.

Question from the audience. If Schwarzschild time freezes at the horizon, should we follow the photon that originally fell in and built up the hole?[a]

Response. For the temperature question, follow the photons emitted outward. They begin at very high local frequency, lose energy while climbing out, and arrive as Hawking radiation. The fate of the original infalling photon is a separate information question.

[a]Lecture timestamp: 00:04:19.

9.3 The Planck Layer and the Stretched Horizon

How close can the local relation in Equation (9.1) be trusted? The lecture takes the Planck length as the shortest meaningful distance in the semiclassical description. This suggests a thin layer, roughly one Planck length outside the mathematical horizon, filled with Planck-scale fluctuations. It emits and absorbs continually.

Most emitted quanta do not escape. Only those directed into a very narrow outward cone avoid falling back. Quanta launched at generic angles return to the horizon. Outgoing and ingoing excitations collide, producing pairs and other particles; in this sense the stationary description contains a thin shell of extremely hot material.

There is a complementary spacetime picture. Away from a horizon, a virtual fluctuation may be represented schematically by a short closed loop: a particle-antiparticle excitation appears and disappears before it can become an asymptotic state. Near a horizon, a fluctuation can straddle the causal boundary. To the exterior observer, the outside portion looks like a particle emitted by the horizon and later reabsorbed. A very small fraction escapes and becomes Hawking radiation.

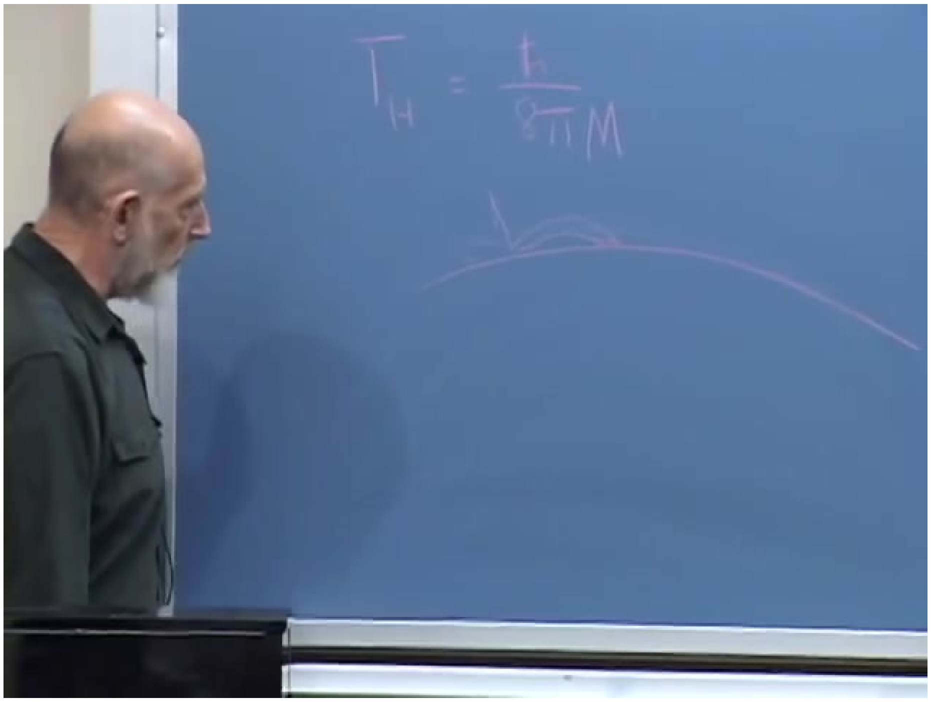

Figure 9.2: The narrow escape cone for near-horizon radiation. Most locally emitted quanta return to the horizon.

Lecture frame: 00:11:38.

Question from the audience. In the exterior diagram, does such a fluctuation extend from $t = -\infty$ to $t = +\infty$?[a]

Response. The exact classical horizon leads to that singular-looking description. In practice we replace it by a stretched horizon, a timelike surface about one Planck length outside. Emission and reabsorption then occur at finite exterior times.

---

[a] Lecture timestamp: 00:15:00.

The stretched horizon is not a new invariant boundary. It is an effective surface placed where semiclassical geometry ceases to be reliable. Quantum fluctuations of matter and of the geometry itself smear the exact location on Planck scales. The construction packages that uncertainty into a surface on which the exterior description remains finite.

Nothing in this local argument knows whether the horizon belongs to a black hole or to an accelerating universe. That is the bridge to de Sitter space.

9.4 de Sitter Space in Expanding Coordinates

The formal lecture begins with the spatially flat slicing of de Sitter spacetime,

$$ds^2 = -dt^2 + e^{2Ht} d\mathbf{x}^2, \qquad d\mathbf{x}^2 = dx^2 + dy^2 + dz^2. \tag{9.6}$$

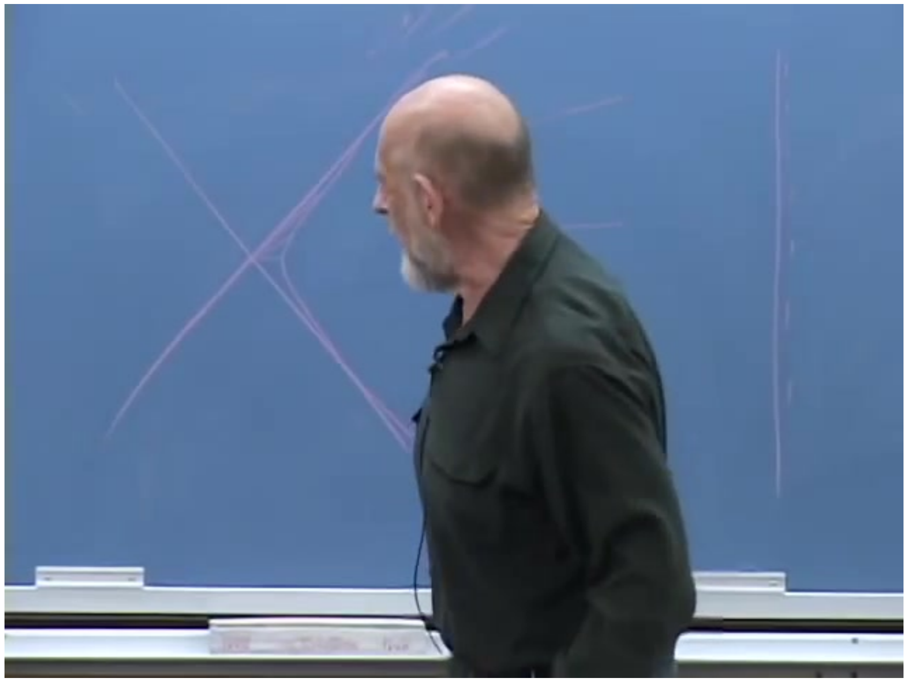

Figure 9.3: The classical horizon and the nearby stretched horizon used in the exterior description.
Lecture frame: 00:15:52.

The constant H is the de Sitter expansion rate. The factor appears as e^{2Ht} because the metric measures squared distance; a fixed coordinate interval has physical length

$$D(t) = e^{Ht}|\Delta \mathbf{x}|. \tag{9.7}$$

These coordinates make expansion transparent but place future infinity infinitely far up the page. To bring it onto the blackboard, introduce conformal time

$$\eta = -\frac{1}{H}e^{-Ht}, \qquad -\infty < \eta < 0. \tag{9.8}$$

Then

$$dt = \frac{d\eta}{-H\eta}, \qquad e^{Ht} = -\frac{1}{H\eta}, \tag{9.9}$$

and Equation (9.6) becomes

$$ds^2 = \frac{1}{H^2\eta^2}\left(-d\eta^2 + d\mathbf{x}^2\right). \tag{9.10}$$

As $t \to +\infty$, $\eta \to 0^-$. Infinite proper time has been compressed to the finite horizontal line $\eta = 0$. [2]

The conformal factor multiplies both the temporal and spatial terms, so it does not alter null directions. For a radial light ray,

$$ds^2 = 0 \quad\Longrightarrow\quad -d\eta^2 + dx^2 = 0 \quad\Longrightarrow\quad \frac{dx}{d\eta} = \pm 1. \tag{9.11}$$

[2]Editorial clarification: The time transformation is corrected several times while being written on the board. Equation (9.8) is the consistent form: early cosmic time maps to $\eta \to -\infty$, and future infinity maps to $\eta \to 0^-$.

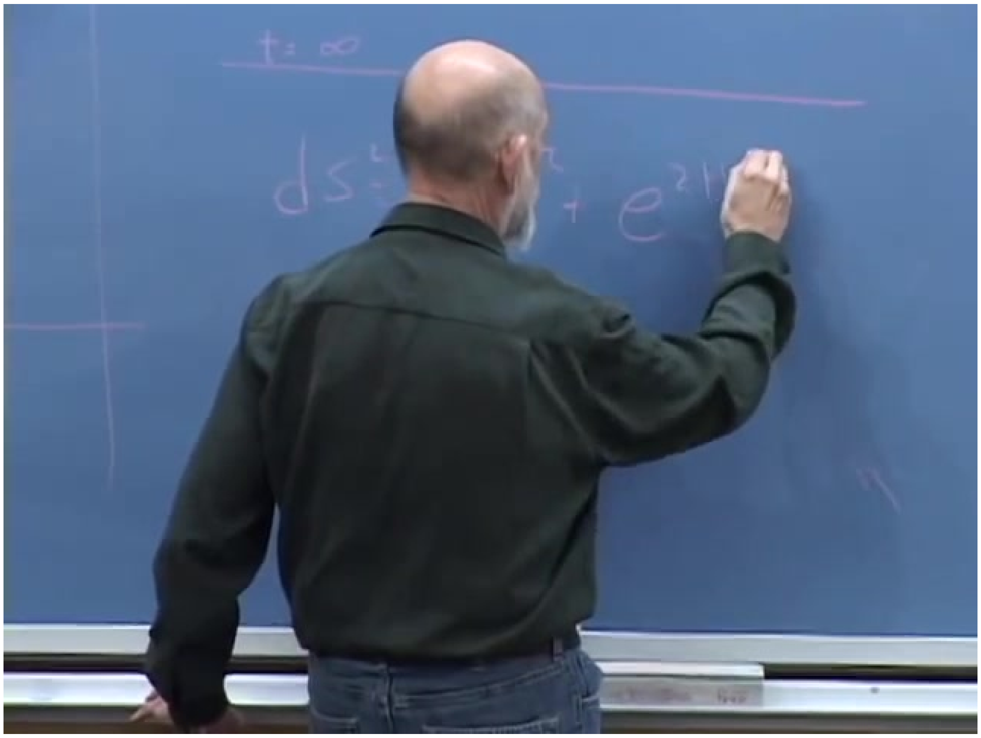

Figure 9.4: The exponentially expanding de Sitter metric in flat coordinates.
Lecture frame: 00:18:50.

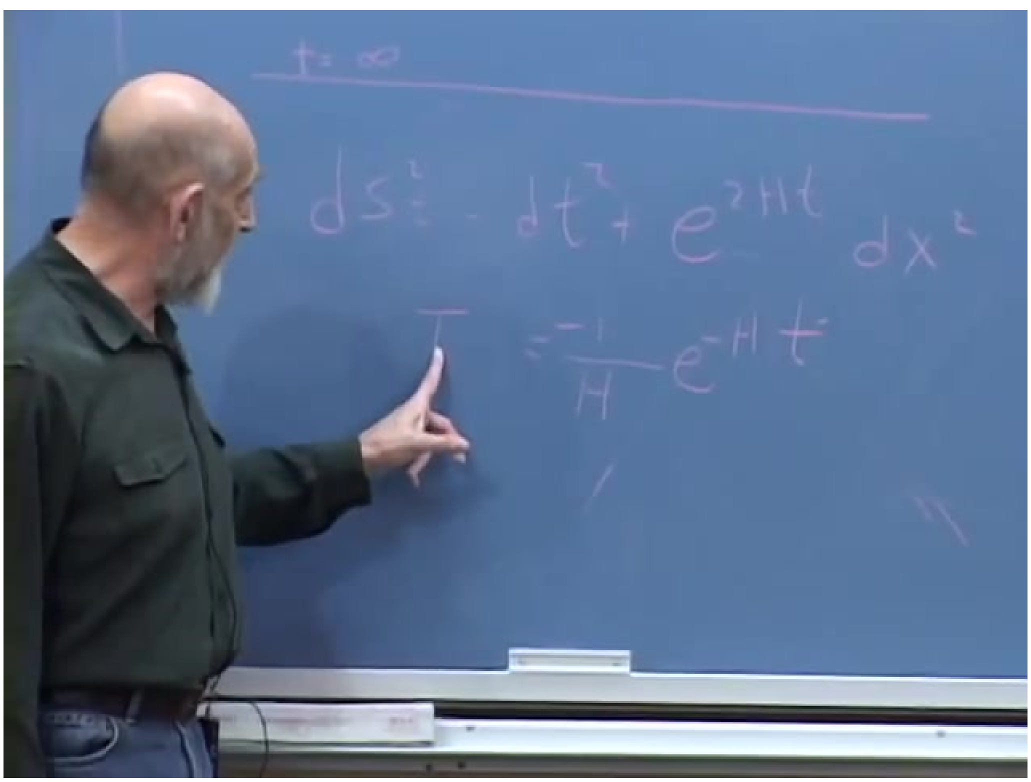

Figure 9.5: The conformal-time transformation and the conformally flat metric.
Lecture frame: 00:21:24.

Light therefore travels at 45° in the conformal diagram. This is why the change of coordinates is useful: causal accessibility becomes visible.

9.5 One Observer's Causal Patch

Imagine an observer whose worldline continues for infinite proper time. On the compactified diagram it ends at one point on $\eta = 0$. Every signal the observer can ever receive lies inside the backward light cone of that endpoint. Events outside the cone can never communicate with the observer, however long the observer waits.

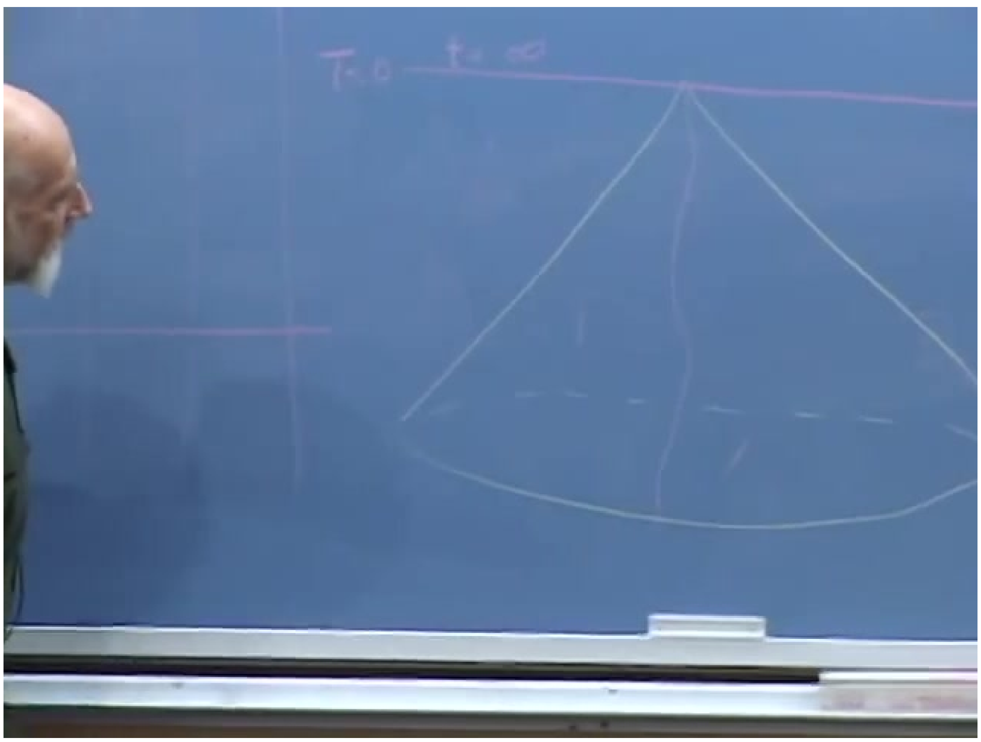

Figure 9.6: An observer's limiting backward light cone in conformal de Sitter coordinates.
Lecture frame: 00:24:38.

A second observer has a different cone. The two accessible regions overlap, but neither is the whole spacetime. For the remainder of the argument we adopt the first observer's point of view and call that backward light cone the observer's causal patch.

The drawing seems to make the patch shrink toward the future, but coordinate width is not proper distance. On a constant-η slice the squared physical distance between the observer at $x = 0$ and the horizon at $x = \Delta x$ is

$$D^2 = \frac{(\Delta x)^2}{H^2\eta^2}. \tag{9.12}$$

The horizon is a 45° null line ending at $\eta = 0$, hence

$$\Delta x = |\eta|. \tag{9.13}$$

Substitution gives

$$D_{\text{hor}}^2 = \frac{1}{H^2}, \qquad D_{\text{hor}} = \frac{1}{H}. \tag{9.14}$$

The coordinate separation shrinks, but the conformal factor grows by exactly the compensating amount. The proper horizon radius is constant.

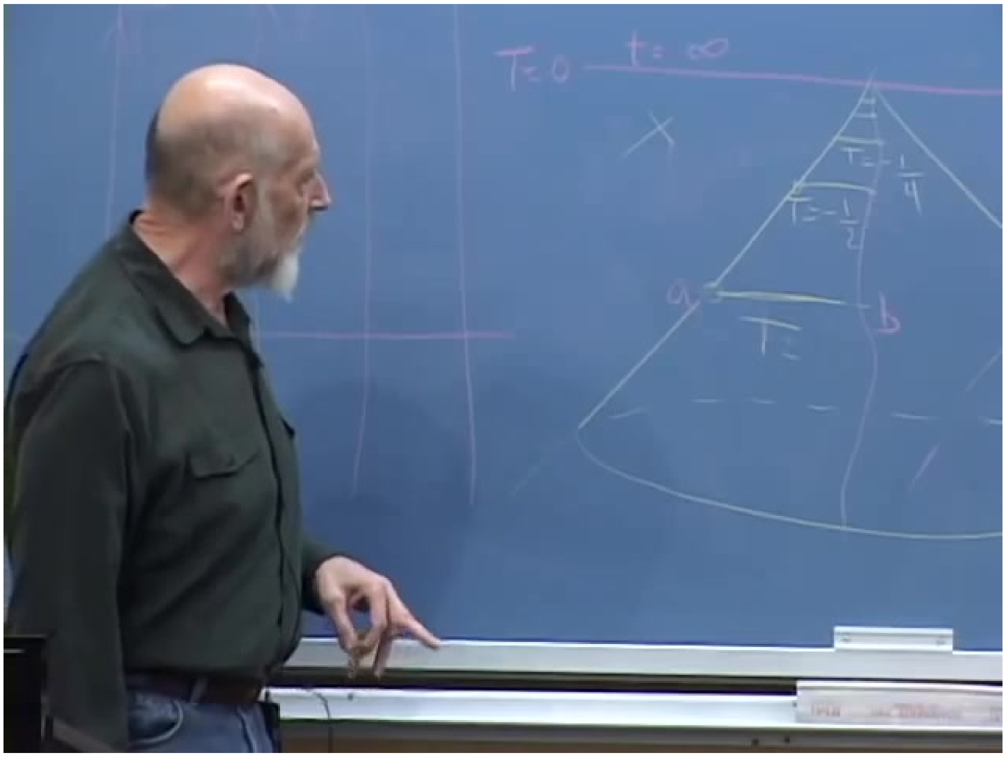

Figure 9.7: The board derivation that converts the shrinking coordinate interval into the fixed proper distance H^{-1}.

Lecture frame: 00:28:42.

9.6 The Static Patch

A fixed horizon radius suggests coordinates adapted to the central observer. Without deriving the coordinate transformation, the lecture writes the result:

$$ds^2 = -\left(1 - H^2r^2\right) d\tau^2 + \frac{dr^2}{1 - H^2r^2} + r^2 d\Omega_2^2. \tag{9.15}$$

The coefficients depend on r, not on τ. This chart covers the observer's static patch,

$$0 \leq r < \frac{1}{H}. \tag{9.16}$$

The resemblance to Schwarzschild is immediate:

$$f_{\mathrm{dS}}(r) = 1 - H^2r^2, \qquad f_{\mathrm{Schw}}(r) = 1 - \frac{2G_NM}{r}. \tag{9.17}$$

In either case the static-coordinate horizon occurs where $f(r) = 0$. For de Sitter,

$$1 - H^2r^2 = 0 \quad \Longrightarrow \quad r_H = H^{-1}. \tag{9.18}$$

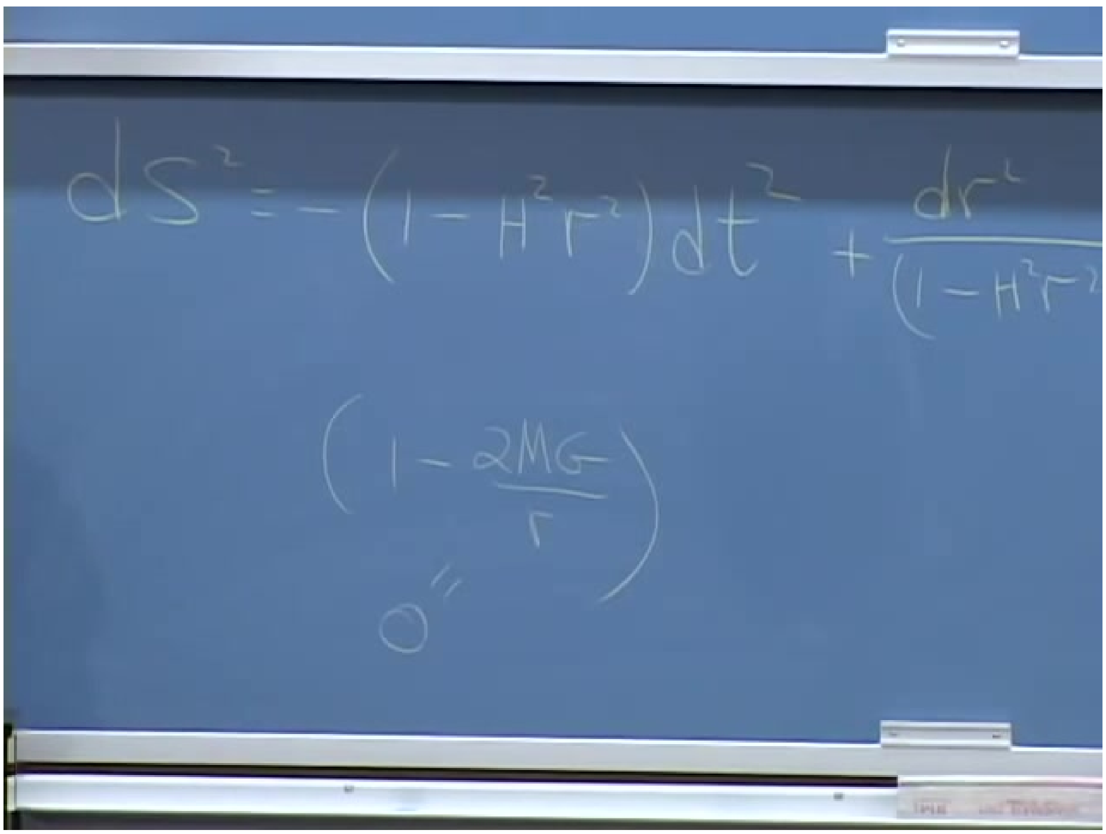

Figure 9.8: The de Sitter static-patch metric and its comparison with the Schwarzschild factor.
Lecture frame: 00:34:22.

The chart is singular there, just as Schwarzschild coordinates are singular at a black-hole horizon. The global geometries nevertheless differ in an important way. A black-hole observer stands outside and looks inward; a de Sitter observer stands inside and looks outward.

Question from the audience. How can the geometry be static when freely released particles move apart in an exponentially expanding universe?[a]

Response. Static does not mean that every particle remains fixed. It means that the metric coefficients and the outcome of a repeated experiment are independent of the start time. Release two particles now and they accelerate apart; repeat the same experiment later and the same motion occurs.

[a]Lecture timestamp: 00:36:06.

Question from the audience. Do the two separating particles see one another redshift?[a]

Response. Yes. Their increasing separation produces a relative redshift, even though the observer-adapted metric is time independent.

[a]Lecture timestamp: 00:38:34.

9.7 Temperature Seen from Inside

The near-horizon structures of Schwarzschild and de Sitter space are locally equivalent. The de Sitter horizon therefore emits and reabsorbs thermal quanta. A rare photon propagates inward to the central observer, losing energy as it climbs the static-patch potential. The centrally measured Gibbons–Hawking temperature is

$$T_{\mathrm{dS}} = \frac{H}{2\pi} \tag{9.19}$$

in natural units. A stationary thermometer at radius r reads the Tolman-redshifted value

$$T_{\mathrm{local}}(r) = \frac{T_{\mathrm{dS}}}{\sqrt{1 - H^2 r^2}}, \tag{9.20}$$

which diverges as $r \to H^{-1}$ and reduces near the horizon to Equation (9.1).
[3]

For the observed late-time acceleration, the lecture uses the period estimate

$$H^{-1} \sim 20 \times 10^9 \text{ light years.} \tag{9.21}$$

A typical horizon photon then has a wavelength comparable to billions of light years. This is not microwave-background radiation; it is vastly colder.

The operational thought experiment is a fishing pole on a cosmological scale. Alice remains at $r = 0$, attaches a thermometer to a very long line that also transmits data, and lets cosmic repulsion carry the instrument outward. She stops it just short of the horizon, waits for the signal to return, and finds a large local temperature. The thermometer beside her at the center reads the tiny value $H/(2\pi)$. One horizon supports both readings because they are made at different gravitational potentials.

[3]Editorial clarification: The lecture obtains the scale from the horizon distance and redshift rather than writing Equations (9.19) and (9.20) explicitly. They state the standard normalization of that argument.

9.8 Crossing the Horizon

The classical outside description resembles the familiar black-hole story. Bob at the center watches Alice recede. Her signals become increasingly redshifted and arrive increasingly slowly; she appears flattened against the horizon and ultimately fades. A later traveler, Jane, appears as another thinner layer in a sedimentary stack near the boundary.

Question from the audience. From our point of view, do objects ever pass through the cosmic horizon?[a]

Response. Not in the static observer's classical description. Their signals redshift and fade while their apparent position approaches the horizon. In their own freely falling description, however, they cross without encountering a locally singular surface.

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[a]Lecture timestamp: 00:45:30.

Question from the audience. Is the horizon hot only for observers who remain close without crossing? An infalling observer is not cooked merely by passing through it, is she?[a]

Response. Alice crosses smoothly in her own frame. The extreme temperature belongs to the stationary description. If Bob insists on resolving Alice ever closer to the horizon, he must illuminate her with ever shorter wavelengths. The attempted Heisenberg microscope then deposits enough energy to cook what it tries to observe.

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[a]Lecture timestamp: 00:48:27.

This is not a contradiction between two local events. The stationary observer and the infaller use radically different measurements. Horizon complementarity asks how both descriptions can be valid without allowing one observer to compare duplicate information directly.

9.9 A Newtonian Model of Vacuum Repulsion

At this point a student asks what replaces the black hole's gravitational well. Susskind answers by constructing a Newtonian model. Let ϕ be the gravitational potential, so the acceleration of a test particle is

$$\mathbf{a} = -\boldsymbol{\nabla}\phi. \tag{9.22}$$

For ordinary nonrelativistic matter, Poisson's equation is

$$\nabla^2\phi = 4\pi G_N\rho. \tag{9.23}$$

Question from the audience. If the black-hole effects come from its gravitational well, what is the corresponding source in de Sitter space?[a]

Response. The vacuum energy itself supplies a spatially uniform gravitational source. In a Newtonian description it produces a quadratic potential rather than a $1/r$ well.

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[a]Lecture timestamp: 00:49:36.

The board suppresses normalization, spherical-measure terms, pressure, and initially even the overall sign, reducing the exercise to

$$\frac{d^2\phi}{dr^2} \sim H^2. \tag{9.24}$$

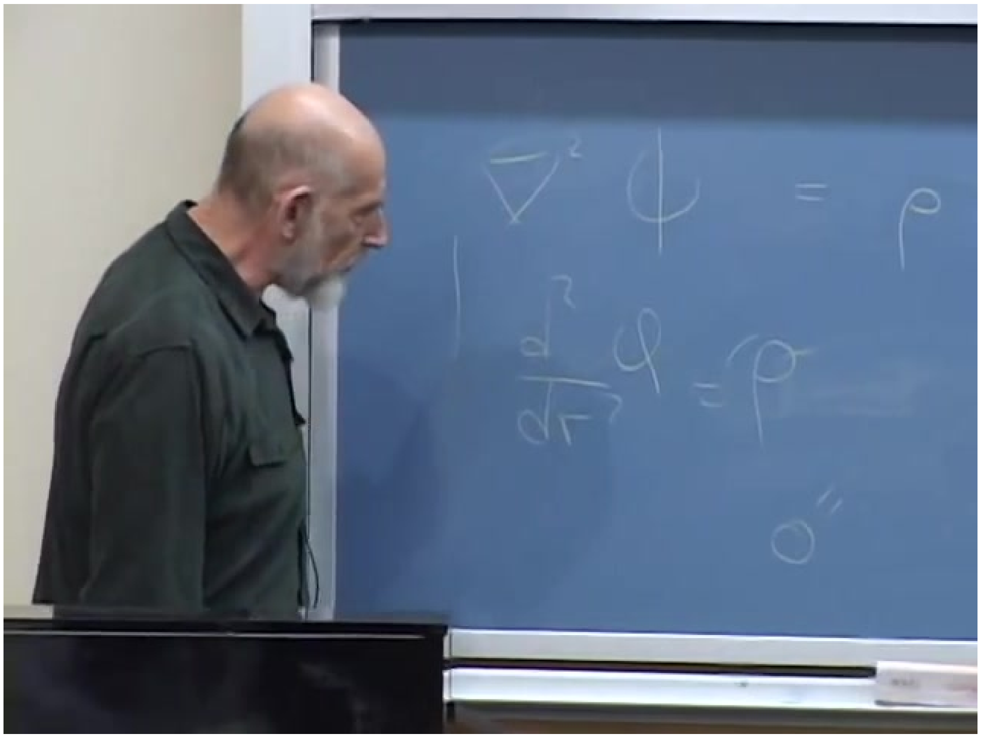

Figure 9.9: Poisson's equation and the lecture's schematic radial second-derivative form.
Lecture frame: 00:51:38.

The essential inference is that a constant source gives a quadratic function. With the sign required for de Sitter repulsion,

$$\phi_{\mathrm{dS}}(r) = -\frac{1}{2}H^2r^2, \qquad a_r = -\frac{d\phi_{\mathrm{dS}}}{dr} = H^2r. \tag{9.25}$$

The acceleration points outward and grows linearly with distance. Solving $\ddot{r} = H^2r$ gives

$$r(\tau) = Ae^{H\tau} + Be^{-H\tau}, \tag{9.26}$$

so the growing branch reproduces exponential separation.
[4]

Place an ordinary mass M at the center. The qualitative potential becomes

$$\phi(r) = -\frac{G_NM}{r} - \frac{1}{2}H^2r^2, \tag{9.27}$$

and the radial acceleration is

$$a_r = -\frac{G_NM}{r^2} + H^2r. \tag{9.28}$$

At small r, local attraction wins; at large r, vacuum repulsion wins. Their balance occurs at

$$r_* = \left(\frac{G_NM}{H^2}\right)^{1/3}. \tag{9.29}$$

[4]Editorial clarification: In general relativity the weak-field acceleration of a homogeneous fluid depends on $\rho + 3p$, not on ρ alone. Vacuum energy has $p = -\rho$, so $\rho + 3p = -2\rho$ and the acceleration is repulsive. Equation (9.25) is the correctly normalized de Sitter result; the board calculation intentionally tracks only the quadratic dependence and sign.

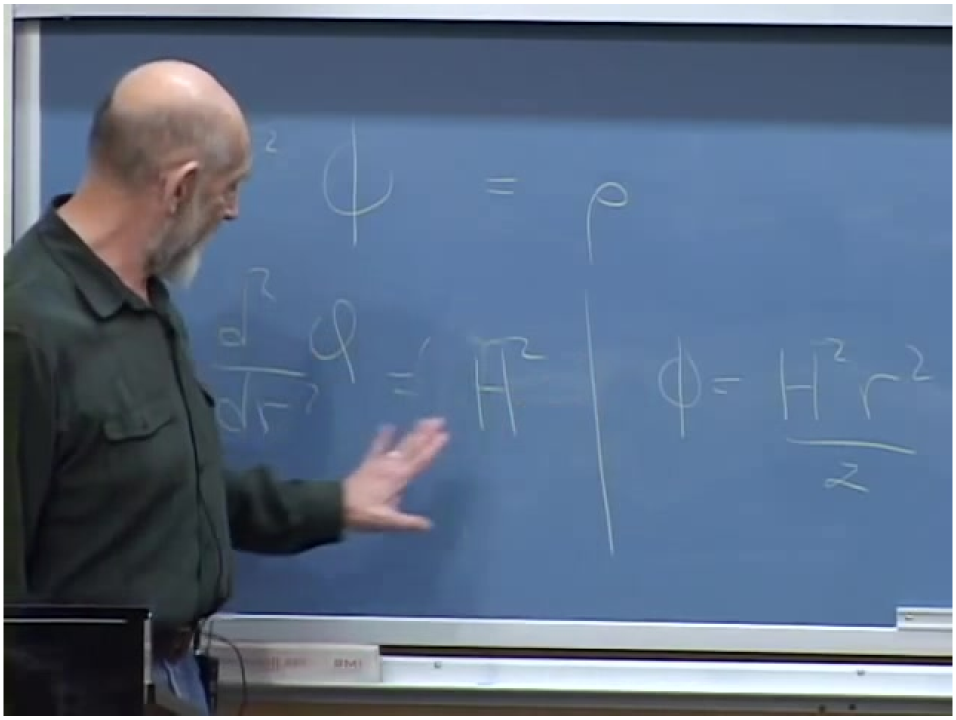

Figure 9.10: The uniform-source equation and the quadratic de Sitter potential.
Lecture frame: 00:53:32.

This is the clean version of the lecture's potential maximum. It explains qualitatively why nearby gravitationally bound systems can remain together while sufficiently distant galaxies accelerate away.

Question from the audience. Do ordinary forces still matter, or does vacuum repulsion eventually tear apart every island of matter?[a]

Response. All forces must be included. On sufficiently small scales, electromagnetic binding and local gravity dominate. Vacuum repulsion becomes important only across enormous distances set by H^{-1} and by the balance scale appropriate to the bound mass.

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[a] Lecture timestamp: 00:57:29.

9.10 One Source, Several Names

The discussion now pauses over terminology. Vacuum energy, a cosmological constant, and dark energy are not three additive substances in the model being used. They are three descriptions of the same $w = -1$ source. One may speak of a constant term in Einstein's equation, a Lorentz-invariant energy density of the vacuum, or the component driving late-time acceleration.

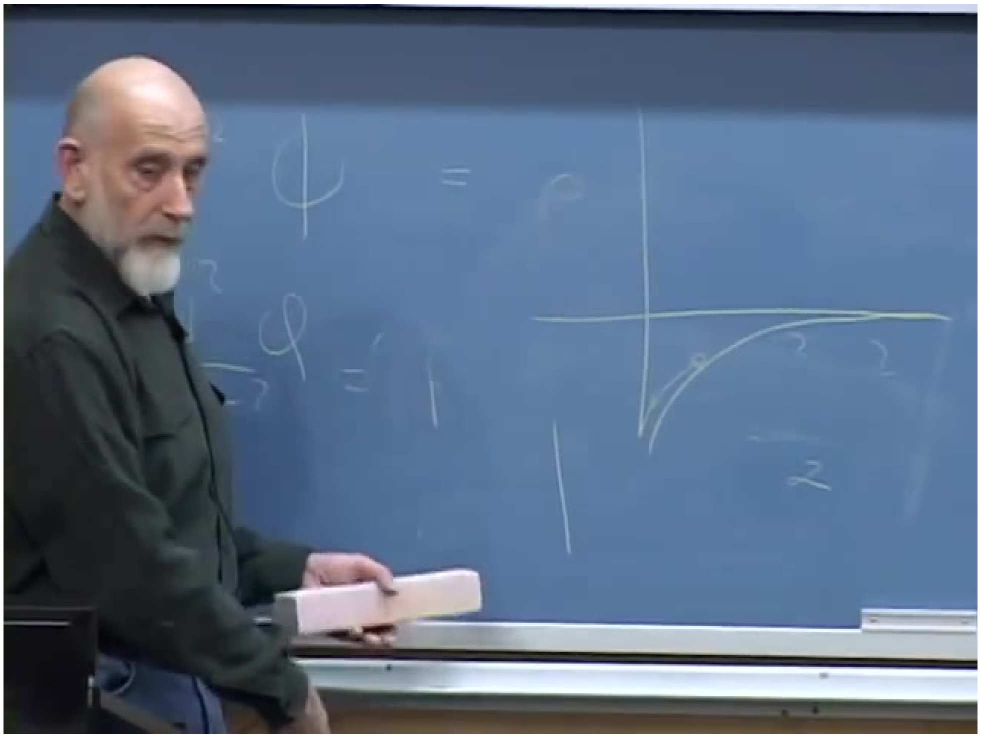

Figure 9.11: Ordinary $1/r$ attraction combined with the upside-down quadratic de Sitter potential.
Lecture frame: 00:58:20.

Question from the audience. Are the uniform Newtonian source and the quantum vacuum energy both present as separate effects?[a]

Response. No. The uniform density is the Newtonian model of the same source. Vacuum energy, cosmological constant, and, in this lecture, dark energy refer to one physical component.

[a] Lecture timestamp: 00:59:14.

Question from the audience. Does the same phenomenon therefore have a classical description and a quantum description?[a]

Response. Its microscopic origin may be quantum mechanical, just as part of an electron's mass has a quantum origin. Once the effective energy density is known, it enters the classical gravitational equations as an ordinary source. We do not know the ultimate origin of the observed value.

[a] Lecture timestamp: 01:00:32.

In Planck units, the observed vacuum-energy density is extraordinarily small,

$$\rho_\Lambda \sim 10^{-123}\rho_{\rm Pl}, \tag{9.30}$$

up to the convention used for the Planck density. That small number is the cosmological-constant problem. It is also why the horizon is so large: in four dimensions,

$$H^2 = \frac{8\pi G_N}{3}\rho_\Lambda, \qquad r_H = H^{-1}. \tag{9.31}$$

If H were larger, the horizon would be smaller, the central de Sitter temperature would be higher, and the repulsive acceleration $H^2 r$ would be stronger.

Question from the audience. Is the tiny cosmological constant related to Planck's constant and quantum fluctuations?[a]

Response. The clean comparison uses dimensionless Planck units, where $G_N = \hbar = c = 1$. In those units the vacuum density is about 123 orders of magnitude below the natural Planck density. Explaining that hierarchy is one of the central puzzles of fundamental physics.

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[a] Lecture timestamp: 01:03:27.

Question from the audience. Must vacuum energy be an exactly constant cosmological constant? Could the accelerated component decrease and eventually settle to zero?[a]

Response. That is a logically possible alternative: a dynamical dark-energy field could vary with time. The lecture adopts the constant case because observations available at the time increasingly favored behavior close to a cosmological constant, not because every alternative had been proved impossible.

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[a] Lecture timestamp: 01:04:52.

9.11 Astronomy in the Distant Future

In an asymptotically de Sitter universe, structures that are not gravitationally bound eventually cross the observer's horizon. The cosmic microwave background redshifts below detectability. Distant galaxies disappear. A surviving astronomer may see only a merged local gravitational island inside an otherwise dark patch.

Question from the audience. Without a telescope, would anyone even notice that the distant galaxies had disappeared?[a]

Response. Probably not. The serious problem is deeper: even with advanced instruments, future observers may lack the distant tracers and microwave background from which we infer expansion today.

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[a] Lecture timestamp: 01:06:17.

The old island-universe picture could then seem compelling. Olbers' paradox would not force a different conclusion because it assumes an infinite, static, uniform population of luminous sources; the future observer would see only the finite stars of the local island. Dark energy does not continually manufacture replacement galaxies from empty space.

The horizon in exact de Sitter space does not shrink around the local galaxy. Its physical radius remains H^{-1}, enormously larger than a galaxy. What disappears is the unbound matter carried beyond that fixed causal boundary.

Question from the audience. Could stars ejected from the galaxy serve as test bodies for detecting the large-scale geometry?[a]

Response. For a while, yes. Ejected objects and long-baseline light signals could probe the geometry. But the effect becomes appreciable only over distances of order a substantial fraction of a billion light years, so the experiment would take cosmological time.

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[a] Lecture timestamp: 01:11:25.

Question from the audience. How do we know a distant horizon has not already hidden most of the universe from us?[a]

Response. We already see enough galaxies, redshifts, and background radiation to measure the expansion and reconstruct the large-scale geometry. Future observers deprived of those tracers would face a much harder inference problem.

[a]Lecture timestamp: 01:11:45.

Question from the audience. If the horizon lay just beyond Andromeda, would its presence be obvious?[a]

Response. Yes. Such a small horizon would require a much larger H, and the resulting repulsion would strongly alter Andromeda's motion. The measured value instead places the asymptotic horizon on a vastly larger scale.

[a]Lecture timestamp: 01:12:19.

Question from the audience. Could new big bangs be occurring beyond our horizon?[a]

Response. Events beyond the horizon are not visible to us or to our descendants. A related but more precise possibility, bubble nucleation in eternal inflation, will be introduced later in the lecture.

[a]Lecture timestamp: 01:13:25.

With rockets, clocks, and several billion years, future observers could send explorers across a large fraction of the patch and reconstruct its metric. On ordinary laboratory or civilizational time scales, however, the effect is almost inaccessible. This distinction corrects the overly strong claim that the expansion would be impossible to discover: it is operationally remote, not forbidden in principle.

Question from the audience. Could future explorers determine the geometry much sooner than waiting for a thermal recurrence?[a]

Response. Certainly. A recurrence is not the proposed measurement. Explorers exchanging light signals across billion-light-year baselines could map the static patch on a vastly shorter, though still cosmological, time scale.

[a]Lecture timestamp: 01:21:40.

Question from the audience. Would redshift erase all information returned by such explorers?[a]

Response. Not if they probe scales well inside the horizon. A baseline of order a billion light years is already sensitive to expansion while producing only a moderate redshift. The experiment is limited chiefly by its immense duration.

[a]Lecture timestamp: 01:23:24.

9.12 Finite Entropy and Recurrence

The static patch is not empty in the thermodynamic sense. Its horizon has area

$$A_H = 4\pi H^{-2}, \tag{9.32}$$

temperature $T_{\mathrm{dS}} = H/(2\pi)$, and entropy

$$S_{\mathrm{dS}} = \frac{A_H}{4G_N} = \frac{\pi}{G_N H^2} \tag{9.33}$$

in units $\hbar = c = k_B = 1$. The entropy is finite. One may therefore picture the static patch, with suitable caution, as a finite-entropy thermal cavity whose excitations are concentrated near the horizon in the stationary description.

An ordinary box of gas at equilibrium appears featureless on short time scales, but statistical mechanics does not forbid fluctuations. If a macrostate has an entropy deficit ΔS relative to equilibrium, its probability is roughly

$$P \sim e^{-\Delta S}. \tag{9.34}$$

The corresponding waiting time grows as

$$t \sim t_0 e^{\Delta S}. \tag{9.35}$$

For a finite system, the time to revisit a state arbitrarily close to an earlier one is the Poincare recurrence time,

$$t_{\mathrm{rec}} \sim H^{-1} e^{S_{\mathrm{dS}}}. \tag{9.36}$$

Since S_{dS} for our horizon is of order 10^{120}, the lecture summarizes the scale as $e^{10^{120}}$. Whether the prefactor is measured in years, Hubble times, or microscopic units is irrelevant beside the double exponential.

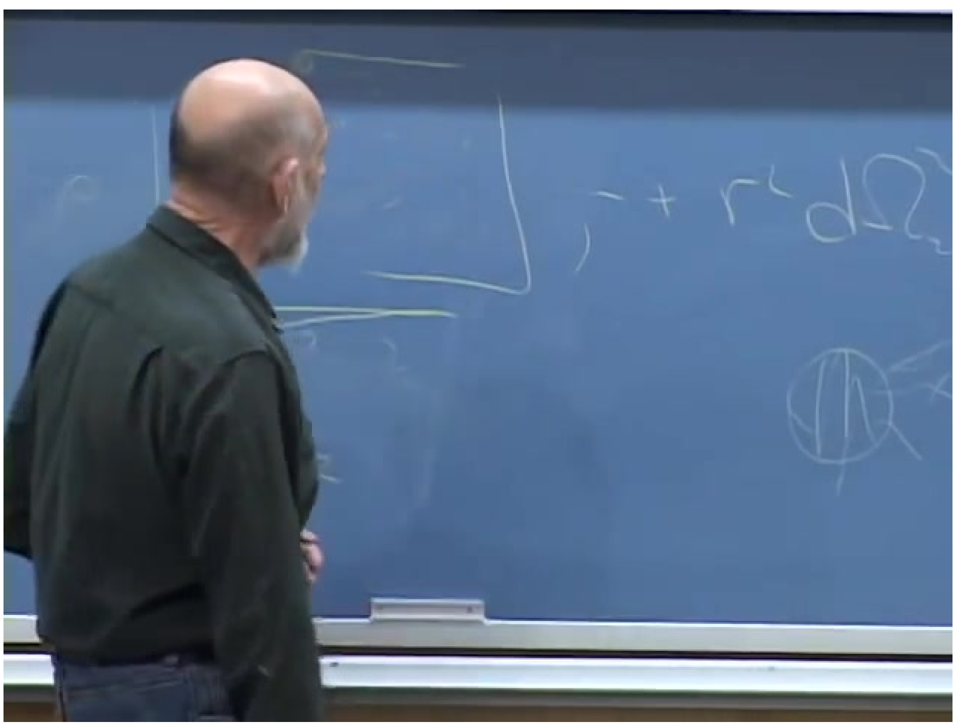

Figure 9.12: The recurrence discussion, with the enormous exponential waiting-time scale on the board.

Lecture frame: 01:16:24.

Before such a recurrence, smaller fluctuations are vastly more common. The molecules in a room might all gather in one corner; a complicated vortex might briefly form; in the lecture's playful hierarchy, thermal motion might assemble an elephant. The smaller the required entropy deficit, the cheaper the fluctuation.

Question from the audience. Are Poincare recurrences specifically a consequence of quantum mechanics?[a]

Response. No. Recurrence and rare fluctuations already arise in classical statistical mechanics. Their application to de Sitter space uses our present physical framework, which might someday be revised, but thermodynamics and statistical reasoning are among the most robust structures we know.

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[a] Lecture timestamp: 01:25:44.

The meaning of probability itself may deepen in a future theory. Susskind does not claim otherwise. His narrower point is that abandoning statistical mechanics merely because its long-time implications are strange would discard one of physics' best-tested organizing principles.

9.13 Boltzmann Brains and Typical Observers

Suppose a thermal fluctuation creates an entire big-bang-like state. That requires an enormous entropy decrease. A fluctuation producing one solar system is cheaper; an already warm Earth is cheaper still. If the only requirement is one observer, the least costly fluctuation might create only a functioning brain complete with apparent memories. Such a hypothetical observer is called a Boltzmann brain.

The issue is not whether one bizarre fluctuation is logically possible. It is whether a cosmological model predicts that almost every observer is such a fluctuation. We reason from our observations by assuming that we are not fantastically atypical. A theory in which ordinary observers with coherent cosmic histories are exponentially outnumbered by isolated thermal brains undermines that inference.

The Boltzmann-brain problem therefore names a quantitative consistency test. One compares the expected abundance of ordinary observers with the abundance of fluctuation-born observers. If the latter dominate, either the vacuum must decay before they proliferate, the measure must be changed, or the cosmological assumptions must be reconsidered.

An earlier question about an ultraviolet–infrared connection is explicitly deferred. The recurrence argument does not depend on resolving that separate issue.

9.14 Eternal Inflation and Bubble Universes

The lecture next introduces, without attempting a full derivation, a second probability problem. A metastable de Sitter vacuum can nucleate a bubble in which the vacuum energy is lower. The bubble wall expands, and the interior can develop into an open Friedmann–Robertson–Walker universe. Meanwhile the surrounding inflating region keeps expanding and nucleates more bubbles.

The crucial feature is exponential population growth. New physical volume is produced faster than bubbles remove it, so the number of bubble regions grows without bound. Different bubbles may settle into different low-energy states and therefore exhibit different effective physics.

To expose the statistical difficulty, Susskind first leaves cosmology and considers an exponentially growing human population with finite but unknown resources. Assume that observers are typical

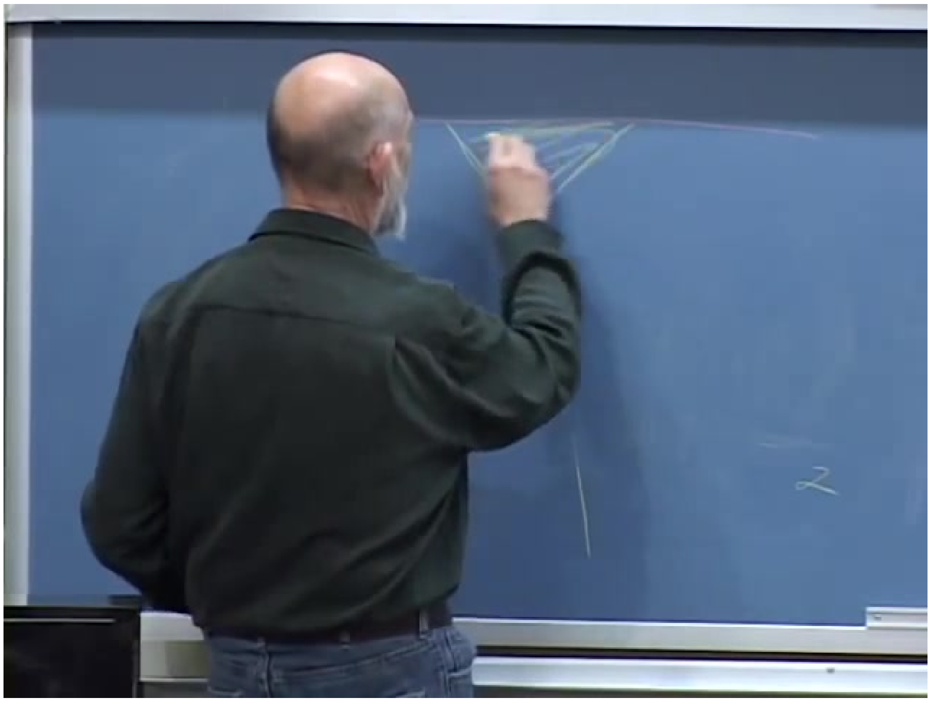

Figure 9.13: A lower-vacuum-energy bubble being drawn inside an inflating de Sitter background.
Lecture frame: 01:29:38.

among all people who ever live. If the population doubles every generation and then ends, half of all people live in the last generation, three quarters in the last two, and seven eighths in the last three:

$$f_n = 1 - 2^{-n}. \tag{9.37}$$

Typicality would then place us uncomfortably near the end, even though we know nothing about the total resource supply. This is not offered as a reliable prediction of human extinction. It is a warning that exponential growth makes counts dominated by the cutoff.

The coin-flipper example states the ordinary typicality assumption more cleanly. If many experimenters each flip a fair coin N times, the overwhelming majority obtain

$$n_H = \frac{N}{2} + O(\sqrt{N}). \tag{9.38}$$

An extreme observer should first suspect a bad model or a bad coin rather than assume without evidence that she occupies a fantastically rare tail. Cosmological probability uses the same typicality instinct, but in an infinite population it needs a well-defined way to count.

9.15 Escher's Boundary and the Measure Problem

Susskind's geometric analogy is M. C. Escher's *Circle Limit IV*, the disk tiled by angels and devils that appear smaller toward the boundary. In the hyperbolic geometry represented by the drawing, the figures are equivalent; their apparent Euclidean size changes only because of the projection.

Imagine that Escher has finite time, ink, and patience. Near the edge he eventually stops, perhaps leaving some figures incomplete. Because the number of figures grows exponentially with hyperbolic

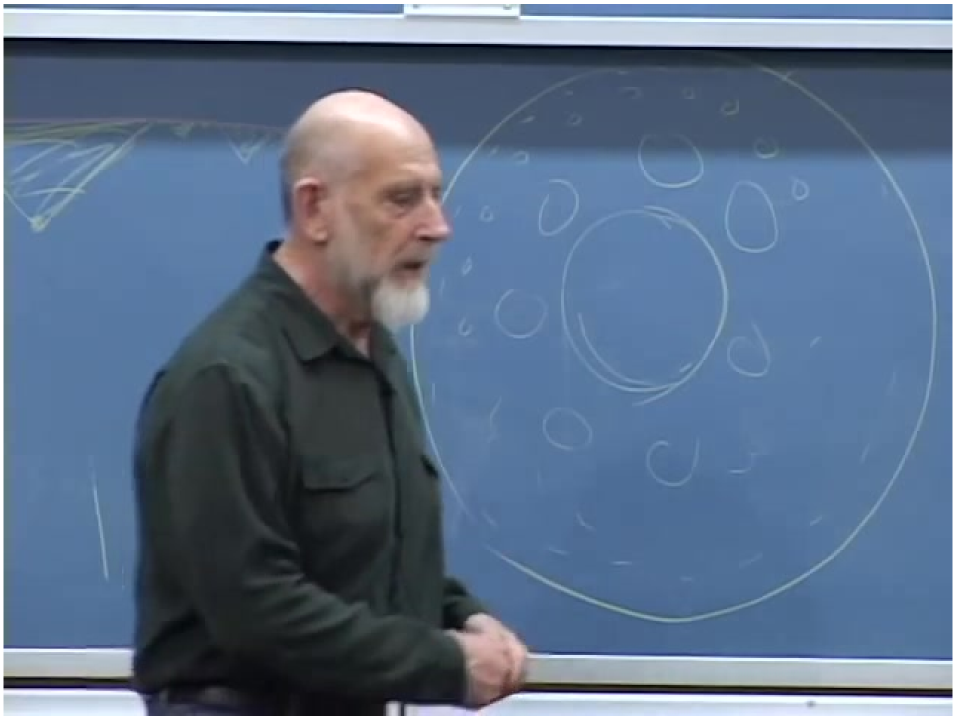

Figure 9.14: The blackboard surrogate for Escher's exponentially proliferating figures near a circular boundary.

Lecture frame: 01:37:54.

radius, a finite outer layer contains a fixed-order fraction of every figure drawn. Roughly half can lie in the final layer, no matter how far outward the cutoff is moved.

Now ask for the ratio of angels to devils. In the ideal infinite pattern symmetry suggests one-to-one equality. In any finite count, however, a slightly different boundary can include many more figures of one type. Since most of the count lies near the boundary, a small deformation of that boundary produces an order-one change in the answer.

The same ambiguity appears in eternal inflation. Count blue and pink bubble universes below some time cutoff, then push the cutoff later. Both counts diverge, and their ratio can depend sensitively on what *time* means and how the cutoff surface is drawn. A probability measure is a prescription that regulates the infinite population and extracts relative probabilities. The failure of different plausible prescriptions to agree is the measure problem.

Question from the audience. Is there a relation between this cutoff problem and renormalization?[a]

Response. There is an analogy. In quantum field theory one introduces a lattice or ultraviolet cutoff, refines it, and seeks quantities insensitive to the regulator's edge details. Cosmologists hope for an equally robust extraction from eternal inflation, but no comparably decisive prescription is yet available in the framework discussed here.

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[a] Lecture timestamp: 01:45:43.

The analogy should not be overstated. Renormalization succeeds when a controlled flow identifies regulator-independent observables. The measure problem is precisely that the analogous invariant questions and limiting procedure remain unclear. Saying that we may be asking the wrong question

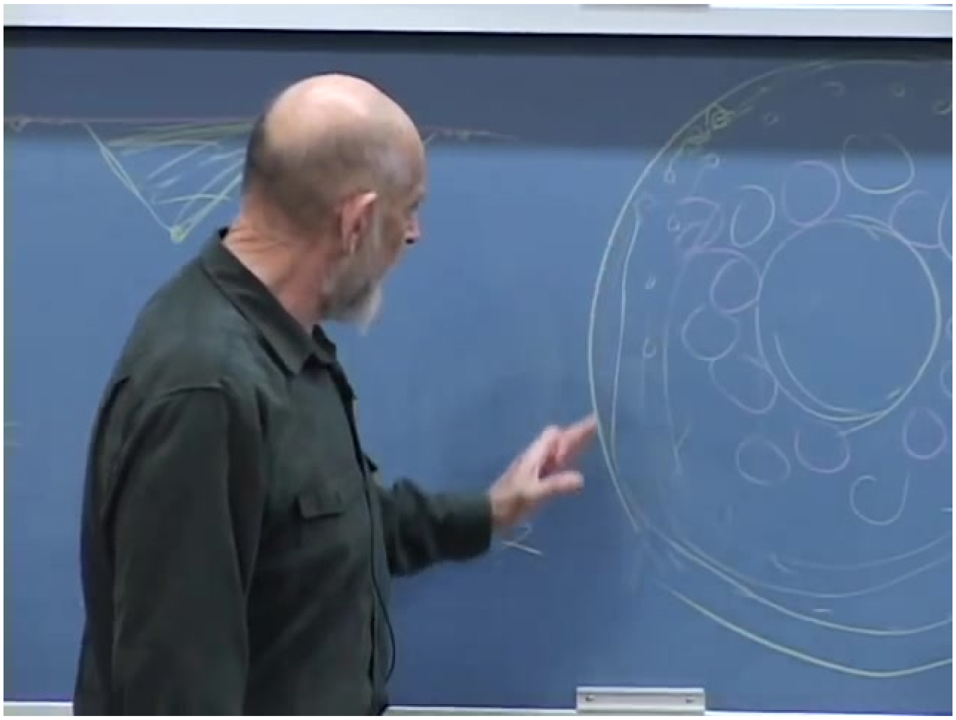

Figure 9.15: The cutoff boundary dominates the count in an exponentially growing pattern.
Lecture frame: 01:43:06.

is not a solution until the right question is supplied.

Question from the audience. Is this bubble picture the same phenomenon as inflation in our early universe?[a]

Response. It is closely connected. In eternal inflation, the parent background continues inflating while local bubble interiors undergo their own cosmological evolution. Our observable universe could lie inside one such region.

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[a]Lecture timestamp: 01:48:19.

9.16 Could Bubble Collisions Be Seen?

Two bubbles do not simply add their vacuum energies when they meet. A domain wall separates regions with different vacuum states, and the geometry of the wall determines what either side can observe. Depending on the collision's location relative to our past light cone, its imprint may be hidden or visible.

Question from the audience. What happens when two vacuum bubbles collide?[a]

Response. The collision creates a domain-wall geometry whose behavior depends on the two vacua and on the wall tension. Under favorable causal circumstances, the disturbance can enter our observable region and leave an anisotropic signature.

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[a]Lecture timestamp: 01:52:07.

The lecture discusses a cold region in the cosmic microwave background as a possible candidate and points to polarization around a circular feature as a discriminant. This is deliberately presented as a possibility, not a detection. A convincing collision signature would be revolutionary because it would supply direct evidence for neighboring bubble regions and make the measure problem experimentally urgent.

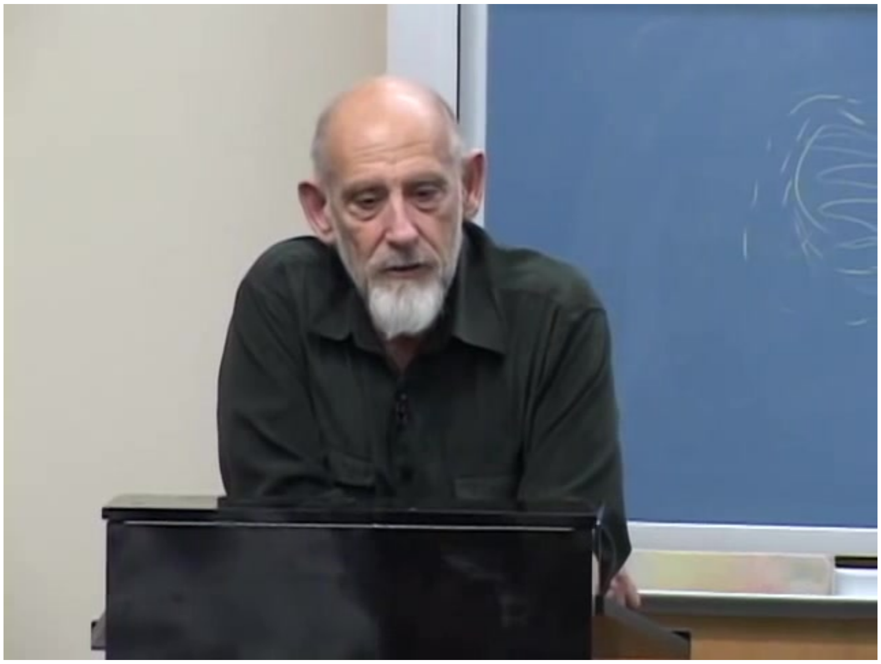

Figure 9.16: A circular sky feature and its possible polarization pattern in the bubble-collision discussion.

Lecture frame: 01:50:22.

[5]

Question from the audience. Are the bubbles four-dimensional, and what spatial curvature do their interiors have?[a]

Response. They are spacetime regions, not three-dimensional soap bubbles evolving in an external time. Their interiors are typically open FRW universes with negative spatial curvature. A collision perturbs that geometry but does not simply turn negative curvature into positive curvature.

[a]Lecture timestamp: 01:53:37.

9.17 What String Theory Does and Does Not Establish

The final discussion begins with an invitation to assign a probability that string theory is correct. Susskind refuses the number because the proposition is too ambiguous. He instead separates two claims.

[5]Editorial clarification: The cold-spot discussion records the observational status and proposals at the time of the lecture. A temperature anomaly by itself is not evidence of a bubble collision; a detailed, correlated signature would be required.

First, there is a highly constrained mathematical structure combining quantum mechanics and gravity. It has generated precise results and deep mathematical identities, and its internal consistency has survived extensive scrutiny. In the lecture's informal vocabulary this is *String Theory* with a capital S. The versions understood most sharply rely heavily on supersymmetry.

Second, the observed world is not supersymmetric at accessible energies. Connecting the exact supersymmetric constructions to our nonsupersymmetric universe requires supersymmetry breaking and a larger framework that is not controlled with the same precision. In the lecture's vocabulary, that hoped-for extension is *string theory* with a small s.

Question from the audience. Can one assign a probability that string theory is the correct theory of nature?[a]

Response. Not responsibly without specifying which claim is meant. The exact mathematical structures exist and include quantum gravity, but the rigorously controlled supersymmetric models are not literally our observed world. Whether a larger extension contains our world is a matter on which physicists can have informed prejudices, not a well-defined numerical probability.

[a] Lecture timestamp: 01:54:18.

The analogy is with mechanics before the full mathematical machinery was available. One might know $F = ma$ and the inverse-square force yet be able to solve only highly symmetric circular orbits. The solvable sector would be genuine but not the whole theory. Likewise, exact supersymmetric string constructions may be a special, unusually tractable region of a larger structure.

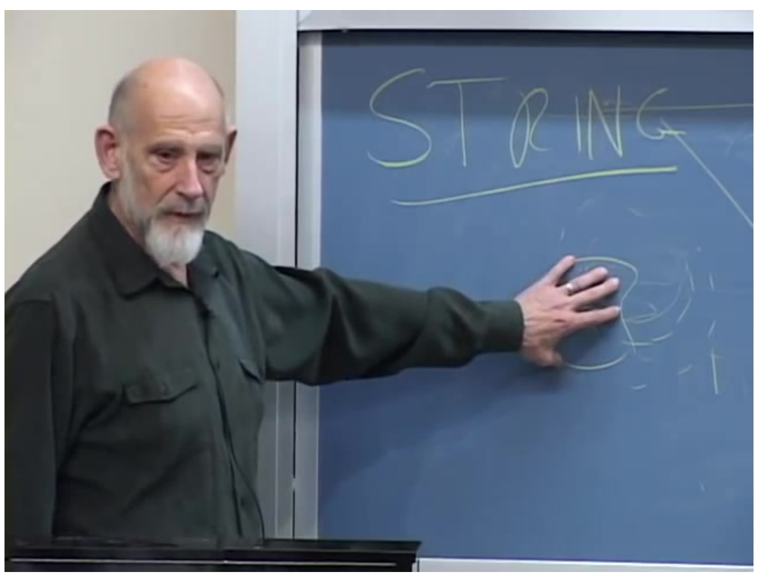

Figure 9.17: The lecture's nested picture: rigorously controlled supersymmetric string constructions inside a conjecturally larger framework.

Lecture frame: 01:57:34.

Question from the audience. Why not simply break supersymmetry in the exact theory?[a]

Response. Many approximate constructions study supersymmetry breaking, but the fully controlled non-supersymmetric backbone is precisely what is missing. One can calculate after making standard approximations without always knowing the exact object those approximations approach.

[a]Lecture timestamp: 01:57:02.

Question from the audience. Does the exact supersymmetric theory rule out a larger, less symmetric extension that contains our world?[a]

Response. No. It establishes that the observed world is not described by the exact supersymmetric sector alone. It neither proves nor disproves that this sector is connected to a larger theory containing realistic physics.

[a]Lecture timestamp: 02:00:33.

Question from the audience. Do string constructions include symmetry breaking analogous to particle-physics symmetry breaking?[a]

Response. There is extensive work on supersymmetry breaking, and much is known about supersymmetric mathematics. The unresolved point is a precise, nonperturbative formulation that combines realistic breaking with the full theory rather than only with controlled approximations.

[a]Lecture timestamp: 02:01:21.

Question from the audience. Is special relativity postulated in string theory, or does it emerge as a prediction?[a]

Response. The underlying rules are Lorentz invariant, even though a particular background full of matter can select a preferred rest frame. Gravity is still more unavoidable: the string spectrum contains a massless spin-two excitation with the interactions of a graviton.

[a]Lecture timestamp: 02:03:02.

Question from the audience. Is string theory consistent with general relativity and background independence?[a]

Response. Its low-energy gravitational dynamics are consistent with general relativity, and local backgrounds can be changed dynamically. The subtlety lies in asymptotic boundary conditions: they label the problem itself and cannot generally be changed by a local operation within that spacetime.

[a]Lecture timestamp: 02:04:11.

The lecture closes on that distinction between a robust mathematical core and an unfinished physical extrapolation. It is not a verdict that the core is useless, nor a claim that the extrapolation is established. It is a statement of where exact knowledge ends.